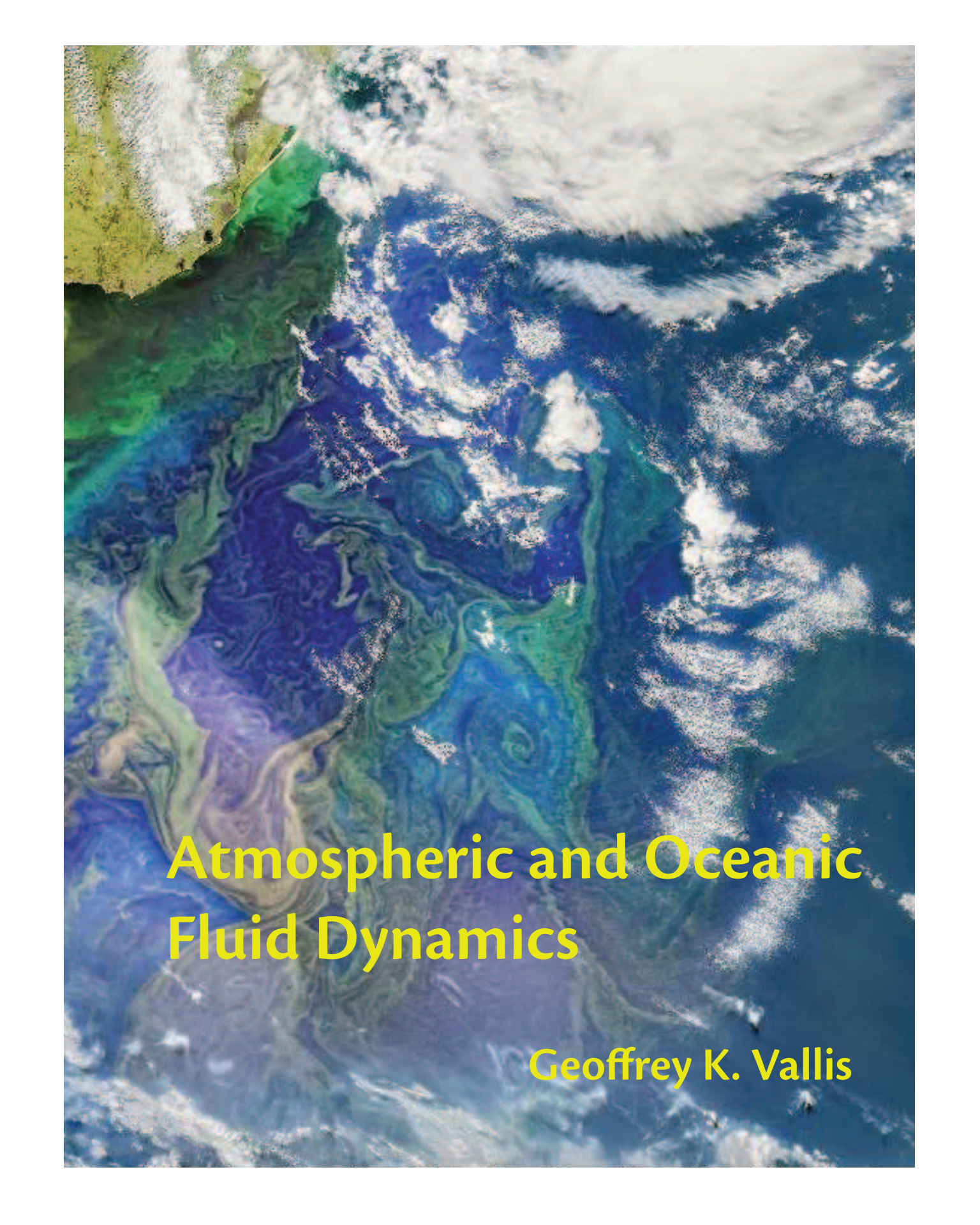


A satellite image of Earth showing swirling ocean currents and clouds. The image is a composite of various colors, including green, blue, and purple, representing different oceanic features. The text "Atmospheric and Oceanic Fluid Dynamics" is overlaid in yellow. The author's name "Geoffrey K. Vallis" is also overlaid in yellow at the bottom right.

Atmospheric and Oceanic Fluid Dynamics

Geoffrey K. Vallis

Placeholder

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In the main text, sections that are more advanced or that contain material that is peripheral to the main narrative are marked, like difficult ski slopes, with a black diamond, **◆**. Sections that contain material that is still not settled or are at the edge of knowledge are marked with a dagger, **†**.

Look again at that dot. That's here. That's home. That's us... Our planet is a lonely speck in the great enveloping cosmic dark. In our obscurity, in all this vastness, there is no hint that help will come from elsewhere to save us from ourselves... Like it or not, for the moment the Earth is where we make our stand.
Carl Sagan, *Pale Blue Dot*, 1994.

Preface

November 11, 2016.

The preface is yet to be written,

Placeholder

Part I

FUNDAMENTALS OF GEOPHYSICAL FLUID DYNAMICS

Part II

WAVES, INSTABILITIES AND TURBULENCE

Part III

LARGE-SCALE ATMOSPHERIC CIRCULATION

I think the causes of the general trade-winds have not been fully explained by any of those who have wrote on that subject... That the action of the Sun is the original cause of these Winds, I think all are agreed.

George Hadley, *Concerning the Cause of the General Trade Winds*, 1735.

CHAPTER 14

The Overturning Circulation: Hadley and Ferrel Cells

WE NOW BEGIN OUR DISCUSSION of the large-scale circulation of the atmosphere, meaning the flow on scales from the synoptic scale — about 1000 km — to the global scale, concentrating on the troposphere. The *general circulation* is virtually synonymous with the large-scale circulation, although the former is sometimes restricted to the time- or ensemble-averaged flow. In this chapter we focus on the dynamics of the Hadley Cell and then, rather descriptively, on the mid-latitude overturning cell or the Ferrel Cell, moving on to the dynamics of the extratropical zonally averaged circulation in chapter 15.

The atmosphere is a terribly complex system, and we cannot hope to fully explain its motion as the analytic solution to a small set of equations. Rather, a full understanding of the atmosphere requires describing it in a consistent way on many levels simultaneously. One of these levels involves simulating the flow by numerically solving the governing equations of motion as completely as possible, for example by using a comprehensive General Circulation Model (GCM). However, such a simulation brings problems of its own, including the problem of understanding the simulation itself and discerning whether it is a good representation of reality, and in this chapter and the ones following we concentrate on simpler, more conceptual models. We begin this chapter with a brief observational overview of some of the pre-eminent large-scale features of the atmosphere, concentrating on the zonally averaged fields.¹

14.1 BASIC FEATURES OF THE ATMOSPHERE

14.1.1 The radiative equilibrium distribution

A gross but informative measure characterizing the atmosphere, and the effects that dynamics have on it, is the pole-to-equator temperature distribution. The *radiative equilibrium* temperature is the hypothetical, three-dimensional, temperature field that would obtain if there were no atmospheric or oceanic motion, given the composition and radiative properties of the atmosphere and surface. The field is a function of the incoming solar radiation at the top of the atmosphere and the atmospheric composition and entails a complicated calculation, especially as the radiative properties of the atmosphere depend heavily on the amount of water vapour and cloudiness in the atmosphere.

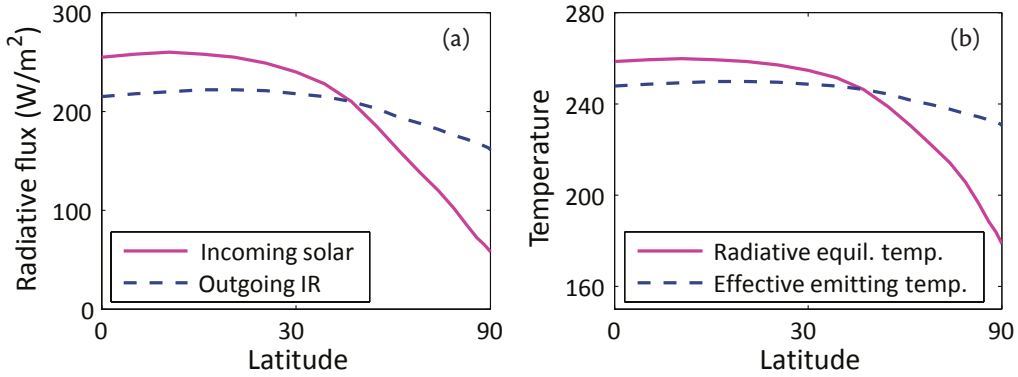


Fig. 14.1 (a) The (approximate) observed net average incoming solar radiation and outgoing infrared radiation at the top of the atmosphere, as a function of latitude (plotted on a sine scale). (b) The temperatures associated with these fluxes, calculated using $T = (R/\sigma)^{1/4}$, where R is the solar flux for the radiative equilibrium temperature and R is the infrared flux for the effective emitting temperature. Thus, the solid line is an approximate radiative equilibrium temperature

(The distribution of absorbers is usually taken to be that which obtains in the observed, moving, atmosphere, in order that the differences between the calculated radiative equilibrium temperature and the observed temperature are due to fluid motion.)

A much simpler calculation that illustrates the essence of the situation is to first note that at the top of the atmosphere the globally averaged incoming solar radiation is balanced by the outgoing infrared radiation. If there is no lateral transport of energy in the atmosphere or ocean then *at each latitude* the incoming solar radiation will be balanced by the outgoing infrared radiation, and if we parameterize the latter using a single latitudinally-dependent temperature we will obtain a crude radiative-equilibrium temperature (the ‘radiative emitting temperature’) for the atmospheric column at each latitude. Specifically, a black body subject to a net incoming radiation of S (watts per square metre) has a radiative-equilibrium temperature T_{rad} given by $\sigma T_{\text{rad}}^4 = S$, this being Stefan’s law with Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$. Thus, for the Earth, we have, at each latitude,

$$\sigma T_{\text{rad}}^4 = S(\theta)(1 - \alpha), \quad (14.1)$$

where α is the albedo of the Earth and $S(\theta)$ is the incoming solar radiation at the top of the atmosphere, and its solution is shown in Fig. 14.1. The solid lines in the two panels show the net solar radiation and the solution to (14.1), T_{rad} ; the dashed lines show the observed outgoing infrared radiative flux, I , and the effective emitting temperature associated with it, $(I/\sigma)^{1/4}$. The emitting temperature does not quantitatively characterize that temperature at the Earth’s surface, nor at any single level in the atmosphere, because the atmosphere is not a black body and the outgoing radiation originates from multiple levels. Nevertheless, the qualitative point is evident: the radiative equilibrium temperature has a much stronger pole-to-equator gradient than does the effective emitting temperature, indicating that there is a poleward transport of heat in the atmosphere–ocean system. More detailed calculations indicate that the atmosphere is further from its radiative equilibrium in winter than summer, indicating a larger heat transport. The transport occurs because polewards moving air tends to have a higher static energy ($c_p T + gz$ for dry air; in addition there is some energy transport associated with water vapour evaporation and condensation) than the equatorwards moving air, most of this movement being associated with the large-scale circulation. The radiative forcing thus seeks to maintain a pole-to-equator temperature gradient, and the ensuing circulation seeks to reduce this gradient.

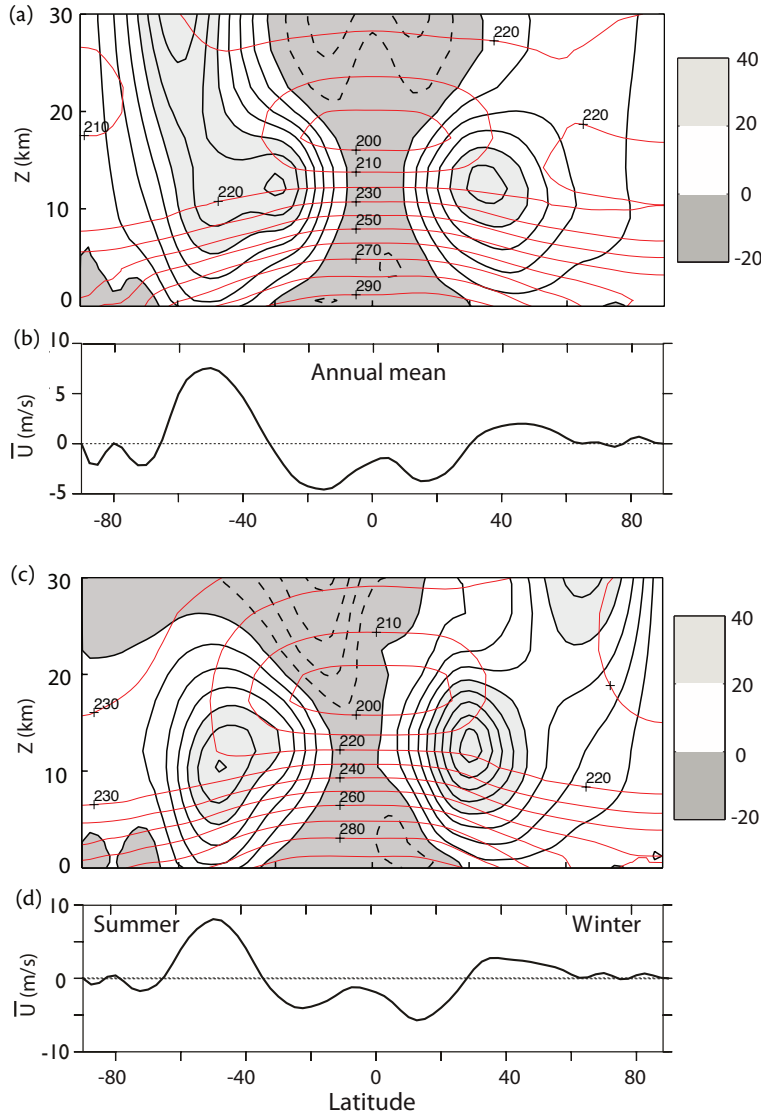


Fig. 14.2 (a) Annual mean, zonally averaged zonal wind (heavy contours and shading) and the zonally averaged temperature (red, thinner contours).

(b) Annual mean, zonally averaged zonal winds at the surface.

(c) and (d) Same as (a) and (b), except for northern hemisphere winter (DJF).

The wind contours are at intervals of 5 m s^{-1} with shading for eastward winds above 20 m s^{-1} and for all westward winds, and the temperature contours are labelled. The ordinate of (a) and (c) is $Z = -H \log(p/p_R)$, where $H = 7.5 \text{ km}$.

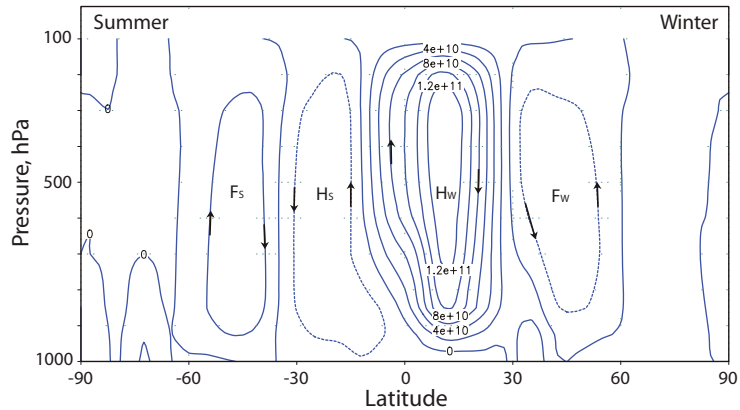
14.1.2 Observed wind and temperature fields

The observed zonally averaged temperature and zonal wind fields are illustrated in Fig. 14.2. The vertical coordinate is log pressure, multiplied by a constant factor $H = RT_0/g = 7.5 \text{ km}$, so that the ordinate is similar to height in kilometres. [In an isothermal hydrostatic atmosphere $(RT_0/g)d \ln p = -dz$, and the value of H chosen corresponds to $T_0 = 256 \text{ K}$.] To a good approximation temperature and zonal wind are related by thermal wind balance, which in pressure coordinates is

$$f \frac{\partial u}{\partial p} = \frac{R}{p} \frac{\partial T}{\partial y}. \quad (14.2)$$

In the lowest several kilometres of the atmosphere temperature falls almost monotonically with latitude and height, and this region is called the *troposphere*. The temperature in the lower troposphere in fact varies more rapidly with latitude than does the effective emitting temperature, T_E , the latter being more characteristic of the temperature in the mid-to-upper troposphere. The merid-

Fig. 14.3 The observed meridional overturning circulation of the atmosphere (kg s^{-1}) averaged over December–January–February. Note the direct *Hadley Cells*, particularly strong in winter (H_W and H_S , in winter and summer) with rising motion near the equator, descending motion in the subtropics, and the weaker, indirect, *Ferrel Cells* (F_W and F_S) at mid-latitudes.



ional temperature gradient is much larger in winter than summer, because in winter high latitudes receive virtually no direct heating from the Sun. The gradient is also strongest at the edge of the subtropics, and here it is associated with a zonal jet, particularly strong in winter. There is no need to ‘drive’ this wind with any kind of convergent momentum fluxes: given the temperature, the flow is a consequence of thermal wind balance, and to the extent that the upper troposphere is relatively frictionless there is no need to maintain it against dissipation. Of course just as the radiative-equilibrium temperature gradient is much larger than that observed, so the zonal wind shear associated with it is much larger than that observed. Thus, the overall effect of the atmospheric and oceanic circulation, and in particular of the turbulent circulation of the mid-latitude atmosphere, is to *reduce* the amplitude of the vertical shear of the eastward flow by way of a poleward heat transport. Observations indicate that about two-thirds of this transport is effected by the atmosphere, and about a third by the ocean, more in low latitudes.²

Above the troposphere is the *stratosphere*, and here temperature typically increases with height. The boundary between the two regions is called the *tropopause*, and this varies in height from about 16 km in the tropics to about 8 km in polar regions. We consider the maintenance of this stratification in section 15.5.

The surface winds typically have, going from the equator to the pole, an E–W–E (easterly–westerly–easterly) pattern, although the polar easterlies are weak and barely present in the Northern Hemisphere. (Meteorologists use ‘westerly’ to denote winds from the west, that is eastward winds; similarly ‘easterlies’ are westward winds.) In a given hemisphere, the surface winds are stronger in winter than summer, and they are also consistently stronger in the Southern Hemisphere than in the Northern Hemisphere, because in the former the surface drag is weaker because of the relative lack of continental land masses and topography. The surface winds are *not* explained by thermal wind balance. Indeed, unlike the upper level winds, they must be maintained against the dissipating effects of friction, and this implies a momentum convergence into regions of surface westerlies and a divergence into regions of surface easterlies. Typically, the maxima in the eastward surface winds are in mid-latitudes and somewhat polewards of the subtropical maxima in the upper-level westerlies and at latitudes where the zonal flow is a little more constant with height. The mechanisms of the momentum transport in the mid-latitudes and the maintenance of the surface westerly winds are the topics of section 15.1.

14.1.3 Meridional overturning circulation

The observed (Eulerian) zonally averaged meridional overturning circulation is shown in Fig. 14.3. The figure shows a streamfunction, Ψ for the vertical and meridional velocities such that, in the

Some Features of the Large-scale Atmospheric Circulation

From Figures 14.1–14.3 we see or infer the following.

1. A pole–equator temperature gradient that is much smaller than the radiative equilibrium gradient.
2. A troposphere, in which temperature generally falls with height, above which lies the stratosphere, in which temperature increases with height. The two regions are separated by a tropopause, which varies in height from about 16 km at the equator to about 6 km at the pole.
3. A monotonically decreasing temperature from equator to pole in the troposphere, but a weakening and sometimes reversal of this above the tropopause.
4. A westerly (i.e., eastward) tropospheric jet. The time and zonally averaged jet is a maximum at the edge or just polewards of the subtropics, where it is associated with a strong meridional temperature gradient. In mid-latitudes the jet has a stronger barotropic component.
5. An E–W–E (easterlies–westerlies–easterlies) surface wind distribution. The latitude of the maximum in the surface westerlies is in mid-latitudes, where the zonally averaged flow is more barotropic.

pressure coordinates used in the figure,

$$\frac{\partial \Psi}{\partial y} = -\bar{\omega}, \quad \frac{\partial \Psi}{\partial p} = \bar{v}. \quad (14.3)$$

where the overbar indicates a zonal average. In each hemisphere there is rising motion near the equator and sinking in the subtropics, and this circulation is known as the *Hadley Cell*.³ The Hadley Cell is a thermally direct cell (i.e., the warmer fluid rises, the colder fluid sinks), much stronger in the winter hemisphere, and extends to about 25–30°. In mid-latitudes the sense of the overturning circulation is apparently reversed, with rising motion in the high-mid-latitudes, at around 60° and sinking in the subtropics, and this is known as the *Ferrel Cell*. However, as with most pictures of averaged streamlines in unsteady flow, this gives a misleading impression as to the actual material flow of parcels of air because of the presence of eddying motion, and we discuss this in the next chapter. At low latitudes the circulation is more nearly zonally symmetric and the picture does give a qualitatively correct representation of the actual flow. At high latitudes there is again a thermally direct cell (although it is weak and not always present), and thus the atmosphere is often referred to as having a three-celled structure.

14.1.4 Summary

Some of the main features of the zonally averaged circulation are summarized in the shaded box above. We emphasize that the zonally averaged circulation is not synonymous with a zonally symmetric circulation, and the mid-latitude circulation is highly asymmetric. Any model of the mid-latitudes that did not take into account the zonal asymmetries in the circulation — of which the weather is the main manifestation — would be seriously in error. This was first explicitly realized in the 1920s, and taking into account such asymmetries is the main task of the dynamical meteorology of the mid-latitudes, and is the subject of the next chapter. On the other hand, the large-scale

tropical circulation of the atmosphere is to a large degree zonally symmetric or nearly so, and although monsoonal circulations and the Walker circulation (a cell with rising air in the Eastern Pacific and descending motion in the Western Pacific) are zonally asymmetric, they are also relatively weaker than typical mid-latitude weather systems. Indeed the boundary between the tropics and mid-latitude may be usefully defined by the latitude at which such zonal asymmetries become dynamically important on the large scale and this boundary, at about 30° on average, roughly coincides with the latitude at which the mean meridional overturning circulation vanishes. We begin our dynamical description with a study of the low-latitude zonally symmetric atmospheric circulation.

14.2 A STEADY MODEL OF THE HADLEY CELL

Ceci n'est pas une pipe.

René Magritte. Title of painting, 1929.

14.2.1 Assumptions

Let us try to construct a zonally symmetric model of the Hadley Cell,⁴ recognizing that such a model is likely applicable mainly to the tropical atmosphere, this being more zonally symmetric than the mid-latitudes. We will suppose that heating is maximum at the equator, and our intuitive picture, drawing on the observed flow of Fig. 14.3, is of air rising at the equator and moving polewards at some height H , descending at some latitude ϑ_H , and returning equatorwards near the surface. We will make three major assumptions:

- (i) that the circulation is steady;
- (ii) that the polewards moving air conserves its axial angular momentum, whereas the zonal flow associated with the near-surface, equatorwards moving flow is frictionally retarded and weak;
- (iii) that the circulation is in thermal wind balance.

We also assume the model is symmetric about the equator (an assumption we relax in section 14.4). These are all reasonable assumptions, but they cannot be rigorously justified; in other words, we are constructing a *model* of the Hadley Cell, schematically illustrated in Fig. 14.4. The model defines a limiting case — steady, inviscid, zonally-symmetric flow — that cannot be expected to describe the atmosphere quantitatively, but that can be analysed fairly completely. Another limiting case, in which eddies play a significant role, is described in section 14.5. The real atmosphere may defy such simple characterizations, but the two limits provide invaluable benchmarks of understanding.

14.2.2 Dynamics

We now try to determine the strength and poleward extent of the Hadley circulation in our steady model. For simplicity we will work with a Boussinesq atmosphere, but this is not an essential aspect. We will first derive the conditions under which conservation of angular momentum will hold, and then determine the consequences of that.

The zonally averaged zonal momentum equation may be easily derived from (2.50a) and/or (2.62) and in the absence of friction it is

$$\frac{\partial \bar{u}}{\partial t} - (f + \bar{\zeta})\bar{v} + \bar{w}\frac{\partial \bar{u}}{\partial z} = -\frac{1}{a \cos^2 \vartheta} \frac{\partial}{\partial \vartheta} (\cos^2 \vartheta \overline{u'v'}) - \frac{\partial \overline{u'w'}}{\partial z}, \quad (14.4)$$

where $\bar{\zeta} = -(a \cos \vartheta)^{-1} \partial_{\vartheta}(\bar{u} \cos \vartheta)$ and the overbars represent zonal averages. If we neglect the vertical advection and the eddy terms on the right-hand side, then a steady solution, if it exists,

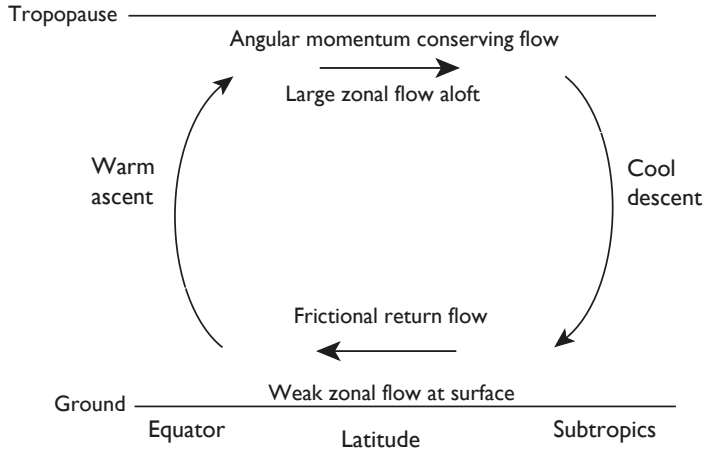


Fig. 14.4 A simple model of the Hadley Cell. Rising air near the equator moves polewards near the tropopause, descending in the subtropics and returning.

The poleward moving air conserves its angular momentum, leading to a shear of the zonal wind that increases away from the equator. By thermal wind the temperature of the air falls as it moves poleward, and to satisfy the thermodynamic budget it sinks in the subtropics.

obeys

$$(f + \bar{\zeta})\bar{v} = 0. \quad (14.5)$$

Presuming that the meridional flow $\bar{v}$ is non-zero (an issue we address in section 14.2.8) then $f + \bar{\zeta} = 0$, or equivalently

$$2\Omega \sin \vartheta = \frac{1}{a} \frac{\partial \bar{u}}{\partial \vartheta} - \frac{\bar{u} \tan \vartheta}{a}. \quad (14.6)$$

At the equator we shall assume that $\bar{u} = 0$, because here parcels have risen from the surface where, by assumption, the flow is weak. Equation (14.6) then has a solution of

$$\bar{u} = \Omega a \frac{\sin^2 \vartheta}{\cos \vartheta} \equiv U_M. \quad (14.7)$$

This gives the zonal velocity of the polewards moving air in the upper branch of the (model) Hadley Cell, above the frictional boundary layer. We can derive (14.7) directly from the conservation of axial angular momentum, m , of a parcel of air at a latitude ϑ . In the shallow atmosphere approximation we have [cf. (2.64) and equations following]

$$\bar{m} = (\bar{u} + \Omega a \cos \vartheta)a \cos \vartheta, \quad (14.8)$$

and if $\bar{u} = 0$ at $\vartheta = 0$ and if $\bar{m}$ is conserved on a polewards moving parcel, then (14.8) leads to (14.7). It also may be directly checked that

$$f + \bar{\zeta} = -\frac{1}{a^2 \cos \vartheta} \frac{\partial \bar{m}}{\partial \vartheta}. \quad (14.9)$$

We have thus shown that, if eddy fluxes and frictional effects are negligible, the poleward flow will conserve its angular momentum, the result of which, by (14.7), is that magnitude of the zonal flow in the Earth's rotating frame will increase with latitude (see Fig. 14.5). (Also, given the absence of eddies our model is zonally symmetric and we shall drop the overbars over the variables.)

If (14.7) gives the zonal velocity in the upper branch of the Hadley Cell, and that in the lower branch is close to zero, then the thermal wind equation can be used to infer the vertically averaged temperature. Although the geostrophic wind relation is not valid at the equator (a more accurate balance is the gradient wind balance, $f\bar{u} + \bar{u}^2 \tan \vartheta/a = -a^{-1} \partial \phi / \partial \vartheta$) the zonal wind is in fact geostrophically balanced until very close to the equator, and at the equator itself the horizontal

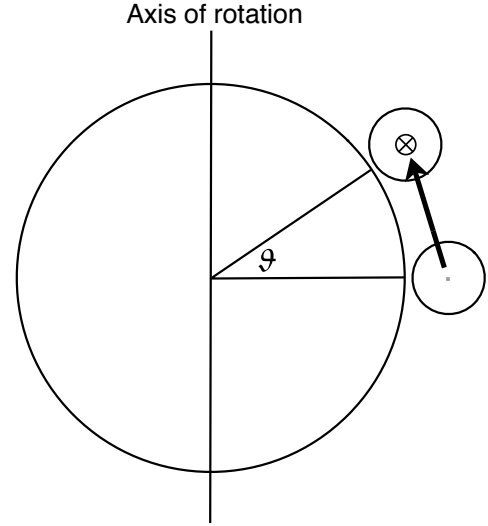


Fig. 14.5 If a ring of air at the equator moves polewards it moves closer to the axis of rotation. If the parcels in the ring conserve their angular momentum their zonal velocity must increase; thus, if $m = (\bar{u} + \Omega a \cos \vartheta)a \cos \vartheta$ is preserved and $\bar{u} = 0$ at $\vartheta = 0$ we recover (14.7).

temperature gradient in our model vanishes, because of the assumed interhemispheric symmetry. Thus, conventional thermal wind balance suffices for our purposes, and this is

$$2\Omega \sin \vartheta \frac{\partial u}{\partial z} = -\frac{1}{a} \frac{\partial b}{\partial \vartheta}, \quad (14.10)$$

where $b = g \delta\theta/\theta_0$ is the buoyancy and $\delta\theta$ is the deviation of potential temperature from a constant reference value θ_0 . (Be reminded that θ is potential temperature, whereas ϑ is latitude.) Vertically integrating from the ground to the height H where the outflow occurs and substituting (14.7) for u yields

$$\frac{1}{a\theta_0} \frac{\partial \theta}{\partial \vartheta} = -\frac{2\Omega^2 a \sin^3 \vartheta}{gH \cos \vartheta}, \quad (14.11)$$

where $\theta = H^{-1} \int_0^H \delta\theta dz$ is the vertically averaged potential temperature. If the latitudinal extent of the Hadley Cell is not too great we can make the small-angle approximation, and replace $\sin \vartheta$ by ϑ and $\cos \vartheta$ by one, then integrating (14.11) gives

$$\theta = \theta(0) - \frac{\theta_0 \Omega^2 y^4}{2gHa^2}, \quad (14.12)$$

where $y = a\vartheta$ and $\theta(0)$ is the potential temperature at the equator, as yet unknown. Away from the equator, the zonal velocity given by (14.7) increases rapidly polewards and the temperature correspondingly drops. How far polewards is this solution valid? And what determines the value of the integration constant $\theta(0)$? To answer these questions we turn to thermodynamics.

14.2.3 Thermodynamics

In the above discussion, the temperature field is slaved to the momentum field in that it seems to follow passively from the dynamics of the momentum equation. Nevertheless, the thermodynamic equation must still be satisfied. Let us assume that the thermodynamic forcing can be represented by a Newtonian cooling to some specified radiative equilibrium temperature, θ_E ; this is a severe simplification, especially in equatorial regions where the release of heat by condensation is important. The thermodynamic equation is then

$$\frac{D\theta}{Dt} = \frac{\theta_E - \theta}{\tau}, \quad (14.13)$$

where τ is a relaxation time scale, perhaps a few weeks. Let us suppose that θ_E falls monotonically from the equator to the pole, and that it increases linearly with height, and a simple representation of this is

$$\frac{\theta_E(\vartheta, z)}{\theta_0} = 1 - \frac{2}{3}\Delta_H P_2(\sin \vartheta) + \Delta_V \left(\frac{z}{H} - \frac{1}{2} \right), \quad (14.14)$$

where Δ_H and Δ_V are non-dimensional constants that determine the fractional temperature difference between the equator and the pole, and the ground and the top of the fluid, respectively. P_2 is the second Legendre polynomial, and it is usually the leading term in the Taylor expansion of symmetric functions (symmetric around the equator) that decrease from pole to equator; it also integrates to zero over the sphere. $P_2(y) = (3y^2 - 1)/2$, so that in the small-angle approximation and at $z = H/2$, or for the vertically averaged field, we have

$$\frac{\theta_E}{\theta_0} = 1 + \frac{1}{3}\Delta_H - \Delta_H \left(\frac{y}{a} \right)^2 \quad \text{or} \quad \theta_E = \theta_{E0} - \Delta\theta \left(\frac{y}{a} \right)^2, \quad (14.15a,b)$$

where θ_{E0} is the equilibrium temperature at the equator, $\Delta\theta$ determines the equator–pole radiative-equilibrium temperature difference, and

$$\theta_{E0} = \theta_0(1 + \Delta_H/3), \quad \Delta\theta = \theta_0\Delta_H. \quad (14.16)$$

Now, let us suppose that the solution (14.12) is valid between the equator and a latitude ϑ_H where $v = 0$, so that within this region the system is essentially closed. Conservation of potential temperature then requires that the solution (14.12) must satisfy

$$\int_0^{Y_H} \theta dy = \int_0^{Y_H} \theta_E dy, \quad (14.17)$$

where $Y_H = a\vartheta_H$ is as yet undetermined. Polewards of this, the solution is just $\theta = \theta_E$. Now, we may demand that the solution be continuous at $y = Y_H$ (without temperature continuity the thermal wind would be infinite) and so

$$\theta(Y_H) = \theta_E(Y_H). \quad (14.18)$$

The constraints (14.17) and (14.18) determine the values of the unknowns $\theta(0)$ and Y_H . A little algebra gives

$$Y_H = \left(\frac{5\Delta\theta gH}{3\Omega^2\theta_0} \right)^{1/2}, \quad (14.19)$$

and

$$\theta(0) = \theta_{E0} - \left(\frac{5\Delta\theta^2 gH}{18a^2\Omega^2\theta_0} \right). \quad (14.20)$$

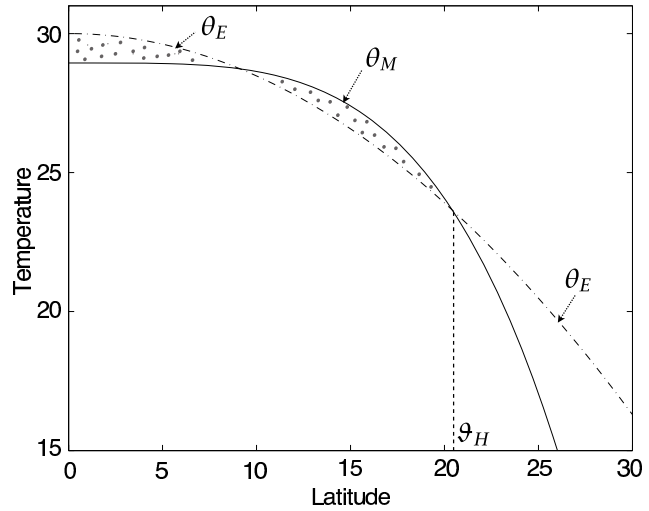
A useful non-dimensional number that parameterizes these solutions is

$$R \equiv \frac{gH\Delta\theta}{\theta_0\Omega^2 a^2} = \frac{gH\Delta_H}{\Omega^2 a^2}, \quad (14.21)$$

which is the square of the ratio of the speed of shallow water waves to the rotational velocity of the Earth, multiplied by the fractional temperature difference from equator to pole. Typical values for the Earth's atmosphere are a little less than 0.1. In terms of R we have

$$Y_H = a \left(\frac{5}{3}R \right)^{1/2}, \quad \theta(0) = \theta_{E0} - \left(\frac{5}{18}R \right) \Delta\theta. \quad (14.22a,b)$$

Fig. 14.6 The radiative equilibrium temperature (θ_E , dashed line) and the angular-momentum-conserving solution (θ_M , solid line) as a function of latitude. The two dotted regions have equal areas. The parameters are: $\theta_{E0} = 303 \text{ K}$, $\Delta\theta = 50 \text{ K}$, $\theta_0 = 300 \text{ K}$, $\Omega = 7.272 \times 10^{-5} \text{ s}^{-1}$, $g = 9.81 \text{ m s}^{-2}$, $H = 10 \text{ km}$. These give $R = 0.076$ and $Y_H/a = 0.356$, corresponding to $\vartheta_H = 20.4^\circ$.



The solution, (14.12) with $\theta(0)$ given by (14.22b) is plotted in Fig. 14.6. Perhaps the single most important aspect of the model is that it predicts that the Hadley Cell has a *finite* meridional extent, *even for an atmosphere that is completely zonally symmetric*. The baroclinic instability that does occur in mid-latitudes is not necessary for the Hadley Cell to terminate in the subtropics, although it may be an important factor, or even the determining factor, in the real world.

14.2.4 Zonal wind

The angular-momentum-conserving zonal wind is given by (14.7), which in the small-angle approximation becomes

$$U_M = \Omega \frac{y^2}{a}. \quad (14.23)$$

This relation holds for $y < Y_H$. The zonal wind corresponding to the radiative-equilibrium solution is given using thermal wind balance and (14.15b), which leads to

$$U_E = \Omega a R. \quad (14.24)$$

That the radiative-equilibrium zonal wind is a constant follows from our choice of the second Legendre function for the radiative equilibrium temperature and is not a fundamental result; nonetheless, for most reasonable choices of θ_E the corresponding zonal wind will vary much less than the angular-momentum-conserving wind (14.23). The winds are illustrated in Fig. 14.7. There is a discontinuity in the zonal wind at the edge of the Hadley Cell, and of the meridional temperature gradient, but not of the temperature itself.

14.2.5 Properties of solution

From (14.22) we can see that the model predicts that the latitudinal extent of the Hadley Cell is:

- proportional to the square root of the meridional radiative equilibrium temperature gradient: the stronger the gradient, the farther the circulation must extend to achieve thermodynamic balance via the equal-area construction in Fig. 14.6;
- proportional to the square root of the height of the outward flowing branch: the higher the outward flowing branch, the weaker the ensuing temperature gradient of the solution (via thermal wind balance), and so the further polewards the circulation must go;

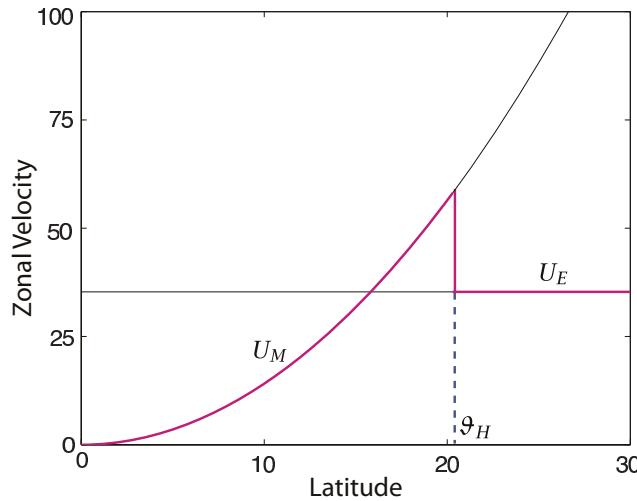


Fig. 14.7 The zonal wind corresponding to the radiative equilibrium temperature (U_E) and the angular-momentum-conserving solution (U_M) as a function of latitude, given (14.23) and (14.24) respectively.

The parameters are the same as those of Fig. 14.6, and the radiative equilibrium wind, U_E is a constant, $\Omega a R$. The actual zonal wind (in the model) follows the thick solid line: $u = U_m$ for $\vartheta < \vartheta_H$ ($y < Y_H$), and $u = U_E$ for $\vartheta > \vartheta_H$ ($y > Y_H$).

- inversely proportional to the rotation rate Ω : the stronger the rotation rate, the stronger the angular-momentum-conserving wind, the stronger the ensuing temperature gradient and so the more compact the circulation.

These precise dependencies on particular powers of parameters are not especially significant in themselves, nor are they robust to changes in parameters. For example, were we to choose a meridional distribution of radiative equilibrium temperature different from (14.14) we might find different exponents in some of the solutions, although we would expect the same qualitative dependencies. However, the dependencies do provide predictions that may be tested with a numerical model. Also, as we have already noted, a key property of the model is that it predicts that the Hadley Cell has a finite meridional extent, even in the absence of mid-latitude baroclinic instability.

Another interesting property of the solutions is a discontinuity in the zonal wind. For tropical latitudes (i.e., $y < Y_H$), then $\bar{u} = U_M$ (the constant angular momentum solution), whereas for $y > Y_H$, $\bar{u} = U_E$ (the thermal wind associated with radiative equilibrium temperature θ_E). There is therefore a discontinuity of $\bar{u}$ at $y = Y_H$, because u is related to the meridional gradient of θ which changes discontinuously, even though θ itself is continuous. No such discontinuity is observed in the real world, although one may observe a baroclinic jet at the edge of the Hadley Cell.

14.2.6 Strength of the circulation

We can make an estimate of the strength of the Hadley Cell by consideration of the thermodynamic equation at the equator, namely

$$w \frac{\partial \theta}{\partial z} \approx \frac{\theta_{E0} - \theta}{\tau}, \quad (14.25)$$

this being a balance between adiabatic cooling and radiative heating. If the static stability is determined largely by the forcing, and not by the meridional circulation itself, then $\theta_0^{-1} \partial \theta / \partial z \approx \Delta_V / H$, and (14.25) gives

$$w \approx \frac{H}{\theta_0 \Delta_V} \frac{\theta_{E0} - \theta}{\tau}. \quad (14.26)$$

Thus, the strength of the circulation is proportional to the distance of the solution from the radiative equilibrium temperature. The right-hand side of (14.25) can be evaluated from the solution itself, and from (14.22b) we have

$$\frac{\theta_{E0} - \theta}{\tau} = \frac{5R\Delta\theta}{18\tau}. \quad (14.27)$$

The vertical velocity is then given by

$$w \approx \frac{5R\Delta\theta H}{18\tau\Delta_V\theta_0} = \frac{5R\Delta_H H}{18\tau\Delta_V}. \quad (14.28)$$

Using mass continuity we can transform this into an estimate for the meridional velocity. Thus, if we let $(v/Y_H) \sim (w/H)$ and use (14.22), we obtain

$$v \sim \frac{R^{3/2}a\Delta_H}{\tau\Delta_V} \propto \frac{\Delta_H^{5/2}}{\Delta_V} \quad \text{and} \quad \Psi \sim vH \sim \frac{R^{3/2}aH\Delta_H}{\tau\Delta_V} \propto (\Delta\theta)^{5/2}. \quad (14.29)$$

where Ψ is the meridional overturning stream function Ψ , which evidently increases fairly rapidly as the gradient of the radiative equilibrium temperature increases. The characteristic overturning time of the circulation, τ_d is then

$$\tau_d = \frac{H}{w} \sim \frac{\tau\Delta_V}{R\Delta_H}. \quad (14.30)$$

We require $\tau_d/\tau \gg 1$ for the effects of the circulation on the static stability to be small and therefore $\Delta_V/(R\Delta_H) \gg 1$, or equivalently, using (14.16),

$$\theta_0\Delta_V \gg R(\theta_{E0} - \theta_0). \quad (14.31)$$

If instead $\tau \gg \tau_d$, then the potential temperature would be nearly conserved as a parcel ascended in the rising branch of the Hadley Cell, and the static stability would be nearly neutral.

14.2.7† Effects of moisture

Suppose now that moisture is present, but that the Hadley Cell remains a self-contained system; that is, it neither imports nor exports moisture. We envision that water vapour joins the circulation by way of evaporation from a saturated surface into the equatorward, lower branch of the Hadley Cell, and that this water vapour then condenses in and near the upward branch of the cell. The latent heat released by condensation is exactly equal to the heat required to evaporate moisture from the surface, and no heat is lost or gained to the system. However, the heating *distribution* is changed from the dry case, becoming a strong function of the solution itself and likely to have a sharp maximum near the equator. Even if we were to try to parameterize the latent heat release by simply choosing a flow dependent radiative equilibrium temperature, the resulting problem would still be quite nonlinear and a general analytic solution seems out of our reach.⁵

Nevertheless, we may see quite easily the qualitative features of moisture, at least within the context of this model. The meridional distribution of temperature is still given by way of thermal wind balance with an angular-momentum-conserving zonal wind, and so is still given by (14.12). We may also assume that the meridional extent of the Hadley Cell is unaltered; that is, a solution exists with circulation confined to $\vartheta < \vartheta_H$ (although it may not be the unique solution). Then, if θ_E^* is the effective radiative equilibrium temperature of the moist solution, we have that $\theta_E^*(Y_H) = \theta_E(Y_H)$ and, in the small-angle approximation,

$$\int_0^{Y_H} \theta dy = \int_0^{Y_H} \theta_E^* dy = \int_0^{Y_H} \theta_E dy, \quad (14.32)$$

where the first equality holds because it defines the solution, and the second equality holds because moisture provides no net energy source. Because condensation will occur mainly in the upward branch of the Hadley Cell, θ_E^* will be peaked near the equator, as schematically sketched in Fig. 14.8. This construction makes it clear that the main difference between the dry and moist solutions is that the latter has a more intense overturning circulation, because, from (14.25), the circulation

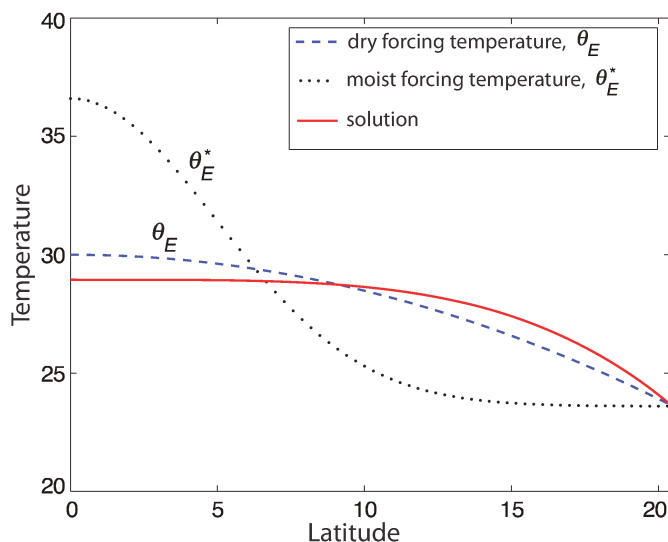


Fig. 14.8 Schematic of the effects of moisture on a model of the Hadley Cell. The temperature of the solution (solid line) is the same as that of a dry model, because this is determined from the angular-momentum-conserving wind. The heating distribution (as parameterized by a forcing temperature) is peaked near the equator in the moist case, leading to a more vigorous overturning circulation.

increases with the temperature difference between the solution and the forcing temperature. Concomitantly, our intuition suggests that the upward branch of the moist Hadley circulation will become much narrower and more intense than the downward branch because of the enhanced efficiency of moist convection, and these expectations are generally confirmed by numerical integrations of the moist equations of motion.

14.2.8 The radiative equilibrium solution

Instead of a solution given by (14.12), could the temperature not simply be in radiative equilibrium everywhere? Such a state would have no meridional overturning circulation and the zonal velocity would be determined by thermal wind balance; that is,

$$v = 0, \quad \theta = \theta_E, \quad f \frac{u}{H} = -g \frac{\partial}{\partial y} \left(\frac{\theta_E}{\theta_0} \right). \quad (14.33)$$

To answer this question we consider the steady zonally symmetric zonal angular momentum equation with viscosity; that is, the zonally averaged, viscous, steady, shallow atmosphere version of (2.68), namely

$$\frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (vm \cos \vartheta) + \frac{\partial(mw)}{\partial z} = \frac{\nu}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} \left(\cos^2 \vartheta \frac{\partial}{\partial \vartheta} \frac{u}{\cos \vartheta} \right) + \nu a \cos \vartheta \frac{\partial^2 u}{\partial z^2}, \quad (14.34)$$

where the variables vary only in the ϑ - z plane. The viscous term on the right-hand side arises from the expansion in spherical coordinates of the Laplacian. Note that it is angular *velocity*, not the angular momentum, that is diffused, because there is no diffusion of the angular momentum due to the Earth's rotation. However, to a very good approximation, the viscous term will be dominated by vertical derivatives and we may then write (14.34) as

$$\nabla_x \cdot (\mathbf{v}m) = \nu \frac{\partial^2 m}{\partial z^2}. \quad (14.35)$$

where $\nabla_x \cdot$ is the divergence in the meridional plane. The right-hand side now has a diffusive form, and in section 13.5.1 we showed that variables obeying equations like this can have no extrema within the fluid. Thus, there can be no maximum or minimum of angular momentum in the

interior of the fluid, a result known as Hide's theorem.⁶ In effect, diffusion always acts to smooth away an isolated extremum, and this cannot be counterbalanced by advection. The result also implies that there can be no interior extrema in a statistically steady state if there is any zonally asymmetric eddy motion that transports angular momentum downgradient.

If the viscosity were so large that the viscous term were dominant in (14.34), and so with the horizontal term now important, then the fluid would evolve toward a state of solid body rotation, this being the fluid state with no internal stresses. In that case, there would be a maximum of angular momentum at the equator — a state of 'super-rotation'. (Related mechanisms have been proposed for the maintenance of super-rotation on Venus.⁷)

Returning now to the question posed at the head of this section, suppose that the radiative equilibrium solution does hold. Then a radiative equilibrium temperature decreasing away from the equator more rapidly than the angular-momentum-conserving solution θ_M implies, using thermal wind balance, a maximum of m at the equator and above the surface, in violation of the no-extremum principle. Of course, we have derived the angular-momentum-conserving solution in the inviscid limit, in which the no-extrema principle does not apply. But any small viscosity will make the radiative equilibrium solution completely invalid, but potentially have only a small effect on the angular-momentum-conserving solution; that is, in the *limit* of small viscosity the angular-momentum-conserving solution can conceivably hold approximately, at least in the absence of boundary layers, whereas the radiative equilibrium solution cannot.

However, if the radiative equilibrium temperature varies more slowly with latitude than the temperature corresponding to the angular momentum conserving solution then a radiative equilibrium solution *can* pertain, without violating Hide's theorem. In particular, this is the case if $\theta_E \propto P_4(\sin \vartheta)$, where P_4 is the fourth Legendre polynomial, and so the possibility exists of two equilibrium solutions for the same forcing; however, P_4 is an unrealistically flat radiative equilibrium temperature for the Earth's atmosphere.

14.3 A SHALLOW WATER MODEL OF THE HADLEY CELL

Although expressed in the notation of the primitive equations, the model described above takes no account of any vertical structure in its stratification and is, *de facto*, a shallow water model. (We discuss how the primitive equations reduce to the shallow water equations in sections 3.4 and 18.7.) Furthermore, the geometric aspects of sphericity play no essential role. Thus, we may transparently express the essence of the model by:

- (i) explicitly using the shallow water equations instead of the stratified equations;
- (ii) using the equatorial β -plane, with $f = f_0 + \beta y$ and $f_0 = 0$.

Let us therefore, if only an exercise, construct a reduced-gravity model with an active upper layer overlying a stationary lower layer.

14.3.1 Momentum balance

The inviscid zonal momentum equation of the upper layer is

$$\frac{Du}{Dt} - \beta yv = 0 \quad (14.36)$$

or

$$\frac{D}{Dt} \left(u - \frac{\beta y^2}{2} \right) = 0, \quad (14.37)$$

which is the β -plane analogue of the conservation of axial angular momentum. (In this section, all variables are zonally averaged, but we omit any notation denoting that.) From (14.37) we obtain

the zonal wind as a function of latitude,

$$u = \frac{1}{2}\beta y^2 + A, \quad (14.38)$$

where A is a constant, which is zero if $u = 0$ at the equator, $y = 0$. The flow given by (14.38) is then analogous to the angular momentum conserving flow in the spherical model, (14.7). Because the lower layer is stationary, the analogue of thermal wind balance in the stratified model is just geostrophic balance, namely

$$fu = -g' \frac{\partial h}{\partial y}, \quad (14.39)$$

where h is the thickness of the active upper layer. Using (14.39) and $f = \beta y$ we obtain

$$g' \frac{\partial h}{\partial y} = -\frac{1}{2}\beta^2 y^3, \quad \text{whence} \quad h = -\frac{1}{8g'}\beta^2 y^4 + h(0), \quad (14.40a,b)$$

where $h(0)$ is the value of h at $y = 0$.

14.3.2 Thermodynamic balance

The thermodynamic equation in the shallow water equations is just the mass conservation equation, which we write as

$$\frac{Dh}{Dt} = -\frac{1}{\tau}(h - h^*), \quad (14.41)$$

where the right-hand side represents heating — h^* is the field to which the height relaxes on a time scale τ . For illustrative purposes we will choose

$$h^* = h_0(1 - \alpha|y|). \quad (14.42)$$

(If we chose the more realistic quadratic dependence on y , the model would be more similar to that of the previous section.) To be in thermodynamic equilibrium we require that the right-hand side integrates to zero over the Hadley Cell; that is

$$\int_0^Y (h - h^*) dy = 0, \quad (14.43)$$

where Y is the latitude of the poleward extent of the Hadley Cell, thus far unknown. Polewards of this, the height field is simply in equilibrium with the forcing — there is no meridional motion and $h = h^*$. Since the height field must be continuous, we require that

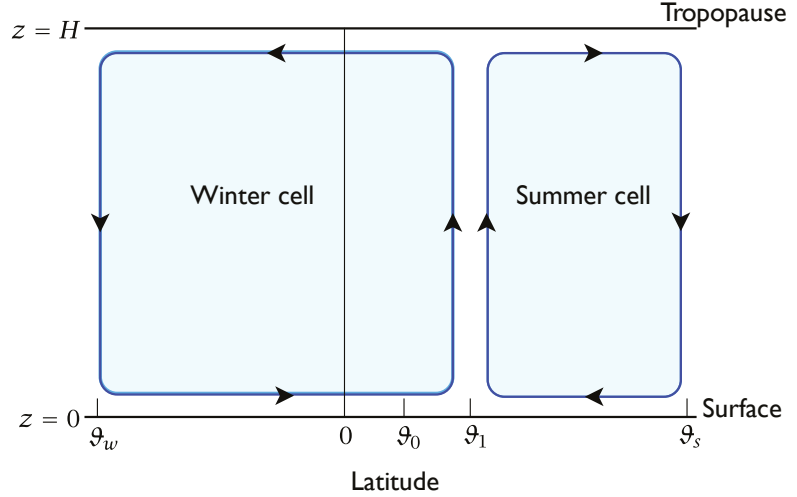
$$h(Y) = h^*(Y). \quad (14.44)$$

The two constraints (14.43) and (14.44) provide values of the unknowns $h(0)$ and Y , and give

$$Y = \left(\frac{5h_0\alpha g'}{\beta^2} \right)^{1/3}, \quad (14.45)$$

which is analogous to (14.19), as well as an expression for $h(0)$ that we leave as a problem for the reader. The qualitative dependence on the parameters is similar to that of the full model, although the latitudinal extent of the Hadley Cell is proportional to the cube root of the meridional thickness gradient α .

Fig. 14.9 A Hadley circulation model in which the heating is centred off the equator, at a latitude ϑ_0 . The lower level convergence occurs at a latitude ϑ_1 that is not in general equal to ϑ_0 . The resulting winter Hadley Cell is stronger and wider than the summer cell.



14.4 † ASYMMETRY AROUND THE EQUATOR

The Sun is overhead at the equator but two days out of the year, and in this section we investigate the effects that asymmetric heating has on the Hadley circulation. Observations indicate except for the brief periods around the equinoxes, the circulation is dominated by a single cell with rising motion centred in the summer hemisphere, but extending well into the winter hemisphere. That is, as seen in Fig. 14.3, the ‘winter cell’ is broader and stronger than the ‘summer cell’, and it behoves us to try to explain this. We will stay in the framework of the inviscid angular-momentum model of section 14.2, changing only the forcing field to represent the asymmetry and being a little more attentive to the details of spherical geometry.⁸

To represent an asymmetric heating we may choose a radiative equilibrium temperature of the form

$$\begin{aligned} \frac{\theta_E(\vartheta, z)}{\theta_0} &= 1 - \frac{2}{3} \Delta_H P_2(\sin \vartheta - \sin \vartheta_0) + \Delta_V \left(\frac{z}{H} - \frac{1}{2} \right) \\ &= 1 + \frac{\Delta_H}{3} [1 - 3(\sin \vartheta - \sin \vartheta_0)^2] + \Delta_V \left(\frac{z}{H} - \frac{1}{2} \right). \end{aligned} \quad (14.46)$$

This is similar to (14.14), but now the forcing temperature falls monotonically from a specified latitude ϑ_0 . If $\vartheta_0 = 0$ the model is identical to the earlier one, but if not we envision a circulation as qualitatively sketched in Fig. 14.9, with rising motion off the equator at some latitude ϑ_1 , extending into the winter hemisphere to a latitude ϑ_w , and into the summer hemisphere to ϑ_s . We will discover that, in general, $\vartheta_1 \neq \vartheta_0$ except when $\vartheta_0 = 0$. Following our procedure we used in the symmetric case as closely as possible, we then make the following assumptions.

- (i) The flow is quasi-steady. That is, at any time of year the flow adjusts to a steady circulation on a time scale more rapid than that on which the solar zenith angle appreciably changes.
- (ii) The flows in the upper branches conserve angular momentum, m . Further assuming that $u = 0$ at $\vartheta = \vartheta_1$ so that $m = \Omega a^2 \cos^2 \vartheta_1$ we obtain

$$u(\vartheta) = \frac{\Omega a (\cos^2 \vartheta_1 - \cos^2 \vartheta)}{\cos \vartheta}. \quad (14.47)$$

Thus, we expect to see westward (negative) winds aloft at the equator. In the lower branches the zonal flow is assumed to be approximately zero, i.e., $u(0) \approx 0$.

- (iii) The flow satisfies hydrostatic and gradient wind balance. The meridional momentum equation is then

$$fu + \frac{u^2 \tan \vartheta}{a} = -\frac{1}{a} \frac{\partial \phi}{\partial \vartheta}, \quad (14.48)$$

and because the flow crosses the equator we cannot neglect the second term on the left-hand side. Combining this with hydrostatic balance ($\partial \phi / \partial z = g\theta / \theta_0$) leads to a generalized thermal wind balance, which may be written as

$$m \frac{\partial m}{\partial z} = -\frac{ga^2 \cos^2 \vartheta}{2\theta_0 \tan \vartheta} \frac{\partial \theta}{\partial \vartheta}. \quad (14.49)$$

If the undifferentiated m is approximated by $\Omega a^2 \cos^2 \vartheta$, this reduces to conventional thermal wind balance, (14.10).

- (iv) Potential temperature in each cell is conserved when integrated over the extent of the cell. Thus,

$$\int_{\vartheta_1}^{\vartheta_s} (\theta - \theta_E) \cos \vartheta d\vartheta = 0, \quad \int_{\vartheta_1}^{\vartheta_w} (\theta - \theta_E) \cos \vartheta d\vartheta = 0, \quad (14.50)$$

for the summer and winter cells, respectively, where θ is the vertically averaged potential temperature.

- (v) Potential temperature is continuous at the edge of each cell, so that

$$\theta(\vartheta_s) = \theta_E(\vartheta_s), \quad \theta(\vartheta_w) = \theta_E(\vartheta_w), \quad (14.51)$$

and is also continuous at ϑ_1 . This last condition must be explicitly imposed in the asymmetric model, whereas in the symmetric model it holds by symmetry. Now, recall from the symmetric model that the value of the temperature at the equator was determined by the integral constraint (14.17) and the continuity constraint (14.18). We have analogues of these in each hemisphere [(14.50) and (14.51)] and thus, if ϑ_1 is set equal to ϑ_0 we cannot expect that they each would give the same temperature at ϑ_0 . Thus, ϑ_1 must be a free parameter to be determined.

Given these assumptions, the solution may be calculated. Using thermal wind balance, (14.49), with $m(H) = \Omega a^2 \cos^2 \vartheta_1$ and $m(0) = \Omega a^2 \cos^2 \vartheta$ we find

$$-\frac{1}{\theta_0} \frac{\partial \theta}{\partial \vartheta} = \frac{\Omega^2 a^2}{gH} \left(\frac{\sin \vartheta}{\cos^3 \vartheta} \cos^4 \vartheta_1 - \sin \vartheta \cos \vartheta \right), \quad (14.52)$$

which integrates to

$$\theta(\vartheta) - \theta(\vartheta_1) = -\frac{\theta_0 \Omega^2 a^2}{2gH} \frac{(\sin^2 \vartheta - \sin^2 \vartheta_1)^2}{\cos^2 \vartheta}. \quad (14.53)$$

The value of ϑ_1 , and the value of $\theta(\vartheta_1)$, are determined by the constraints (14.50) and (14.51). It is not in general possible to obtain a solution analytically, but one may be found numerically by an iterative procedure and one such is illustrated in Fig. 14.10. The zonal wind of the solution is always symmetric around the equator, because it is determined solely by angular momentum conservation. The temperature is therefore also symmetric, as (14.53) explicitly shows. However, the width of the solution in each hemisphere will, in general, be different.

Furthermore, because the strength of the circulation increases with difference between the temperature of the solution and the radiative equilibrium temperature, the circulation in the winter hemisphere will also be much stronger than that in the summer, a prediction that is consistent with the observations (see Fig. 14.3). More detailed calculations show that, because the strength of

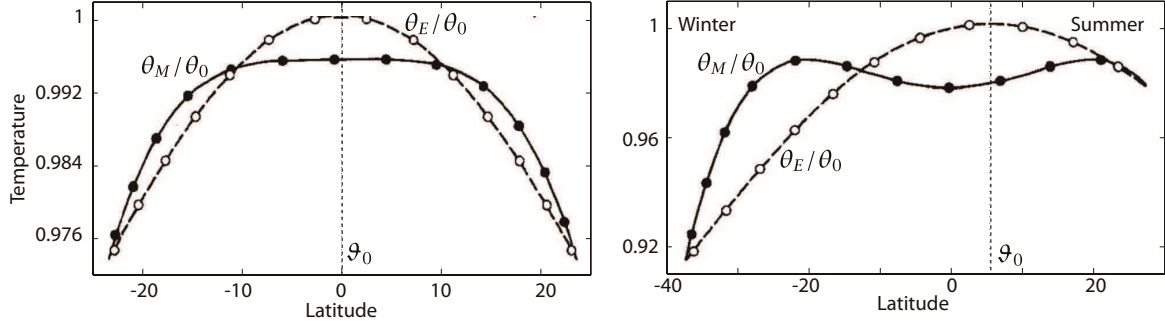


Fig. 14.10 Solutions of the Hadley Cell model with heating centred at the equator ($\vartheta_0 = 0^\circ$, left) and off the equator ($\vartheta_0 = +6^\circ$ N, right), with $\Delta_H = 1/6$. The dashed line is the radiative equilibrium temperature and the solid line is the angular-momentum-conserving solution. In the right-hand panel, $\vartheta_1 \approx +18^\circ$, and the circulation is dominated by the cell extending from $+18^\circ$ to -36° .⁹

the model Hadley Cell increases nonlinearly with ϑ_0 , the time-average strength of the Hadley Cell with seasonal forcing is stronger than that produced by annually averaged forcing. However, this does not appear to be a feature of either the observations or more complete numerical simulations, suggesting that an angular-momentum-conserving model has some deficiencies.¹⁰

The lack of consideration of zonal asymmetries (such as monsoons) and the lack of angular momentum conservation because of the effects of baroclinic eddies are issues that are shared with the steady model with hemispheric symmetry. A problem that is unique to the asymmetric model is the quasi-steady assumption, given the presence of a temporally progressing seasonal cycle. Because the latitude of the upward branch of the Hadley Cell varies with season, the value of the angular momentum entering the system also varies with time, and so a homogenized value of angular momentum is hard to achieve. Nonetheless, the overall picture that the model paints, and its qualitative explanation of the strengthened and extended winter Hadley Cell, are useful notions.

14.5 EDDY EFFECTS ON THE HADLEY CELL

So far, we have ignored the effects of baroclinic eddies on the Hadley circulation — ‘ignored’ rather than ‘neglected’, because we have no reason to believe that their effects will be negligible. In fact, as the upper-level flow moves poleward the shear of the zonal wind increases, as described above, and at some point the flow will become baroclinically unstable. We first describe a simple model of this before considering eddy fluxes more generally.

14.5.1 A Hadley Cell limited by baroclinic instability

Suppose that the flow moving polewards conserves its angular momentum, and to keep matters simple we consider flow on a beta-plane. The flow is given by (14.7), which in the small angle approximation implies a shear of

$$\Lambda_M \approx \frac{\Omega a \vartheta^2}{H} = \frac{\beta y^2}{2H} \quad (14.54)$$

where H is the height of the outflow and the last two expressions hold in the small angle approximation. Now, in a quasi-geostrophic two-level model, the flow becomes unstable when it reaches a critical velocity shear between upper and lower levels given by

$$\Lambda_C \equiv \frac{U_1 - U_2}{H/2} = \frac{1}{2H} \beta L_d^2 \quad (14.55)$$

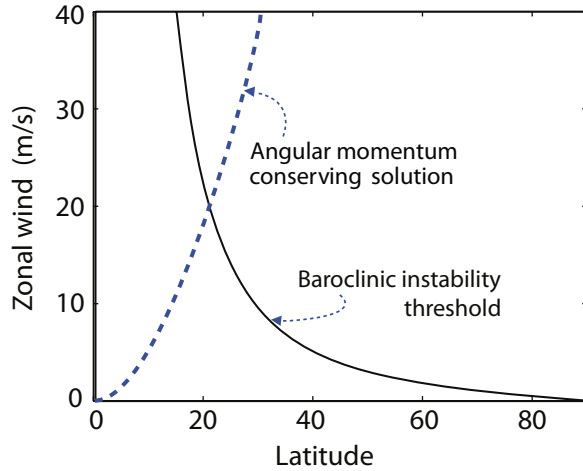


Fig. 14.11 Upper tropospheric zonal winds. The dashed curve shows the angular momentum conserving wind, with $u = 0$ at the equator. The solid curve shows the threshold for baroclinic instability of the upper-level flow using a two-layer quasi-geostrophic calculation, (14.55), with f varying with latitude in the deformation radius.

where $L_d = NH/f$ is the baroclinic deformation radius, and on the sphere $\beta = 2\Omega \cos \phi/a$. Both β factor and the f hiding in L_d make Λ_C grow towards the equator. Equating (14.54) and (14.55) suggests that the angular-momentum conserving flow will become unstable at a latitude ϑ_C given by, in the small angle approximation,

$$\vartheta_C \approx \left(\frac{NH}{2\Omega a} \right)^{1/2}. \quad (14.56)$$

The value of ϑ_C above should not be taken literally — the real atmosphere is not a two-level quasi-geostrophic model! But it does capture an essential truth that the angular momentum conserving solution will become baroclinically unstable at some latitude, as sketched in Fig. 14.11. It is a quantitative issue as to whether the Hadley flow becomes strongly unstable before it reaches its natural poleward extent. However, even if baroclinic instability itself is weak over much of the tropics, baroclinic instability further polewards will have an effect and lead to the nonconservation of angular momentum, as we now discuss.

14.5.2 Qualitative and observational considerations

The zonally averaged zonal momentum equation, (14.4), may be written as an equation for angular momentum, $\bar{m}$. Referring back to section 2.2 if needs be, the equation may be written as

$$\begin{aligned} \frac{\partial \bar{m}}{\partial t} + \frac{1}{\cos \vartheta} \frac{\partial}{\partial y} (\bar{v} \bar{m} \cos \vartheta) + \frac{\partial}{\partial z} (\bar{w} \bar{m}) &= -\frac{1}{\cos \vartheta} \frac{\partial}{\partial y} (\overline{m'v'} \cos \vartheta) - \frac{\partial}{\partial z} (\overline{m'w'}) \\ &= -\frac{1}{\cos \vartheta} \frac{\partial}{\partial y} (\overline{u'v'} a \cos^2 \vartheta) - \frac{\partial}{\partial z} (\overline{u'w'} a \cos \vartheta), \end{aligned} \quad (14.57)$$

where $\bar{m} = (\bar{u} + \Omega a \cos \vartheta) a \cos \vartheta$, $m' = u' a \cos \vartheta$, $y = a\vartheta$, and the vertical and meridional velocities are related by the mass continuity relation

$$\frac{1}{\cos \vartheta} \frac{\partial}{\partial y} (\bar{v} \cos \vartheta) + \frac{\partial \bar{w}}{\partial z} = 0. \quad (14.58)$$

In the angular-momentum-conserving model the eddy fluxes were neglected and (14.57) was approximated by the simple expression $\partial \bar{m} / \partial \vartheta = 0$, and by construction the Rossby number is $\mathcal{O}(1)$, because $\zeta = -f$.

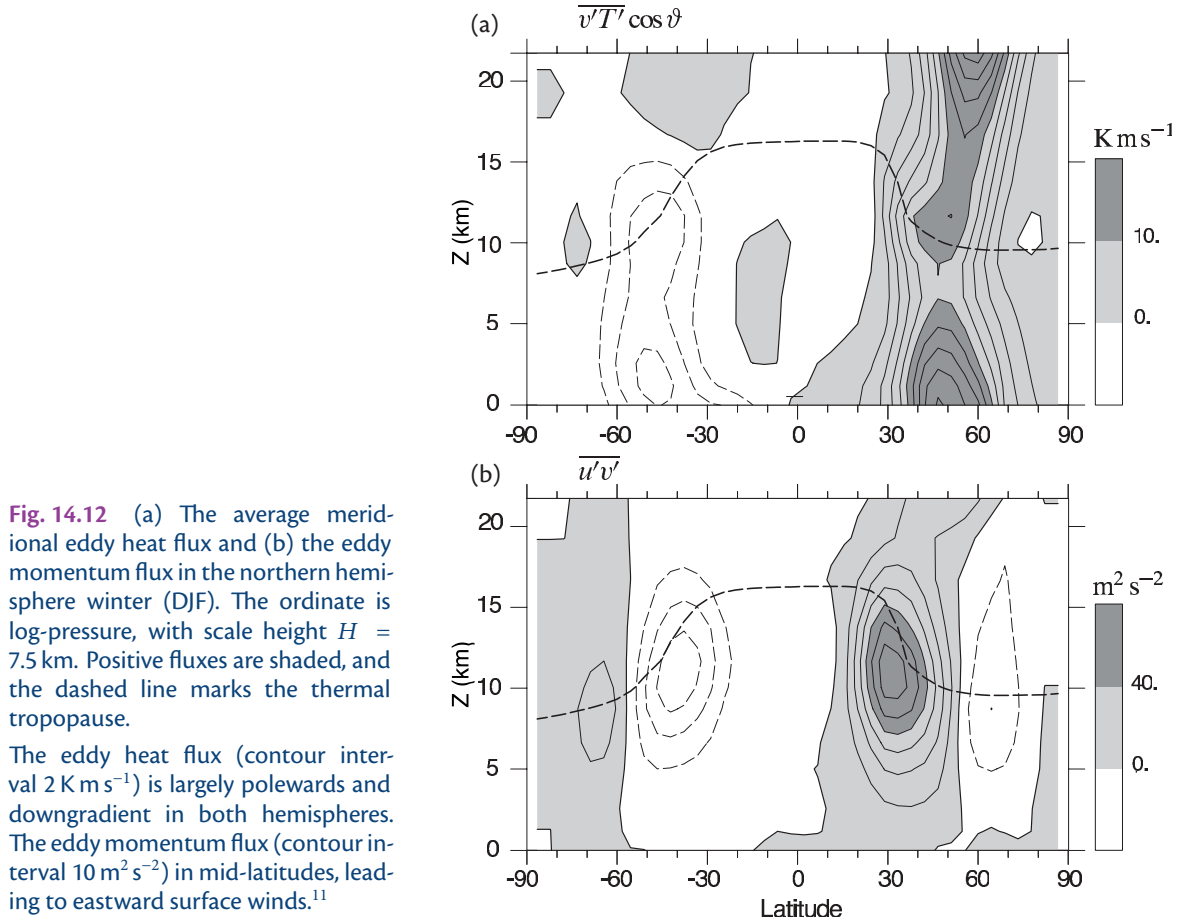


Fig. 14.12 (a) The average meridional eddy heat flux and (b) the eddy momentum flux in the northern hemisphere winter (DJF). The ordinate is log-pressure, with scale height $H = 7.5$ km. Positive fluxes are shaded, and the dashed line marks the thermal tropopause.

The eddy heat flux (contour interval 2 K m s^{-1}) is largely polewards and downgradient in both hemispheres. The eddy momentum flux (contour interval $10 \text{ m}^2 \text{s}^{-2}$) in mid-latitudes, leading to eastward surface winds.¹¹

The observed eddy heat and momentum fluxes are shown in Fig. 14.12. The eddy momentum flux is generally polewards, converging in the region of the mid-latitude surface westerlies. Its magnitude, and more particularly its meridional gradient, is as large or larger than the momentum flux associated with the mean flow. Neglecting vertical advection and vertical eddy fluxes, and using (14.58), (14.57) may be written as

$$\frac{\partial \bar{m}}{\partial t} + \bar{v} \frac{\partial \bar{m}}{\partial y} = -\frac{1}{\cos \vartheta} \frac{\partial}{\partial y} (\overline{u'v'} a \cos^2 \vartheta). \quad (14.59)$$

The quantity $\overline{u'v'}$ increases to a maximum at about 30° , so the right-hand side is negative in the tropics and lower subtropics. Thus, if $\bar{v} > 0$ (as in the upper branch of the Northern Hemisphere Hadley Cell) and the flow is steady, the observed eddy fluxes are such as to cause the angular momentum of the zonal flow to *decrease* as it moves polewards, and the zonal velocity is lower than it would be in the absence of eddies. (In the Southern Hemisphere the signs of v and the eddy momentum flux are reversed, but the dynamics are equivalent.)

The eddy flux of heat will also affect the Hadley Cell, although in a different fashion. We see from Fig. 14.12 that the eddy flux of temperature is predominantly polewards, and therefore that eddies export heat from the subtropics to higher latitudes. Now, the zonally averaged thermodynamic equation may be written

$$\frac{\partial \bar{b}}{\partial t} + \frac{1}{\cos \vartheta} \frac{\partial}{\partial y} (\bar{v} \bar{b} \cos \vartheta) + \frac{\partial}{\partial z} (\bar{w} \bar{b}) = -\frac{1}{\cos \vartheta} \frac{\partial}{\partial y} (\overline{v'b'} \cos \vartheta) - \frac{\partial}{\partial z} (\overline{w'b'}) + Q_b. \quad (14.60)$$

where Q_b represents the heating. After vertical averaging, the vertical advection terms vanish and the resulting equation is the thermodynamic equation implicitly used in the angular-momentum-conserving model, with the addition of the meridional eddy flux on the right-hand side. A diverging eddy heat flux in the subtropics (as in Fig. 14.12) is evidently equivalent to increasing the meridional gradient of the radiative equilibrium temperature, and therefore will increase the intensity of the overturning circulation.

14.5.3 An idealized eddy-driven model

Consider now the extreme case of an ‘eddy-driven’ Hadley Cell. (The driving for the Hadley Cell, and the atmospheric circulation in general, ultimately comes from the differential heating between equator and pole. Recognizing this, ‘eddy driving’ is a convenient way to refer to the role of eddies in producing a zonally averaged circulation. See also endnote 4 on page 858.) The model is over-simple, but revealing. Neglecting vertical derivatives the zonally averaged zonal momentum equation (14.4) may be written

$$\frac{\partial \bar{u}}{\partial t} - (f + \bar{\zeta})\bar{v} = -\frac{1}{a \cos^2 \vartheta} \frac{\partial}{\partial \vartheta} (\cos^2 \vartheta \overline{u'v'}). \quad (14.61)$$

If the Rossby number is sufficiently low this becomes simply

$$\frac{\partial \bar{u}}{\partial t} - f\bar{v} = M, \quad (14.62)$$

where $M = -\partial_y(\overline{u'v'})$, and for simplicity we now use Cartesian geometry. This approximation is not quantitatively accurate but it will highlight the role of the eddies. (Note the contrast between this model and the angular-momentum-conserving model. In the latter we assumed $f + \bar{\zeta} \approx 0$, and $Ro = \mathcal{O}(1)$; now we are neglecting $\bar{\zeta}$ and assuming the Rossby number is small.) At a similar level of approximation let us write the thermodynamic equation, (14.60), as

$$\frac{\partial \bar{b}}{\partial t} + N^2 \bar{w} = J, \quad (14.63)$$

where $J = Q_b - \partial_y(\overline{v'b'})$ represents the diabatic terms and eddy forcing. We are assuming, as in quasi-geostrophic theory, that the mean stratification, N^2 is fixed, and now $\bar{b}$ represents only the (zonally averaged) deviations from this. The mass continuity equation allows us to define a meridional streamfunction Ψ ; that is

$$\frac{\partial \bar{v}}{\partial y} + \frac{\partial \bar{w}}{\partial z} = 0 \quad \text{allows} \quad \bar{w} = \frac{\partial \Psi}{\partial y}, \quad \bar{v} = -\frac{\partial \Psi}{\partial z}. \quad (14.64a,b)$$

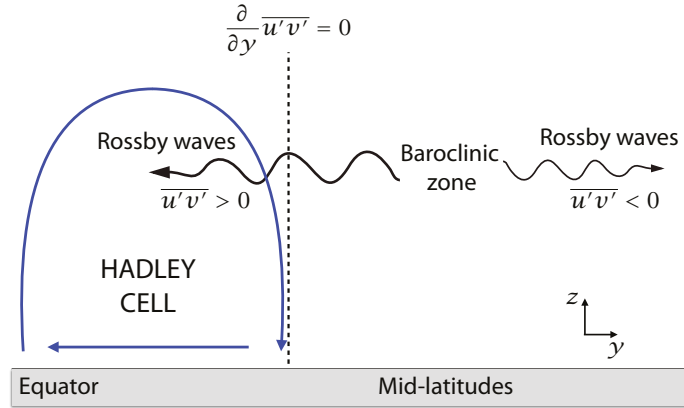
We may then use the thermal wind relation, $f\partial \bar{u}/\partial z = -\partial \bar{b}/\partial y$, to eliminate time derivatives in (14.62) and (14.63), giving¹²

$$f^2 \frac{\partial^2 \Psi}{\partial z^2} + N^2 \frac{\partial^2 \Psi}{\partial y^2} = f \frac{\partial M}{\partial z} + \frac{\partial J}{\partial y}. \quad (14.65)$$

This is a linear equation for the overturning streamfunction, one that holds even if the flow is not in a steady state, and a positive value of Ψ , in the Northern Hemisphere, corresponds to rising at the equator. The equation is equally (or in fact more) valid in mid-latitudes as in the tropics. We see that the overturning circulation is forced by eddy fluxes of heat and momentum, as well as heating and other terms that might appear on the right-hand sides of (14.62) and (14.63). If we rescale the vertical coordinate by the Prandtl ratio (i.e., let $z = z' f/N$) then (14.65) is a Poisson equation for the streamfunction. A few other germane points are as follows.

Fig. 14.13 Sketch of how eddy fluxes may affect the Hadley Cell even when the baroclinic zone is well poleward of the Cell.

Rossby waves are generated through baroclinic instability at mid-latitudes, well poleward of the Hadley Cell. Some propagate equatorward, and deposit westward momentum, $\partial(\overline{u'v'})/\partial y > 0$ inside the Hadley cell. At some latitude the Rossby wave momentum flux is neither convergent nor divergent, $\partial(\overline{u'v'})/\partial y = 0$, corresponding to the edge of the Hadley cell.



- (i) The horizontal gradient of the thermodynamic forcing partially drives the circulation, and both the heating term and the horizontal eddy flux divergence act in the same sense. A thermodynamically forced overturning circulation with warm fluid rising and cold fluid sinking, is called a 'direct cell'.
- (i) The vertical gradient of the horizontal eddy momentum divergence also partially drives the circulation, and because $\partial_y(\overline{u'v'}) > 0$ over the tropics and subtropics (for the Northern Hemisphere, see Fig. 14.12) these fluxes will intensify the circulation. The same term both weakens the zonal flow aloft and strengthens the overturning circulation. In the momentum equation — the Coriolis term $f\bar{v}$ is balanced by the eddy momentum flux convergence.
- (i) If N is small, then the circulation will become stronger if the other terms remain the same. That is, a dry atmosphere with a lapse rate close to that of a dry adiabat (i.e., $N = 0$) may have a stronger overturning circulation than otherwise, because the air can circulate without transporting any heat.
- (i) In winter, the increased strength of eddy momentum and buoyancy fluxes will drive a stronger Hadley Cell. This constitutes a different mechanism from that given in section 14.4 for the increased strength of the winter cell.

14.6 THE MECHANISM OF EDDY DRIVING

We saw above that eddy fluxes will strengthen the Hadley Cell and weaken the zonal winds. It is important to realise that these eddy fluxes can be important even if the baroclinic instability is well poleward of the Hadley Cell termination, because Rossby waves can propagate equatorward from the baroclinic zone into the subtropics. This propagation will be discussed more in sections 15.1 and 16.2, but suffice it to say here that equatorward propagating Rossby waves produce a polewards momentum flux, $\overline{u'v'} > 0$. (We will use Northern Hemisphere and Cartesian notation.) This flux has a maximum at about 30° in winter (varying somewhat with season) and diminishes equatorward of that, so that $\partial_y(\overline{u'v'}) > 0$. As noted previously, this flux divergence both strengthens the overturning circulation and weakens the zonal wind in the Hadley Cell.

In this model the edge of the Hadley Cell will coincide with the latitude at which the poleward eddy flux divergence is zero, since the momentum equation is approximately

$$-(f + \bar{\zeta})\bar{v} = -\frac{\partial}{\partial y}(\overline{u'v'}). \quad (14.66)$$

At the edge of the Hadley Cell we have $\bar{v} = 0$ and thus $\partial_y(\overline{u'v'}) = 0$. We emphasize that this is *not* necessarily the latitude where the poleward flow in the Hadley Cell becomes baroclinically

unstable.

The question of what determines the latitude at which the right-hand side of (14.66) is zero is a difficult problem in wave–mean-flow interaction. An equatorward propagating Rossby wave may encounter a critical latitude where it will break and accelerate the zonal wind westward, with the wave activity then diminishing equatorward of this. The Rossby wave momentum flux convergence $\partial_y(\overline{u'v'})$ is then negative in the mid-latitudes and positive in the low-latitudes, requiring a zero crossing $\partial_y(\overline{u'v'}) = 0$ at some latitude in between, which becomes the edge of the Hadley cell (as sketched in Fig. 14.13). The actual latitude where this occurs, and $\bar{v} = 0$, will be established through an interaction between the eastward acceleration by angular momentum conservation and westward acceleration by Rossby wave breaking, although it is not necessarily the critical latitude.

14.6.1 Numerical Solutions

Some illustrative results from two idealized numerical experiments with a GCM are shown in Figs. 14.14 and 14.15. The GCM has no explicit representation of moisture, except that the lapse rate is adjusted to a value close to the moist adiabatic lapse rate if it exceeds that value. In one experiment the model is constrained to produce an axisymmetric solution (top panels of the figures), and the zonal wind produced by the model in the Hadley Cell outflow is fairly close to being angular-momentum-conserving. In a three-dimensional version of the model, in which baroclinic eddies are allowed to form, the zonal wind is significantly reduced from its angular-momentum-conserving value, and correspondingly the overturning circulation is much stronger. Indeed, the strength of the Hadley Cell increases roughly linearly with the strength of the eddies in a sequence of numerical integrations similar to those shown, as suggested by (14.65). Qualitatively similar results are found in a model with no convective parameterization. In this case, the lapse rate is closer to neutral, N^2 is small, and the overturning circulation is generally stronger, as also expected from (14.65).

14.6.2 Final remarks

Is the real Hadley circulation ‘eddy-driven’, as in section 14.5.3 and 14.6, or is it a largely zonally symmetric structure constrained by angular momentum conservation, as in section 14.2 and its hemispherically asymmetric extensions? Or something else?

Observations of the overturning flow in summer and winter provide a guide. Figure 14.16 shows the thickness weighted transport overturning circulation in isentropic coordinates, and (as discussed more in chapter 10) this circulation includes both the Eulerian mean transport and the transport due to eddies. In winter there is considerable recirculation within the Hadley Cell, most of it coming from the zonally symmetric flow, and it is a quite distinct structure from the mid-latitude circulation. This suggests that the poleward extent of the winter Hadley Cell is strongly influenced by axisymmetric dynamics, for if it were solely a response to eddy heat fluxes one might expect it to join more smoothly with the mid-latitude Ferrel Cell. It seems likely that the effects of condensation and the concomitant concentration of the thermodynamic source (as in section 14.2.7) are acting to strengthen the axisymmetric circulation and give it a significant role. In summer, there is virtually no recirculation within the Hadley Cell and it does not appear as a self-contained structure, and this is suggestive of eddy effects and/or a strong mid-latitude influence. Zonally *asymmetric* circulations are also important — the Hadley Cell over India and South East Asia is tied up with monsoonal circulations, but tying the monsoon circulation into a theory of the Hadley Cell is a task for another day.

14.7 THE FERREL CELL

In this section we give a descriptive introduction to the Ferrel Cell, taking the eddy fluxes of heat and momentum to be given and viewing the circulation from a zonally averaged and Eulerian perspective. We investigate the associated dynamics in the next chapter.

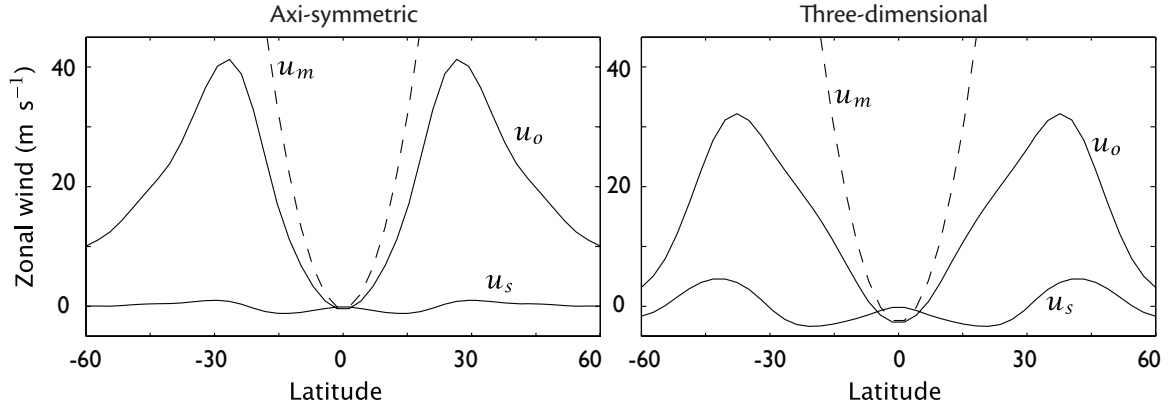


Fig. 14.14 The zonal wind in two numerical simulations. The right panel is from an idealized dry, three-dimensional atmospheric GCM, and the left panel is an axisymmetric version of the same model. Plotted are the zonal wind at the level of the Hadley Cell outflow, u_o ; the surface wind, u_s ; and the angular-momentum-conserving value, u_m .¹³

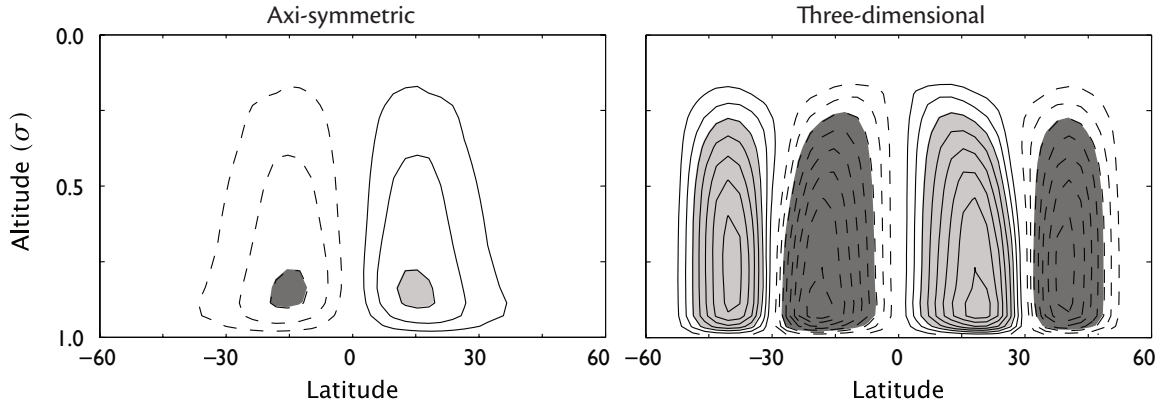


Fig. 14.15 As for Fig. 14.14, but now showing the streamfunction of the overturning circulation. ‘Altitude’ is $\sigma = p/p_s$, where p_s is surface pressure, and contour interval is 5 Sv (i.e., $5 \times 10^9 \text{ kg s}^{-1}$). The effects of eddies may be exaggerated because the model is dry.

The Ferrel Cell is an indirect meridional overturning circulation in mid-latitudes (see Fig. 14.3) that is apparent in the zonally averaged v and w fields, or the meridional overturning circulation defined by (14.3) or (14.64b). It is ‘indirect’ because cool air apparently rises in high latitudes, moves equatorwards and sinks in the subtropics. Why should such a circulation exist? The answer, in short, is that it is there to balance the eddy momentum convergence of the mid-latitude eddies and it is effectively driven by those eddies. To see this, consider the zonally averaged zonal momentum equation in mid-latitudes; at low Rossby number, and for steady flow this is just

$$-f\bar{v} = -\frac{1}{\cos^2 \vartheta} \frac{\partial}{\partial \vartheta} (\cos^2 \vartheta \overline{u'v'}) + \frac{1}{\rho} \frac{\partial \tau}{\partial z}. \quad (14.67)$$

This is a steady version of (14.62) with the addition of a frictional term $\partial \tau / \partial z$ on the right-hand side. At the surface we might approximate the stress by a drag, $\tau = r\bar{u}_s$, where r is a constant, with the stress falling away with height so that it is important only in the lowest kilometre or so of the atmosphere, in the atmospheric Ekman layer. Above this layer, the eddy momentum flux convergence is balanced by the Coriolis force on the meridional flow. In mid-latitudes (from about 30° to

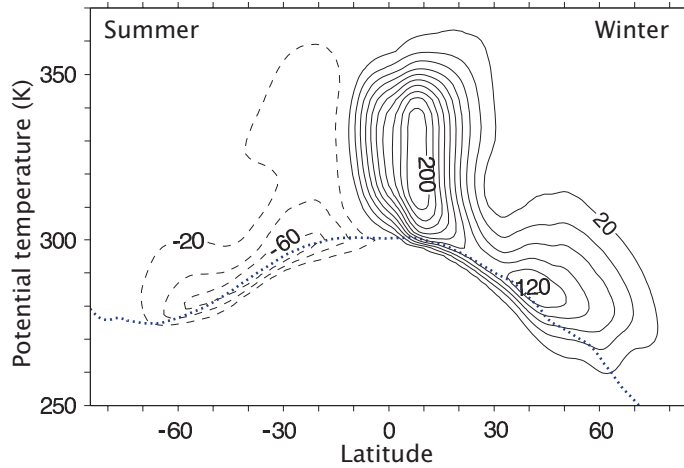


Fig. 14.16 The observed mass transport streamfunction in isentropic coordinates in northern hemisphere winter (DJF). The dotted line is the median surface temperature.

The return flow is nearly all in a layer near the surface, much of it at a lower temperature than the median surface temperature. Note the more vigorous circulation in the winter hemisphere.¹⁴

70°) the eddy momentum flux divergence is negative in both hemispheres (Fig. 14.12) and therefore, from (14.67), the averaged meridional flow must be equatorwards, as illustrated schematically in Fig. 14.17.

The flow cannot be equatorward everywhere, simply by mass continuity, and the return flow occurs largely in the Ekman layer, of depth d say. Here the eddy balance is between the Coriolis term and the frictional term, and integrating over this layer gives

$$-fV \approx -r\bar{u}_s, \quad (14.68)$$

where $V = \int_0^d \rho \bar{v} dz$ is the meridional transport in the boundary layer, above which the stress vanishes. The return flow is polewards (i.e., $V > 0$ in the Northern Hemisphere) producing an eastward Coriolis force. This can be balanced by a westward frictional force provided that the surface flow has an eastward component. In this picture, then, the mid-latitude eastward zonal flow at the surface is a proximate consequence of the polewards flowing surface branch of the Ferrel Cell, this poleward flow being required by mass continuity given the equatorward flow in the upper branch of the cell. In this way, the Ferrel Cell is responsible for bringing the mid-latitude eddy momentum flux convergence to the surface where it may be balanced by friction (refer again to Fig. 14.17).

A more direct way to see that the surface flow must be eastwards, given the eddy momentum flux convergence, is to vertically integrate (14.67) from the surface to the top of the atmosphere. By mass conservation, the Coriolis term vanishes (i.e., $\int_0^\infty f \rho \bar{v} dz = 0$) and we obtain

$$\int_0^\infty \frac{1}{\cos^2 \vartheta} \frac{\partial}{\partial \vartheta} (\cos^2 \vartheta \overline{u'v'}) \rho dz = [\tau]_0^\infty = -r\bar{u}_s. \quad (14.69)$$

That is, the surface wind is proportional to the vertically integrated eddy momentum flux convergence. Because there is a momentum flux convergence, the left-hand side is negative and the surface winds are eastwards.

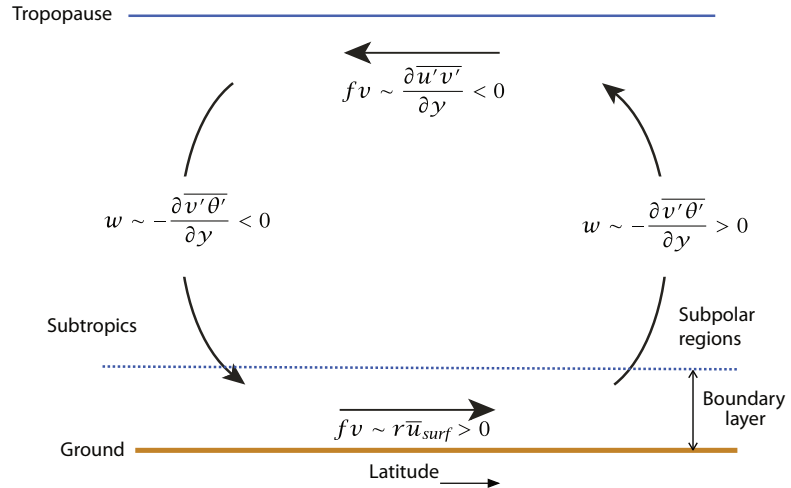
The eddy heat flux also plays a role in the Ferrel Cell, for in a steady state we have, from (14.63)

$$w = \frac{1}{N^2} \left[Q_b - \frac{1}{\cos \vartheta} \frac{\partial (\overline{v'b'} \cos \vartheta)}{\partial y} \right] \quad (14.70)$$

and inspection of Fig. 14.12 shows that the observed eddy heat flux produces an overturning circulation in the same sense as the observed Ferrel Cell (again see Fig. 14.17).

Fig. 14.17 The eddy-driven Ferrel Cell, from an Eulerian point of view.

Above the planetary boundary layer the mean flow is largely in balance with the eddy heat and momentum fluxes. The lower branch of the Ferrel Cell is largely confined to the boundary layer, where it is in a frictional–geostrophic balance.



Is the circulation produced by the heat fluxes *necessarily* the same as that produced by the momentum fluxes? In a non-steady state the effects of both heat and momentum fluxes on the Ferrel Cell are determined by (14.65) (an equation which applies more accurately at mid-latitudes than at low ones because of the low-Rossby number assumption), and there is no particular need for the heat and momentum fluxes to act in the same way. But in a steady state they must act to produce a consistent circulation. To see this, for simplicity let us take f and N^2 to be constant, let us suppose the fluid is incompressible and work in Cartesian coordinates. Take the y -derivative of (14.67) and the z -derivative of (14.70) and use the mass continuity equation. Noting that $\overline{v'\zeta'} = -\partial \overline{u'v'}/\partial y$ we obtain

$$\frac{\partial}{\partial y} \left(\overline{v'\zeta'} + \frac{f_0}{N^2} \frac{\partial \overline{v'b'}}{\partial z} \right) = \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} Q[b] \right) + \frac{\partial}{\partial y} \left(\frac{1}{\rho_0} \frac{\partial \tau}{\partial z} \right). \quad (14.71)$$

The expression on the left-hand side is the divergence of the eddy flux of quasi-geostrophic potential vorticity! That the heat and momentum fluxes act to produce a consistent overturning circulation is thus equivalent to requiring that the terms in the quasi-geostrophic potential vorticity equation are in a steady-state balance. The eddy fluxes of heat and momentum evidently play a huge role in the midlatitude circulation, and in the next chapter we examine the fluid dynamics giving rise to these eddy fluxes.

Notes

- 1 Many of the observations presented here are so-called *reanalyses*, prepared by the National Centers for Environmental Prediction (NCEP) and the European Centre for Medium-Range Weather Forecasts (ECMWF), described in Kalnay (1996) and Dee *et al.* (2011). Reanalysis products are syntheses of observations and model results and so are not wholly accurate representations of the atmosphere. However, especially in data-sparse regions of the globe and for poorly measured fields, they are likely to be more accurate representations of the atmosphere than could be achieved using only the raw data. Of course, this in turn means they contain biases introduced by the models. A reanalysis is just a particular form of *state estimate*, which is the more general name given to similar products and which is used in oceanography.
- 2 For example, Trenberth & Caron (2001).
- 3 George Hadley (1685–1768) was a British meteorologist who formulated the first dynamical theory for the trade winds, presented in a paper (Hadley 1735) entitled ‘Concerning the cause of the general trade winds.’ (At that time, trade winds referred to any large-scale prevailing wind, and not

just tropical winds. Some etymologists have associated the name with the commercial [i.e., trade] exploitation of the wind by mariners on long ocean journeys, but such an origin has been disputed: trade also means customary, and the winds customarily blow in one direction. Relatedly, in Middle English the word trade means path or track — hence the phrase ‘the wind blows trade’, meaning the wind is on track.) Hadley realized that in order to account for the zonal winds, the Earth’s rotation makes it necessary for there also to be a meridional circulation. His vision was of air heated at low latitudes, cooled at high latitudes, giving rise to a single meridional cell between the equator and each pole. Although he thought of the cell as essentially filling the hemisphere, and he did not account for the instability of such a flow, it was nevertheless a foundational contribution to meteorology. The thermally direct cell in low latitudes is now named after him.

A three-celled circulation was proposed by William Ferrel (1817–1891), an American school teacher and meteorologist, and the middle of these cells is now named for him. His explanation of the cell (Ferrel 1856a) was not correct, but this is hardly surprising because the eddy motion producing the angular momentum convergence that drives the Ferrel Cell was not understood for another 100 years or so. Ferrel in fact abandoned his three-celled picture in favour of a something more akin to a two-celled picture (Ferrel 1859), similar to that proposed by J. Thomson in 1857. The history of these ideas, and those of Hadley, is discussed by Thomson (1892). Ferrel did however give the first essentially correct description of the role of the Coriolis force and the geostrophic wind in the general circulation (Ferrel 1858, a paper with a quite modern style), a key development in the history of geophysical fluid dynamics. Ferrel also contributed to tidal theory [in Ferrel (1856b) he noted, for example, that the tidal force due to the moon would slow the Earth’s rotation] and to ocean dynamics. (See <http://www.history.noaa.gov/giants/ferrel2.html>).

Although Hadley’s single-celled viewpoint was in part superseded by the three-celled and two-celled structures, the modern view of the overturning circulation is, ironically, that of a single cell of ‘residual circulation’, which, although having distinct tropical and extratropical components, in some ways qualitatively resembles Hadley’s original picture. See chapter 15.

- 4 Drawing from Schneider (1977) and Held & Hou (1980).
- 5 Nevertheless, Fang & Tung (1996) find some analytic solutions in the presence of moisture.
- 6 After Hide (1969).
- 7 Gierasch (1975).
- 8 Largely following Lindzen & Hou (1988).
- 9 Solutions from Lindzen & Hou (1988).
- 10 See Dima & Wallace (2003) for some relevant observations. They noted that the asymmetry of the Hadley Cell is affected by monsoonal circulations (which are, of course, not accounted for in the model presented here). Fang & Tung (1999) investigated the effects of time dependence and note that quasi-steadiness is not well satisfied, although this alone was unable to limit the nonlinear amplification effect. The effects of baroclinic eddies are also important in the nonconservation of angular momentum (Walker & Schneider 2005, Frierson *et al.* 2007), as discussed in section 14.5.
- 11 Figure courtesy of M. Juckes, using an ECMWF reanalysis.
- 12 A simple generalization of (14.65) is to replace (14.62) by (14.59) and then to use the thermal wind equation in the form

$$\frac{f}{a \cos \vartheta} \frac{\partial \bar{m}}{\partial z} = -\frac{\partial \bar{b}}{\partial y}. \quad (14.72)$$

An equation very similar to, but a little more general than, (14.65) may then be derived. A still more general, usually elliptic equation for the overturning circulation may be derived from the zonally averaged primitive equations, assuming only that the zonally averaged zonal wind is in gradient wind balance with the pressure field (Vallis 1982).

- 13 Simulations kindly performed by C. Walker. See also Walker & Schneider (2005).
- 14 Figure courtesy of T. Schneider, using an ECMWF reanalysis.

*Further reading**Atmospheric dynamics and circulation*

- Andrews, D. G., Holton, J. R. & Leovy, C.B., 1987. *Middle Atmosphere Dynamics*.
Discusses the theory and observations of the middle atmosphere in some detail.
- Green, J. S. A., 1999. *Atmospheric Dynamics*.
A personal view of the subject, with numerous interesting insights on how the atmosphere works.
- Lorenz, E. N., 1967. *The Nature and Theory of the General Circulation of the Atmosphere*.
A classic monograph on the atmospheric general circulation.
- Marshall, J. & Plumb, R. A. *Atmosphere, Ocean and Climate Dynamics: An Introductory Text*.
Discusses the dynamics and circulation of both atmosphere and ocean at an advanced undergraduate, beginning graduate level.
- Peixoto, J. P. & Oort, A. H., 1992. *Physics of Climate*.
A descriptive but physical discussion of the general circulation, with an emphasis on observations.
- Randall, D. 2015. *An Introduction to the Global Circulation of the Atmosphere*.
Discusses both the observations and mechanisms of the large-scale atmospheric circulation.
- Schneider, T. & Sobel, A., 2007. *The Global Circulation of the Atmosphere: Phenomena, Theory, Challenges*.
Contains several useful review articles on the large-scale atmospheric circulation.

Atmospheric physics and thermodynamics

- Ambaum, M., 2010. *Thermal Physics of the Atmosphere*.
Distinguished by treating the subject as a branch of classical thermodynamics, which indeed it is.
- Andrews, D. G., 2010. *An Introduction to Atmospheric Physics*.
Readable discussions of radiation and thermodynamics at the undergraduate level.
- Bohrens, C. F., & Albrecht, B., 1998. *Atmospheric Thermodynamics*.
A well-known, and rather lively, introduction to the field.
- Cabellero, R. 2014. *Physics of the Atmosphere*.
A short, readable introduction to the subject of its title.
- Emanuel, K., 1994. *Atmospheric Convection*.
A comprehensive discussion of convection, progressing from basic theory to an advanced level.
- Pierrehumbert, R. T., 2010. *Principles of Planetary Climate*.
Pierrehumbert's book is an ideal complement to the one you are currently looking at.
- Wallace, J. M. & Hobbs, P. V., 2006. *Atmospheric Science: An Introductory Survey*.
An introduction to a broad range of topics in atmospheric sciences, for scientists at all levels.

Any theory of the atmospheric circulation must be based on a theory of (large-scale) atmospheric turbulence.

Eric Eady, *The cause of the general circulation of the atmosphere*, 1950.

CHAPTER 15

Zonally Averaged Mid-latitude Atmospheric Circulation

OUR GOAL IN THIS CHAPTER is to gain a gross understanding of the zonally *averaged* circulation of the mid-latitude atmosphere. Because of the presence of strong zonal asymmetries — that is, eddies, in particular baroclinic eddies — this circulation differs considerably from the zonally *symmetric* circulation that would exist if eddies did not develop at all. (The angular-momentum-conserving model of the Hadley Cell discussed in chapter 14 is an example of a zonally symmetric circulation.) Let us explain more.

When studying some aspects of the large-scale ocean circulation, or the low-latitude atmospheric circulation, we can make a great deal of progress by treating the large-scale flow as if it were absolutely steady with the eddies having only a perturbative effect. However, this approach fails badly for the mid-latitude atmosphere: the large-scale mid-latitude circulation is intrinsically unsteady on the large-scale to the extent that the associated eddies essentially *are* the circulation. The eddies are also unpredictable and chaotic; that is to say, *the large-scale mid-latitude circulation of the atmosphere is a turbulent flow*. This turbulence involves the large-scale, geostrophically and hydrostatically balanced flow — that is, it is geostrophic turbulence — and so has different properties than the smaller-scale, more nearly three-dimensional turbulence that may occur in boundary layers and the like. Further, this large-scale turbulence (sometimes called ‘macro-turbulence’) is neither fully-developed nor isotropic — it interacts with the Rossby waves that arise from the planetary rotation and might be best regarded as a form of weak turbulence. Why is the large-scale flow turbulent? Why is it zonally asymmetric at all? There are two potential sources for zonal asymmetries.

- (i) The zonal asymmetries that exist in the underlying boundary conditions and forcing: mountains, land–sea contrasts, the diurnal cycle, and so on.
- (ii) Hydrodynamic instability: even if the surface and the forcing were exactly zonally symmetric, the corresponding zonally symmetric solutions of the equations of motion would have a large shear in the zonal wind and this might be baroclinically unstable to zonally asymmetric perturbations.

If the flow were not unstable, then we might expect that the zonal asymmetries of item (i) would give rise to corresponding, steady, zonal asymmetries in the resulting circulation, and this process is discussed in the next chapter. However, the flow *is* unstable, primarily via baroclinic instability

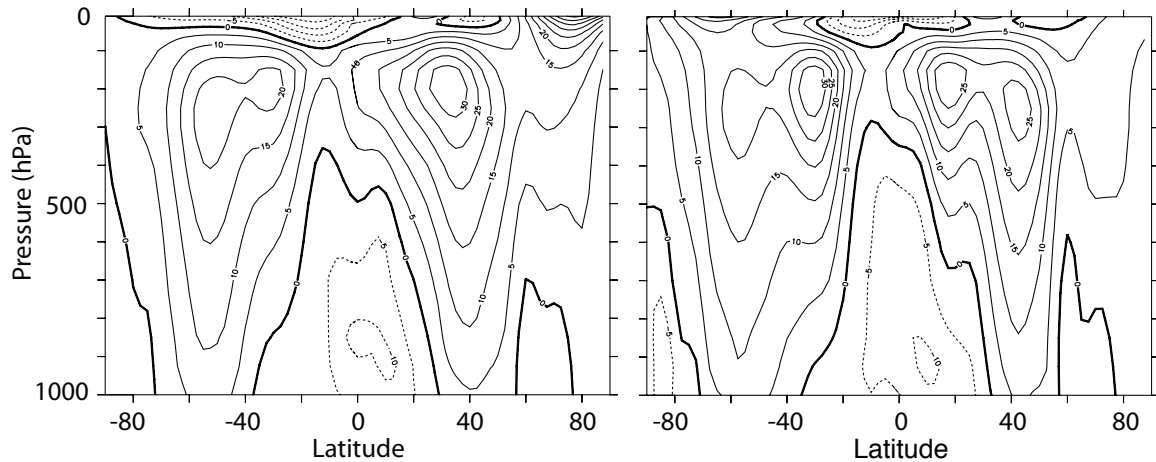


Fig. 15.1 The time averaged zonal wind at 150° W (in the mid-Pacific) in December–January–February (DJF, left), March–April–May (MAM, right). The contour interval is 5 m s^{-1} . Note the double jet in each hemisphere, one in the subtropics and one in mid-latitudes. The subtropical jets are associated with a strong meridional temperature gradient, whereas the mid-latitude, eddy-driven, jets have a stronger barotropic component and are associated with westerly winds at the surface.

(as discussed in chapter 9), and this leads to eddy growth and ultimately to geostrophic turbulence; this is, essentially, what gives rise to *weather*. The large-scale circulation is not, however, so turbulent that such things as Rossby waves cease to have meaning. Indeed, the most important nonlinear interactions are those involving Rossby waves and the zonally-averaged flow, and ideas of wave–mean–flow interaction go a long way in explaining the generation of the zonally-averaged flow. In particular, it is this interaction that produces the momentum convergence that gives rise to surface winds pattern.

In this chapter we focus on the effects of item (ii), and try to understand the mid-latitude circulation of an atmosphere with zonally symmetric forcing and boundary conditions. We assume that the zonal asymmetries due to the boundary conditions do not *qualitatively* affect our arguments, and that the circulation of the atmosphere with a perfectly smooth surface would resemble that of the real circulation. We will present our arguments semi-independently of earlier chapters, and in particular we will develop some of the results of wave–mean flow interaction *ab initio*, so that a reader with just a little experience can start here and refer back to these chapters as needed. We begin with a discussion of the mechanisms that maintain the surface westerlies¹

15.1 SURFACE WESTERLIES AND THE MAINTENANCE OF A BAROTROPIC JET

15.1.1 Observations and motivation

The atmosphere above the surface has a generally eastward flow, with a broad maximum about 10 km above the surface at around 40° in either hemisphere. But if we look a little more at the zonally average wind in Fig. 14.2(a) we see hints of there being two jets — one (the subtropical jet) at around 30°, and another somewhat polewards of this, especially apparent in the Southern Hemisphere. Such a jet is particularly noticeable in certain regions of the globe, when a zonal average is not taken, as in Fig. 15.1. The subtropical jet is associated with a strong meridional temperature gradient at the edge of the Hadley Cell, and is quite baroclinic. On the other hand, the mid-latitude jet (sometimes called the subpolar jet) is more barotropic (it has little vertical structure, with less shear than the subtropical jet) and lies above an eastward surface flow. This flow feels the effect of friction and so there must be a momentum *convergence* into this region,

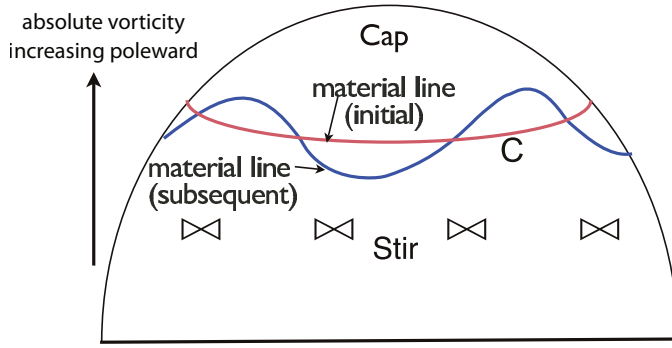


Fig. 15.2 Sketch of the effects of a mid-latitude disturbance on the circulation around the latitude line C.

If initially the absolute vorticity increases monotonically polewards, then the disturbance will bring fluid with lower absolute vorticity into the cap region. Then, using Stokes theorem, the velocity around the latitude line C will become more westwards.

as is seen in Fig. 14.12. We will find that this momentum convergence occurs largely in transient eddies, and the jet is known as the *eddy-driven jet*. Although the eddies are a product of baroclinic instability, the essential mechanism of jet production is present in barotropic dynamics, so we first consider how an eastward jet can be maintained in a turbulent two-dimensional flow on the surface of a rotating sphere.

In barotropic turbulence, alternating east–west jets can be maintained if β is non-zero, as described in section 12.1. However, that case was homogeneous, with no preferred latitude for a particular direction of jet, whereas in the atmosphere there appears to be but one mid-latitude jet, and although it meanders it certainly has a preferred average location. In the subsections that follow we give four explications as to how the jet is maintained; the first has a different flavour from the others, but they are all really just different perspectives on the same mechanism; they are all the same explanation.²

15.1.2 The mechanism of jet production

1. The vorticity budget

Suppose that the absolute vorticity normal to the surface (i.e., $\zeta + 2\Omega \sin \vartheta$) increases monotonically polewards. (A sufficient condition for this is that the fluid is at rest.) By Stokes' theorem, the circulation around a line of latitude circumscribing the polar cap, I , is equal to the integral of the absolute vorticity over the cap. That is,

$$I_i = \int_{\text{cap}} \boldsymbol{\omega}_{ia} \cdot d\mathbf{A} = \oint_C u_{ia} dl = \oint_C (u_i + \Omega a \cos \vartheta) dl, \quad (15.1)$$

where $\boldsymbol{\omega}_{ia}$ and u_{ia} are the initial absolute vorticity and velocity, respectively, u_i is the initial zonal velocity in the Earth's frame of reference, and the line integrals are around the line of latitude. For simplicity let us take $u_i = 0$ and suppose there is a disturbance equatorwards of the polar cap, and that this results in a distortion of the material line around the latitude circle C (Fig. 15.2). Since we are supposing the source of the disturbance to be distant from the latitude of interest, then if we neglect viscosity the circulation along the material line is conserved, by Kelvin's circulation theorem. Thus, vorticity with a lower value is brought into the region of the polar cap — that is, the region polewards of the latitude line C. Using Stokes' theorem again the circulation around the latitude circle C must therefore fall; that is, denoting values after the disturbance with a subscript f ,

$$I_f = \int_{\text{cap}} \boldsymbol{\omega}_{fa} \cdot d\mathbf{A} < I_i \quad (15.2)$$

so that

$$\oint_C (u_f + \Omega a \cos \vartheta) dl < \oint_C (u_i + \Omega a \cos \vartheta) dl \quad (15.3)$$

and

$$\bar{u}_f < \bar{u}_i \quad (15.4)$$

with the overbar indicating a zonal average. Thus, there is a tendency to produce *westward* flow polewards of the disturbance. By a similar argument westward flow is also produced equatorwards of the disturbance — to see this one might apply Kelvin's theorem over all of the globe south of the source of the disturbance (taking care to take the dot-product correctly between the direction of the vorticity vector and the direction normal to the surface). Finally, note that the overall situation is the same in the Southern Hemisphere. Thus, on the surface of a rotating sphere, external stirring will produce westward flow *away* from the region of the stirring.

Now suppose, furthermore, that the disturbance imparts no net angular momentum to the fluid. Then the integral of $ua \cos \vartheta$ over the entire hemisphere must be constant. But the fluid is accelerating westwards away from the disturbance. Therefore, the fluid in the region of the disturbance must accelerate *eastwards*; that is, angular momentum must converge into the stirred region, producing an eastward flow. This simple mechanism is the essence of the production of eastward eddy-driven jets in the atmosphere, and of the eastward surface winds in mid-latitudes. The stirring that here we have externally imposed comes, in reality, from baroclinic instability.

If the stirring subsides then the flow may reversibly go back to its initial condition, with a concomitant reversal of the momentum convergence that caused the zonal flow. Thus, we must have some form of dissipation and irreversibility in order to produce permanent changes, and in particular we need to irreversibly mix vorticity. (This result is closely related to the non-acceleration results of chapter 10.) If the fluid is continuously mixed, then of course we also need a source that restores the absolute vorticity gradient, otherwise we will completely homogenize the vorticity over the hemisphere.

II. Rossby waves and momentum flux

We saw above that a mean gradient of vorticity is an essential ingredient in the mechanism whereby a mean flow is generated by stirring. Given such, we expect Rossby waves to be excited, and we now show how Rossby waves are intimately related to the momentum flux maintaining the mean flow.

If a stirring is present in mid-latitudes then we expect that Rossby waves will be generated there, propagate away and break and dissipate. To the extent that the waves are quasi-linear and do not interact, then just away from the source region each wave has the form

$$\psi = \text{Re } C e^{i(kx+ly-\omega t)} = \text{Re } C e^{i(kx+ly-ckt)}, \quad (15.5)$$

where C is a constant, with dispersion relation

$$\omega = ck = \bar{u}k - \frac{\beta k}{k^2 + l^2} \equiv \omega_R, \quad (15.6)$$

provided that there is no meridional shear in the zonal flow. The meridional component of the group velocity is given by

$$c_g^y = \frac{\partial \omega}{\partial l} = \frac{2\beta kl}{(k^2 + l^2)^2}. \quad (15.7)$$

Now, the direction of the group velocity must be *away* from the source region; this is a radiation condition (discussed more in the next subsection), demanded by the requirement that Rossby waves transport energy *away* from the disturbance. Thus, northwards of the source kl is positive and southwards of the source kl is negative. That the product kl can be positive or negative arises because for each k there are two possible values of l that satisfy the dispersion relation (15.6), namely

$$l = \pm \left(\frac{\beta}{\bar{u} - c} - k^2 \right)^{1/2}, \quad (15.8)$$

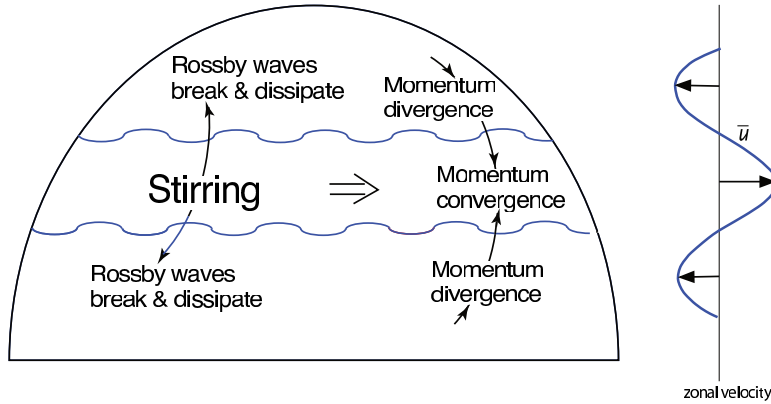


Fig. 15.3 Generation of zonal flow on a rotating sphere.

Stirring in mid-latitudes (by baroclinic eddies) generates Rossby waves that propagate away. Momentum converges in the region of stirring, producing eastward flow there and weaker westward flow on its flanks.

assuming that the quantity in parentheses is positive.

The velocity variations associated with the Rossby waves are

$$u' = -\text{Re } C i l e^{i(kx+ly-\omega t)}, \quad v' = \text{Re } C i k e^{i(kx+ly-\omega t)}, \quad (15.9a,b)$$

and the associated momentum flux is

$$\overline{u'v'} = -\frac{1}{2}C^2kl. \quad (15.10)$$

Thus, given that the sign of kl is determined by the group velocity, northwards of the source the momentum flux associated with the Rossby waves is southward (i.e., $\overline{u'v'}$ is negative), and southwards of the source the momentum flux is northward (i.e., $\overline{u'v'}$ is positive). That is, the momentum flux associated with the Rossby waves is *toward* the source region. Momentum converges in the region of the stirring, producing net eastward flow there and westward flow to either side (Fig. 15.3).

Another way of describing the same effect is to note that if kl is positive then lines of constant phase ($kx + ly = \text{constant}$) are tilted north-west/south-east, and the momentum flux associated with such a disturbance is negative ($\overline{u'v'} < 0$). Similarly, if kl is negative then the constant-phase lines are tilted north-east/south-west and the associated momentum flux is positive ($\overline{u'v'} > 0$). The net result is a convergence of momentum flux into the source region. In physical space this is reflected by having eddies that are ‘bow-shaped’, as in Fig. 15.4.

♦ The radiation condition and Rayleigh friction

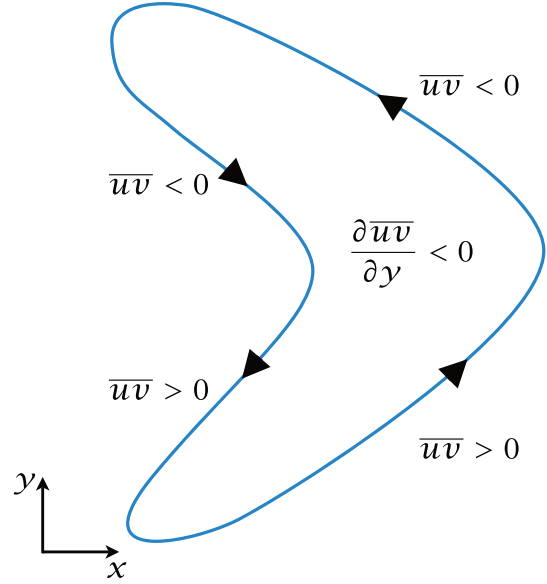
Why is the group velocity directed away from the source region? It is because the energy flux travels at the group velocity, and the energy flux must be directed away from the source region; the reader comfortable with that statement may stop here. (See section 6.7 for more on group velocity.) Another way to determine the direction of the group velocity is to employ a common trick in problems of wave propagation, that of adding a small amount of friction to the inviscid problem.³ The solution of the ensuing problem in the limit of small friction will often make clear which solution is physically meaningful in the inviscid problem, and therefore which solution nature chooses. Consider the linear barotropic vorticity equation with linear friction,

$$\frac{\partial \zeta}{\partial t} + \beta \frac{\partial \psi}{\partial x} = -r\zeta \quad (15.11)$$

where r is a small friction coefficient. The dispersion relation is

$$\omega = -\frac{\beta k}{K^2} - ir = \omega_R(k, l) - ir, \quad (15.12)$$

Fig. 15.4 The momentum transport in physical space, caused by the propagation of Rossby waves away from a source in mid-latitudes. The ensuing bow-shaped eddies are responsible for a convergence of momentum, as indicated. If the arrows were reversed the momentum transport would still have the same sign.



where ω_R is defined by (15.6), with $\bar{u} = 0$, and so the wave decays with time. Now suppose a wave is generated in some region, and that it propagates meridionally away, decaying as it moves away. Then, instead of an imaginary frequency, we may suppose that the frequency is real and the y -wavenumber is imaginary. Specifically, we take $l = l_0 + l'$, where $l_0 = \pm[\beta/(\bar{u} - c) - k^2]^{1/2}$ for some zonal wavenumber k , as in (15.8), and $\omega = \omega_R(k, l_0)$. For small friction, we obtain l' by Taylor-expanding the dispersion relation around its inviscid value, $\omega_R(k, l_0)$, giving

$$\omega + ir = \omega_R(k, l) \approx \omega_R(k, l_0) + \left. \frac{\partial \omega_R(k, l)}{\partial l} \right|_{l=l_0} l', \quad (15.13)$$

and therefore

$$l' = \frac{ir}{c_g^y}, \quad (15.14)$$

where $c_g^y = \partial_l \omega_R(k, l)|_{l=l_0}$ is the y -component of the group velocity. The wavenumber is imaginary, so that the wave either grows or decays in the y -direction, and the wave solution obeys

$$\psi \approx \text{Re } C \exp[i(kx - \omega_R t)] \exp(il_0 y - r y/c_g^y). \quad (15.15)$$

We now demand that the solution decay away from the source, because any other choice is manifestly unphysical, even as we let r be as small as we please. Thus, with the source at $y = 0$, c_g^y must be positive for positive y and negative for negative y . In other words, the group velocity must be directed *away* from the source region, and so momentum flux converges on the source region.

III. The pseudomomentum budget

The kinematic relation between vorticity flux and momentum flux for non-divergent two-dimensional flow is

$$v\zeta = \frac{1}{2} \frac{\partial}{\partial x} (v^2 - u^2) - \frac{\partial}{\partial y} (uv). \quad (15.16)$$

After zonal averaging this gives

$$\overline{v'\zeta'} = -\frac{\partial \overline{u'v'}}{\partial y}, \quad (15.17)$$

noting that $\bar{v} = 0$ for two-dimensional incompressible (or geostrophic) flow. For reference, in spherical coordinates this expression becomes

$$\overline{v'\zeta'} \cos \vartheta = -\frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (\cos^2 \vartheta \overline{u'v'}). \quad (15.18)$$

If (15.17) [or (15.18)] is integrated with respect to y between two quiescent latitudes then the right-hand side vanishes. That is the zonally averaged meridional vorticity flux vanishes when integrated over latitude.

Now, the barotropic zonal momentum equation is (for horizontally non-divergent flow)

$$\frac{\partial u}{\partial t} + \frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y} - fv = -\frac{\partial \phi}{\partial x} + F_u - D_u, \quad (15.19)$$

where F_u and D_u represent the effects of any forcing and dissipation. Zonal averaging, with $\bar{v} = 0$, gives

$$\frac{\partial \bar{u}}{\partial t} = -\frac{\partial \bar{u}\bar{v}}{\partial y} + \bar{F}_u - \bar{D}_u, \quad \text{or} \quad \frac{\partial \bar{u}}{\partial t} = \overline{v'\zeta'} + \bar{F}_u - \bar{D}_u. \quad (15.20)$$

using (15.17). Thus, the zonally averaged wind is maintained by the zonally averaged vorticity flux. On average there is little if any direct forcing of horizontal momentum and we may set $\bar{F}_u = 0$, and if the dissipation is parameterized by a linear drag (15.20) becomes

$$\frac{\partial \bar{u}}{\partial t} = \overline{v'\zeta'} - r\bar{u}, \quad (15.21)$$

where the constant r is an inverse frictional time scale.

Now consider the maintenance of this vorticity flux. The barotropic vorticity equation is

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + v\beta = F_\zeta - D_\zeta, \quad (15.22)$$

where F_ζ and D_ζ are forcing and dissipation of vorticity. Linearize about a mean zonal flow to give

$$\frac{\partial \zeta'}{\partial t} + \bar{u} \frac{\partial \zeta'}{\partial x} + \gamma v' = F'_\zeta - D'_\zeta, \quad (15.23)$$

where

$$\gamma = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} \quad (15.24)$$

is the meridional gradient of absolute vorticity. (We use γ rather than β^* to denote this quantity because of a later extension to layered models, where $\gamma = \partial \bar{q} / \partial y$.) Now multiply (15.23) by ζ' / γ and zonally average, assuming that $\bar{u}_{yy}$ is small compared to β or varies only slowly, to form the pseudomomentum equation,

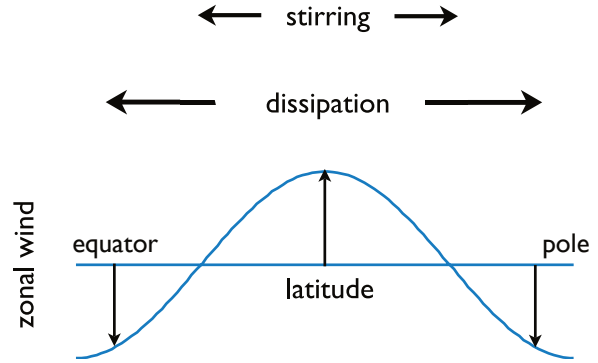
$$\frac{\partial \mathcal{P}}{\partial t} + \overline{v'\zeta'} = \frac{1}{\gamma} (\overline{\zeta' F'_\zeta} - \overline{\zeta' D'_\zeta}), \quad (15.25)$$

where

$$\mathcal{P} = \frac{1}{2\gamma} \overline{\zeta'^2} \quad (15.26)$$

is a wave activity density, equal to the pseudomomentum for this problem [see also section 10.2 and (10.29b)]. The parameter γ is positive if the average absolute vorticity increases monotonically northwards, and this is usually the case in both Northern and Southern Hemispheres.

Fig. 15.5 Mean flow generation by a meridionally confined stirring. Because of Rossby wave propagation away from the source region, the distribution of pseudomomentum dissipation is broader than that of pseudomomentum forcing, and the sum of the two leads to the zonal wind distribution shown, with positive (eastward) values in the region of the stirring. See also Fig. 15.8.



In the absence of forcing and dissipation, (15.21) and (15.25) imply an important relationship between the change of the mean flow and the pseudomomentum, namely

$$\frac{\partial \bar{u}}{\partial t} + \frac{\partial \mathcal{P}}{\partial t} = 0. \quad (15.27)$$

Now if for some reason $\mathcal{P}$ increases, perhaps because a wave enters an initially quiescent region because of stirring elsewhere, then mean flow must decrease. However, because the vorticity flux integrates to zero, the zonal flow cannot decrease everywhere. Thus, if the zonal flow decreases in regions away from the stirring, it must *increase* in the region of the stirring. In the presence of forcing and dissipation this mechanism can lead to the production of a statistically steady jet in the region of the forcing, since (15.21) and (15.25) combine to give

$$\frac{\partial \bar{u}}{\partial t} + \frac{\partial \mathcal{P}}{\partial t} = -r\bar{u} + \frac{1}{\gamma} \left(\overline{\zeta' F_{\zeta}'} - \overline{\zeta' D_{\zeta}'} \right), \quad (15.28)$$

and in a statistically steady state

$$r\bar{u} = \frac{1}{\gamma} \left(\overline{\zeta' F_{\zeta}'} - \overline{\zeta' D_{\zeta}'} \right). \quad (15.29)$$

The terms on the right-hand side represent the stirring and dissipation of vorticity, and integrated over latitude their sum will vanish, or otherwise the pseudomomentum budget cannot be in a steady state. However, let us suppose that forcing is confined to mid-latitudes. In the forcing region, the first term on the right-hand side of (15.29) will be larger than the second, and an eastward mean flow will be generated. Away from the direct influence of the forcing, the dissipation term will dominate and westward mean flows will be generated, as sketched in Fig. 15.5. Thus, *on a β -plane or on the surface of a rotating sphere an eastward mean zonal flow can be maintained by a vorticity stirring that imparts no net momentum to the fluid.* In general, stirring in the presence of a vorticity gradient will give rise to a mean flow, and on a spherical planet the vorticity gradient is provided by differential rotation.

It is crucial to the generation of a mean flow that the dissipation has a broader latitudinal distribution than the forcing: if all the dissipation occurred in the region of the forcing then from (15.29) no mean flow would be generated. However, Rossby waves are generated in the forcing region, and these propagate meridionally before dissipating thus broadening the dissipation distribution and allowing the generation of a mean flow.

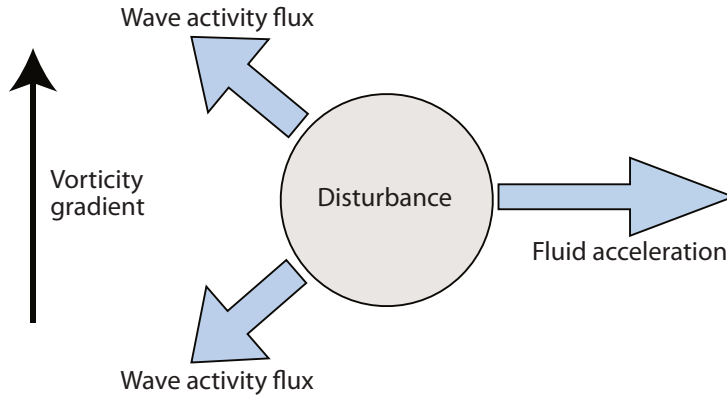


Fig. 15.6 If a region of fluid on the β -plane or on a rotating sphere is stirred, then Rossby waves propagate away from the disturbance, and this is the direction of the wave-activity flux vector. Thus, there is divergence of wave activity in the stirred region, and using (15.34) this produces an eastward acceleration.

IV. The Eliassen–Palm flux

The Eliassen–Palm (EP) flux (section 10.2) provides a convenient framework for determining how waves affect the mean flow, and the barotropic case is a particularly simple and instructive example. In the unforced case, the zonally averaged momentum equation may be written as

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \nabla_x \cdot \mathcal{F}, \quad (15.30)$$

where $\bar{v}^*$ is the residual meridional velocity and $\mathcal{F}$ is the Eliassen–Palm flux, and $\nabla_x \cdot$ is the divergence in the meridional plane. In the barotropic case $\bar{v}^* = 0$ and

$$\mathcal{F} = -j \overline{u'v'}. \quad (15.31)$$

If the momentum flux is primarily the result of interacting nearly monochromatic waves, then the EP flux obeys the group velocity property (section 6.7), namely that the flux of wave activity density is equal to the group velocity multiplied by the wave activity density. Thus,

$$\mathcal{F}^y \equiv j \cdot \mathcal{F} \approx c_g^y \mathcal{P}, \quad (15.32)$$

where $\mathcal{P}$ the wave activity density, or pseudomomentum, given by

$$\mathcal{P} = \frac{\overline{\zeta'^2}}{2\bar{q}_y} = \frac{\overline{\zeta'^2}}{2\gamma}, \quad (15.33)$$

and, if $\gamma > 0$, $\mathcal{P}$ is a positive-definite quantity. The zonal momentum equation and the Eliassen–Palm relation (10.29a) become respectively

$$\frac{\partial \bar{u}}{\partial t} = \frac{\partial}{\partial y}(c_g^y \mathcal{P}), \quad \frac{\partial \mathcal{P}}{\partial t} + \frac{\partial}{\partial y}(c_g^y \mathcal{P}) = 0, \quad (15.34a,b)$$

and so

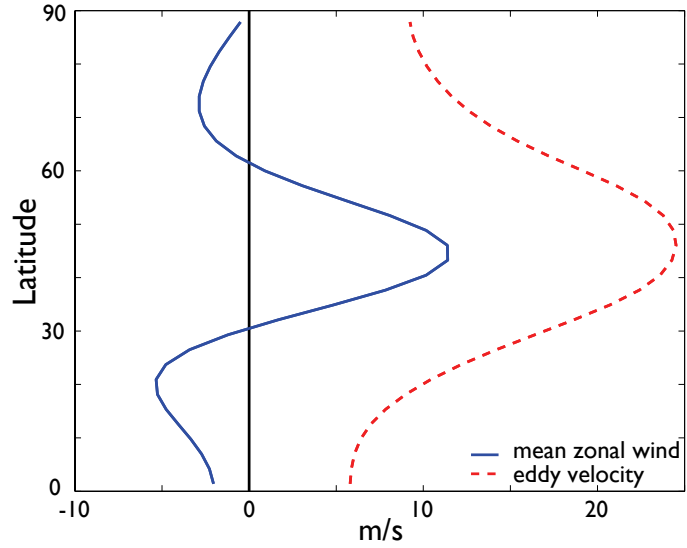
$$\frac{\partial \bar{u}}{\partial t} = -\frac{\partial \mathcal{P}}{\partial t}, \quad (15.35)$$

as in (15.27).

Now suppose that we initiate a disturbance at some latitude, and then let the fluid evolve freely. The disturbance generates Rossby waves whose group velocity will be directed away from the region of disturbance, and from (15.34b) the wave activity density $\mathcal{P}$ will diminish in the region of

Fig. 15.7 The time and zonally averaged wind (solid line) obtained by an integration of the barotropic vorticity equation (15.36) on the sphere.

The fluid is stirred in mid-latitudes by a random wavemaker that is statistically zonally uniform, acting around zonal wavenumber 8, and that supplies no net momentum. Momentum converges in the stirring region leading to an eastward jet with a westward flow to either side, and zero area-weighted spatially integrated velocity. The dashed line shows the r.m.s. (eddy) velocity created by the stirring.



the disturbance (and increase elsewhere). However, from (15.34a) the zonal velocity will *increase* in the region of the disturbance, and an eastward flow will be generated. That is, momentum converges in the region of the disturbance and an eastward jet is generated, as sketched in Fig. 15.6.

This EP flux argument, the pseudomomentum argument and the Rossby wave argument are, in essence, just different expressions of the same physical process, different explications but the same explanation.⁴ Indeed, the result of (15.10) can be regarded as illustrating the group velocity property of the EP flux for barotropic Rossby waves. The vorticity budget argument is a little more general than these, because it does not depend on linearization or small amplitude disturbances. At the same time, none of these arguments is truly turbulent in nature, and although they are all involve nonlinear interactions, it is the presence of β -effect — a linear term in the vorticity equation — that is crucial in the development of a mean flow.

15.1.3 A numerical example

We conclude from the above arguments that momentum will converge into a rapidly rotating flow that is stirred in a meridionally localized region. To illustrate this, we numerically integrate the barotropic vorticity equation on the sphere, with a meridionally localized stirring term; explicitly, the equation that is integrated is

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta) + \beta \frac{\partial \psi}{\partial x} = -r\zeta + \kappa \nabla^4 \zeta + F_\zeta. \quad (15.36)$$

The first term on the right-hand side is a linear drag, parameterizing momentum loss in an Ekman layer. The second term removes enstrophy that has cascaded to small scales; it has a negligible impact at large scales. The forcing term F_ζ is a wavemaker confined to a zonal strip of about 15° meridional extent, centred at about 45° N, that is statistically zonally uniform and that spatially integrates to zero. Within that region it is a random stirring with a temporal decorrelation scale of a few days and a spatial decorrelation scale corresponding to about wavenumber 8, thus mimicking weather scales. Thus, it provides no net source of vorticity or momentum, but it is a source of pseudomomentum because $\overline{F_\zeta \zeta} > 0$.

The results of a numerical integration of (15.36) are illustrated in Figs. 15.7 and 15.8. An eastward jet forms in the vicinity of the forcing, with westward flow on either side. The pseudomomentum stirring and dissipation that produce this flow are shown in Fig. 15.8. As expected, the

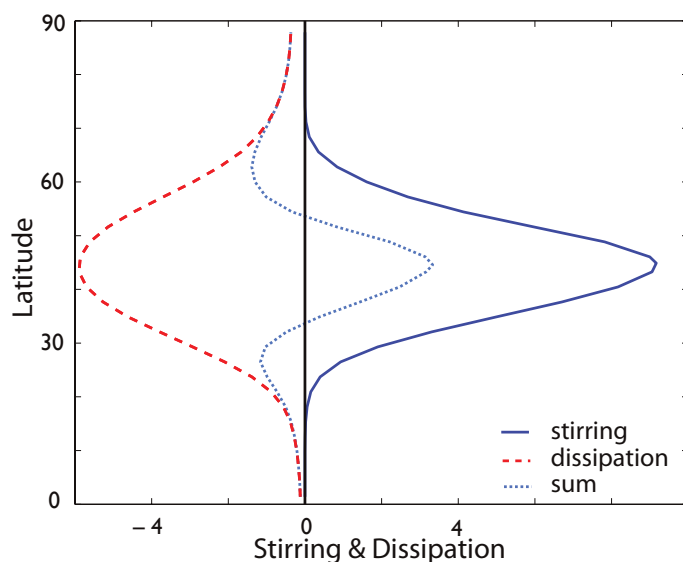


Fig. 15.8 The pseudomomentum stirring (solid line, $\overline{F'_\zeta \zeta'}$), dissipation (dashed line, $\overline{D'_\zeta \zeta'}$) and their sum (dot-dashed), for the same integration as Fig. 15.7.

Because Rossby waves propagate away from the stirred region before breaking, the distribution of dissipation is broader than the forcing, resulting in an eastward jet where the stirring is centred, with westward flow on either side.

dissipation has a broader distribution than the forcing, and the sum of the two (the dot-dashed line in the figure) has the same meridional distribution as the zonal flow itself.

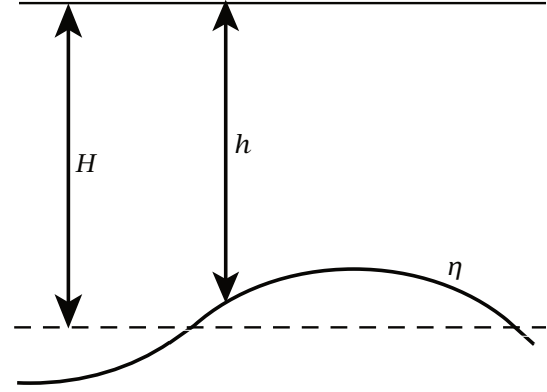
15.2 LAYERED MODELS OF THE MID-LATITUDE CIRCULATION

Let us now extend our barotropic model in the direction of increasing realism. So far we have shown that localized stirring can give rise to an eastward acceleration as Rossby waves propagate away from the disturbance. The source of the disturbance baroclinic instability, and to incorporate these effects we will need to introduce some vertical structure into the problem; we do this by way of layered models of the circulation.⁵ That is, we will consider an atmosphere to consist of one or more *isentropic layers*, as described in section 3.10. The equations describing such layers are virtually isomorphic to the shallow water equations and, for the sake of familiarity and simplicity, and with no loss of essential dynamics, we will use the Boussinesq shallow water equations. We begin with a model of a single layer.

15.2.1 A single-layer model

We first consider a single layer obeying the shallow water equations. We further restrict the flow by supposing that it is constrained by two rigid surfaces: an upper flat lid and a lower, wavy (but stationary) surface (Fig. 15.9). We may imagine the fluid layer to crudely represent the upper troposphere, with the (given) lower wavy surface corresponding to the undulating mid-atmosphere interface of a two-layer model. (This section is in some ways an exercise, and too much realism should not be ascribed to the model.) Thus frictional effects are small in the momentum equation, and in particular there is no Ekman layer and no drag on the velocity field. However, there may be some dissipative effects in the vorticity equation, arising from the cascade of enstrophy to small scales. We also suppose that the Rossby number is small, that the variations in layer thickness are small compared to the mean layer thickness, and that variations in Coriolis parameter are small. Let the initial flow be a uniform zonal current, passing over the wavy lower boundary. The boundary is waviest in mid-latitudes, creating a disturbance from which Rossby waves emanate. Our questions are: (i) How does the wavy interface affect the mean zonal flow? (ii) What if any meridional circulation is induced?

Fig. 15.9 A model atmosphere with an active layer of mean thickness H , local thickness h , and a variable lower surface of height displacement η , lying above a stationary layer with a slightly larger potential density.



Equations of motion

The zonal momentum equation for the layer may be written as

$$\frac{\partial u}{\partial t} - (f + \zeta)v = -\frac{\partial B}{\partial x}, \quad (15.37)$$

where $B = \phi + u^2/2$ is the Bernoulli function for the problem and ϕ is the kinematic pressure, p/ρ_0 . The zonal average of the equation is

$$\frac{\partial \bar{u}}{\partial t} - f\bar{v} = \bar{\zeta}\bar{v} + \overline{\zeta'v'}. \quad (15.38)$$

Note that $\bar{v}$ is wholly ageostrophic ($\bar{v}_g = \overline{\partial_x \psi} = 0$). Now, using $\partial u/\partial x + \partial v/\partial y = 0$, the vorticity flux is related to the momentum flux by

$$v\zeta = -\frac{\partial}{\partial y}(uv) + \frac{1}{2}\frac{\partial}{\partial x}(v^2 - u^2), \quad (15.39)$$

so that, under quasi-geostrophic scaling, (15.38) simplifies to

$$\frac{\partial \bar{u}}{\partial t} - f_0\bar{v} = \overline{\zeta'v'} = -\frac{\partial}{\partial y}\overline{u'v'}. \quad (15.40)$$

Note that although $\bar{v}$ is small and ageostrophic, mass conservation does not demand that it be zero, because the thickness of the layer is not constant [look ahead to (15.43)]. Thus, as $\bar{v}$ is multiplied by the large term f_0 , the term $f_0\bar{v}$ term should be retained (whereas $\bar{\zeta}\bar{v}$ is dropped). If the flow is statistically steady and there are no sources or sinks of momentum (15.40) becomes

$$f_0\bar{v} = \frac{\partial}{\partial y}\overline{u'v'}. \quad (15.41)$$

To complete the model we use the zonally averaged mass conservation equation, namely

$$\frac{\partial \bar{h}}{\partial t} + \frac{\partial}{\partial y}\overline{vh} = 0. \quad (15.42)$$

In the situation here $\partial \bar{h}/\partial t = 0$, because the flow is confined between two rigid surfaces, and so $\partial \overline{vh}/\partial y = 0$. If the mass flux vanishes somewhere, for example at a meridional boundary, it therefore vanishes everywhere and we have

$$\overline{vh} + \overline{v'h'} = 0. \quad (15.43)$$

Using (15.41) and (15.43) gives

$$\frac{1}{f_0} \frac{\partial}{\partial y} \overline{u'v'} + \frac{\overline{v'h'}}{h} = 0, \quad \text{or} \quad \overline{v'\zeta'} - f_0 \frac{\overline{v'h'}}{h} = 0. \quad (15.44a,b)$$

Because thickness variations are assumed to be small we may write this as

$$\overline{v'\zeta'} - f_0 \frac{1}{H} \overline{v'h'} = 0, \quad (15.45)$$

where H is the reference thickness of the layer, which may be taken as its mean thickness. The left-hand side of (15.45) is actually just the potential vorticity flux for this problem — look ahead to (15.51). The potential vorticity equation for the layer is

$$\frac{DQ}{Dt} = \frac{D}{Dt} \left[\frac{\zeta + f}{h} \right] = 0, \quad (15.46)$$

where h is the fluid layer thickness. For small variations in layer thickness and Coriolis parameter this becomes

$$\frac{Dq}{Dt} = \frac{\partial q}{\partial t} + u \frac{\partial q}{\partial x} + v \frac{\partial q}{\partial y} = 0, \quad q = \zeta + \beta y + f_0 \frac{\eta}{H}, \quad (15.47a,b)$$

where $\eta = H - h$ is the height of the lower interface (Fig. 15.9) and this is a function of x and y but not, in this model, time. Using the horizontal non-divergence of the flow, the zonally averaged potential vorticity equation is

$$\frac{\partial \bar{q}}{\partial t} = -\frac{\partial \bar{v} \bar{q}}{\partial y} - \frac{\partial \overline{v'q'}}{\partial y}. \quad (15.48)$$

The term involving $\bar{v}$ is very small, and omitting it and using (15.47a) we obtain the perturbation potential vorticity equation

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = -D' \quad (15.49)$$

where we include a term, D' , to represent dissipative processes. Multiplying by $q' / (\partial \bar{q} / \partial y)$ and zonally averaging we obtain the pseudomomentum equation for this system, namely

$$\frac{\partial \mathcal{P}}{\partial t} = \frac{\partial}{\partial t} \left(\frac{\overline{q'^2}}{2\gamma} \right) = -\overline{v'q'} - \frac{\overline{D'q'}}{\gamma}, \quad (15.50)$$

where $\gamma = \partial \bar{q} / \partial y$. This equation is the equivalent of (15.25), but now for the layered system. In a turbulent fluid we cannot, in general, demand that $D' = 0$, even as the viscosity goes to zero, because of the presence of an enstrophy flux to smaller scales and a concomitant dissipation. But in regions where D' is zero (where there is no wave breaking) then the potential vorticity flux must also be zero in a steady state. For our argument let us assume that, in fact, $D' = 0$.

Using (15.47b), the eddy potential vorticity flux is

$$\overline{v'q'} = \overline{v'\zeta'} + \frac{f_0}{H} \overline{v'\eta'} = \overline{v'\zeta'} - \frac{f_0}{H} \overline{v'h'}, \quad (15.51)$$

where η' is the topography and h' is the layer thickness perturbation. Using this in the zonal momentum equation (15.40) gives

$$\frac{\partial \bar{u}}{\partial t} = \overline{v'q'} + \frac{f_0}{H} \overline{v'h'} + f_0 \bar{v}. \quad (15.52)$$

Informal Summary of the Single-Layer Arguments

The zonally averaged momentum equation is

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v} = \overline{v' \zeta'} = -\frac{\partial \overline{u' v'}}{\partial y}. \quad (\text{SL.1})$$

A region that is the source of Rossby waves will generally be a region where there is momentum flux convergence, where $\partial \overline{u' v'} / \partial y < 0$. In this region $\bar{v}$ will be directed *equatorwards* if $\bar{u}$ is steady, and this flow is the upper branch of the Ferrel Cell. To think about this in terms of potential vorticity, first define the residual meridional velocity by

$$\bar{v}^* = \frac{\overline{v' h'}}{\bar{h}} + \bar{v}. \quad (\text{SL.2})$$

This is proportional to the total meridional mass flux in a layer, and is zero in this one-layer model. The momentum equation is then

$$\frac{\partial \bar{u}}{\partial t} = f_0 \bar{v}^* - \frac{f_0}{\bar{h}} \overline{v' h'} + \overline{v' \zeta'} \quad (\text{SL.3a})$$

$$= f_0 \bar{v}^* + \overline{v' q'}. \quad (\text{SL.3b})$$

using $\overline{v' q'} = \overline{v' \zeta'} - (f_0 / \bar{h}) \overline{v' h'}$, where q is the potential vorticity. The second term on the right-hand side of (SL.3a) is the *form drag* exerted by the topography on the flow, and in a steady state this balances the momentum flux convergence of the Rossby waves. Because of the presence of Rossby waves we expect $\overline{v' \zeta'} > 0$. If there is no dissipation then in steady flow $\overline{v' q'} = -f_0 \bar{v}^* = 0$ and the eddy mass flux is polewards (positive if $f_0 > 0$) and a meridional flow is generated as in Fig. 15.10.

We can infer the potential vorticity flux more directly using the pseudomomentum equation:

$$\frac{\partial \mathcal{P}}{\partial t} = \frac{\partial}{\partial t} \left(\frac{\overline{q'^2}}{2\gamma} \right) = -\overline{v' q'} - \frac{\overline{D' q'}}{\gamma}. \quad (\text{SL.4})$$

where $\gamma = \partial \bar{q} / \partial y$. If dissipation is identically zero, then the potential vorticity flux is zero if the waves are steady. Then, using (SL.3b), there is no acceleration of the zonal flow — an example of the *non-acceleration theorem*.

More generally (and in the real atmosphere) there *will* be some dissipation away from the source region: Rossby waves will preferentially break in critical layers (near where $\bar{u} = c$) and/or more generally Rossby waves will interact producing an enstrophy cascade. These processes give $\overline{D' q'} > 0$ and (for $\gamma > 0$) a negative potential vorticity flux, $\overline{v' q'} < 0$. In these regions, a balance in the momentum equation (SL.3b) can be achieved either by balancing the PV flux with a friction term, as in the barotropic model of section 15.1, or by a Coriolis force on a poleward residual meridional velocity. That is, $f_0 \bar{v}^* \approx -\overline{v' q'} > 0$, so generating a poleward residual flow.

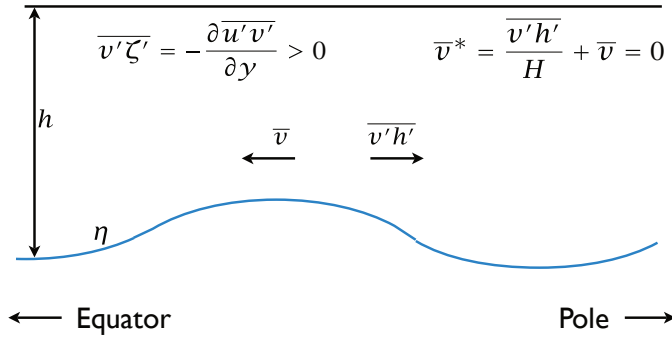


Fig. 15.10 Dynamics of a single layer, with no dissipation. The force on the active layer arises from the form drag exerted by the interface. Vorticity dynamics demands that this produce a converging eddy momentum flux ($\partial_y \overline{u'v'} < 0$), which in turn produces a poleward eddy mass flux ($\overline{v'h'} > 0$), and so an equatorward Eulerian flow.

But the last two terms on the right-hand side constitute the total mass flux so we finally write

$$\frac{\partial \bar{u}}{\partial t} = \overline{v'q'} + f_0 \bar{v}^*, \quad \bar{v}^* = \bar{v} + \frac{\overline{v'h'}}{H}. \quad (15.53a,b)$$

The quantity $\bar{v}^*$ is the *residual circulation* for this problem; it is proportional to the sum of mass flux from the mean flow and the eddies (see section 10.3 for more discussion). Now, $\bar{v}^*$ is proportional to the total meridional mass flux and therefore, because the flow is confined between rigid lids and if there are no sources or sinks of mass, $\bar{v}^* = 0$ everywhere [see (15.43) with $\bar{h} = H$].

Dynamics

When the flow passes over the wavy boundary, Rossby waves will, as in the barotropic case, cause momentum flux to converge in the generation region. If the flow is steady and dissipation-free then from the momentum equation

$$f_0 \bar{v} = \frac{\partial \overline{u'v'}}{\partial y}, \quad (15.54)$$

and, in regions of momentum flux convergence (i.e., where $\partial \overline{u'v'}/\partial y < 0$) the *mean meridional velocity is equatorwards*. Thus, whereas frictional forces balance the vorticity flux in a constant-thickness barotropic model (because in that case $\bar{v} = 0$) in the free atmosphere a meridional circulation may be generated, and this is basis of the equatorial flow in the upward branch of the Ferrel cell. However, this does not imply that the *total* mass flux is equatorwards; in fact, for this single-layer model it must be zero and therefore

$$\overline{v'h'} = -\bar{h}\bar{v} > 0. \quad (15.55)$$

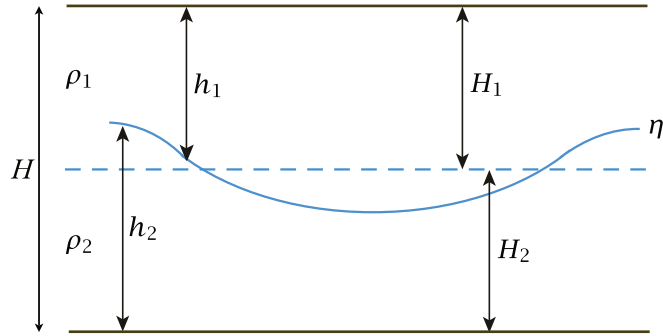
That is, the *eddy mass flux is polewards*, balancing the equatorward mean flow. These balances are sketched in Fig. 15.10.

Another way to arrive at this result is to utilize potential vorticity fluxes directly. For steady, dissipation-free flow the pseudomomentum equation (15.50) reveals that the potential vorticity flux vanishes. Then using (15.53) and noting that $\bar{v}^* = 0$ we have $\partial \bar{u}/\partial t = 0$ — an example of the non-acceleration theorem that steady non-dissipative waves do not induce a change in the zonal momentum. Then, using (15.51) we find

$$\overline{v'\zeta'} = \frac{f_0}{H} \overline{v'h'} \quad (15.56)$$

and using (15.43) we recover (15.54).

Fig. 15.11 An atmosphere with two homogeneous (or isentropic) layers of mean thickness H_1 and H_2 , local thickness h_1 and h_2 , and interface η , contained between two flat, rigid surfaces.



From the point of view of the momentum equation, the momentum flux convergence is balanced by the *form drag* caused by the flow over the wavy boundary. To see this we use (15.53b) to write the zonal momentum equation as

$$\frac{\partial \bar{u}}{\partial t} = f_0 \bar{v} + \overline{v' \zeta'} = f_0 \bar{v}^* - \frac{f_0}{H} \overline{v' h'} + \overline{v' \zeta'}, \quad (15.57)$$

where here $\bar{v}^*$ (but not $\bar{v}$) is zero. The term $-(f_0/H) \overline{v' h'}$ represents the force on the fluid layer coming from the wavy boundary — the form drag, as described in section 3.6. Specifically, the average force per unit area exerted on the layer by the sloping surface is given by

$$F = -f_0 \rho_0 \overline{v' \eta'} = f_0 \rho_0 \overline{v' h'}, \quad (15.58)$$

and dividing by $\rho_0 H$ provides the acceleration on the active fluid layer. The atmosphere also exerts an equal and opposite force on the wavy surface, an effect we consider in the next section. A steady state is achieved, without dissipation, when the form drag is balanced by the eddy momentum flux convergence. From (15.51) or (15.53), this is the same condition that the potential vorticity flux vanishes.

Final remarks on the one-layer model

A summary of the single-layer arguments is given in the shaded box on page 552. In the single-layer model, as in the barotropic model, the zonal flow is proximately driven by eddy fluxes of potential vorticity, and in the model the eddy fluxes must be zero if a steady state is to be achieved. *Vis-à-vis* the real atmosphere this is a little unrealistic, because from the pseudomomentum equation (15.50) we expect these fluxes to be negative, and there is then nothing to balance them in the momentum equation, (15.53), if $\bar{v}^* = 0$. In the real atmosphere, there are effectively sources and sinks in the mass conservation equation that arise from the thermodynamics that allow $\bar{v}^*$ to be non-zero; we then expect $\bar{v}^* > 0$, but to explore this requires a two-layer model, in which the single layer of the one-layer model will correspond to the upper layer of the two-layer model.

15.2.2 A two-layer model

We now consider a model with two active layers, constructing what is probably the simplest model that can capture the dynamics of the mid-latitude tropospheric general circulation without undue approximation. Indeed virtually all of the phenomenology that we associate with the circulation — a thermal wind, mid-latitude surface westerly winds, the Ferrel cell, breaking Rossby waves — is present. A three-layer model introduces no new physics, although a continuously stratified model does lead to some differences of interpretation. The physical model we have in mind is one of two isentropic layers of a compressible ideal gas, virtually equivalent to a two-layer shallow water model illustrated in Fig. 15.11, and our presentation will be in terms of the latter. The upper layer may be

thought of as being forced by an undulating interface between the lower and upper layers, a crude representation of stratification. We will continue to assume that quasi-geostrophic scaling holds; that is, the flow is in near geostrophic balance, variations in layer thickness are small compared to their mean thickness and variations in the Coriolis parameter are small. We also assume that the two fluid layers are held between two flat rigid lids — topography is an unnecessary complication at this stage.

Equations of motion

The equations of motion are those of a two-layer Boussinesq shallow water model confined between two rigid flat surfaces, and readers comfortable with these dynamics (see sections 3.3 and 3.5) may quickly skip through this section, glancing at the boxed equations. The momentum equations of each layer are

$$\frac{D\mathbf{u}_1}{Dt} + \mathbf{f} \times \mathbf{u}_1 = -\nabla\phi_1 \quad (15.59a)$$

$$\frac{D\mathbf{u}_2}{Dt} + \mathbf{f} \times \mathbf{u}_2 = -\nabla\phi_2 - r\mathbf{u}_2, \quad (15.59b)$$

where $\phi_1 = p_T/\rho_0$ and $\phi_2 = p_T/\rho_0 + g'\eta$, with p_T being the pressure at the lid at the top, η the interface displacement (see Fig. 15.11) and $g' = g(\rho_2 - \rho_1)/\rho_0$ the reduced gravity, and we may take $\rho_0 = \rho_1$. We have also included a simple representation of surface drag, $-r\mathbf{u}_2$, in the lowest layer, and r is a constant. We will use a constant value of the Coriolis parameter except where it is differentiated, and on zonal averaging the zonal components of (15.59) become

$$\frac{\partial \bar{u}_1}{\partial t} - f_0 \bar{v}_1 = \overline{v'_1 \zeta'_1} \quad (15.60a)$$

$$\frac{\partial \bar{u}_2}{\partial t} - f_0 \bar{v}_2 = \overline{v'_2 \zeta'_2} - r \bar{u}_2. \quad (15.60b)$$

Geostrophic balance in each layer implies

$$f_0 \mathbf{u}_{g1} = \mathbf{k} \times \nabla\phi_1, \quad f_0 \mathbf{u}_{g2} = \mathbf{k} \times \nabla\phi_1 + g' \mathbf{k} \times \nabla\eta, \quad (15.61a,b)$$

where the subscript g denotes geostrophic. Subtracting one equation from the other gives

$$f_0(\mathbf{u}_1 - \mathbf{u}_2) = -g' \mathbf{k} \times \nabla\eta, \quad (15.62)$$

dropping the subscripts g on $\mathbf{u}$. This is thermal wind balance (or the Margules relation) for this system. A temperature gradient thus corresponds to a slope of the interface height, with the interface sloping upwards toward lower temperatures, analogous to isentropes sloping up toward the pole in the real atmosphere.

The quasi-geostrophic potential vorticity for each layer is

$$q_i = \zeta_i + f - f_0 \frac{h_i}{H_i}, \quad (15.63)$$

where H_i is the reference thickness of each layer, which we take to be its mean thickness. The potential vorticity flux in each layer is then

$$\overline{v'_i q'_i} = \overline{v'_i \zeta'_i} - \frac{f_0}{H_i} \overline{v'_i h'_i}. \quad (15.64)$$

Phenomenology of a Two-layer Mid-latitude Atmosphere

A radiative forcing that heats low latitudes and cools high latitudes will lead to an interface that slopes upward with increasing latitude, and a poleward total mass flux in the upper layer and an equatorward flux in the lower layer. The interface implies a thermal wind shear between the two layers. Neglecting relative vorticity, the potential vorticity gradients in each layer are given by

$$\frac{\partial \bar{q}_1}{\partial y} = \beta - \frac{f_0}{H_1} \frac{\partial \bar{h}_1}{\partial y} > 0 \quad \text{and} \quad \frac{\partial \bar{q}_2}{\partial y} = \beta - \frac{f_0}{H_2} \frac{\partial \bar{h}_2}{\partial y} \lesssim 0. \quad (\text{TL.1})$$

The gradient generally is large and positive in the upper layer and small and negative in the lower layer — the gradient must change sign if there is to be baroclinic instability as we assume to be the case. This baroclinic instability generates eddy fluxes that largely determine the surface winds and the meridional overturning circulation. The zonal momentum equation in each layer is

$$\frac{\partial \bar{u}_1}{\partial t} = f_0 \bar{v}_1 + \overline{v_1' \zeta_1'} = f_0 \bar{v}_1^* + \overline{v_1' q_1'} \quad (\text{TL.2a})$$

$$\frac{\partial \bar{u}_2}{\partial t} = f_0 \bar{v}_2 + \overline{v_2' \zeta_2'} - r \bar{u}_2 = f_0 \bar{v}_2^* + \overline{v_2' q_2'} - r \bar{u}_2. \quad (\text{TL.2b})$$

where

$$\bar{v}_i^* = \bar{v}_i + \frac{\overline{v_i' h_i'}}{H_i} \quad (\text{TL.3})$$

is the residual meridional flow. In steady state the potential vorticity flux will be equatorward in the upper layer and polewards in the lower layer. Because the mass flux in each layer is equal and opposite, the surface (i.e., lower-layer) wind is given by the vertical integral of the vorticity or potential vorticity fluxes, namely

$$r H_2 \bar{u}_2 = H_1 \overline{v_1' q_1'} + H_2 \overline{v_2' q_2'} = H_1 \overline{v_1' \zeta_1'} + H_2 \overline{v_2' \zeta_2'}. \quad (\text{TL.4})$$

The vorticity flux is positive in the upper layer and negative in the lower layer. However, because the potential vorticity gradient in the upper layer is large, this layer is more linear than the lower layer and Rossby waves are better able to transport momentum. The magnitude of the vorticity flux is thus larger in the upper layer than in the lower layer and, using (TL.4), the surface winds are positive (eastwards) in the mid-latitude baroclinic zone (see Fig. 15.14).

To balance the upper-layer mid-latitude momentum flux convergence a meridional overturning circulation (a Ferrel cell) is generated. In a steady state $f_0 \bar{v}_1 = -\overline{v_1' \zeta_1'}$ so that the zonally averaged upper level flow is equatorwards. However, the total mass flux in the upper level is polewards; thus, the equatorward meridional velocity in the upper branch of the Ferrel cell is the consequence of an Eulerian zonal average and does not correspond to a net equatorward mass transport.

In the real atmosphere, the equatorward residual flow occurs close to the surface. Thus, for more realism, we might think of the lower layer as representing a near-surface layer and choose $H_1 \gg H_2$, or even construct a three-layer model with a shallow near-surface layer and two interior layers

Using this in (15.60) gives

$$\frac{\partial \bar{u}_1}{\partial t} = \overline{v'_1 q'_1} + f_0 \bar{v}_1^*, \quad \frac{\partial \bar{u}_2}{\partial t} = \overline{v'_2 q'_2} + f_0 \bar{v}_2^* - r \bar{u}_2, \quad (15.65a,b)$$

where

$$\bar{v}_i^* = \bar{v}_i + \frac{\overline{v'_i h'_i}}{H_i} \quad (15.66)$$

is the meridional component of the residual velocity in each layer, proportional to the *total* meridional mass flux in each layer. These are the transformed Eulerian mean (TEM) forms of the equations, first encountered in section 10.3.

In the barotropic model of section 15.1 the mean meridional velocity vanished at every latitude, a consequence of mass conservation in a single layer between two rigid flat surfaces. In the single-layer model of section 15.2.1 the mean meridional velocity was in general non-zero, but the total meridional mass flux (i.e., the meridional component of the residual velocity) was zero if the domain is bounded laterally by solid walls. In the two-layer model we will allow a transformation of mass from one layer to another, which is the equivalent of heating: a conversion of mass from the lower layer to the upper layer is heating, and conversely for cooling. Thus, heating at low latitudes and cooling at high latitudes leads to the interface sloping upwards toward the pole. In the two-layer model the constraint that mass conservation supplies is that, assuming a statistically steady state, the total poleward mass flux summed over both layers must vanish.

The mass conservation equation for each layer is

$$\frac{\partial h_i}{\partial t} + \nabla \cdot (h_i \mathbf{u}_i) = S_i, \quad (15.67)$$

where S_i is the mass source term and we may suppose that $S_1 + S_2 = 0$ everywhere. A zonal average gives

$$\frac{\partial \bar{h}_i}{\partial t} + \frac{\partial \bar{h}_i \bar{v}_i}{\partial y} = \bar{S}_i \quad (15.68)$$

or, setting $\bar{h}_i = H_i$ and using (15.66),

$$\frac{\partial \bar{h}_i}{\partial t} + H_i \frac{\partial \bar{v}_i^*}{\partial y} = \bar{S}_i. \quad (15.69)$$

The mass source term in these equations is equivalent to heating, and let us suppose that this is such as to provide heating at low latitudes and cooling at high ones. This is equivalent to conversion of an upper-layer mass to a lower-layer mass at high latitudes, and the reverse at low latitudes; such a conversion can only be balanced by a poleward mass flux in the upper layer and an equatorward mass flux in the lower layer (Fig. 15.12). That is to say, an Earth-like radiative forcing between equator and pole implies that *the total mass flux in the upper layer will be polewards*. This is the opposite of the mean meridional circulation of the Ferrel cell shown in Fig. 14.3! What's going on? Before we can answer that, let us manipulate the equations of motion and obtain a couple of useful preliminary results.

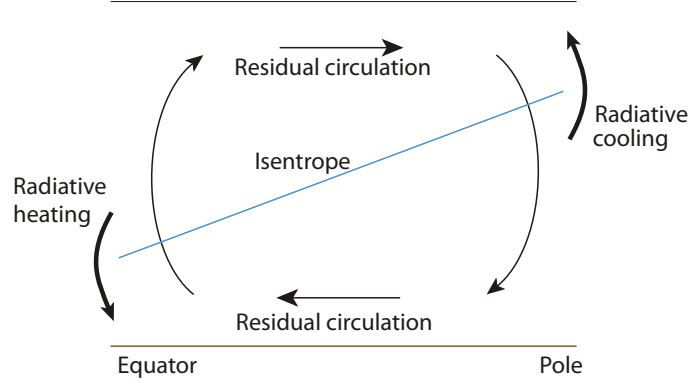
Manipulating the equations

Because the total depth of the fluid is fixed, the mass conservation equations in each layer, (15.67), may each be written as an equation for the interface displacement, namely

$$\frac{\partial \eta}{\partial t} + \nabla \cdot (\eta \mathbf{u}_1) = -S_1, \quad \text{or} \quad \frac{\partial \eta}{\partial t} + \nabla \cdot (\eta \mathbf{u}_2) = S_2, \quad (15.70a,b)$$

Fig. 15.12 Thermodynamics of a two-layer model, with an isentrope (or an interface between two layers) sloping up toward a cold pole, caused by cooling at high latitudes and heating at low.

The heating is balanced by a net mass flux — the meridional overturning circulations. In the tropics this circulation is the Hadley Cell, and is nearly all in the mean flow. In mid-latitudes the circulation is largely in the residual flow, and the Eulerian mean flow (the Ferrell Cell) is in the opposite sense.



where $\eta = H_1 - h_1 = h_2 - H_2$ and $h_1 + h_2 = H_1 + H_2$. Because of the thermal wind equation, (15.62), (15.70a) and (15.70b) are identical: $\mathbf{u}_1 \cdot \nabla \eta = \mathbf{u}_2 \cdot \nabla \eta$ and $S_1 = -S_2$. (If $S_1 \neq -S_2$ the flow would not remain balanced and the thermal wind equation could not be satisfied.) The zonally averaged interface equation may be written as

$$\frac{\partial \bar{\eta}}{\partial t} - H_1 \frac{\partial \bar{v}_1^*}{\partial y} = \bar{S}, \quad \text{or} \quad \frac{\partial \bar{\eta}}{\partial t} + H_2 \frac{\partial \bar{v}_2^*}{\partial y} = \bar{S}, \quad (15.71)$$

where $\bar{S} = -\bar{S}_1 = +\bar{S}_2$, consistent with the mass conservation statement

$$H_1 \bar{v}_1^* + H_2 \bar{v}_2^* = 0, \quad (15.72)$$

which states that the vertically integrated total mass flux vanishes at each latitude.

Now, whereas (15.72) is a kinematic statement about the total mass flux, the dynamics provides a constraint on the *eddy* mass flux in each layer. Using the thermal wind relationship we have

$$f_0 \overline{(v'_1 - v'_2)\eta'} = g' \overline{\frac{\partial \eta'}{\partial x} \eta'} = 0. \quad (15.73)$$

Hence, if the upper and lower surfaces are both flat, we have that

$$\overline{v'_1 h'_1} = -\overline{v'_2 h'_2}, \quad (15.74)$$

and the eddy meridional mass fluxes in each layer are equal and opposite. If the bounding surfaces are not flat, we have

$$\overline{v'_1 \eta'_T} - \overline{v'_1 h'_1} = \overline{v'_2 \eta'_B} + \overline{v'_2 h'_2} \quad (15.75)$$

instead, where η_T and η_B are the topographies at the top and the bottom. Equations (15.74) and (15.75) are *dynamical* results, and not just kinematic ones; they are equivalent to noting that the form drag on one layer due to the interface displacement is equal and opposite to that on the other, namely

$$\overline{v'_1 \eta'} = -[\overline{v'_2 \eta'}], \quad (15.76)$$

where the minus sign inside the square brackets arises because the interface displacement is into layer one but out of layer two.

Using (15.64) and (15.75) the eddy potential vorticity fluxes in the two layers are related by

$$H_1 \overline{v'_1 q'_1} + H_2 \overline{v'_2 q'_2} = H_1 \overline{v'_1 \zeta'_1} + H_2 \overline{v'_2 \zeta'_2} - f_0 \overline{v'_1 \eta'_T} + f_0 \overline{v'_2 \eta'_B}, \quad (15.77)$$

which is the layered version of the continuous result [eq. (10.26) on page 383]

$$\int_B^T \overline{v'q'} dz = \int_B^T \overline{v'\zeta'} dz + \frac{f_0}{N^2} [v'b']_B^T. \quad (15.78)$$

For flat upper and lower surfaces, and using $\overline{v_i \zeta_i} = -\partial \overline{u_i v_i} / \partial y$, (15.77) becomes

$$H_1 \overline{v'_1 q'_1} + H_2 \overline{v'_2 q'_2} = -H_1 \frac{\partial}{\partial y} \overline{u'_1 v'_1} - H_2 \frac{\partial}{\partial y} \overline{u'_2 v'_2}, \quad (15.79)$$

and integrating with respect to y between quiescent latitudes gives

$$\int [H_1 \overline{v'_1 q'_1} + H_2 \overline{v'_2 q'_2}] dy = 0. \quad (15.80)$$

That is, the total meridional flux of potential vorticity must vanish. This is a consequence of the fact that the potential vorticity flux is the divergence of a vector field; in the continuous case

$$\overline{v'q'} = -\frac{\partial \overline{u'v'}}{\partial y} + f_0 \frac{\partial}{\partial z} \frac{\overline{v'b'}}{N^2}, \quad (15.81)$$

which similarly vanishes when integrated over a volume if there are no boundary contributions.

15.2.3 Dynamics of the two-layer model

We now consider the climate, or the time averaged statistics, of our two-layer model. The equations of motion are (15.60) or (15.65), and (15.68) or (15.69). These equations are not closed because of the presence of eddy fluxes, and in this section we make some phenomenological and rather general arguments about how these behave in order to get a sense of the general circulation. In the next section we use a specific closure to address the same problem.

Let us summarize the physical situation. The two layers of our model are confined in the vertical direction between two flat, rigid surfaces, and they are meridionally confined between slippery walls at high and low latitudes (the ‘pole’ and ‘equator’). The circulation is driven thermodynamically by heating at low latitudes and cooling at high ones, which translates to a conversion of layer 1 fluid to layer 2 fluid at high latitudes, and the converse at low latitudes (see Fig. 15.12). This sets up an interface that slopes upwards toward the pole and, by thermal wind, a shear. This situation is baroclinically unstable, and this sets up a field of eddies, most vigorous in mid-latitudes where the temperature gradient (or interface slope) is largest. Three fields encapsulate the dynamics — the lower-layer wind field, the meridional circulation, and the meridional temperature gradient, and our goal is to understand their qualitative structure. We note from the outset that the residual circulation is polewards in the upper layer, equatorwards in the lower layer, and that this is a thermodynamic result, a consequence of heating at low latitudes and cooling at high latitudes.

From (15.65), the steady-state lower-layer wind is given by

$$rH_2 \overline{u}_2 = H_1 \overline{v'_1 q'_1} + H_2 \overline{v'_2 q'_2} = H_1 \overline{v'_1 \zeta'_1} + H_2 \overline{v'_2 \zeta'_2}, \quad (15.82)$$

where the second equality uses (15.79). That is, *the lower-layer wind is determined by the vertical integral of either the vorticity flux or the potential vorticity flux.*

Neglecting contributions due to the mean horizontal shear (which are small if the *beta-Rossby number*, $U/\beta L^2$, is small) the potential vorticity gradient in each layer is given by

$$\frac{\partial \overline{q}_1}{\partial y} = \beta - \frac{f_0}{H_1} \frac{\partial \overline{h}_1}{\partial y} \gg 0 \quad \text{and} \quad \frac{\partial \overline{q}_2}{\partial y} = \beta - \frac{f_0}{H_2} \frac{\partial \overline{h}_2}{\partial y} \leq 0. \quad (15.83a,b)$$

In the upper layer $\partial \bar{h}_1 / \partial y$ is negative so that the total potential vorticity gradient is positive and larger than β itself. In the lower layer $\partial \bar{h}_2 / \partial y$ is positive and indeed if there is to be baroclinic instability it must be as large as β in order for $\partial \bar{q} / \partial y$ to change sign somewhere. Thus, although negative the potential vorticity gradient is much weaker in the lower layer. Thus, Rossby waves (meaning waves that exist because of a background gradient in potential vorticity) will propagate further in the upper layer, and this asymmetry is the key to the production of surface winds.

Now, the potential vorticity flux must be negative (and downgradient) in the upper layer, and there are various ways to see this. One is from the upper-layer momentum equation (15.65a) which in a steady state gives

$$\overline{v_1' q_1'} = -f_0 \bar{v}_1^*. \quad (15.84)$$

Because $\bar{v}_1^*$ is polewards, $f_0 \bar{v}_1^*$ is positive and the potential vorticity flux is negative in both Northern and Southern Hemispheres. Equivalently, in the upper layer the radiative forcing is increasing the potential vorticity gradient between the equator and the pole, so there must be an equatorward potential vorticity flux to compensate. Finally, the perturbation enstrophy or pseudomomentum equations tell us that in a steady state the potential vorticity flux is downgradient (also see section 15.3.2). This is not an independent argument, since it merely says that the enstrophy budget may be balanced through a balance between production proportional to the potential vorticity gradient and the dissipation. For similar reasons we expect the potential vorticity flux to be positive (polewards) in the lower layer.

Now, (15.80) tells us that the latitudinally integrated potential vorticity flux is equal and opposite in the two layers. If the potential vorticity flux in the lower layer were everywhere equal and opposite to that in the upper layer, then using (15.82) there would be no surface wind, in contrast to the observations. In fact, the potential vorticity flux is more uniformly distributed in the upper layer, and this gives rise to the surface wind observed. Let us give a couple of perspectives (on the same argument) as to why this should be so. The argument centres around the fact that the potential vorticity gradient is stronger in the upper layer, as we can see from (15.83).

I. Rossby waves and the vorticity flux

The stronger potential vorticity gradient of the upper layer is better able to support linear Rossby waves than the lower layer. Thus, the vorticity flux in the region of Rossby-wave genesis in mid-latitudes will be large and positive in the upper layer, and small and negative in the lower layer. Thus, there will be more momentum convergence into the source region in the upper layer than in the lower layer, and the vertical integral of the vorticity flux will largely be dominated by that of the upper layer. This is positive in mid-latitudes and, to ensure that its latitudinal integral is zero, it is negative on either side. Using (15.82), a surface wind has the same pattern as the net vorticity flux, and so is eastwards in the mid-latitude source region and westwards on either side.

II. Potential vorticity flux

Rossby waves are generated in the region of baroclinic instability, at approximately the same latitude in both upper and lower layers. However, because the potential vorticity gradient is larger in the upper layer than in the lower layer, Rossby waves are able to propagate more efficiently and breaking and associated dissipation will tend to be further from the source region in the upper layer than in the lower layer. Now, the pseudomomentum equation for each layer is, similar to (15.50) for the one-layer case,

$$\frac{\partial P_i}{\partial t} = \frac{\partial}{\partial t} \left(\frac{\bar{q_i'^2}}{2\gamma_i} \right) = -\overline{v_i' q_i'} - \frac{\overline{D_i' q_i'}}{\gamma_i}, \quad i = 1, 2, \quad (15.85)$$

where γ_i , the potential vorticity gradient, has opposite signs in each layer. In a statistically steady state, the region of strongest dissipation is the region where the potential vorticity

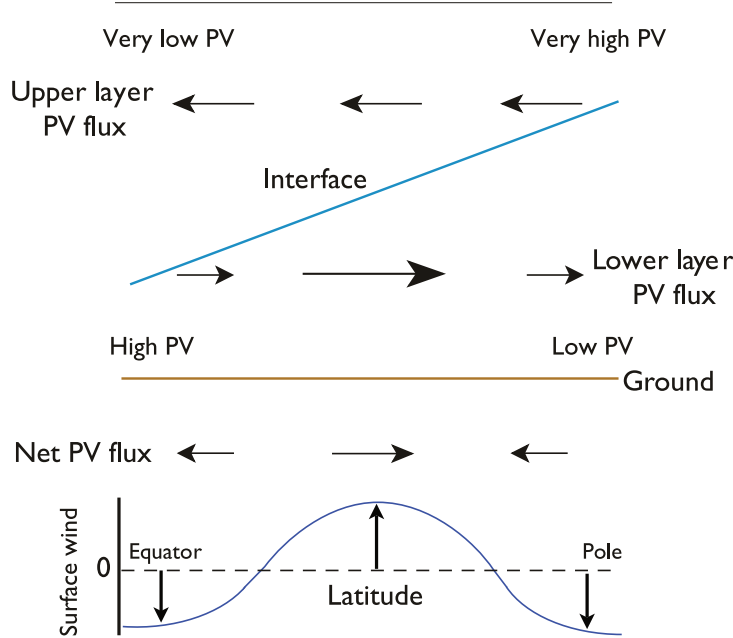


Fig. 15.13 Sketch of the potential vorticity fluxes in a two-layer model. The surface wind is proportional to their vertical integral.

The PV fluxes are negative (positive) in the upper (lower) layer, but are more uniformly distributed at upper levels. The lower panel shows the net (vertically integrated) PV fluxes and the associated surface winds.

flux is largest. In the upper layer, Rossby-wave propagation allows the dissipation region to spread out from the source, whereas in the lower layer the dissipation region will be concentrated near the source. The distribution of the potential vorticity flux then becomes as illustrated in Fig. 15.13. The surface winds, being the vertical integral of the potential vorticity fluxes, are westerly in the baroclinic region and easterly to either side.

Momentum balance and the overturning circulation

From thermodynamic arguments we deduced that the residual circulation is ‘direct’, meaning that warm fluid rises at low latitudes, moves polewards aloft, and returns near the surface. At low latitudes where eddy effects are small the zonally averaged Eulerian circulation circulates in the same way, giving us the Hadley Cell. In mid-latitudes, we may determine the Eulerian circulation from the momentum equation, (15.60). In the upper layer the balance is between the vorticity flux and the Coriolis term, namely

$$f_0 \bar{v}_1 = -\overline{v'_1 \zeta'_1} < 0. \quad (15.86)$$

That is, *the mean Eulerian flow is equatorwards*, and this is the upper branch of the Ferrel cell. Note that the Eulerian circulation is in the opposite sense to the residual circulation.

In the lower layer the vorticity fluxes are weak and the balance is largely between the Coriolis force on the meridional wind and the frictional force on the zonal wind (as in Fig. 14.17). If the upper-layer flow is equatorwards, the lower-layer flow must be polewards by mass conservation, and so the zonal wind is positive (eastwards); that is

$$r \bar{u}_2 \approx f_0 \bar{v}_2 = -\frac{H_1}{H_2} f_0 \bar{v}_1 > 0, \quad (15.87a,b)$$

where the second equality follows by mass conservation of the Eulerian flow.

In terms of the TEM form of the equations, (15.65), the corresponding balances in the centre of the domain are

$$f_0 \bar{v}_1^* = -\overline{v'_1 q'_1} > 0 \quad (15.88a)$$

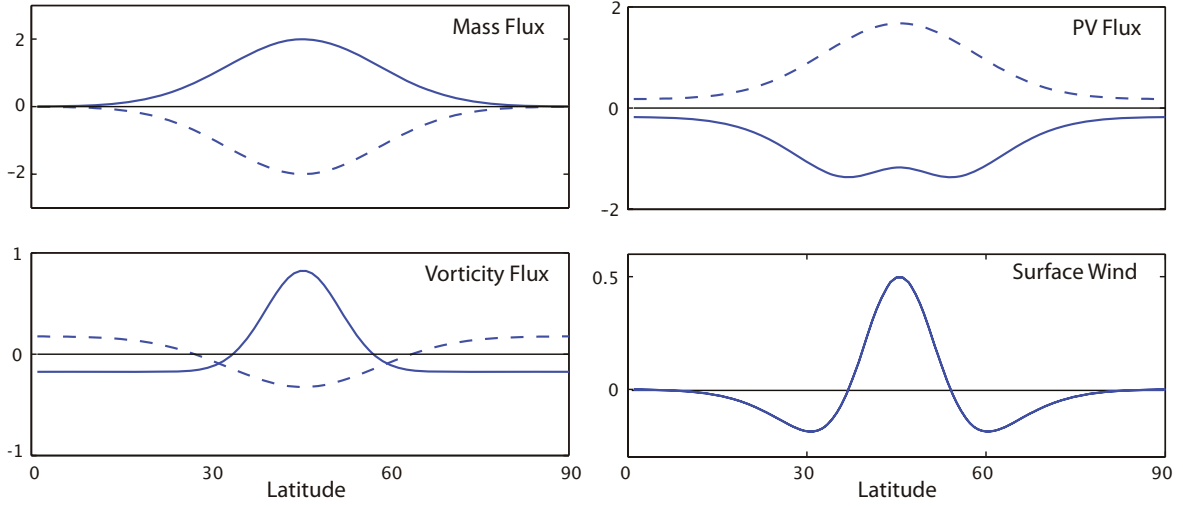


Fig. 15.14 Eddy fluxes in a two-layer model of an atmosphere with a single mid-latitude baroclinic zone. The upper-layer fluxes are solid lines and the lower-layer fluxes are dashed. The bottom-right panel shows the sum of the lower- and upper-layer vorticity fluxes (or, equivalently, the sum of the potential vorticity fluxes), which is proportional (when the surface friction is a linear drag) to the surface wind. The fluxes satisfy the various relationships and integral constraints of section 15.2.2 but are otherwise idealized.

and

$$r\bar{u}_2 = f_0\bar{v}_2^* + \overline{v_2'q_2'} = -f_0\frac{H_1}{H_2}\bar{v}_1^* + \overline{v_2'q_2'} = \frac{H_1}{H_2}\overline{v_1'q_1'} + \overline{v_2'q_2'} > 0, \quad (15.88b)$$

using mass conservation and the fact that the lower-layer potential vorticity fluxes are larger than those of the upper layer. Illustrations of the dynamical balances of the two-layer model are given in Figs. 15.13 and 15.14.

15.3 † EDDY FLUXES AND AN EXAMPLE OF A CLOSED MODEL

The arguments above are heuristic and phenomenological and, although quite plausible they are not wholly systematic. In this section we give a more axiomatic calculation, and make certain closure assumptions that relate the eddy fluxes to the mean fields; specifically, we invoke the diffusion of potential vorticity, and then calculate the zonal winds and meridional circulation. The main purpose of this section is explicitly show that if we do invoke a potential vorticity closure then a complete solution of the flow follows. However, too much credence should not be given to the particular closure we do invoke, for it cannot be rigorously justified.

15.3.1 Equations for a closed model

With quasi-geostrophic scaling, the equations of motion are the momentum equations

$$\frac{\partial \bar{u}_1}{\partial t} = f_0\bar{v}_1^* + \overline{v_1'q_1'} \quad (15.89a)$$

$$\frac{\partial \bar{u}_2}{\partial t} = f_0\bar{v}_2^* + \overline{v_2'q_2'} - r\bar{u}_2 \quad (15.89b)$$

and the mass conservation equation for each layer which may be written as an equation for the interface height,

$$\frac{\partial \bar{\eta}}{\partial t} - H_1 \frac{\partial \bar{v}_1^*}{\partial y} = S, \quad (15.90)$$

where $\eta = H_1 - h_1 = h_2 - H_2$, with the notation of the previous sections. This can be written in terms of $\bar{v}_2^*$ because

$$H_1 \bar{v}_1^* + H_2 \bar{v}_2^* = 0. \quad (15.91)$$

The velocities and thickness of the layers are related by the thermal wind relation

$$f_0(\bar{u}_1 - \bar{u}_2) = g' \frac{\partial \bar{\eta}}{\partial y} = -g' \frac{\partial \bar{h}_1}{\partial y}. \quad (15.92)$$

Using this to eliminate time derivatives between (15.89) and (15.90) reveals that the residual circulation satisfies

$$f_0^2 \frac{H}{H_2} \bar{v}_1^* - H_1 g' \frac{\partial^2 \bar{v}_1^*}{\partial y^2} = +g' \frac{\partial S}{\partial y} - f_0 (\overline{v_1' q_1'} - \overline{v_2' q_2'}) - f_0 r \bar{u}_2, \quad (15.93)$$

where $H = H_1 + H_2$. Thus, *the residual circulation is driven by the potential vorticity fluxes*, plus the diabatic terms. We may derive a similar expression for the Eulerian mean meridional flow, namely

$$f_0^2 \frac{H}{H_2} \bar{v}_1 - H_1 g' \frac{\partial^2 \bar{v}_1}{\partial y^2} = g' \frac{\partial S}{\partial y} + g' \frac{\partial^2}{\partial y^2} \overline{v_1' h_1'} - f_0 (\overline{v_1' \zeta_1'} - \overline{v_2' \zeta_2'}) - f_0 r \bar{u}_2. \quad (15.94)$$

However, the right-hand side now involves *both* the eddy vorticity fluxes and the eddy mass fluxes. The above equations illustrate the natural way in which the potential vorticity fluxes proximately 'drive' the extratropical atmosphere (see also the box on page 566).

Potential vorticity equation

A single prognostic equation for each layer is obtained by eliminating the residual circulation from (15.89) and (15.90), giving

$$\frac{\partial \bar{q}_1}{\partial t} = -\frac{\partial \overline{v_1' q_1'}}{\partial y} + \frac{f_0}{H_1} S \quad (15.95a)$$

$$\frac{\partial \bar{q}_2}{\partial t} = -\frac{\partial \overline{v_2' q_2'}}{\partial y} - \frac{f_0}{H_2} S + r \frac{\partial \bar{u}_2}{\partial y}, \quad (15.95b)$$

where q_i are the quasi-geostrophic potential vorticities of each layer given by

$$\bar{q}_1 = -\frac{\partial \bar{u}_1}{\partial y} + f_0 \frac{\bar{\eta}}{H_1}, \quad \bar{q}_2 = -\frac{\partial \bar{u}_2}{\partial y} - f_0 \frac{\bar{\eta}}{H_2}. \quad (15.96a,b)$$

Closure

If the potential vorticity fluxes can be expressed in terms of the mean fields then (15.95) is a closed set of equations. We can then solve for the potential vorticity in each layer and, using (15.93), for the residual circulation. One simple and rational closure is to assume that potential vorticity flux is transferred downgradient so that

$$\overline{v_i' q_i'} = -K_i \frac{\partial \bar{q}_i}{\partial y}, \quad (15.97)$$

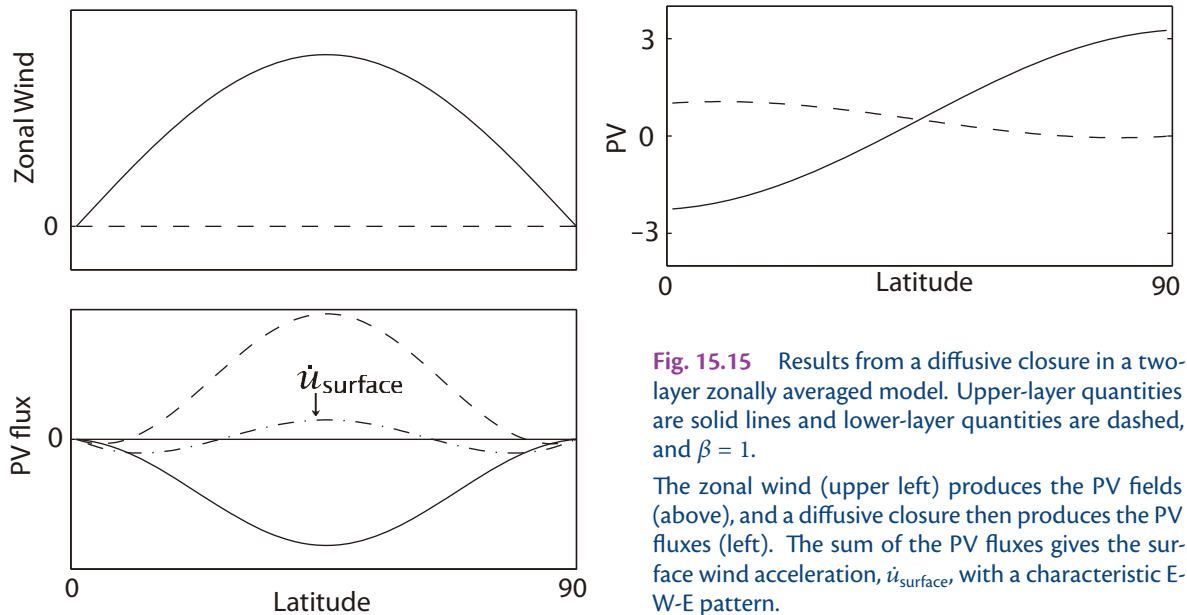


Fig. 15.15 Results from a diffusive closure in a two-layer zonally averaged model. Upper-layer quantities are solid lines and lower-layer quantities are dashed, and $\beta = 1$.

The zonal wind (upper left) produces the PV fields (above), and a diffusive closure then produces the PV fluxes (left). The sum of the PV fluxes gives the surface wind acceleration, $\dot{u}_{\text{surface}}$, with a characteristic E-W-E pattern.

where K_i is an eddy diffusivity, or transfer coefficient, which here is just a scalar quantity.⁶ Note that the model demands a closure of the potential vorticity flux — not momentum, vorticity or the mass flux — and potential vorticity, being a materially conserved variable, is also that field for which a diffusive closure is most applicable.

Such a closure has all of the features and problems associated with diffusive closures discussed in chapter 13, plus some of its own. One such is that a diffusive closure will not automatically respect the kinematic constraint that the volume integral of the potential vorticity flux must vanish, which for the two-layer model is expressed by (15.80). We may *choose* the vertical structure of the diffusivity in such a way that this constraint is satisfied, and in that case the model produces the results illustrated in Fig. 15.15.

The diffusive closure does indeed then produce potential vorticity fluxes similar to the observed westward–eastward–westward surface wind pattern, and a residual circulation of the same sense as in Fig. 15.12, and constitutes perhaps the simplest closed model of the zonally averaged atmospheric circulation. Note that the surface wind is produced by the integral of the potential vorticity flux and, because the fluxes are quite different in the two layers, two layers are needed to produce a realistic pattern of surface wind without oversimplification, as well as to represent the meridional overturning and residual circulations. However, the model is a little ad hoc and the results depend on the structure of the transfer coefficients and the boundary conditions chosen.

15.3.2 ♦ Necessary conditions for a statistically steady state

In linear baroclinic instability problems, a necessary condition for instability (the Charney–Stern–Pedlosky, or CSP, condition) is that the potential vorticity change sign in the interior of the fluid, or that the potential vorticity gradient in the interior has a particular sign with respect to the buoyancy gradient at horizontal bounding surfaces, as discussed in chapters 9 and 10. These conditions do not apply in the statistical steady state of the forced-dissipative problem, but we may derive related conditions that do, although they are not completely general. We will focus on the interior condition and not the boundary conditions, as is appropriate in a layered model, but the argument could be extended to cover boundary issues explicitly.

The linear perturbation potential vorticity equation is

$$\frac{\partial q'}{\partial t} = -\bar{u} \frac{\partial q'}{\partial x} - v' \frac{\partial \bar{q}}{\partial y} - D', \quad (15.98)$$

where D' represents dissipative processes. From this we form the enstrophy equation

$$\frac{1}{2} \frac{\partial \overline{q'^2}}{\partial t} + \overline{D'q'} = -\overline{v'q'} \frac{\partial \bar{q}}{\partial y}, \quad (15.99)$$

where an overbar is a zonal average and we generally assume $\overline{D'q'} > 0$, as with a linear drag on q . In the standard linear problem we take $D = 0$ and then for growing waves ($\partial_t \overline{q'^2} > 0$) the right-hand side must be positive. But the integral of $\overline{v'q'}$ over latitude and height is zero, and thus $v'q'$ must take both positive and negative signs. Hence, for growing waves, $\partial \bar{q} / \partial y$ must also take both positive and negative signs, and we recover the CSP condition that $\partial \bar{q} / \partial y$ must change sign for an instability. (Note that we need not assume that the instabilities have normal form. A very similar argument was given in section 10.6.)

In a statistically steady state the production of variance by the terms on the right-hand side is balanced by a cascade of variance to small scales. The quantity $\overline{v'q'}$ is nonzero, but its spatial integral still vanishes. We therefore have

$$\int \frac{\overline{D'q'}}{\partial \bar{q} / \partial y} dy dz = - \int \overline{v'q'} dy dz = 0. \quad (15.100)$$

Now, $\overline{D'q'} > 0$, and therefore to satisfy the equation $\partial \bar{q} / \partial y$ must change sign somewhere. That is, in a statistically steady state with dissipation, $\partial \bar{q} / \partial y$ must be both positive and negative somewhere in the domain. This is an analogue of the CSP result for wave-mean-flow interaction. Furthermore, if (as we have assumed) the left-hand side of (15.99) is positive everywhere, then eddy flux must be downgradient everywhere.

We can go a little further and obtain a result with the nonlinear terms. The zonally averaged perturbation enstrophy equation is then

$$\frac{1}{2} \frac{\partial \overline{q'^2}}{\partial t} = -\overline{v'q'} \frac{\partial \bar{q}}{\partial y} - \frac{1}{2} \frac{\partial}{\partial y} \overline{v'q'^2} - \overline{D'q'}. \quad (15.101)$$

On integrating in y the third-order term vanishes and we obtain

$$\int \left(\frac{1}{2} \frac{\partial}{\partial t} \overline{q'^2} + \overline{D'q'} \right) dy = - \int \overline{v'q'} \frac{\partial \bar{q}}{\partial y} dy, \quad (15.102)$$

and so, if the left-hand side is positive, the flux must still be downgradient in the integrated sense that

$$\int \overline{v'q'} \frac{\partial \bar{q}}{\partial y} dy < 0. \quad (15.103)$$

Suppose that the flux is *locally* downgradient, meaning that $\partial \bar{q} / \partial y$ and $\overline{v'q'}$ have opposite signs, and in the nonlinear case this is an additional physical assumption. Then, because $\overline{v'q'}$ has both positive and negative values (because its integral is zero), then so must the potential vorticity gradient, $\partial \bar{q} / \partial y$. That is, *when dissipation is present and if the potential vorticity fluxes are downgradient, a statistically steady state can be maintained only if the potential vorticity gradient changes sign somewhere*. In the continuously stratified case, this condition is replaced by ones involving a combination of the interior potential vorticity gradient and the buoyancy gradient at the boundary, the conditions being the same as necessary conditions for instability.

Potential Vorticity Fluxes and the Extratropical Atmosphere

The extratropical circulation of the atmosphere is driven by the differential heating between equator and pole, mediated by fluxes of potential vorticity. Thus, in a layered model we have the following.

- (i) *Zonal winds.* At each level the acceleration of the zonal winds is governed by the potential vorticity fluxes:

$$\frac{\partial \bar{u}_i}{\partial t} = \overline{v'_i q'_i} + f_0 \bar{v}_i^* + F_i. \quad (\text{PV.1})$$

where $\bar{v}_i^*$ is the residual meridional flow and F_i represents friction.

- (ii) *Surface winds.* In steady state, the surface winds are produced by the vertically integrated potential vorticity fluxes:

$$r H_s \bar{u}_s = \sum_i H_i \overline{v'_i q'_i}, \quad (\text{PV.2})$$

where u_s is the surface wind, H_s the thickness of the lowest layer, and r is a frictional coefficient.

- (iii) *Meridional transport.* The total (or residual) meridional transport is, proximately, forced by the potential vorticity fluxes. For example, in a two-layer model

$$f_0^2 \frac{H}{H_2} \bar{v}_1^* - H_1 g' \frac{\partial^2 \bar{v}_1^*}{\partial y^2} = +g' \frac{\partial S}{\partial y} - f_0 (\overline{v'_1 q'_1} - \overline{v'_2 q'_2}) - f_0 (F_1 - F_2), \quad (\text{PV.3})$$

where S is proportional to the diabatic forcing, and this equation holds at all times. In a steady state the momentum equation gives simply

$$f_0 \bar{v}_i^* = -\overline{v'_i q'_i} - F_i. \quad (\text{PV.4})$$

Above the surface layer friction is negligible and the meridional transport responds almost solely to the potential vorticity fluxes.

15.4 A STRATIFIED MODEL AND THE REAL ATMOSPHERE

In the previous section we introduced the effects of stratification by way of a two-layer model. Let us now discuss, albeit rather qualitatively, the dynamics of a continuously stratified model more relevant to the real atmosphere. These dynamics are generally similar to that of the two-layer model, although a number of differences in interpretation do arise. In particular, rather than the potential vorticity flux in the two layers, it is the potential vorticity flux in the interior and the buoyancy flux near the boundary that are the key aspects in producing the mean circulation.

15.4.1 Potential vorticity and its fluxes

The observed zonally averaged potential vorticity field is shown in Fig. 15.16. Of interest to us is the fact that over most of the atmosphere, over most of the year, the potential vorticity gradient is monotonic, with the potential vorticity increasing northwards. (The potential vorticity in the troposphere also increases moving northwards along isentropes.) How, then, can the atmosphere be baroclinically unstable? It is because the surface buoyancy (or temperature) decreases

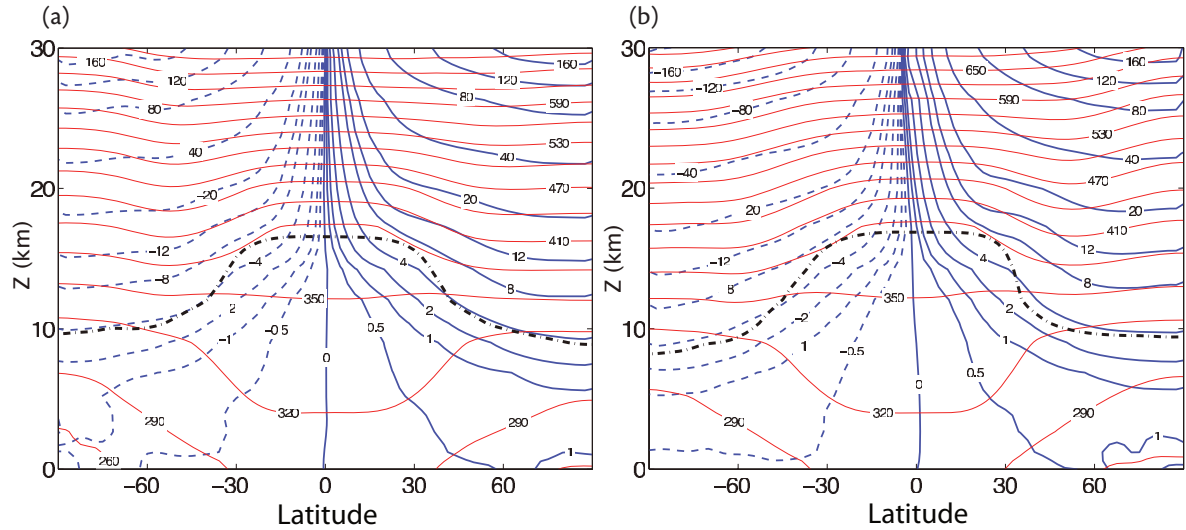


Fig. 15.16 The observed zonally averaged Ertel potential vorticity distribution (dark solid and dashed lines, peaking up at the equator) and the potential temperature (lighter, red lines) for (a) annual mean, (b) December–January–February. Also shown is the position of the thermal tropopause (black, dot-dashed line). The potential vorticity is in ‘PV units’, $1 \text{ PVU} \equiv 1.0 \times 10^{-6} \text{ m}^2 \text{ K s}^{-1} \text{ kg}^{-1}$, and has uneven contour intervals. The vertical coordinate is log pressure, with $Z = -H \log(p/p_R) \text{ km}$, where $p_R = 10^5 \text{ Pa}$ and $H = 7.5 \text{ km}$.

polewards, and thus the atmosphere becomes unstable via the interaction of a surface edge wave with an interior Rossby wave (see the conditions on page 351 or page 404). This baroclinic instability may then excite Rossby waves which propagate meridionally, producing a momentum convergence and westward surface flow, and an associated meridional circulation or Ferrel Cell, much as described in section 15.1. Let us explore these phenomena in a little more detail.

Surface winds

Consider the zonally averaged, continuously stratified momentum equations with quasi-geostrophic scaling,

$$\frac{\partial \bar{u}}{\partial t} = \bar{v}' \zeta' + f_0 \bar{v} + F = \bar{v}' q' + f_0 \bar{v}^* + F, \quad (15.104)$$

where F represents frictional effects and the residual velocity $\bar{v}^*$ is given by

$$\bar{v}^* = -\frac{\partial \psi^*}{\partial z} = \bar{v} - \frac{\partial}{\partial z} \left(\frac{1}{N^2} \overline{v' b'} \right). \quad (15.105)$$

The friction is given by the vertical gradient of a stress, $F = \partial \tau / \partial z$, and at the surface we may parameterize the stress, following (5.210) on page 208, by $\tau = r \bar{u}$ where r is a constant. Then, vertically integrating (15.104) from the surface to the top of the atmosphere (where frictional stresses and the buoyancy flux both vanish) we find, in steady state,

$$r \bar{u}(0) = \langle \bar{v}' \zeta' \rangle = \langle \bar{v}' q' \rangle + \frac{f_0}{N^2} \overline{v' b'}(0), \quad (15.106)$$

where the angle brackets denote a vertical integral and (0) denotes surface values. Thus, the surface winds are determined, analogously to (15.82), by the vertically integrated relative vorticity fluxes, or equivalently by the integral of the interior potential vorticity fluxes and the buoyancy fluxes at

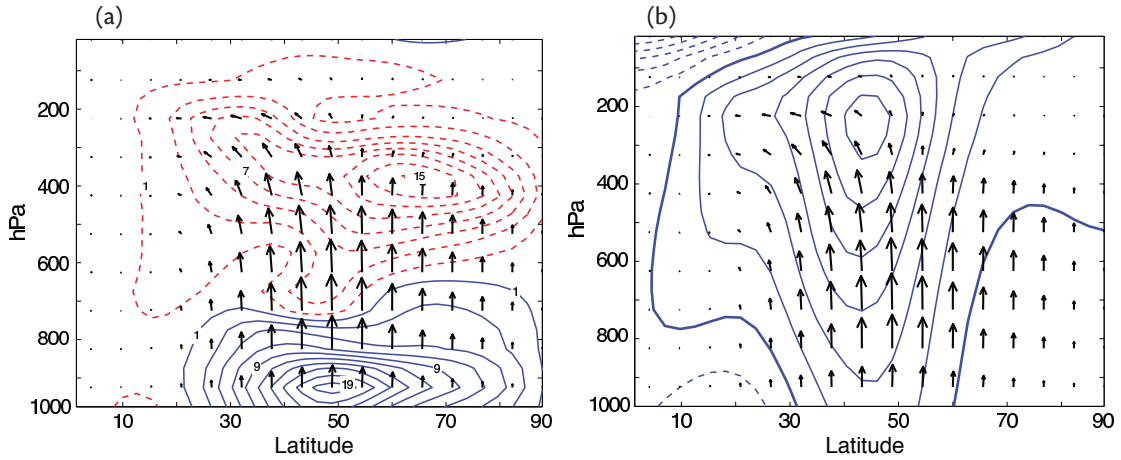


Fig. 15.17 The Eliassen–Palm flux in an idealized primitive equation of the atmosphere. (a) The EP flux (arrows) and its divergence (contours, with intervals of $2 \text{ m s}^{-1}/\text{day}$). The solid contours denote flux divergence, a positive PV flux, and eastward flow acceleration; the dashed contours denote flux convergence and deceleration. (b) The EP flux (arrows) and the time and zonally averaged zonal wind (contours). See the appendix for details of plotting EP fluxes.

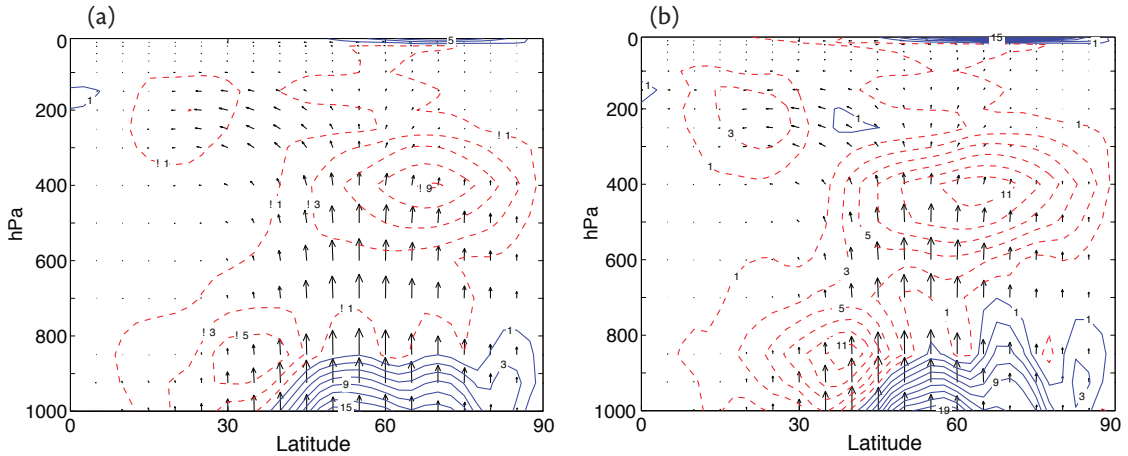


Fig. 15.18 The observed Eliassen–Palm flux (arrows) and its divergence (contours, with intervals of $2 \text{ m s}^{-1}/\text{day}$, zero contour omitted) in the Northern Hemisphere. Solid contours denote divergence, a positive (eastward) torque on the flow, and dashed contours denote convergence, a westward torque. (a) Annual mean, (b) DJF (December–January–February).

the surface. The advantage of the latter representation is that both potential vorticity and buoyancy are materially conserved variables and it may be easier to deduce some properties of their fluxes than of the fluxes of relative vorticity. Compared to the two-layer formulation, the interior fluxes are analogous to those of the upper layer whereas the surface fluxes are analogous to those of the lower layer, especially as the lower layer becomes thin.

Potential vorticity and Eliassen–Palm fluxes

As in section 10.2, the quasi-geostrophic potential vorticity flux may be written as the divergence of the Eliassen–Palm (EP) vector,

$$\overline{v'q'} = \nabla_x \cdot \mathcal{F} \quad (15.107)$$

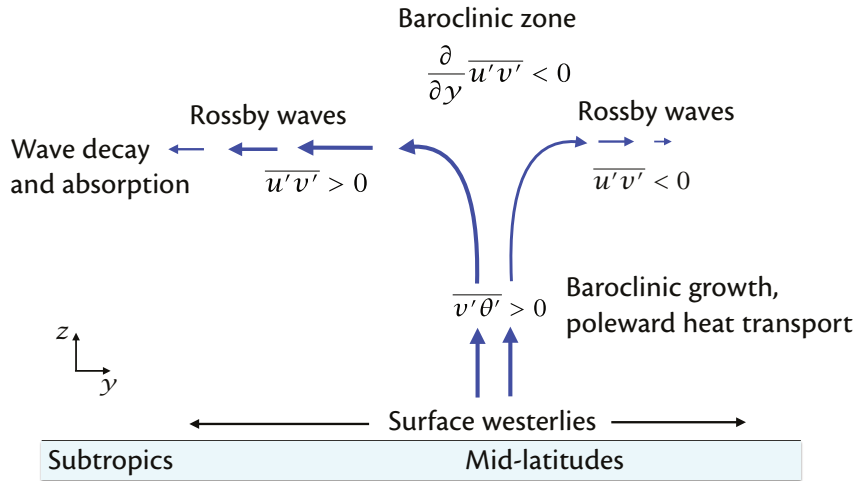


Fig. 15.19 Schematic of Eliassen-Palm fluxes in a baroclinic atmosphere. The vertical component, $\overline{v'\theta'}$ (or $\overline{v'b'}$) occurs primarily during the growing phase of the baroclinic lifecycle and corresponds to a poleward heat flux. The meridional fluxes in the upper atmosphere are associated with Rossby waves, and the vertical integral of the momentum flux divergence gives rise to surface westerlies.

where $\nabla_x \cdot \equiv \mathbf{j} \partial / \partial y + \mathbf{k} \partial / \partial z$ and

$$\mathcal{F} \equiv -\overline{u'v'} \mathbf{j} + \frac{f_0}{N^2} \overline{v'\theta'} \mathbf{k}. \quad (15.108)$$

(See the appendix to this chapter for primitive equation and spherical coordinate versions; in practice the quasi-geostrophic expression qualitatively captures the dominant terms in the primitive equation expressions.) The EP vector as obtained from an idealized primitive equation general circulation model integration is shown in Fig. 15.17, and the EP vector from observations is shown in Fig. 15.18, and both show qualitatively similar properties — a generally upwards-pointing vector in mid-latitude, veering equatorward aloft (and sometimes there is some poleward propagation) as schematically illustrated in Fig. 15.19.

The upward component represents the meridional transfer of heat, and this occurs during the growth phase of the baroclinic lifecycle and is qualitatively captured by linear models — for example, in the Eady problem the EP flux is directed purely vertically (Fig. 10.2), and this aspect resembles the vertical components of Figs. 15.17 and 15.18. But why should the average over a complete baroclinic lifecycle (which Fig. 15.19 schematically represents) even approximately resemble that of the growing phase of the baroclinic lifecycle? After all, the eddies must subsequently decay, and one might imagine that the fluxes would then reverse themselves. In fact, this is not the case: the baroclinic lifecycle is not reversible, because of two effects. First, there is transfer of baroclinic energy to barotropic modes (as described in chapter 12) followed by a barotropic decay. Thus, over the complete cycle, there is no downwelling branch of EP fluxes that would correspond to equatorward heat transfer, and on average the poleward heat transfer (the $\overline{v'\theta'}$ branch) balances the net atmospheric heating. Second, there is an irreversible absorption and decay of the Rossby Waves emanating from the baroclinic zone. These Rossby waves give rise to the lateral component of the EP flux, much as was described in section 15.1), although the wave activity is predominantly in the upper troposphere where the potential vorticity gradient is larger. This propagation is an irreversible process since the Rossby waves break and dissipate some distance from their source; this dissipation breaks the non-acceleration conditions and provides the mean flow acceleration and, consequentially, the observed zonal wind,

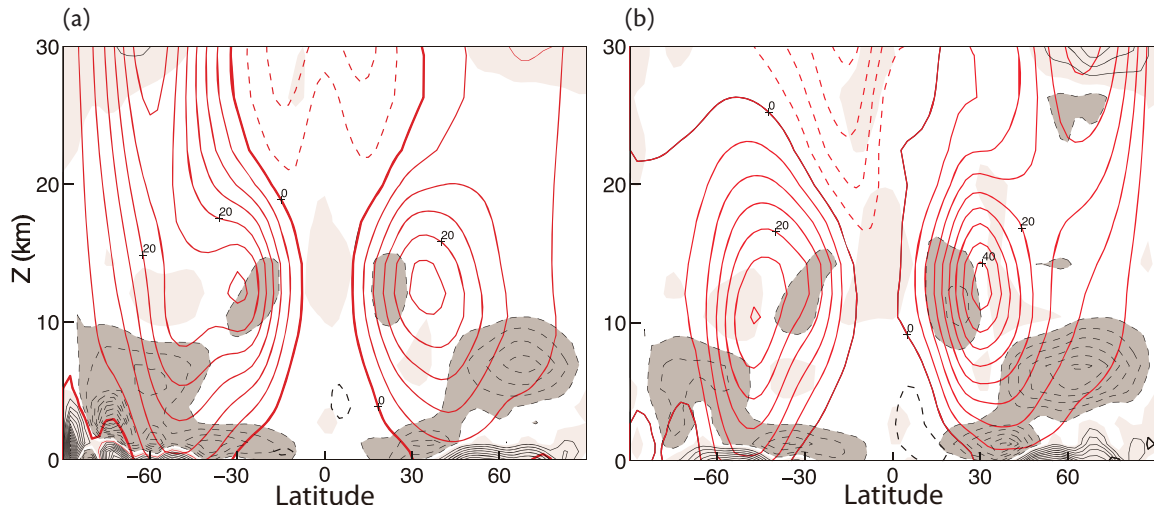


Fig. 15.20 The observed zonally averaged zonal wind (thicker, red, contours, interval 5 m s^{-1}), and the Eliassen–Palm flux divergence (contour interval $2 \text{ m s}^{-1}/\text{day}$, zero contour omitted). Regions of positive EP flux divergence (eastward acceleration) are lightly shaded; regions less than $-2 \text{ m s}^{-1}/\text{day}$ are more darkly shaded. (a) Annual mean, (b) DJF (December–January–February).

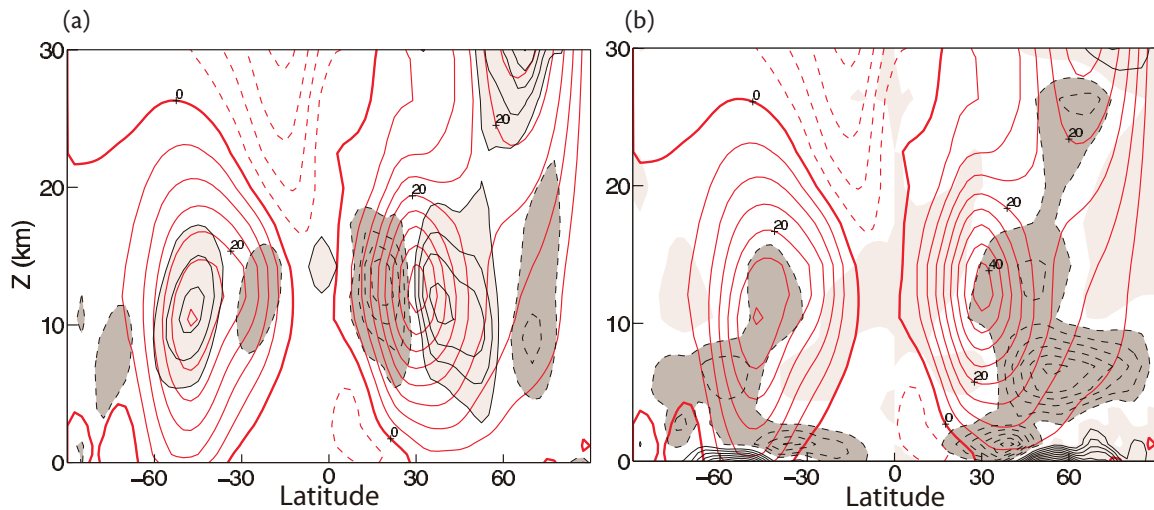


Fig. 15.21 The divergence of the two components of the EP flux (shaded), and the zonally averaged zonal wind (thicker, red, contours) for DJF. (a) The momentum fluxes, $-\partial_y \overline{u'v'}$, contour interval is $1 \text{ m s}^{-1}/\text{day}^{-1}$, light shaded for positive values > 1 , dark shaded for negative values < -1 . (b) The buoyancy flux, $f \partial_z (\overline{v'b'}/N^2)$, with contour interval and shading convention as in Fig. 15.20.

The divergence of the EP flux — that is, the potential vorticity flux — accelerates (or decelerates) the mean flow, as can be seen from (15.104) and Fig. 15.20. Broadly speaking, the EP flux decelerates the flow aloft (where it is balanced by the Coriolis force on the poleward residual flow) but provides an eastward acceleration at the surface (where it is largely balanced by friction). However, the two components of the flux (Fig. 15.21) have rather different effects on the mean flow. The horizontal component acts to extract momentum from the subtropics and deposit it in mid-latitudes, and so accelerate the flow producing a fairly barotropic eastward jet. (It is this component

that gives rise to so-called negative viscosity, in which the eddies transfer momentum upgradient.) The vertical component of the EP flux arises from the meridional buoyancy flux and acts to reduce the intensity of the mid-latitude westerlies aloft, transferring momentum to the surface where it may be balanced by friction, and producing the surface westerlies. Two questions spring to mind.

- (i) Why is the meridional wave-activity propagation predominantly in the upper atmosphere? That is to say, why do the EP vectors only veer laterally above about 400 hPa, and not in the lower atmosphere?
- (ii) Why is the wave-activity propagation (the direction of the EP flux vectors) predominantly equatorward?

As regards item (i), the propagation is mainly in the upper atmosphere because it is here that the potential vorticity gradient is strongest, as can be seen from Fig. 15.16. In the upper troposphere the beta effect is reinforced by the thermodynamic vortex stretching term, whereas the two effects partial cancel in the lower troposphere, an effect that is seen most clearly in a two-layer model (for example, Fig. 15.15). Thus, wave propagation is more efficient in the upper troposphere, whereas the lower troposphere is more nonlinear and so here the enstrophy cascade, and wavebreaking, occur locally and closer to the region of baroclinic instability itself. Regarding item (ii), the proximate reason is that waves predominantly break on the equatorial side of the instability, and this in turn is for two possible reasons. One is that β increases towards the equator, so that linear propagation is more dominant. The other is that there is often a critical layer in the subtropics, where the speed of the waves equals that of the flow itself ($\bar{u} = c$), and here breaking can efficiently occur.

15.4.2 Overturning circulation

The Eulerian overturning circulation (meaning the circulation from a conventional zonal average at constant height) in mid-latitudes is a single indirect cell, the Ferrel Cell, with rising motion at high latitudes and sinking in the subtropics (top panel of Fig. 15.22). The residual circulation is direct and, consistent with the theory of section 10.3.3, resembles closely the thickness-weighted circulation (Fig. 15.22). Qualitatively these features are captured by the two-layer dynamics of section 15.2.2, but the continuously stratified case differs somewhat.

The main difference between the continuous and two-layer cases is that in the former the return flows — both the lower branch of the Ferrel Cell and the equatorial branch of the residual circulation — are not distributed over the lower troposphere, but are confined to a relatively thin layer. In the lower branch of the Ferrel Cell the dynamical balance is between friction and the Coriolis force on the meridional flow, so that its thickness is that of a turbulent Ekman layer and about a kilometre. To understand this better, let us take a quasi-geostrophic perspective. The mean potential vorticity gradient in the free atmosphere is nearly everywhere polewards and the potential vorticity flux is largely downgradient and equatorwards. This means that here the residual circulation is largely polewards, satisfying the balance

$$f\bar{v}^* \approx -\overline{v'q'}. \quad (15.109)$$

In a multi-layer quasi-geostrophic model, with friction acting only in the lowest layer, the circulation is closed by return flow in the lowest layer; thus, as the number of layers increases the return flow is carried in an ever-thinner layer, this becoming a delta-function in the continuous limit, just as in the example of residual flow in the Eady problem (section 10.5). In the real atmosphere, the return flow cannot be confined to a delta-function, but this argument suggests that it will occur close to the surface and this expectation is borne out in the lower panels of Fig. 15.22 and in Fig. 14.16. In fact, much of the equatorial return flow occurs in isentropic layers that have a potential temperature below the mean value at the surface — that is, in cold air outbreaks.

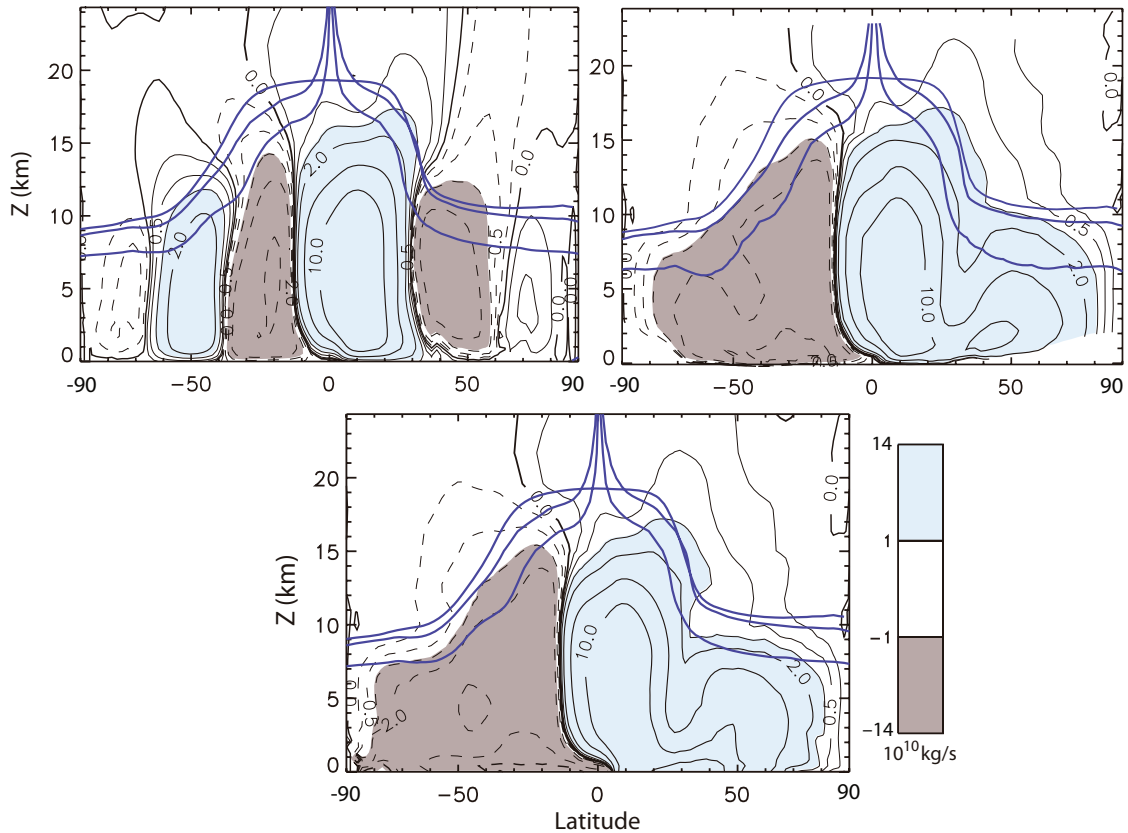


Fig. 15.22 Top left: The observed zonally averaged, Eulerian-mean, streamfunction in northern hemisphere winter (DJF, 1994–1997). Negative contours are dashed, and values greater or less than $10^{10} \text{ kg s}^{-1}$ (10 Sv) are shaded, darker for negative values. The circulation is clockwise around the lighter shading. The three thick solid lines indicate various measures of the tropopause: the two that peak at the equator are iso-lines of potential vorticity, $Q = \pm 1.5, \pm 4$ PV units, and the flatter one is the thermal tropopause. Top right: The thickness-weighted meridional mass streamfunction. After calculation in isentropic coordinates, the streamfunction is projected back on to log-pressure coordinates for display. Bottom: the residual streamfunction calculated from the Eulerian circulation and the eddy fluxes.⁷

15.5 † THE TROPOPAUSE AND THE STRATIFICATION OF THE ATMOSPHERE

In the previous sections we have, by and large, taken the stratification of the atmosphere as given. Let us now explore the physical processes that determine this stratification, beginning with a brief observational overview.

The atmosphere may be divided by stratification into certain distinct regions, illustrated in Fig. 15.23 and Fig. 15.24. The left panel of Fig. 15.23 shows the so-called ‘U.S. standard atmosphere’, a rough average temperature profile and a sometimes-useful standard, as well as actual observed values in the lower atmosphere. In the lower 10 km or so of the atmosphere we have the *troposphere*, a dynamically active region wherein most of the weather and the vast predominance of heat transport occurs. The troposphere is capped by the *tropopause*, above which lies the *stratosphere*, a region of stable stratification extending upwards to about 50 km. (Troposphere means ‘turning sphere’, appropriately so as within it dynamical overturning is prevalent. Stratosphere means ‘layered sphere’, and here there is much less vertical motion.) The stratosphere is capped by the *stratopause*, above which are the mesosphere, thermosphere and exosphere, regions of the

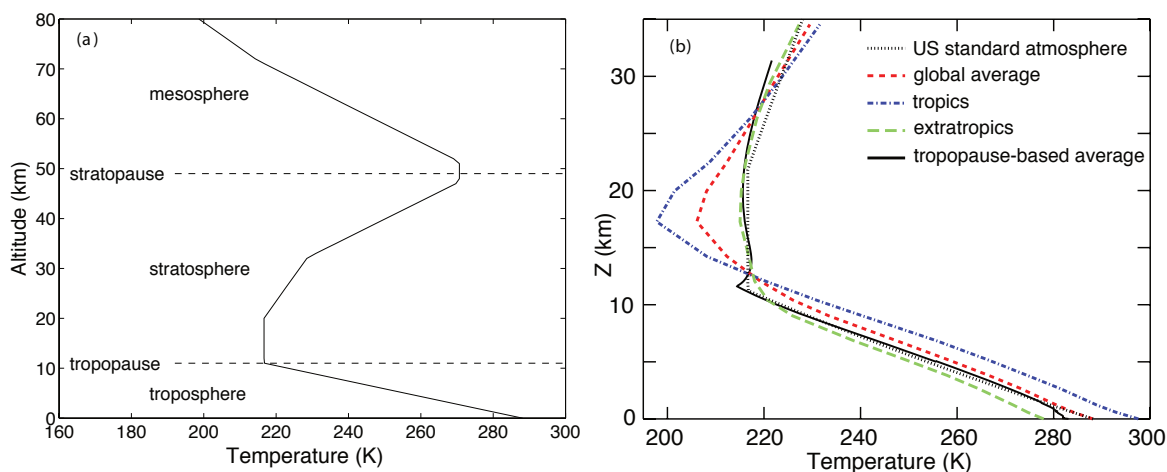


Fig. 15.23 (a) The temperature profile of the ‘U.S. standard atmosphere’, marking the standard regions of the atmosphere below 80 km. In addition to the regions shown, the top of the mesosphere is marked by the mesopause, at about 80 km, above which lies the ‘thermosphere’, in which temperatures rise again into the ‘exosphere’, extending a few thousand kilometres and where the atmospheric temperature ceases to have a useful meaning. (b) Observed, annually averaged profiles of temperature in the atmosphere, where the ordinate is log-pressure. ‘Tropics’ is the average from 30° S to 30° N, and the extratropics is the average over the rest of the globe. The observations are from a reanalysis over 1958–2003 that extends upwards to about 35 km. See text for the meaning of ‘tropopause-based average’.

upper atmosphere that do not concern us here. Our focus will be on the processes that determine the stratification of the lower atmosphere and the height of the tropopause.⁸

In the troposphere temperature generally falls with height, whereas in the stratosphere it increases with height, and this gives rise to a thermal definition of the tropopause:⁹ *The tropopause is the lowest level at which the lapse rate decreases to 2 K km^{-1} or less, provided also that the average lapse rate between this level and all higher levels within 2 km does not exceed 2 K km^{-1} .* At any particular time there might also be a second tropopause: if above the first tropopause the average lapse rate between any level and all higher levels within 1 km exceed 3 K km^{-1} , then a second (higher) tropopause is defined by that same criterion. Finally, such definitions are presumed not to apply if they are satisfied below 500 mb. As so defined, the thermal tropopause typically varies in height from about 16 km at low latitudes to about 8 km near the poles.

The above statements are a practical definition of the tropopause appropriate for today’s climate on Earth — they would not hold on another planet or in a changed climate. Regardless, the tropopause is a distinct boundary separating two differently stratified regions, the troposphere and the stratosphere. The thermal tropopause is marked in Fig. 15.22 and, as we see there and in Fig. 15.16, in the extratropics it is almost parallel to isolines of potential vorticity, and sometimes an isoline of potential vorticity (say $Q = 3$ or 4 PV units) is used as a somewhat ad hoc definition of the extratropical tropopause.

Finally we note that the tropopause appears as a rather sharp feature when viewed instantaneously, although this sharpness is often blurred when time or spatial averages are taken. The solid line in Fig. 15.23, denoted ‘tropopause-based average’, shows the profile obtained when the tropopause height itself is taken as a common reference level, using data from individual radiosonde ascents over the United States.¹⁰

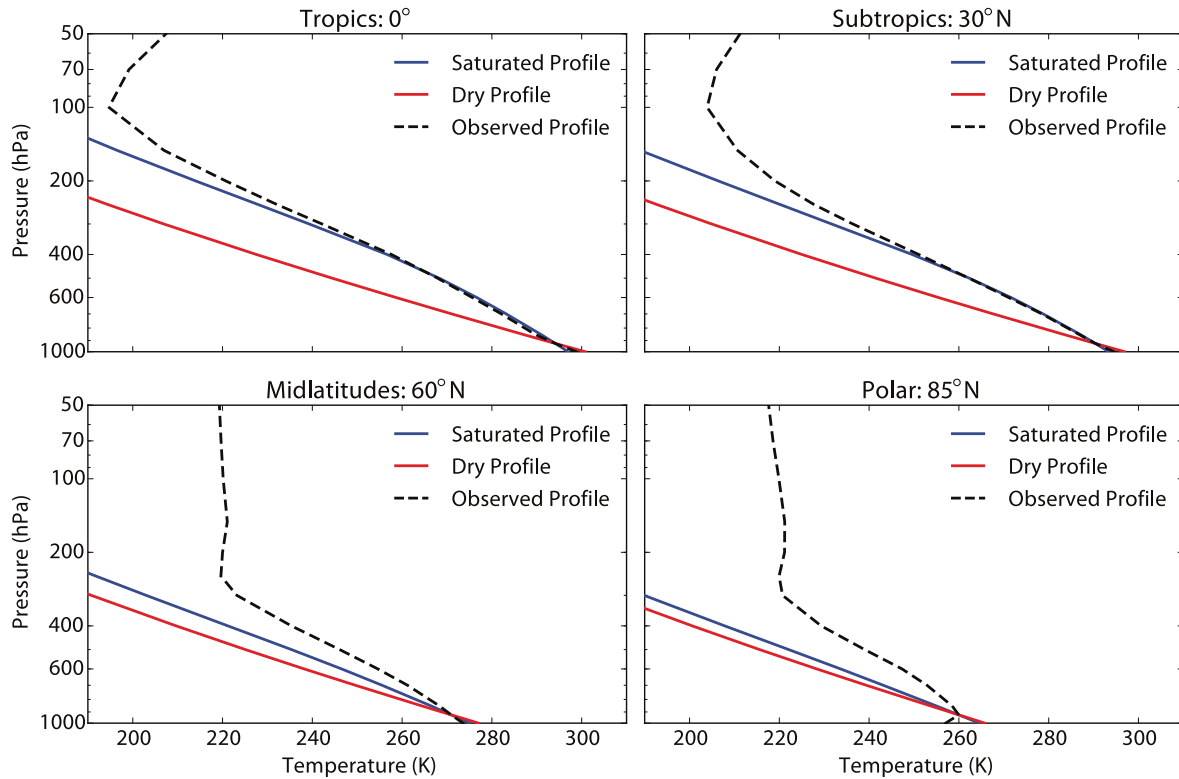


Fig. 15.24 The annually- and zonally-averaged observed (from reanalysis) temperature profiles at various latitudes, along with profiles constructed by integrating the saturated and dry adiabatic lapse rates, chosen to coincide with the observed temperature at 925 hPa, or about 750 m.

15.5.1 Baroclinic eddies and the maintenance of stratification

The atmosphere is largely heated at the surface – the ground is heated by the Sun and the ground heats the atmosphere. If the air near the surface becomes too warm it will become convectively unstable and the resulting convection (here meaning predominantly vertical convection occurring on relatively small scales) will transport heat upwards, stabilizing the temperature profile. If we look at the temperature profiles in Fig. 15.24 we see that in the tropics and subtropics the lapse rate is very close to the saturated adiabatic lapse rate, which is the lapse rate that is neutrally stable to convection in the presence of moisture (as we consider further in chapter 18), suggesting that the stratification is indeed maintained by convection. But in mid- and high latitudes the lapse rate is quite stable with respect to convection, suggesting that some process other than convection is transporting heat upwards. The obvious candidate is baroclinic instability because that has the property of transporting heat both poleward and upward, both of which occur in the atmosphere, and if the vertical transport is efficient enough convection will then not be needed.

The baroclinic eddies do not extend infinitely upwards, and so we expect a boundary — a tropopause — between a dynamical troposphere and a stratosphere that is more nearly in radiative equilibrium. That is to say, the troposphere is that region in which a dynamical distribution of energy takes place; in the tropic this redistribution is by convection, and in mid-latitudes the redistribution is effected by baroclinic eddies. Given this picture, two questions present themselves.

- (i) What determines the stratification (i.e., the lapse rate) of the mid-latitude troposphere?
- (ii) What determines the height of the tropopause?

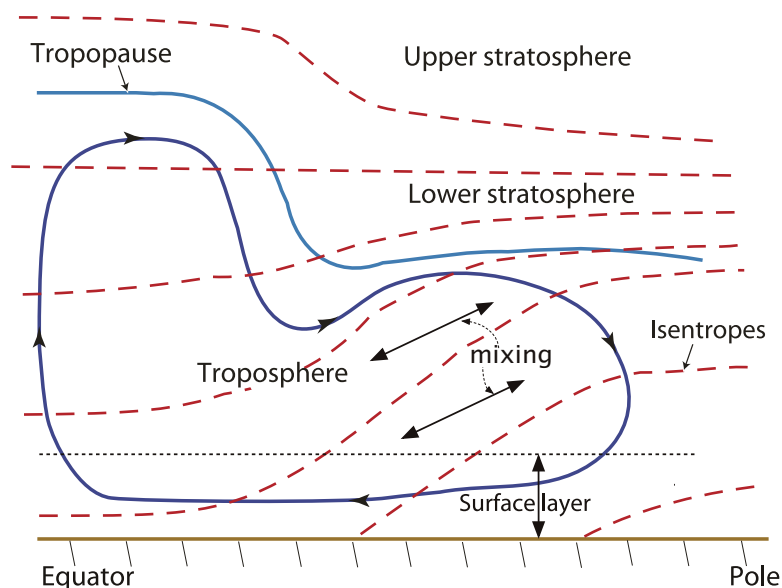


Fig. 15.25 The stratification and residual overturning circulation in the lower atmosphere. The overturning circulation has two distinct parts, a tropical Hadley Cell where the large-scale flow is largely zonally-symmetric and a shallower extratropical cell where the heat and momentum transfer occurs in eddying motion. The equatorward return flow is mostly confined to a shallow surface layer. The lower stratosphere is ventilated by the troposphere along isentropic surfaces, whereas in the upper stratosphere isentropes do not intersect the tropopause. The tropopause is the boundary between the partially mixed troposphere and the near-radiative equilibrium stratosphere. Mixing tends to occur along slopes somewhat shallower than the isentropes.

The full answer to these questions involves both radiation and dynamics. In the following sections, we will focus more on dynamics, and in section 15.6 we will bring radiation into the mix. However, we won't discuss radiation itself until chapter 18, so the reader who desires a full understanding will have to skip back and forth a little.

15.5.2 † Mixing potential vorticity and baroclinic adjustment

Baroclinic eddies grow from small beginnings to finite amplitude, and it is the finite amplitude eddies that stir the atmosphere and define the troposphere. However, they do not forget their linear properties, so let us remind ourselves of one.

A linear height scale

On the β -plane, linear baroclinic instability can produce a height scale that is different from the height of any pre-existing 'lid' or from the density scale height. This height scale is (section 9.9.1)

$$h = \frac{\Lambda f^2}{\beta N^2}, \quad (15.110)$$

where $\Lambda = \partial \bar{u} / \partial z$. That is to say, if $h < H$, where H is the density scale height (or the height of some lid in a Boussinesq model), then the baroclinic eddies will extend upwards to a height h , and this will be the vertical extent of significant heat fluxes. Thus (one might argue), below h the thermal structure is determined by the dynamical effects of baroclinic instability, whereas above h the atmosphere is more nearly in radiative equilibrium. Using $\Lambda = (15 \text{ m s}^{-1}) / (10 \text{ km})$,

$\beta = 1.6 \times 10^{-11} \text{ s}^{-1} \text{ m}^{-1}$, $f = 1 \times 10^{-4} \text{ s}^{-1}$ and $N = 10^{-2} \text{ s}^{-1}$ gives $h \approx 10 \text{ km}$, which approximates the height of the tropopause in mid-latitudes. This is not a prediction because we have taken observed values for the stratification and shear. The amplitude of meridional heat transfer, which determines the meridional temperature gradient and so the shear Λ , is really determined by nonlinear effects.

Mixing

Let us now suppose that baroclinic eddies grow to finite amplitude and mix potential vorticity (see section 13.5 for a general discussion of potential vorticity mixing). Such mixing will try to homogenize potential vorticity, or equivalently to expel potential vorticity gradients to a boundary, and if so the (extratropical) tropopause will occur at an isoline of potential vorticity, and be marked by a near-discontinuity in the potential vorticity distribution. Because $Q \approx (f/\rho)\partial\theta/\partial z$, the tropopause would also correspond to a discontinuity in stratification. These ideas are, at least qualitatively, in accord with observations: the potential vorticity distribution in the troposphere is somewhat more homogeneous than in the stratosphere — see Fig. 15.16, noting the unequal contour intervals of PV, and there is a near-discontinuity in stratification (by definition) at the tropopause. What are the effects of that mixing?

Potential vorticity mixing will occur only so far as needed in order to stabilize the mean flow. We also know that the meridional surface temperature gradient remains negative, so that if the flow is stabilized it must involve changes in the interior potential vorticity distribution. If horizontal scales are sufficiently large we may neglect the contribution of relative vorticity and the quasi-geostrophic potential vorticity and its gradient become

$$q = \beta y + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right) \quad \text{and} \quad \frac{\partial \bar{q}}{\partial y} = \beta - \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \Lambda \right). \quad (15.111)$$

We might hypothesize that the vertical extent to which mixing occurs is just sufficient to make the two terms on the right-hand side a similar size in order that the potential vorticity can become homogeneous, or that it can change sign and be just unstable. This gives $\beta \sim f_0^2 \Lambda / (N^2 H_T)$, where H_T is the vertical extent of the instability and, we assume, the height of the tropopause; that is

$$H_T \sim \frac{f_0^2 \Lambda}{N^2 \beta}. \quad (15.112)$$

Put another way, in this model *the troposphere extends vertically as far as baroclinic waves can alter the potential vorticity from its planetary value*. (A similar depth scale occurs when evaluating the depth of the wind's influence in an ocean circulation model, section 20.2.1.) If the lapse rate (and the shear) is known, this height determines the tropopause. Note the similarity of (15.112) to (15.110) — a similarity that is unsurprising given that we are constructing a height scale from a shear, f , β and N^2 using potential vorticity dynamics in both cases. We still do not have a prediction for H_T , because the stratification N^2 is unknown.

Baroclinic adjustment

Equilibration by potential vorticity mixing is closely related to a process known as *baroclinic adjustment*, by analogy with convective adjustment. The essential idea is that baroclinic eddies are sufficiently efficient that they can stabilize the mean flow by transferring heat poleward and upward until the necessary condition for instability (the Charney–Stern–Pedlosky condition, in so far as the flow is quasi-geostrophic and inviscid) is barely satisfied, and the atmosphere is marginally supercritical to baroclinic instability. The adjustment might conceivably proceed predominantly by changes in the static stability, N^2 , or predominantly by changes in the horizontal temperature gradient. The arguments for such an adjustment process are a little ad hoc, and the final state to which the atmosphere putatively adjusts is not well-defined since there is no critical shear for instability

in a continuously stratified model. In a two-level model the critical shear for instability is given by (9.121), which may be put in the form

$$\Lambda_{\text{crit}} = \frac{\beta N^2 H}{f_0^2}, \quad (15.113)$$

which is of the same form as (15.112). In a continuously stratified model, (15.113) may be regarded as a value of the shear above which the baroclinic eddies become deep (all eddies are deep in the two-level model) and transport heat efficiently.

Allowing the parameters f and β to vary with latitude, and using the thermal wind relation, (15.112) or (15.113) may be written in the form

$$H_T = - \left(\frac{\partial_y \theta}{\partial_z \theta} \right) \frac{f}{\beta} = s \frac{f}{\beta} \sim sa \quad (15.114)$$

where $s = -(\partial_y \theta)/(\partial_z \theta)$ is the isentropic slope and a is the Earth's radius. This equation suggests that the isentropic slope is roughly such that isentropes extend from the surface in the subtropics to the tropopause at the poles, and this is more-or-less true in the present atmosphere (Fig. 15.16). That is to say, if adjustment-like arguments do hold, the isentropic slope will remain roughly constant even as other parameters change — for example, the horizontal temperature gradient may change with season but the stratification changes such as to keep s the same. The numerical and observational evidence for such an adjustment is mixed, although it is plausible that a weaker version may hold in which potential vorticity is imperfectly homogenized and (15.112) provides a plausible scaling, but not a precise prediction.¹¹

Even if the above scalings were correct, they provide a closed prediction of neither the tropopause height nor stratification; they only provide a relation between the two. We will close the problem in section 15.6, but first we consider another point of view.

15.5.3 † Extratropical Convection and the Ventilated Troposphere

A point of view that differs *qualitatively* from the potential vorticity one is to suppose that the mid-latitude tropospheric lapse rate is maintained by convection, the convection occurring predominantly in the warm sector of mature baroclinic waves.¹² In reality such convection will involve moisture, but first consider a dry atmosphere. In a given baroclinic zone the minimum potential temperature difference between the tropopause and the surface, $\Delta_z \theta = \theta_T - \theta_S$ is approximately zero: if the tropopause were colder than this the column would be convectively unstable, and the difference would become zero. The essential assumption that we make is that within a baroclinic zone there generally *does* exist a region that is convectively unstable, and that convection then ensues with sufficient efficiency to partially fill the troposphere with air with that surface value of potential temperature. The process differs from convection in the tropics because it is organized by baroclinic waves and, if we imagine a succession of baroclinic waves around a latitude band the mean value of $\Delta_z \theta$ will be approximately, we assume, its minimum (zero) plus a fraction of its standard deviation. The standard deviation in turn is a consequence of the pre-existing meridional temperature gradient and meridional advection across that gradient, and therefore

$$\text{standard deviation}(\Delta_z \theta) \propto \Delta_y \theta, \quad (15.115)$$

where the term on the right-hand side is the meridional temperature difference at the surface across the baroclinic zone. The mean potential temperature difference between the surface and tropopause is then simply proportional to the meridional temperature gradient at that latitude, with an undetermined constant of proportionality and so

$$\Delta_z \theta \propto \Delta_y \theta. \quad (15.116)$$

Finally, if moisture is present (as it is!) the vertical potential temperature difference should be replaced by the equivalent potential temperature difference — the equivalent potential temperature being essentially the potential temperature achieved when all the water vapour in a parcel of air condenses and the latent heat of condensation is used to heat the parcel (section 18.3.2).

The physical hypothesis is essentially that within a baroclinic wave the advection of warm air into a cold region *necessarily* leads to convection, and that this convection then efficiently fills the available volume, to the extent possible, with the warmest possible fluid. Oceanographers will find this a comfortable concept, for they are used to the notion of convection filling the domain with the densest available fluid (densest in the oceanic case because oceanic convection usually occurs from the top, with cold, dense water sinking). However, unlike the ocean in which the bottom of the container limits the volume of dense water that can be made, here it is the tropopause that provides the upper lid; the height of this is determined by the dynamics itself, in conjunction with the requirement of radiative balance. Thus, the baroclinic zone becomes, in oceanographic parlance, *ventilated* by the warmest air at the surface. However, the entire baroclinic zone does not completely fill with this warm air because the convection is maintained by a meridional temperature gradient and it is necessarily intermittent: baroclinic instability would shut off if the entire baroclinic zone were filled with homogeneous warm fluid, and the zone would then meridionally restratify. It is this continual maintenance of variance that leads to (15.116).

The ultimate consequence of these convection arguments is that the moist isentropic slope is proportional to that slope which would take an isentrope at the surface to the tropopause across a baroclinic zone. On Earth, this is similar to the potential vorticity mixing ideas, which suggest that the isentropic slope is proportional to the slope that goes from the ground to the tropopause over a horizontal scale f/β , the equator-to-pole scale, but the reasons are different.

15.6 † A MODEL FOR BOTH STRATIFICATION AND TROPOPAUSE HEIGHT

In the previous section we discussed various ideas regarding the effects of baroclinic eddies on the midlatitude stratification. The discussion was incomplete on two grounds. First, the ideas themselves are heuristic. Second, even if true, they don't give us a complete picture; they just give us a relation between stratification and tropopause height. A second relation is needed, and this is provided by the radiative-dynamical arguments of sections 18.6 and 18.C, which the reader may now read with profit. In those sections we showed that, if the lapse rate in the lower atmosphere is assumed constant, it can extend upward only to a certain height in order to maintain an overall radiative balance — that is, in order that the outgoing longwave radiation equal the incoming solar radiation. Specifically, we obtained approximate analytic expression for the tropopause height, H_T , (18.172), namely

$$8\Gamma H_T^2 - C H_T T_T - \tau_s H_a T_T = 0. \quad (15.117)$$

where $C = 2 \log 2 \approx 1.39$. In this expression Γ is the lapse rate, $-\partial T/\partial z$, T_T is the temperature at the tropopause, τ_s is the surface optical depth and H_a is the scale height of the main infra-red absorber. The radiative parameters are determined by the composition of the atmosphere and we regard them as given, and the temperature at the tropopause is, to a decent approximation, given by radiative balance, namely $2\sigma T_T^4 \approx S_{net}$ where S_{net} is the net incoming solar radiation. (If there is a lateral convergence of heat in the atmosphere then S_{net} should be modified appropriately. Alternatively, we regard (15.117) as a theory for the global mean tropopause height.) The main assumptions leading to (15.117) are that the atmosphere is radiatively grey in the infra-red and that the lapse rate is uniform up to H_T .

In mid-latitudes, (15.117) does not provide a closed prediction for tropopause height because the lapse rate is not known (whereas in the tropics we might take the lapse rate to be determined by convection). However, if we combine (15.117) with the ideas about baroclinic transport we obtain a closed model. We do not have a good model of baroclinic transport, but for the purposes

The Stratification of the Troposphere and Stratosphere

- The troposphere is that region of the atmosphere in which a dynamical redistribution of heat occurs. The stratosphere above it more nearly in radiative equilibrium (although in winter in the lower stratosphere, radiative equilibrium is not a very good approximation). The tropopause is the change in stratification between the two regions.
- The tropospheric lapse rate and the height of the tropopause are determined by a combination of dynamics and radiation. The tropopause temperature is, to a fair approximation, determined by the requirements of radiative balance and the height of the tropopause then follows if the lapse rate is known, and vice versa. Dynamical and radiative consistency leads to a state in which the tropopause height is the height to which dynamical effects extend and equal to the height needed for radiative balance.
- In Earth's atmosphere, there are two dynamical processes important for the maintenance of stratification.
 - (i) *Convection, especially moist convection.* It is generally thought that moist convection plays the dominant role in determining tropical stratification, leading to a lapse rate that is approximately neutral to moist convection (discussed more in chapter 18).
 - (ii) *Baroclinic eddies.* In mid-latitudes baroclinic eddies transport heat polewards and upwards, so determining in the mid-latitude meridional temperature gradient and stratification. Various theories for this have been proposed, but none are accepted as having wide applicability and accuracy. In one incarnation ('baroclinic adjustment') baroclinic activity is so efficient that the atmosphere becomes only marginally supercritical to baroclinic instability, a process related to homogenization of potential vorticity. In a variation on this theme, potential vorticity and surface potential temperature are diffused downgradient, but potential vorticity is not necessarily homogenized. A complete theory would require estimates of the structure and magnitude of the eddy diffusivities.

In the extratropics moist convection is less dominant, but convection in the warm sector of baroclinic eddies may act to produce a stratification that is related to the meridional temperature difference across a baroclinic zone.

- If we have a theory for the stratification produced by baroclinic eddies then we can use that theory in conjunction with radiative calculations to make predictions of the tropopause height, as in section 15.6. For example, various dynamical arguments suggest the importance of the height scale $H \sim (f^2 \Lambda)/(N^2 \beta)$, where we associate H with the tropopause height. Radiative arguments also lead to an expression for tropopause height in terms of the stratification. Using the two together gives predictions both tropopause height and the stratification.
- In the mid-stratosphere, ozone absorbs solar radiation and this gives rise to an increase in temperature with height from the lower to the mid stratosphere. This effect makes the stratosphere still more stable to baroclinic instability and may sharpen the tropopause, but it is not the root cause of the tropopause.

here let us suppose that the isentropic slope is such that (15.114) roughly holds. Noting that, in a hydrostatic atmosphere, $(T/\theta) \partial\theta/\partial z = \Gamma_d - \Gamma$ where Γ_d is the dry adiabatic lapse rate, we rewrite (15.114) as

$$\Gamma = \Gamma_d + \left(\frac{\partial_y T}{H_T} \right) \frac{f}{\beta}. \quad (15.118)$$

If we take $H_T \approx 10$ km and $\partial T/\partial y \approx 40$ K/(7000 km) we obtain not-unreasonable values for Γ of around 4 K km^{-1} .

If we suppose that both (15.117) and (15.118) are true we obtain predictions for the height of the tropopause in mid-latitudes and the tropospheric stratification, as a function of the horizontal temperature gradient and the radiative properties of the atmosphere. In doing so we are saying the following, in rather general terms. *The height of the tropopause is the height to which baroclinic eddies extend, and it is also the height demanded by radiative balance.* The simultaneous satisfaction of these two conditions provides predictions for both the tropopause height and the lapse rate. The theory is not complete, because the lateral temperature gradient is not predicted, and this is a function of the efficiency of baroclinic eddies. It is also the case that we have chosen a particular closure for baroclinic eddies that may not have general applicability. If we had a better theory we would use it instead, and the reader is invited to explore other options.

15.6.1 Some calculations

If we combine (15.118) and (15.117) we obtain

$$8\Gamma_d H_T^2 - H_T \left[C T_T - \left(\frac{8f\partial_y T}{\beta} \right) \right] - \tau_s H_a T_T = 0, \quad (15.119)$$

the solution of which is

$$H_T = \frac{1}{16\Gamma_d} \left(A + \sqrt{A^2 + 32\Gamma_d \tau_s H_a T_T} \right), \quad (15.120)$$

where $A = C T_T - 8f\partial_y T/\beta$. The solutions are plotted in Fig. 15.26, but it may help interpret these if we consider the optically thin and thick cases, namely

$$\text{Optically thick:} \quad \tau_s H_a \gg \frac{A^2}{32\Gamma_d T_T} \quad \text{whence} \quad H_T \approx \sqrt{\frac{T_T \tau_s H_a}{8\Gamma_d}}, \quad (15.121)$$

$$\text{Optically thin:} \quad \tau_s H_a \ll \frac{A^2}{32\Gamma_d T_T} \quad \text{whence} \quad H_T \approx \frac{C T_T - 8\partial_y T f/\beta}{8\Gamma_d}. \quad (15.122)$$

The optically thin case shows the effects of baroclinic eddies. If the horizontal heat transport is very strong and $\partial T/\partial y$ is small then the lapse rate will approach the dry adiabat and the tropopause height will diminish, and conversely for a weak heat transport. Solutions to the full problem show that behaviour, as well as a slight increase in height with optical depth, because with bigger optical depth the effective emitting level is higher, and the tropopause height tracks the emitting level.

The solutions are qualitatively reasonable, but we repeat some caveats. On the radiative side the atmosphere is not grey, although that is a quantitative rather than qualitative deficiency. On the dynamical side we have used an ansatz — that the isentropic slopes are fixed — that we believe not be generally true and even this is not a complete treatment of baroclinic activity, since it does not give us the horizontal temperature gradient. Notwithstanding these points, the calculation does capture a fundamental principle, that the tropopause height and stratification are determined by the joint effects of dynamics and radiation. Finally, the stratosphere is quite stably stratified and this strongly inhibits baroclinic instability. Thus, once formed by the mechanisms above, the tropopause provides a natural lid on baroclinic instability, keeping it tropospherically confined.

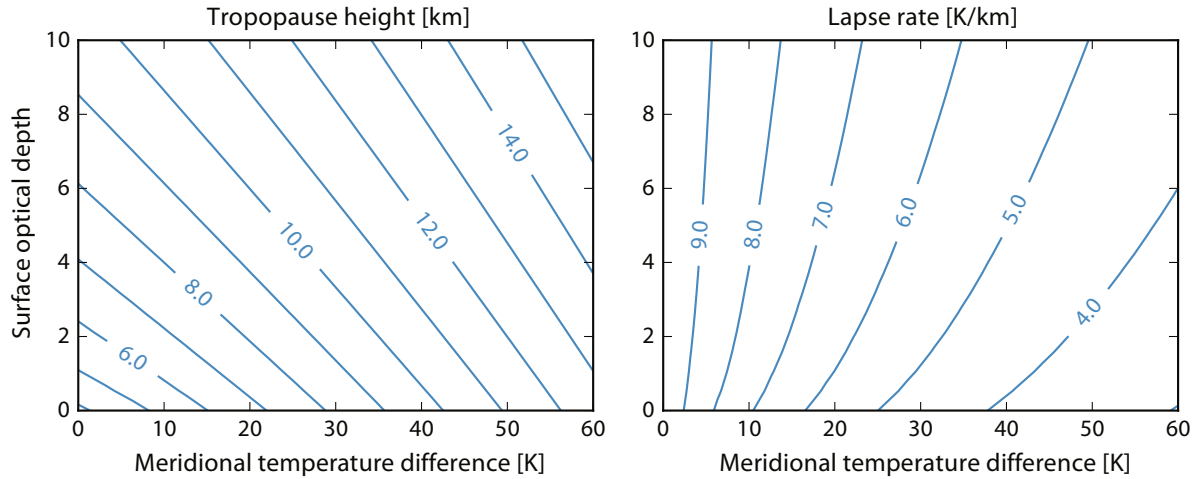


Fig. 15.26 Left: Contours of tropopause height as a function of the temperature difference between the subtropics and pole, and the optical depth. Right: Corresponding lapse rate, $-\partial T/\partial y$. The calculations use (15.120) and (15.118) with $H_a = 2$ km and a tropopause temperature of 215 K.

15.A APPENDIX: TEM FOR THE PRIMITIVE EQUATIONS IN SPHERICAL COORDINATES

In spherical and log-pressure coordinates let us define the residual streamfunction for the ideal-gas primitive equations by¹³

$$\psi^* \equiv \psi + \frac{\overline{v'\theta'}}{\partial_Z \bar{\theta}}. \quad (15.123)$$

Here, an overbar denotes a conventional (Eulerian) zonal average, ψ is the streamfunction of the zonally averaged flow, $Z = -H \ln(p/p_R)$ where p_R is a reference pressure and H is a scale height, and $\rho_R = \rho_0 \exp(-Z/H)$ where ρ_0 is a constant. The associated transformed, or residual, velocities are:

$$\bar{v}^* = -\frac{1}{\rho_R} \frac{\partial}{\partial Z} (\psi^* \rho_R), \quad \bar{w}^* = \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (\psi^* \cos \vartheta), \quad (15.124)$$

with an equivalent expression for $\bar{v}$ and $\bar{w}$ in terms of ψ . (The notation for log-pressure coordinates follows section 2.6.3 on page 82, except here we use a lowercase w for the vertical velocity.) If we write the equations of motion in terms of the residual velocities instead of the Eulerian velocities we obtain the ‘transformed Eulerian mean’, or TEM, equations (section 10.3). The TEM forms of the zonally averaged thermodynamic and zonal momentum equations are:

$$\frac{\partial \bar{\theta}}{\partial t} + \frac{\bar{v}^*}{a} \frac{\partial \bar{\theta}}{\partial \vartheta} + \bar{w}^* \frac{\partial \bar{\theta}}{\partial Z} = \frac{1}{\rho_R} \frac{\partial G}{\partial z}, \quad (15.125a)$$

$$\frac{\partial \bar{u}}{\partial t} + \bar{v}^* \left(\frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (\bar{u} \cos \vartheta) - f \right) + \bar{w}^* \frac{\partial \bar{u}}{\partial Z} = \frac{1}{\rho_R \cos \vartheta} \nabla \cdot \mathcal{F}. \quad (15.125b)$$

The transformed equations of motion are completed by the meridional momentum, mass continuity and hydrostatic equations:

$$\bar{u} \left(f + \frac{\bar{u}}{a} \tan \vartheta \right) = -\frac{1}{a} \frac{\partial \Phi}{\partial \vartheta} + \hat{S}, \quad (15.126a)$$

$$\frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (\bar{v}^* \cos \vartheta) + \frac{1}{\rho_R} \frac{\partial}{\partial Z} (\rho_R \bar{w}^*) = 0, \quad (15.126b)$$

$$\frac{\partial \Phi}{\partial Z} = \frac{RT}{H} T = \frac{R}{H\theta} e^{-\kappa Z/H}. \quad (15.126c)$$

In (15.125a), $\mathcal{F} = (\mathcal{F}^\vartheta, \mathcal{F}^Z)$ is the Eliassen–Palm flux, given by

$$\mathcal{F}^\vartheta = \rho_R \cos \vartheta \left[\bar{u}_Z \frac{\overline{v'\theta'}}{\partial_Z \bar{\theta}} - \overline{u'v'} \right], \quad (15.127a)$$

$$\mathcal{F}^Z = \rho_R \cos \vartheta \left[\left(f - \frac{\partial_\vartheta(\bar{u} \cos \vartheta)}{a \cos \vartheta} \right) \frac{\overline{v'\theta'}}{\partial_Z \bar{\theta}} - \overline{u'w'} \right], \quad (15.127b)$$

with

$$\nabla \cdot \mathcal{F} = \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (\mathcal{F}^\vartheta \cos \vartheta) + \frac{\partial}{\partial Z} \mathcal{F}^Z. \quad (15.127c)$$

In (15.125a)

$$G = \frac{\rho_R}{\partial_Z \bar{\theta}} \left(\overline{v'\theta'} \frac{1}{a} \frac{\partial \bar{\theta}}{\partial \vartheta} + \overline{w'\theta'} \frac{\partial \bar{\theta}}{\partial Z} \right), \quad (15.127d)$$

and $\hat{S}$ in (15.126a) contains various, generally small, terms that lead to departures from gradient-wind balance between $\bar{u}$ and the geopotential Φ . Expressions very similar to the ones above also arise in pressure coordinates.

In many circumstances, the EP flux is well approximated by

$$\mathcal{F} = \left(-\rho_R \cos \vartheta \overline{u'v'}, f \rho_R \cos \vartheta \frac{\overline{v'\theta'}}{\partial_Z \bar{\theta}} \right), \quad (15.128)$$

in which case the zonal flow is accelerated by the EP flux according to

$$\frac{\partial \bar{u}}{\partial t} + \dots = \frac{1}{a \cos^2 \vartheta} \frac{\partial}{\partial \vartheta} (-\overline{u'v'} \cos^2 \vartheta) + \frac{1}{\rho_R} \frac{\partial}{\partial Z} \left(\rho_R f \frac{\overline{v'\theta'}}{\partial_Z \bar{\theta}} \right). \quad (15.129)$$

With $f = f_0$ (15.128) becomes the quasi-geostrophic EP flux, and in this limit G is also neglected.

In the figures that show the EP vectors, the horizontal and vertical components of the EP flux are scaled by a (the Earth's radius) and by $H = 1000$ hPa (the pressure depth of the atmosphere), respectively. The scaling determines the direction of the arrows and makes it possible to see the divergence by eye, and which component dominates in producing that divergence. In the figures that show the EP flux divergence, we plot the right-hand side of (15.125b), namely the EP flux divergence divided by $\rho_R \cos \vartheta$, this being the quantity that directly contributes to the acceleration of the zonal flow.¹⁴

Notes

- 1 The modern view of the mid-latitude general circulation — the largely zonally asymmetric motion that provides the bulk of the meridional transport of heat and momentum in the extratropics — began to take form in the 1920s in papers by Defant (1921) and Jeffreys (1926). Defant regarded the mid-latitude circulation as turbulence on a large scale (albeit without realizing the important organizing effects of waves), and calculated the horizontal eddy-diffusivities using Prandtl-like mixing length arguments. Soon after, Jeffreys presciently wrote of ‘the dynamical necessity for a continual exchange of air between high and low latitude’ and that ‘no general circulation of the atmosphere without cyclones is dynamically possible when friction is taken into account.’ This point of view

slowly gained ground, with, for example, Starr (1948) advocating the point of view that large-scale eddies were responsible for the bulk of the meridional transport of momentum in mid-latitudes, and Rossby (1949) eventually noting in a review article that ‘One is forced to conclude that there no longer exists a compelling reason to build the theory of the maintenance of the general circulation exclusively on *meridional* solenoidal circulations’.

Following this work came a pair of discussion papers by Eady (1950, 1954), that, setting the stage for the modern viewpoint, struggle with the turbulent transport of mid-latitude eddies and the maintenance of the surface currents — the importance of the enstrophy budget is discussed, for example, and Eady comes close to deriving wave activity conservation. Around that time Kuo (1951) discussed the maintenance of zonal flows by the mechanism of vorticity transfer in a state with a meridional background gradient, similar to mechanism I of section 15.1.2. Another landmark is the influential monograph by Lorenz (1967) that summarized and clarified the progress to that date, noting (in his last paragraph) that the cause of the poleward eddy momentum transport across mid-latitudes (and hence the cause of the surface eastward winds) had not at that time been rigorously explained. If perhaps not rigorous, we do now have a qualitative explanation of these dynamics by way of potential vorticity dynamics and the momentum transport in Rossby waves, as described in this chapter and in chapter 10.

- 2 The mechanism producing a westerly jet in the atmosphere, and the associated surface westerlies, used to be referred to as ‘negative viscosity’ (Starr 1968), because it is associated with an upgradient transfer of momentum. The generation of a zonal flow by rearrangements of vorticity on a background state with a meridional gradient was noted by Kuo (1951). Dickinson (1969) and Thompson (1971, 1980) calculated the momentum transport by Rossby waves; in particular Thompson noted that the zonal momentum flux was in the opposite direction to the group velocity. There is thus a potential for upgradient transfer and mean flow generation, as experimentally verified by Whitehead (1975). These ideas were developed further by Green (1970), Rhines & Holland (1979) and others since.
- 3 This technique is noted by Lighthill (1965), who remarks that the idea goes back to Rayleigh.
- 4 For an explanation of explication, see endnote 8 on page 471.
- 5 Following Held (2000).
- 6 Models of the general circulation of this ilk were introduced by Green (1970). Dickinson (1969) also considered the potential vorticity transport in planetary waves.
- 7 Adapted from Jukes (2001).
- 8 Early evidence that the temperature increases above about 11 km came from the balloon measurements of Tesserenc De Bort (1902), who also suggested the names tropopause and stratosphere, and Assmann (1902). See Hoinka (1997) for a historical account. More recently, radiative and dynamical issues relevant to this topic are discussed by, among others, Stone (1972), Held (1982), Jukes (2000) and some of the articles in Schneider & Sobel (2007), and we draw on many of them.
- 9 Paraphrasing World Meteorological Organization (1957); see also Lewis (1991).
- 10 Thomas Birner calculated the tropopause-based averages. See also Birner *et al.* (2002) and Birner (2006).
- 11 In some but not all circumstances it seems that rotating fluids seek to become marginally supercritical, in some sense, to baroclinic instability. The original suggestion (‘baroclinic adjustment’) was due to Stone (1978), and Stone & Nemet (1996) found that the isentropic slope of the real atmosphere does not vary strongly with season, even though the heat flux does, a result supportive of baroclinic adjustment ideas. Related to this (although their interpretation and reasoning were different) Schneider & Walker (2006) found that an idealized model atmosphere was only marginally supercritical over a broad parameter regime. However, other simulations have found examples of supercritical flows. Salmon (1980) and Vallis (1988b) found that quasi-geostrophic flow could be strongly supercritical, and Thuburn & Craig (1997) and Zurita-Gotor & Vallis (2009, 2011) found results that were not supportive of baroclinic adjustment in primitive equation models. In Earth’s atmosphere linear calculations show that the mean atmospheric state, certainly in winter, is baro-

clinically unstable, with growth rates of about 0.2 day^{-1} or more (Valdes & Hoskins 1988), and the Eady growth rate is similarly positive over a large fraction of the midlatitudes. See Zurita-Gotor & Lindzen (2007) for a review of some of these ideas. Jansen & Ferrari (2012, 2013) discussed and modelled the reasons for the various differences, and concluded that marginal criticality is not a general property of a rotating stratified system. It is likely that much of Earth's ocean is supercritical.

- 12 Following Jukes (2000).
- 13 For more detail see Edmon *et al.* (1980) or Andrews (1987).
- 14 E. Gerber kindly constructed these figures.

Catch a wave and you're sitting on top of the world.
Brian Wilson and Mike Love (The Beach Boys), *Catch a Wave*, 1963.

CHAPTER 16

Planetary Waves and Zonal Asymmetries

PLANETARY WAVES ARE LARGE-SCALE ROSSBY WAVES in which the potential vorticity gradient is provided by differential rotation (i.e., the beta-effect). They are ubiquitous in Earth's atmosphere and almost certainly in other planetary atmospheres. They propagate horizontally over the two Poles, and they propagate vertically into the stratosphere and beyond. In the previous chapter we saw that it is the propagation of Rossby waves away from their mid-latitude source that gives rise to the mean eastward eddy-driven jet. In this chapter we will see that the dynamics of such waves also largely determines the large-scale *zonally asymmetric* circulation of the mid-latitude atmosphere. In the first few sections we discuss the properties and propagation of planetary waves themselves, and in many ways these sections are a continuation of chapter 6. We then look more specifically at planetary waves forced by surface variations in topography and thermal properties, for it is these waves that give rise to the zonally asymmetric circulation.

In proceeding this way we are dividing our task of constructing a theory general circulation of the extratropical atmosphere into two. The first task (chapters 14 and 15) was to understand the zonally averaged circulation and the transient zonal asymmetries by supposing that, to a first approximation, this circulation is qualitatively the same as it would be if the boundary conditions were zonally symmetric, with no mountains or land-sea contrasts. Given the statistically zonally symmetric circulation, the second task is to understand the zonally asymmetric circulation. We may do this by supposing that the latter is a perturbation on the former, and using a theory linearized about the zonally symmetric state. It is by no means obvious that such a procedure will be successful, for it depends on the nonlinear interactions among the zonal asymmetries being weak. We might make some a priori estimates that suggest that this might be the case, but the ultimate justification for the approach lies in its a posteriori success. In our discussion of stationary waves we will focus first on the response to orography at the lower boundary, and then consider thermodynamic forcing — arising, for example, from an inhomogeneous surface temperature field, and our focus is the mid-latitudes.

16.1 ROSSBY WAVE PROPAGATION IN A SLOWLY VARYING MEDIUM

In chapters 6 and 7 we looked at wave propagation using linearized equations of motion. We now focus and extend this discussion by looking at Rossby wave propagation in a medium in which the parameters (such as the zonal wind and the stratification) vary spatially — as occurs in the real

atmosphere. If the parameters do vary then waves may propagate into a region in which they amplify, perhaps violating the initial assumption of linearity, so let us first look at what the conditions for linearity are.¹

16.1.1 Linear dynamics

If the linear equations are to be an accurate representation of the dynamics then the perturbation quantities need to be small compared to the background state, or at least the nonlinear terms must be small. In reality this is not always the case and indeed it may be that in course of propagation the waves amplify and may even *break*. Wave breaking is familiar to anyone who has been to the beach and watched water waves move toward the shore and crash in the ‘surf zone’ as the mean depth becomes too shallow to support laminar surface waves. Manifestly, the linear approximation breaks down at this point. More generally, wave breaking simply refers to an irreversible deformation of material surfaces, generally leading to dissipation. Since Rossby waves generally grow in amplitude as they propagate up (because density falls) we can expect Rossby wave breaking to occur somewhere in the atmosphere, but waves can also break as they propagate laterally, if and when they grow in size to such an extent that the nonlinear terms in the equations of motion become important.

To examine this consider the quasi-geostrophic potential vorticity equation,

$$\left(\frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla \right) q = 0, \quad q = \beta y + \nabla^2 \psi' + \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \psi}{\partial z} \right). \quad (16.1a,b)$$

The derivation of this equation was given in chapter 5 and all the terms are defined there. In brief, q is the quasi-geostrophic potential vorticity and ψ the streamfunction, f_0 is the Coriolis parameter and ρ_R is a density profile, a function of z only. Breaking the above equation up into mean and perturbation quantities in the usual way we obtain

$$\left(\frac{\partial}{\partial t} + \bar{u}(y, z) \frac{\partial}{\partial x} \right) q' + v' \frac{\partial \bar{q}}{\partial y} = - \left(\frac{\partial u' q'}{\partial x} + \frac{\partial v' q'}{\partial y} \right). \quad (16.2)$$

In the linear approximation we neglect the terms on the right-hand side and, seeking wave-like solutions of the form $\psi = F(x - ct)$ we obtain

$$(\bar{u} - c) \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0. \quad (16.3)$$

For the linear approximation to be valid the terms in this equation must be larger than the nonlinear terms in (16.2), and this will be the case if

$$|\bar{u} - c| \gg |u'| \quad \text{and} \quad \left| \frac{\partial \bar{q}}{\partial y} \right| \gg \left| \frac{\partial q'}{\partial y} \right|. \quad (16.4a,b)$$

Although it is common to only treat the case in which $\bar{u}$ is a constant, we may also consider the case in which $\bar{u}$ varies slowly, either in latitude or height or both, and (16.3) then approximately holds locally. But if a wave propagates into a region in which $\bar{u} = c$ then the linear criterion *must* break down. Regions where $\bar{u} = c$ are called *critical lines*, *critical surfaces*, *critical heights* or *critical latitudes*, depending on context, and in many circumstances a *critical layer* of finite width will surround the critical line, in which frictional and/or nonlinear effects are important. The location of a critical surface does not depend on the frame of reference used to measure the velocities.

For reference we first write down a few results for the simplest case when $\partial \bar{q} / \partial y$, $\bar{u}$, N^2 and ρ_R are all constant, referring to section 6.5 as needed. We look for solutions of the form

$$\psi' = \text{Re } \tilde{\psi} e^{i(kx + ly + mz - \omega t)}, \quad (16.5)$$

and obtain the dispersion relation

$$\omega = \bar{u}k - \frac{k\beta}{k^2 + l^2 + Pr^2m^2}, \quad (16.6)$$

where $Pr = f_0/N$ is the Prandtl ratio. The components of the group velocity are given by

$$c_g^x = \bar{u} + \frac{(k^2 - l^2 - Pr^2m^2)\beta}{(k^2 + l^2 + Pr^2m^2)^2}, \quad c_g^y = \frac{2kl\beta}{(k^2 + l^2 + Pr^2m^2)^2}, \quad c_g^z = \frac{2kmPr^2\beta}{(k^2 + l^2 + Pr^2m^2)^2}. \quad (16.7a,b,c)$$

16.1.2 Conditions for wave propagation

Suppose that the zonal wind varies slowly with latitude and height, but that, for simplicity, the density, ρ_R , is a constant. The equation of motion is

$$\left(\frac{\partial}{\partial t} + \bar{u}(y, z) \frac{\partial}{\partial x} \right) q' + v' \frac{\partial \bar{q}}{\partial y} = 0. \quad (16.8)$$

Because the coefficients of the equation are not constant we cannot assume harmonic solutions in the y and z directions; rather, we seek solutions of the form

$$\psi' = \tilde{\psi}(y, z) e^{ik(x-ct)}. \quad (16.9)$$

If the parameters in (16.8) are varying slowly compared to the wavelength of the waves then a dispersion relation still exists (as discussed in section 6.3), but the relation will be of the form $\omega = \Omega(\mathbf{k}; \mathbf{x}, t)$; where the function Ω varies slowly in space. Now, if the medium is not an explicit function of x or of time the x -wavenumber and the frequency will be a constant, and hence c is constant too, and we can use the dispersion relation to find what are effectively the other wavenumbers in the problem. Using (16.9) in (16.8) we find (with N^2 constant)

$$\frac{\partial^2 \tilde{\psi}}{\partial y^2} + \frac{f_0^2}{N^2} \frac{\partial^2 \tilde{\psi}}{\partial z^2} + n^2(y, z) \tilde{\psi} = 0, \quad \text{where} \quad n^2(y, z) = \frac{\partial \bar{q} / \partial y}{\bar{u} - c} - k^2. \quad (16.10a,b)$$

Equation (16.10a) is similar to the Rayleigh or Rayleigh-Kuo equation encountered in chapter 9, but now c is given and is not an eigenvalue; rather, the frequency is known and the dispersion relation gives the quantity n . The quantity n is the *refractive index* and it greatly affects how the waves propagate: solutions are wavelike when n^2 is positive and evanescent when n^2 is negative. To see this in a simple case, suppose there is no z -variation so that $\partial^2 \tilde{\psi} / \partial y^2 + n^2 \tilde{\psi} = 0$, whereas if n is constant and real we have harmonic solutions in the y -direction of the form $\exp(iny)$. If $n^2 < 0$ the solutions will evanesce. Waves tend to propagate toward regions of large n^2 and turn away from regions of negative n^2 , as we will see in the examples to follow.

The value of n^2 will become very large if and as $\bar{u}$ approaches c from above and the waves, being very short, will tend to break. If $\bar{u}$ continues to diminish and becomes smaller than c then n^2 switches from being large and positive to large and negative. If n^2 diminishes because $\partial \bar{q} / \partial y$ diminishes then it will transition smoothly to a negative value. The location where $\bar{u} = c$ is called a critical surface (or line). The location where n^2 passes through zero is called a turning surface (or line).

The bounds on n^2 can be translated into bounds on the zonal phase speed c . Given a zonal wind $\bar{u}$, wave propagation requires that c is bounded by

$$\bar{u} - \frac{\partial \bar{q} / \partial y}{k^2 + \gamma^2} < c < \bar{u}. \quad (16.11)$$

At the upper bound (a critical surface) the wavelength is small and wave breaking is likely to occur. At the lower bound (a turning surface) the refractive index tends to zero and the wavelength tends to infinity. Waves will tend to propagate away from regions with a small n and be refracted toward regions of large n . The bounds can also be expressed in terms of the zonal velocity:

$$0 < \bar{u} - c < \frac{\partial \bar{q} / \partial y}{k^2 + \gamma^2}. \quad (16.12)$$

This form is useful when considering a situation in which the wave speed is given, for example by boundary conditions; equation (16.12) then tells us under what configurations of zonal velocity wave propagation can occur. The lower bound corresponds to a critical surface and the upper bound to a turning surface.

It is algebraically complicated to continue our analysis in the three-dimensional case, so let us consider the cases in which the inhomogeneities in the medium occur separately in the horizontal and vertical.

16.2 HORIZONTAL PROPAGATION OF ROSSBY WAVES

Consider the purely horizontal problem for which linearized equation of motion is

$$\left(\frac{\partial}{\partial t} + \bar{u}(y) \frac{\partial}{\partial x} \right) q' + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad (16.13)$$

where $q' = \nabla^2 \psi'$, $v' = \partial \psi' / \partial x$ and $\partial \bar{q} / \partial y = \beta - \bar{u}_{yy}$, which we will denote β^* . If $\bar{u}$ and β^* do not vary in space then we may obtain wavelike solutions in the usual way and obtain the dispersion relation

$$\omega \equiv ck = \bar{u}k - \frac{\partial \bar{q} / \partial y}{k^2 + l^2} \quad (16.14)$$

where k and l are the x - and y -wavenumbers.

If the parameters do vary in the y -direction then we seek a solution $\psi' = \tilde{\psi}(y) \exp[ik(x - ct)]$ and obtain, analogous to (16.10),

$$\frac{\partial^2 \tilde{\psi}}{\partial y^2} + l^2(y) \tilde{\psi} = 0, \quad \text{where} \quad l^2(y) = \frac{\beta^*}{\bar{u} - c} - k^2 \quad (16.15a,b)$$

If the parameter variation is sufficiently small, occurring on a spatial scale longer than the wavelength of the waves, then we may expect that the disturbance will propagate locally as a plane wave. The solution is then of WKB form (see appendix on page 247), namely

$$\tilde{\psi}(y) = A_0 l^{-1/2} \exp\left(i \int l dy\right) \quad (16.16)$$

where A_0 is a constant. The phase of the wave in the y -direction, θ , is evidently given by $\theta = \int l dy$, so that the local wavenumber is given by $d\theta/dy = l$. The group velocity is calculated in the normal way using the dispersion relation (16.14) and we obtain

$$c_g^x = \bar{u} + \frac{(k^2 - l^2)\beta^*}{(k^2 + l^2)^2}, \quad c_g^y = \frac{2kl\beta^*}{(k^2 + l^2)^2}. \quad (16.17a,b)$$

where l is given by (16.15b), with both quantities varying slowly in the y -direction.

16.2.1 Wave amplitude

As a Rossby wave propagates its amplitude is not necessarily constant because, in the presence of a shear, the wave may exchange energy with the background state and the WKB solution, (16.15), tells us that the variation goes like $l^{-1/2}(y)$. This variation can be understood from somewhat more general considerations. As discussed in chapter 10 (specifically section 10.2.1) an inviscid, adiabatic wave will conserve its wave activity, and specifically its pseudomomentum, meaning that

$$\frac{\partial \mathcal{P}}{\partial t} + \nabla \cdot \mathcal{F} = 0, \quad (16.18)$$

where $\mathcal{P}$ is quadratic in the wave amplitude and $\mathcal{F}$ is the flux of $\mathcal{P}$, and the two are related by the group velocity property $\mathcal{F} = \mathbf{c}_g \mathcal{P}$. In the zonally-averaged case the pseudomomentum and flux for the stratified quasi-geostrophic equations are given by

$$\mathcal{P} = \frac{\overline{q'^2}}{2\beta^*}, \quad \mathcal{F} = -\overline{u'v'} \mathbf{j} + \frac{f_0}{N^2} \overline{v'b'} \mathbf{k}, \quad (16.19)$$

with $\mathcal{F}$ is the Eliassen–Palm (EP) flux. If the waves are steady then $\nabla \cdot \mathcal{F} = 0$, and in the two-dimensional case under consideration $b' = 0$ and $\partial \overline{u'v'}/\partial y = 0$. Thus $u'v' = kl|\tilde{\psi}|^2 = \text{constant}$, and since k is constant along a ray the amplitude of a wave varies like

$$|\tilde{\psi}| = \frac{A_0}{\sqrt{l(y)}}, \quad (16.20)$$

as in the WKB solution. The energy of the wave then varies like

$$\text{Energy} = (k^2 + l^2) \frac{A_0^2}{l}. \quad (16.21)$$

16.2.2 Two examples

To illustrate the above ideas in a concrete fashion we consider two examples, one with a turning line and one with a critical line. Very close to the turning line and critical line more detailed analysis is needed to obtain a complete solution, but we can obtain a sense of the behaviour with an elementary treatment.

Waves with a turning latitude

A turning line arises where $l = 0$ and it corresponds to the lower bound of c in (16.11). The line arises if the potential vorticity gradient diminishes to such an extent that $l^2 < 0$ and the waves then cease to propagate in the y -direction. This may happen even in unsheared flow as a wave propagates polewards and the magnitude of beta diminishes.

As a wave packet approaches a turning latitude then n goes to zero so the amplitude and the energy of the wave approach infinity. However, the wave will never reach the turning latitude because the meridional component of the group velocity is zero, as can be seen from the expressions for the group velocity, (16.17). As a wave approaches the turning latitude $c_g^x \rightarrow (\beta - \bar{u}_{yy})/k^2$ and $c_g^y \rightarrow 0$, so the group velocity is purely zonal and indeed as $l \rightarrow 0$

$$\frac{c_g^x - \bar{u}}{c_g^y} = \frac{k}{2l} \rightarrow \infty. \quad (16.22)$$

Because the meridional wavenumber is small the wavelength is large, so we do not expect the waves to break. Rather, we intuitively expect that a wave packet will turn — hence the eponym ‘turning latitude’ — and be reflected.

Rossby Wave Propagation in a Slowly Varying Medium

The linear equation of motion is, in terms of streamfunction,

$$\left(\frac{\partial}{\partial t} + \bar{u}(y, z) \frac{\partial}{\partial x} \right) \left[\nabla^2 \psi' + \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \psi'}{\partial z} \right) \right] + \frac{\partial \psi'}{\partial x} \frac{\partial \bar{q}}{\partial y} = 0. \quad (\text{RP.1})$$

We suppose that the parameters of the problem vary slowly in y and/or z but are uniform in x and t . The frequency and zonal wavenumber are therefore constant. We seek solutions of the form $\psi' = \tilde{\psi}(y, z) e^{ik(x-ct)}$ and find (if, for simplicity, N^2 and ρ_R are constant)

$$\frac{\partial^2 \tilde{\psi}}{\partial y^2} + \frac{f_0^2}{N^2} \frac{\partial^2 \tilde{\psi}}{\partial z^2} + n^2(y, z) \tilde{\psi} = 0, \quad \text{where} \quad n^2(y, z) = \frac{\partial \bar{q} / \partial y}{\bar{u} - c} - k^2. \quad (\text{RP.2})$$

The value of n^2 must be positive in order that waves can propagate, and so waves cease to propagate when they encounter either

- (i) A *turning surface*, where $n^2 = 0$, or
- (ii) A *critical surface*, where $\bar{u} = c$ and n^2 becomes infinite.

For a given wave speed, the location of the turning surface, but not that of the critical surface, depends on wavenumber. The condition for wave propagation may be expressed as bounds on the zonal flow, to wit

$$0 < \bar{u} - c < \frac{\partial \bar{q} / \partial y}{k^2}. \quad (\text{RP.3})$$

If the length scale over which the parameters of the problem vary is much longer than the wavelengths themselves we can expect the solution to look locally like a plane wave and a WKB analysis can be employed. In the purely horizontal problem we assume a solution of the form $\psi' = \tilde{\psi}(y) e^{ik(x-ct)}$ and find

$$\frac{\partial^2 \tilde{\psi}}{\partial y^2} + l^2(y) \tilde{\psi} = 0, \quad l^2(y) = \frac{\partial \bar{q} / \partial y}{\bar{u} - c} - k^2. \quad (\text{RP.4})$$

The WKB solution is of the form

$$\tilde{\psi}(y) = A l^{-1/2} \exp \left(\pm i \int l \, dy \right). \quad (\text{RP.5})$$

Thus, $l(y)$ is the local y -wavenumber, and the amplitude of the solution varies like $l^{-1/2}$. However, the WKB condition fails at both a critical line and a turning line.

Approaching a critical line the amplitude of the wave diminishes (in the WKB approximation) because l is large. In the critical layer wave amplitude is in fact nearly constant but the vorticity becomes very large, and either nonlinearity and/or dissipation become important, and since wavelength is small the waves may break. At a turning line the amplitude and energy will both be large, but since the wavelength is long the waves will not necessarily break; rather, they are reflected.

A similar analysis may be employed for vertically propagating Rossby waves, and either a turning level or a critical level will prevent the upward propagation of waves into the stratosphere — this is the ‘Charney–Drazin’ condition, discussed in section 16.5.3.

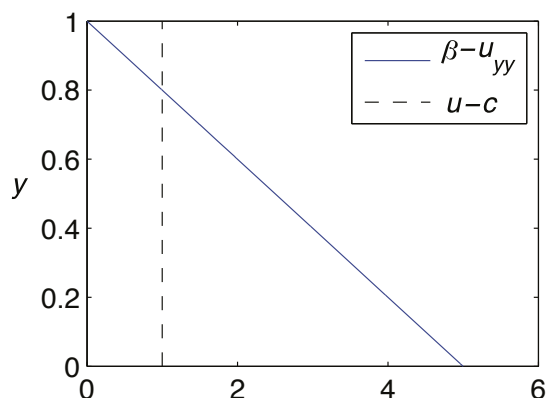


Fig. 16.1 Parameters for the first example considered in section 16.2.2, with all variables nondimensional. The zonal flow is uniform with $u = 1$ and $c = 0$ (so that $\bar{u}_{yy} = 0$) and β diminishes linearly as y increases polewards as shown. With zonal wavenumber $k = 1$ there is a turning latitude at $y = 0.8$, and the wave properties are illustrated in Fig. 16.2.

To illustrate this, consider waves propagating in a background state that has no horizontal shear but with a beta effect that diminishes polewards. To be concrete suppose that $\beta = 5$ at $y = 0$, diminishing linearly to $\beta = 0$ at $y = 0$, and that $\bar{u} - c = 1$ everywhere. There is no critical line but depending on the x -wavenumber there may be a turning line, and if we choose $k = 1$ then the turning line occurs when $\beta = 1$ and so at $y = 0.8$. The turning latitude depends on the value of the x -wavenumber — if the zonal wavenumber is larger then waves will turn further south. The parameters are illustrated in Fig. 16.1.

For a given zonal wavenumber ($k = 1$ in this example) the value of l^2 is computed using (16.15b), and the components of the group velocity using (16.17), and these are illustrated in Fig. 16.2. We may choose either a positive or a negative value of l , corresponding to northward or southward oriented waves, and we illustrate both in the figure. The value of l^2 becomes zero at $y = 0.8$, and this corresponds to a turning latitude. The values of the wave amplitude and energy are computed using (16.20) and (16.21) (with an arbitrary amplitude at $y = 0$) and these both become infinite at the turning latitude.

What is happening physically? We may suppose that at some location in the domain there is a source of waves — baroclinic disturbances for example. Waves propagate away from the source (since the waves must carry energy away), and this determines the sign of n of any particular wave packet. The disturbance may in general consist of many zonal wavenumbers and many frequencies (or phase speeds, c), but the dispersion relation must be satisfied for each pair and this determines the meridional wavenumber via an equation like (16.15). As the wave packet propagates away from the source then, as we noted in section 6.3 on ray theory, if the medium is zonally symmetric the x -wavenumber, k , is preserved. If the medium is not time-varying then the frequency, and therefore the wave speed c , are also preserved. We may approximately construct a ray by following the arrows in Fig. 16.2, and we see that a ray propagating polewards will bend eastward as it approaches the turning latitude. Although its amplitude will become large it will not necessarily break because the wavelength is large; in fact, the packet may be reflected southward. We may heuristically construct a ray trajectory by drawing a line that is always parallel to the arrows marking the group velocity. Indeed, the entire procedure might be thought of as an Eulerian analogue of ray theory; rather than following a wave packet we just evaluate the field of group velocity, and if there is no explicit time dependence in the problem a ray follows the arrows.

The above argument suggests but does not demonstrate reflection at the turning latitude. The arrows of Fig. 16.2 are all zonal at the turning line and do not actually turn back. One might imagine using a WKB analysis but the WKB approximation fails in the vicinity of a turning latitude: the meridional wavenumber l tends to zero but dl/dy does not, and the WKB condition (6.164)

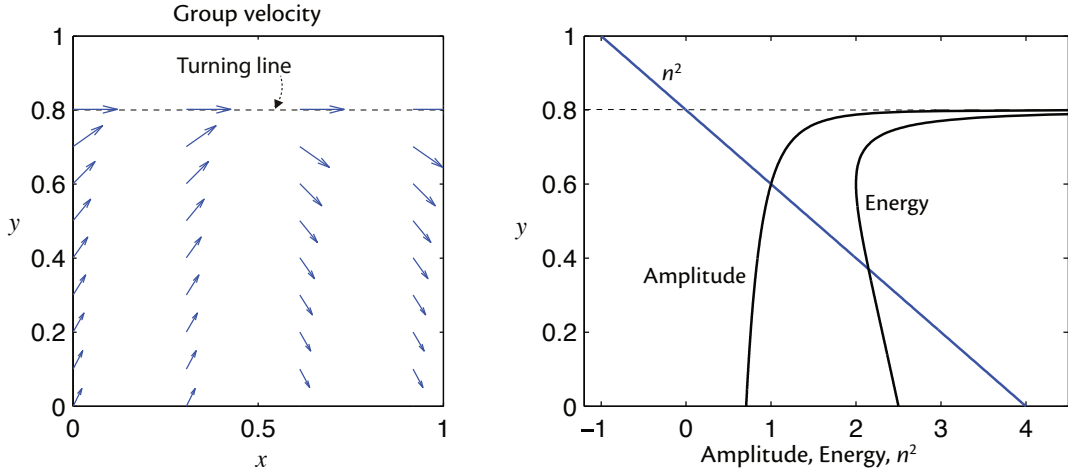


Fig. 16.2 Left: The group velocity evaluated using (16.17) for the parameters illustrated in Fig. 16.1, which give a turning latitude at $y = 0.8$. For $x < 0.5$ we choose positive values of n , and a northward group velocity, whereas for $x > 0.5$ we choose negative values of n . Right panel: Values of refractive index squared (n^2), the energy and the amplitude of a wave. n^2 is negative for $y > 0.8$. See text for more description.

cannot be satisfied (even though the wave equation itself is not singular). However, a momentum flux argument shows that the reflection is in fact perfect in the absence of dissipation. Suppose there is a wave source in mid- or low latitudes (say at $y = 0$ in Fig. 16.2) producing poleward propagating waves. Poleward of the turning line the waves evanesce and the zonally-averaged poleward momentum flux is zero at large y . However, this decay has occurred in the absence of friction, and therefore that momentum flux is zero *everywhere*. To see this with equations, away from forcing regions the inviscid barotropic pseudomomentum conservation equation, (16.18), becomes

$$\frac{\partial \mathcal{P}}{\partial t} + \overline{v' \zeta'} = 0, \quad \text{or} \quad \frac{\partial \mathcal{P}}{\partial t} - \frac{\partial}{\partial y} \overline{u' v'} = 0. \quad (16.23)$$

Thus, in a statistically steady state, $\overline{u' v'}$ is a constant, and that constant is zero if the flux is zero at large y . Since the forcing is producing a poleward propagating Rossby wave (with $\overline{u' v'} < 0$ in the Northern Hemisphere) there must be a reflected wave with $\overline{u' v'} > 0$, and that reflected wave must come from the vicinity of the turning line.

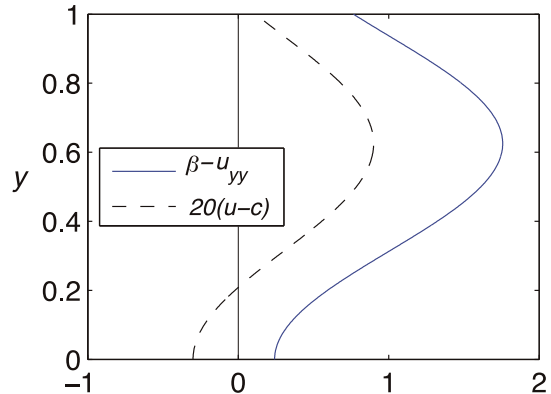


Fig. 16.3 Parameters for the second example considered in section 16.2.2, with all variables nondimensional. The zonal flow has a broad eastward jet and β is constant. There is a critical line at $y = 0.2$, and with zonal wavenumber $k = 5$ the wave properties are illustrated in Fig. 16.4.

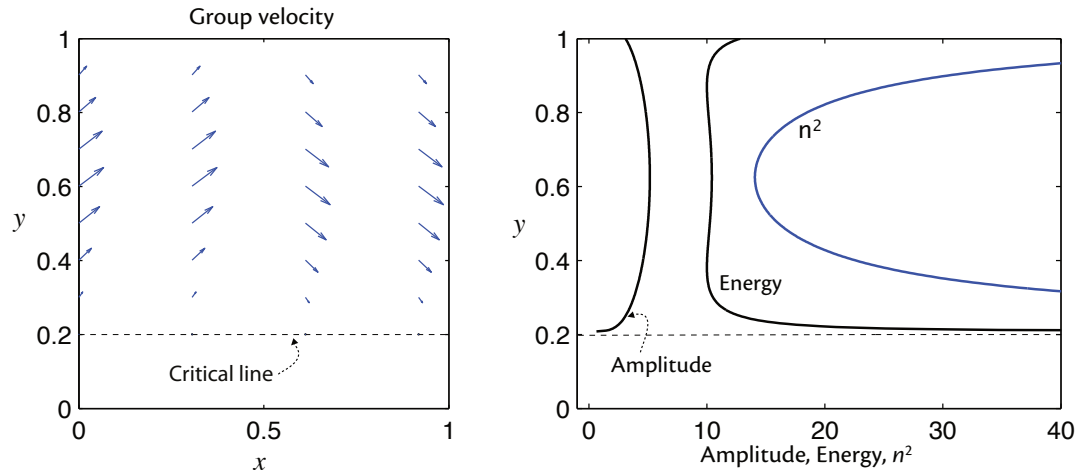


Fig. 16.4 Left: The group velocity evaluated using (16.17) for the parameters illustrated in Fig. 16.1, which give a critical line at $y = 0.2$. For $x < 0.5$ we choose positive values of n , and a northward group velocity, whereas for $x > 0.5$ we choose negative values of n . Right panel: Values of refractive index squared, the energy and the amplitude of a wave. The value of n^2 becomes infinite at the critical line and the linear theory breaks down in its vicinity. See text for more description.

Waves with a critical latitude

A critical line occurs when $\bar{u} = c$, corresponding to the upper bound of c in (16.11), and from (16.15) we see that at a critical line the meridional wavenumber approaches infinity. From (16.17) we see that both the x - and y -components of the group velocity are zero — a wave packet approaching a critical line just stops (at least according to ray theory). Specifically, as l becomes large

$$c_g^x - \bar{u} \rightarrow 0, \quad c_g^y \rightarrow 0, \quad \frac{c_g^x - \bar{u}}{c_g^y} \rightarrow -\frac{l}{k} \rightarrow -\infty. \quad (16.24)$$

From (16.20) the amplitude of the wave packet also approaches zero, but its energy approaches infinity. Since the wavelength is very small we expect the waves to *break* and deposit their momentum, and this situation commonly arises when Rossby waves excited in midlatitudes propagate equatorward and encounter a critical latitude in the subtropics.

To illustrate this let us construct a background state that has an eastward jet in midlatitudes becoming westward at low latitudes, with β constant chosen to be large enough so that $\beta - \bar{u}_{yy}$ is positive everywhere. (Specifically, we choose $\beta = 1$ and $\bar{u} = -0.03 \sin(8\pi y/5 + \pi/2) - 0.5$), but the precise form is not important.) If $c = 0$ then there is a critical line when $\bar{u}$ passes through zero, which in this example occurs at $x = 0.2$. (The value of $\bar{u} - c$ is small at $y = 1$, but no critical line is actually reached.) These parameters are illustrated in Fig. 16.3. We also choose $k = 5$, which results in a positive value for l^2 everywhere.

As in the previous example we compute the value of l^2 using (16.15b) and the components of the group velocity using (16.17), and these are illustrated in Fig. 16.4, with northward propagating waves shown for $x < 0.5$ and southward propagating waves for $x > 0.5$. The value of n^2 increases considerably at the northern and southern edges of the domain, and is actually infinite at the critical line at $y = 0.2$. Using (16.20) the amplitude of the wave diminishes as the critical line approaches, but the energy increases rapidly, suggesting that the linear approximation will break down. The waves (in the linear approximation) tend to stall before reaching the critical line, because both the x and the y components of the group velocity become very small. This is a little misleading because ray theory breaks down and a disturbance can reach the critical line, and in the region

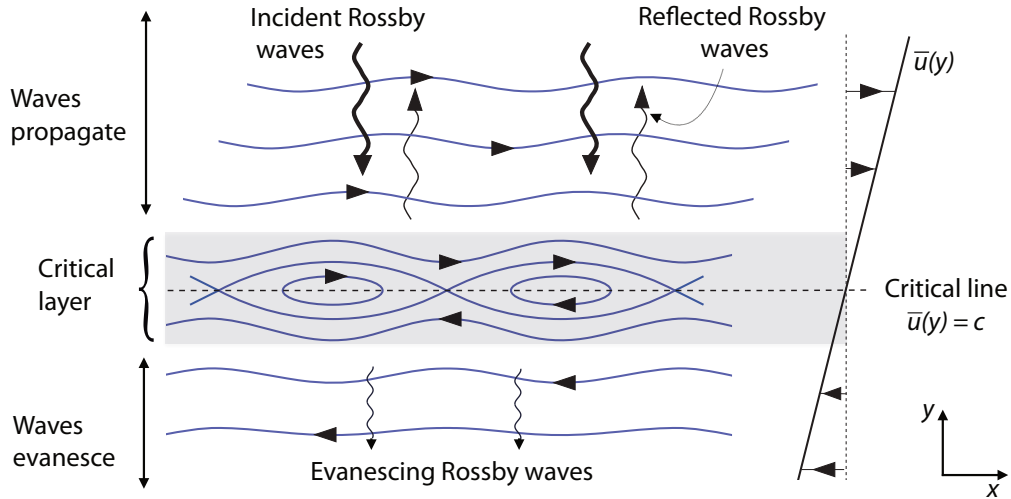


Fig. 16.5 Sketch of a Rossby-wave critical layer. Incident Rossby waves from midlatitudes propagate in a horizontally sheared flow toward a critical line at which $\bar{u}(y) = c$. Surrounding the critical line is a critical layer, in which either nonlinear or frictional effects, or both, are important. If nonlinear effects are important then the critical layer may reflect, but there is still likely to be dissipation in the critical layer. Equatorward of the critical layer the waves evanesce.³

of the critical line — that is, in the critical layer — either frictional or nonlinear effects, or both, become important, as we see in the next section. The situation illustrated in this example is of particular relevance to the maintenance of the zonal wind structure in the troposphere: waves are generated in midlatitudes and propagate equatorward, and as they approach a critical line in the subtropics they break, deposit westward momentum and retard the flow.

16.3 ♦ WAVES NEAR A CRITICAL LINE

We now look a little more closely at the behaviour of Rossby waves near a critical line and then, more briefly, at gravity wave behaviour. Critical layer theory is extensive and technical, and our discussion only scratches the surface and is mostly linear; readers wishing for more should go to the literature.² We use Northern Hemisphere conventions, envisioning waves propagating southward toward a subtropical critical line.

16.3.1 Preliminaries

Consider horizontally propagating Rossby waves obeying the linear barotropic vorticity equation on the beta-plane (vertically propagating waves may be considered using similar techniques). The equation of motion is

$$\left(\frac{\partial}{\partial t} + \bar{u} \frac{\partial}{\partial x} \right) \nabla^2 \psi' + \beta^* \frac{\partial \psi'}{\partial x} = -r \nabla^2 \psi', \quad (16.25)$$

where $\beta^* = \beta - \partial_y^2 \bar{u}$. The parameter r is a drag coefficient that acts directly on the relative vorticity, and we shall assume that it is small compared to the Doppler-shifted frequency of the waves, except possibly near a critical line. It is not a particularly realistic form of dissipation but it is simple and captures the essential process. We seek solutions of the form

$$\psi'(x, y, t) = \tilde{\psi}(y) e^{i(k(x-ct))}. \quad (16.26)$$

Substituting into (16.25) we find, after a couple of lines of algebra, that $\tilde{\psi}$ satisfies, analogously to (16.15),

$$\frac{d^2\tilde{\psi}}{dy^2} + l^2(y)\tilde{\psi} = 0, \quad \text{where} \quad l^2(y) = \frac{\beta^*}{\bar{u} - c - ir/k} - k^2. \quad (16.27a,b)$$

If the zonal wind has a lateral shear then β^* may vary with y , and thus so does l . If r is non-zero then l has an imaginary component so that the wave decays away from its source region, and as $\bar{u} \rightarrow c$ the decay will be particularly strong.

Suppose that the friction is zero. Near the critical line the y -wavenumber will be much larger than k and the streamfunction will obey an equation of the form

$$\frac{d^2\tilde{\psi}}{dy^2} \approx -\frac{\beta^*}{\bar{u} - c}\tilde{\psi} = -\frac{A}{y - y_c}\tilde{\psi}. \quad (16.28)$$

where A is a constant, and by shifting the origin we will take $y_c = 0$. By inspection an approximate solution to this equation for small y is

$$\frac{d\tilde{\psi}}{dy} = B \ln y, \quad \tilde{\psi} = B \left(y \ln y - y - \frac{1}{A} \right) \approx -\frac{B}{A}. \quad (16.29)$$

where B is a constant. (Equation (16.28) actually has Bessel function solutions.) The vorticity, $\tilde{\psi}_{yy}$ then goes as $1/y$ and the velocity goes as $\ln y$ near the critical line. Both quantities blow up, *but the streamfunction itself does not*. The pseudomomentum also blows up as the critical layer. To see this, multiply (16.25) by ζ/β^* and zonally average to give

$$\frac{\partial \mathcal{P}}{\partial t} + \frac{\partial \mathcal{F}}{\partial y} = -\alpha \mathcal{P}, \quad (16.30)$$

where $\mathcal{P} = \overline{\zeta'^2}/2\beta^*$ is the pseudomomentum, $\partial \mathcal{F}/\partial y = -\partial_y(\overline{u'v'}) = \overline{v'\zeta'}$ is its flux divergence, and $\alpha = 2r$. If $\alpha = 0$ then $\mathcal{P}$ must blow up at the critical line because ζ does. Finally, the WKB approximation fails approaching a critical line: equations (16.27) and (16.28) tell us that

$$l^2 \sim \frac{1}{y} \quad \text{and} \quad \frac{dl}{dy} \sim \frac{1}{y^{3/2}}. \quad (16.31)$$

Thus, for small y , $dl/dy > l^2$ because $y^{-3/2} > y^{-1}$ for small y . The WKB condition that the wavenumber, l , varies more slowly than l^2 [requiring $dl/dy \ll l^2$, as in (6.164)] is not satisfied.

Behaviour near a critical line

A detailed analysis is required to determine what does happen at the critical line, but in the linear problem we can make some useful headway. As in Fig. 16.5 we imagine there is a small region surrounding the critical line — the critical layer — in which the either nonlinear terms or frictional terms, or both, are important. In the critical layer y derivatives are much larger than x derivatives so that $\zeta' \approx -\partial u'/\partial y = \partial^2 \psi'/\partial y^2$. Furthermore, since $\bar{u} - c$ is small, a Taylor expansion gives

$$\bar{u} - c \approx y \left(\frac{\partial \bar{u}}{\partial y} \right)_{\bar{u}=c}, \quad (16.32)$$

and we will denote the derivative as $\partial_y \bar{u}_c$. Using (16.27) with $l^2 \gg k^2$ then gives

$$\tilde{\zeta} = \frac{-\beta^* \tilde{\psi}}{\bar{u} - c - ir/k} \approx \frac{-\beta^* \tilde{\psi}}{y \partial_y \bar{u}_c - ir/k} = \frac{-\hat{\beta} \tilde{\psi}}{y - i\hat{r}} = \frac{-\hat{\beta}(y + i\hat{r})\tilde{\psi}}{y^2 + \hat{r}^2}, \quad (16.33)$$

where $\hat{\beta} = \beta^* / \partial_y \bar{u}_c$ and $\hat{r} = r / (k \partial_y \bar{u}_c)$. The vorticity flux in the critical layer is given by

$$\overline{v' \zeta'} = \frac{1}{2} \text{Re} \overline{ik \tilde{\psi} \tilde{\zeta}^*} = \frac{-k \hat{\beta} \hat{r}}{2(y^2 + \hat{r}^2)} |\tilde{\psi}^2|. \quad (16.34)$$

The factor of 2 comes from the averaging, and only the part of $\tilde{\zeta}$ proportional to $i\hat{r}$ contributes because of its phase relative to $i\tilde{\psi}$. There are two interesting aspects to (16.34).

- (i) The vorticity flux is negative. (All the individual terms, including k , are positive.)
- (ii) The stream function is almost constant in the critical layer, so that for small friction (16.34) is sharply peaked around $y = 0$. It tends to a delta-function as $\hat{r} \rightarrow 0$, because

$$\delta(y) = \frac{1}{\pi} \lim_{\hat{r} \rightarrow 0} \frac{\hat{r}}{y^2 + \hat{r}^2}. \quad (16.35)$$

We can see that the thickness of the critical layer, δ_r is just $\hat{r}$.

The first property above tells us that the critical layer is dissipative. The second property tells us that the eddy momentum flux has a finite jump across the critical line. For small r we have

$$[\overline{u'v'}]_-^+ = - \int_-^+ \overline{\zeta'v'} dy = \int_-^+ \frac{-k \hat{\beta} \hat{r}}{2(y^2 + \hat{r}^2)} |\tilde{\psi}^2| dy = - \int_-^+ \frac{1}{2} k \hat{\beta} \pi |\tilde{\psi}^2| \delta(y) dy = - \frac{1}{2} k \hat{\beta} \pi |\tilde{\psi}^2|, \quad (16.36)$$

where the integrals are over the critical layer, and the momentum flux therefore diminishes across it. The last term on the right-hand side has no dependence on r , which means *there is finite absorption, even as friction tends to zero*. There will be no transmission through the critical layer, as we can see using a wave activity argument. For $y < 0$ (south of the critical line) the waves evanesce, but if (as we assume) friction there is negligible then in a statistically steady $\partial_y \overline{u'v'} = 0$. Thus $\overline{u'v'} = 0$ at the southern edge of the critical line, and therefore pseudomomentum must fall to zero across the critical line. Put simply, the eddy momentum fluxes are zero far from the critical layer (because waves evanesce), but the flux does not vary with y so it is zero at the southern edge of the critical layer. In fact, in such a layer there is also no reflection, and absorption is complete.

The zonal velocity itself also jumps across the layer. Since the streamfunction, and hence $v' = \partial \psi' / \partial x$, vary only weakly across the critical layer, $\zeta' \approx -\partial u' / \partial y$. The jump in velocity across the critical layer is then given by

$$[u']_-^+ = - \int_-^+ \zeta' dy, \quad (16.37a,b)$$

which also may be calculated.

Nonlinear effects

We will say only a few words about this problem. First consider the critical layer thickness. Another way to evaluate the thickness in the linear problem is to directly note that the relative sizes of the the advection term and the frictional term in the equation of motion is $y \partial_y \bar{u}_c / r$ so that an estimate for the frictional critical layer thickness is $\delta_r \sim r / (k \partial_y \bar{u}_c)$, as before. We can use a similar argument in the nonlinear case, and if we suppose a balance between the nonlinear terms and advection of the mean flow then the critical layer thickness, δ_{nl} , can be estimated from

$$y \frac{\partial \bar{u}_c}{\partial y} \frac{\partial \zeta'}{\partial x} \sim v' \frac{\partial \zeta'}{\partial y} \sim k \psi' \frac{\zeta'}{\delta} \quad \text{whence} \quad \delta_{nl} \sim \left| \frac{v'}{k \partial_y \bar{u}_c} \right|^{1/2} \sim \left| \frac{\psi'}{\partial_y \bar{u}_c} \right|^{1/2}, \quad (16.38)$$

where v' is the meridional velocity near the critical layer. In Earth's atmosphere the nonlinear terms typically *are* important (and $\delta_{nl} > \delta_r$). Nevertheless, dissipation can and normally will occur in a nonlinear critical layer, either directly by the drag term in (16.25) or because the nonlinear

interactions lead to wave breaking and a nonlinear cascade to dissipation within the layer itself, and that dissipation will retard the mean flow (i.e., make it more westward). This effect is important in the subtropics, where Rossby waves propagating equatorward from midlatitudes in the upper troposphere encounter a critical level, and generate a critical layer. Rossby waves propagating upwards toward and into the stratosphere can also encounter a critical level, contributing to the QBO phenomenon described in section 17.6.

If the critical layer is thin, then the total flow is a superposition of the background flow, $\bar{u}(y) \approx y \partial \bar{u}_c / \partial y$ and the disturbance field. The disturbance streamfunction, ψ' varies like $\exp(ikx)$, with no y variation because it is continuous across the critical line, so that the total streamfunction varies like

$$\psi \approx -\frac{1}{2} \frac{\partial \bar{u}_c}{\partial y} y^2 + \text{Re } \psi'(0) e^{ikx}. \quad (16.39)$$

This gives a pattern like that illustrated in Fig. 16.5, known as a Kelvin cat's eye pattern.

The dynamics of the pattern determines the value of $\int \bar{u}' \zeta'$ which in turn determines the value of $\overline{u'v'}$ is at the edges, as in (16.36). If this value is reduced from the value of the incoming waves then the critical layer is reflecting, but determining whether this is so requires a more detailed analysis than we can provide here.

16.3.2 Internal gravity wave critical layers

Gravity wave critical layers have a rather different character than Rossby wave critical layers because of the nature of the dispersion relation. Referring back to (7.53) on page 259, we linearize the Boussinesq equations about a sheared mean flow $\bar{u}(z)$. Confining attention to two dimensions, x and z , then a little algebra results in the 'Taylor–Goldstein' equation,

$$\frac{d^2 \tilde{\Psi}}{dz^2} + \left[\frac{N^2}{(\bar{u} - c)^2} - \frac{\bar{u}_{zz}}{\bar{u} - c} - k^2 \right] \tilde{\Psi} = 0. \quad (16.40)$$

Here, $\tilde{\Psi}$ is the amplitude of the streamfunction for the streamfunction in the vertical plane, with $(u, w) = (-\partial \Psi / \partial z, \partial \Psi / \partial x)$. The presence of first term in square brackets gives the equation a different nature than the Rossby problem, since if we omit the second term the equation has the form

$$\frac{d^2 \tilde{\Psi}}{dz^2} \sim -\frac{Ri \tilde{\Psi}}{z^2} \quad (16.41)$$

where $Ri = N^2 / (\partial_z \bar{u}_c)^2$ is the Richardson number. This means that, if the flow is stable, the waves oscillate *extremely* rapidly as the critical line is approached — in fact an infinite number of times — and the group velocity also diminishes rapidly. (Solutions to (16.41) have the form $\tilde{\Psi} = z^\alpha$ where $\alpha = (1 \pm \sqrt{1 - 4 Ri})/2$ and are oscillatory if $Ri > 1/4$, which is the condition for flow stability.) The slowdown gives dissipation more time to act, and commonly a wave will completely dissipate before the critical line is reached. If nonlinearity is allowed, the flow will break down into turbulence near the critical layer with a rapid cascade to dissipative scales.

The wkb approximation and notions of group velocity remain valid to a much greater degree than in the Rossby wave case. From (16.41) the vertical wavenumber, m , obeys

$$m^2 \sim \frac{1}{z^2}, \quad \frac{dm}{dz} \sim \frac{1}{z^2} \sim m^2, \quad (16.42)$$

Thus, the wkb condition is neither obviously well satisfied nor badly violated, and we can expect group velocity and wkb theory to provide useful information much closer to the critical line than in the Rossby wave case. In particular, a Rossby wave will typically reach a critical layer even as the group velocity stalls, whereas a gravity wave may be absorbed before that. Aspects of the gravity wave analysis are continued in the discussion of the stratosphere in section 17.3.3.

16.4 ♦ A WKB WAVE–MEAN-FLOW PROBLEM FOR ROSSBY WAVES

Let us now assume that the background properties do vary slowly and see how far we can get with a WKB approximation. As we saw above, WKB theory fails approaching a critical line, so we cannot determine what happens when $\bar{u} - c$ is very small, but the analysis is nevertheless instructive. If the friction is small and $r \ll k(\bar{u} - c)$, and if the meridional wavenumber l is larger than the zonal wavenumber k then l is given by

$$l^2(y) \approx \left[\frac{\beta^*(\bar{u} - c + ir/k)}{(\bar{u} - c)^2 + r^2/k^2} \right] \approx \frac{\beta^*}{\bar{u} - c} \left[1 + \frac{ir}{k(\bar{u} - c)} \right] \quad (16.43)$$

whence

$$l(y) \approx \left(\frac{\beta^*}{\bar{u} - c} \right)^{1/2} \left[1 + \frac{ir}{2k(\bar{u} - c)} \right], \quad (16.44)$$

The solution for the streamfunction is given by, in the WKB approximation,

$$\tilde{\psi} = A l^{-1/2} \exp \left(\pm i \int^y l dy' \right), \quad (16.45)$$

just as in (16.16), but now the wave will decay as it moves away from its source and deposit momentum into the mean flow. Let us calculate this.

The momentum flux, F_k , associated with an x -wavenumber of k is given by

$$F_k(y) = \overline{u'v'} = -ik \left(\psi \frac{\partial \psi^*}{\partial y} - \psi^* \frac{\partial \psi}{\partial y} \right), \quad (16.46)$$

and using (16.44) and (16.45) in (16.46) we obtain

$$F_k(y) = F_0 \exp \left(\int_0^y \frac{\pm r \beta^{*1/2}}{k(\bar{u} - c)^{3/2}} dy' \right). \quad (16.47)$$

In deriving this expression we use the fact that the amplitude of $\tilde{\psi}$ (i.e., $l^{-1/2}$) varies only slowly with y so that when calculating $\partial \tilde{\psi} / \partial y$ the derivative of l may be ignored. In (16.47) F_0 is the value of the flux at $y = 0$ and the sign of the exponent must be chosen so that the group velocity is directed away from the wave source region. Clearly, if $r = 0$ then the momentum flux is constant.

The integrand in (16.47) is the attenuation rate of the wave and it has a straightforward physical interpretation. Using the real part of (16.44) in (16.17b), and assuming $|l| \gg |k|$, the meridional component of the group velocity is given by

$$c_g^y = \frac{2kl\beta^*}{(k^2 + l^2)^2} \approx \frac{2k\beta^*}{l^3} = \frac{2k(\bar{u} - c)^{3/2}}{\beta^{*1/2}}. \quad (16.48a,b)$$

Thus we have

$$\text{Wave attenuation rate} = \frac{r\beta^{*1/2}}{k(\bar{u} - c)^{3/2}} = \frac{2 \times \text{Dissipation rate } (2r)}{\text{Meridional group velocity } (c_g^y)}. \quad (16.49)$$

This result is of some generality in wave dynamics, and its simple interpretation is that as the group velocity diminishes the dissipation has more time to act and so the wave is preferentially attenuated.

How does this attenuation affect the mean flow? The mean flow is subject to many waves and so obeys the equation

$$\frac{\partial \bar{u}}{\partial t} = - \sum_k \frac{\partial F_k}{\partial y} + \text{viscous terms}. \quad (16.50)$$

Because the amplitude varies only slowly compared to the phase, the amplitude of $\partial F_k / \partial y$ varies mainly with the attenuation rate (16.49). Consider a Rossby wave propagating away from some source region with a given frequency and x -wavenumber. Because k is negative a Rossby wave always carries westward (or negative) momentum with it. That is, F_k is always negative and increases (becomes more positive) as the wave is attenuated; that is to say, if $r \neq 0$ then $\partial F_k / \partial y$ is positive and from (16.50) the mean flow is accelerated *westward* as the wave dissipates. The dissipation, and attendant acceleration, will be particularly strong as the wave approaches a critical line where $\bar{u} = c$, although here the quantitative aspects of analysis begin to fail.

The situation arises when Rossby waves, generated in mid-latitudes, propagate equatorward. As the waves enter the subtropics $\bar{u} - c$ becomes smaller and the waves dissipate, producing a westward force on the mean flow. Globally, momentum is conserved because there is an equal and opposite (and therefore eastward) wave force at the wave source producing an eastward eddy-driven jet, as discussed in the previous chapter.

Interpretation using wave activity

We can derive and interpret the above results by thinking about the propagation of wave activity, specifically the pseudomomentum given by (16.30). Referring as needed to the discussion in sections 10.2.1 and 10.2.2, the flux obeys the group velocity property so that

$$\frac{\partial \mathcal{P}}{\partial t} + \frac{\partial}{\partial y}(c_g \mathcal{P}) = -2r\mathcal{P}. \quad (16.51)$$

Let us suppose that the wave is in a statistical steady state and that the spatial variation of the group velocity occurs on a longer spatial scale than the variations in wave activity, consistent with the WKB assumption that group velocity varies slowly. We then have

$$c_g^y \frac{\partial \mathcal{P}}{\partial y} = -2r\mathcal{P}. \quad (16.52)$$

which integrates to give

$$\mathcal{P}(y) = \mathcal{P}_0 \exp\left(-\int^y \frac{2r}{c_g^y} dy'\right). \quad (16.53)$$

That is, the attenuation rate of the wave activity is the dissipation rate of wave activity divided by the group velocity, as in (16.47) and (16.49). The wave-activity method of derivation suggests that this result is a general one, not restricted to Rossby waves, and indeed in section 17.3.2 we will find that the attenuation rate of vertically propagating gravity waves is given by a similar expression.

The divergence of wave activity will lead to a force on the mean zonal flow, much as discussed in section 15.1. For definiteness, suppose that waves propagate away from a mid-latitude source in the Northern Hemisphere. South of the source c_g^y is negative and north of the source c_g^y is positive. In either case, from (16.53) the wave activity density decreases away from the source and, with reference to (15.34a), the ensuing force on the mean flow is negative, or westward.

16.5 VERTICAL PROPAGATION OF ROSSBY WAVES

We now consider the vertical propagation of Rossby waves in a stratified atmosphere. The vertical propagation is important both because it must be taken into account to obtain an accurate picture of the tropospheric response to topographic and thermal forcing, and because it can excite motion in the stratosphere, as considered in chapter 17. We will continue to use the stratified quasi-geostrophic equations, but we now allow the model to be compressible and semi-infinite, extending from $z = 0$ to $z = \infty$. It is simplest to first consider the problem slightly generally, without regard to boundary conditions; section 16.5.2 we will consider the lower boundary conditions

and the requirements for waves propagate vertically into the stratosphere. Our governing equation is the quasi-geostrophic potential vorticity equation and with applications to the stratosphere in mind we will use log-pressure co-ordinates so that the equation of motion is

$$\frac{\partial q}{\partial t} + J(\psi, q) = 0, \quad q = \nabla^2 \psi + \beta y + \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \psi}{\partial z} \right), \quad (16.54)$$

where $z = H \ln(p/p_0)$ and $\rho_R = \rho_0 e^{-z/H}$ with H being a specified density scale height, typically $RT(0)/g$.

16.5.1 Conditions for wave propagation

Let us linearize (16.54) about a zonal wind that depends only on z ; that is, we let

$$\psi = -\bar{u}(z)y + \psi', \quad (16.55)$$

and obtain

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad \frac{\partial \bar{q}}{\partial y} = \beta - \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \bar{u}}{\partial z} \right), \quad (16.56)$$

or equivalently, in terms of streamfunction,

$$\left(\frac{\partial}{\partial t} + \bar{u} \frac{\partial}{\partial x} \right) \left[\nabla^2 \psi' + \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \psi'}{\partial z} \right) \right] + \frac{\partial \psi'}{\partial x} \left[\beta - \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \bar{u}}{\partial z} \right) \right] = 0. \quad (16.57)$$

The first term in square brackets is the perturbation potential vorticity, q' and the second term equals $\partial \bar{q}/\partial y$. Seeking solutions of the form $\psi' = \text{Re } \tilde{\psi}(z) \exp[i(kx + ly - kct)]$ gives

$$\left[\frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \tilde{\psi}}{\partial z} \right) \right] = \tilde{\psi} \left(K^2 - \frac{\partial \bar{q}/\partial y}{\bar{u} - c} \right) \quad (16.58)$$

Let us simplify by assuming that both $\bar{u}$ and N^2 are constants so that $\partial \bar{q}/\partial y = \beta$. Equation (16.58) further simplifies if we define

$$\Phi(z) = \tilde{\psi}(z) \left(\frac{\rho_R}{\rho_R(0)} \right)^{1/2} = \tilde{\psi}(z) e^{-z/2H} \quad (16.59)$$

whence we obtain

$$\frac{d^2 \Phi}{dz^2} + m^2 \Phi = 0, \quad \text{where} \quad m^2 = \frac{N^2}{f_0^2} \left(\frac{\beta}{\bar{u} - c} - K^2 - \gamma^2 \right), \quad (16.60a,b)$$

where $\gamma^2 = f_0^2/(4N^2H^2) = 1/(2L_d)^2$, where L_d is the deformation radius as sometimes defined (i.e., $L_d = NH/f_0$). The above equation has the same form as (16.10b). If the parameters on the right-hand side of (16.60b) are constant then so is m , and (16.60) has solutions of the form $\Phi(z) = \Phi_0 e^{imz}$, so that the streamfunction itself varies as

$$\psi' = \text{Re } \Phi_0 \exp [i(kx + ly + mz - kct) + z/2H]. \quad (16.61)$$

In the (more realistic) case in which m varies with height then, if the variation is slow enough, the solution looks locally like a plane wave with m being a slowly varying vertical wavenumber, and WKB techniques may be used to find a solution, as we discuss further in section 16.6. But even then

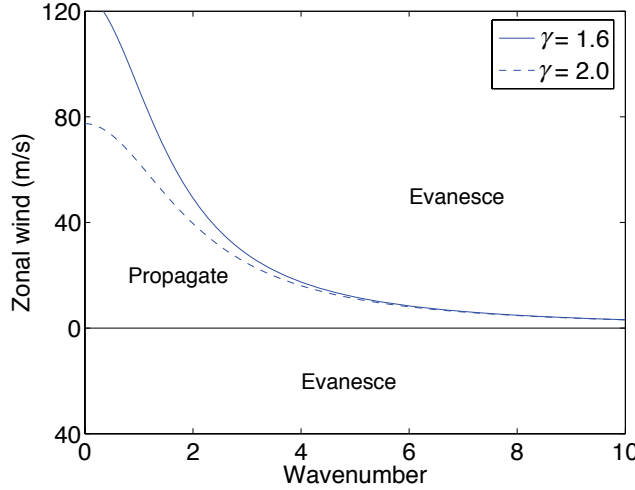


Fig. 16.6 The boundary between propagating waves and evanescent waves as a function of zonal wind and wavenumber, using (16.63), for a couple of values of γ . With $N = 2 \times 10^{-2} \text{ s}^{-1}$, $\gamma = 1.6$ ($\gamma = 2$) corresponds to a scale height of 7.0 km (5.5 km) and a deformation radius NH/f of 1400 km (1100 km).

essentially the same condition for propagation applies, namely that $m^2 > 0$ and, using (16.60b), this condition is satisfied if

$$0 < \bar{u} - c < \frac{\beta}{K^2 + \gamma^2}. \quad (16.62)$$

This condition is obviously similar to (16.12). For waves of some given frequency ($\omega = kc$) the above expression provides a condition on $\bar{u}$ for the vertical propagation of planetary waves. For stationary waves $c = 0$ the condition becomes

$$0 < \bar{u} < \frac{\beta}{K^2 + \gamma^2}. \quad (16.63)$$

That is, in words, stationary, vertically oscillatory modes can exist only for zonal flows that are eastwards and that are less than the critical velocity $U_c = \beta/(K^2 + \gamma^2)$. This criterion, known as the *Charney–Drazin condition*,⁴ is illustrated in Fig. 16.6 and we return to it in section 16.5.3. The critical velocity for stationary waves evidently depends on the scale of the wave. For waves of a non-zero frequency the criterion is less severe, but stationary waves have a particular importance because they can be readily generated by surface topography.

Dispersion relation and group velocity

Noting that $\omega = ck$ and rearranging (16.60b) we obtain the dispersion relation for three-dimensional Rossby waves

$$\omega = \bar{u}k - \frac{\beta k}{K^2 + \gamma^2 + m^2 f_0^2/N^2}. \quad (16.64)$$

The three components of the group velocity for these waves are then:

$$c_g^x = \bar{u} + \frac{\beta[k^2 - (l^2 + m^2 f_0^2/N^2 + \gamma^2)]}{(K^2 + m^2 f_0^2/N^2 + \gamma^2)^2}, \quad (16.65a)$$

$$c_g^y = \frac{2\beta kl}{(K^2 + m^2 f_0^2/N^2 + \gamma^2)^2}, \quad c_g^z = \frac{2\beta km f_0^2/N^2}{(K^2 + m^2 f_0^2/N^2 + \gamma^2)^2}. \quad (16.65b,c)$$

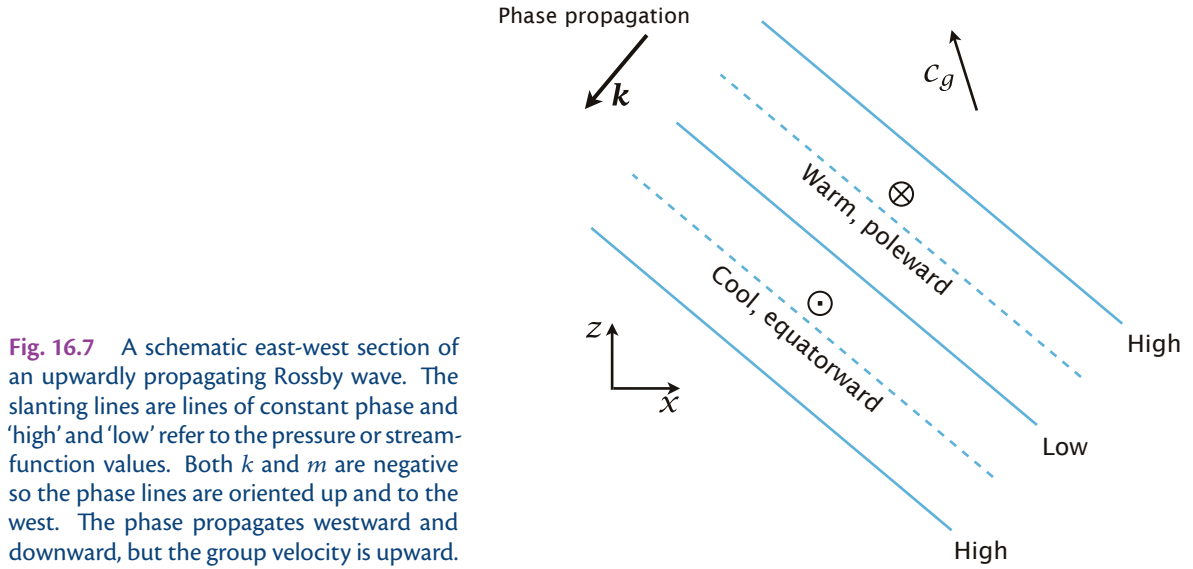


Fig. 16.7 A schematic east-west section of an upwardly propagating Rossby wave. The slanted lines are lines of constant phase and 'high' and 'low' refer to the pressure or stream-function values. Both k and m are negative so the phase lines are oriented up and to the west. The phase propagates westward and downward, but the group velocity is upward.

The propagation in the horizontal is analogous to the propagation in a shallow water model [c.f. (6.66b)]; note also that higher baroclinic modes (bigger m) will have a more westward group velocity. The vertical group velocity is proportional to m , and for waves that propagate signals upward we must choose m to have the same sign as k so that c_g^z is positive. If there is no mean flow then the zonal wavenumber k is negative (in order that frequency is positive) and m must then also be negative. Energy then propagates upward but the phase propagates downward.

16.5.2 ♦ A solution for topographically-excited waves

We now derive some explicit solutions for Rossby waves excited at a lower boundary by topography. Rossby waves may also be excited by thermal anomalies at the lower boundary, although in Earth's atmosphere their amplitude is somewhat smaller, and the treatment of such waves is left to the reader.⁵ The lower boundary is obtained using the thermodynamic equation,

$$\frac{\partial}{\partial t} \left(\frac{\partial \psi}{\partial z} \right) + J \left(\psi, \frac{\partial \psi}{\partial z} \right) + \frac{N^2}{f_0} w = 0, \quad (16.66)$$

along with an equation for the vertical velocity, w , at the lower boundary. This is

$$w = \mathbf{u} \cdot \nabla h_b + r \zeta, \quad (16.67)$$

where the two terms respectively represent the kinematic contribution to vertical velocity due to flow over topography and the contribution from Ekman pumping, with r a constant, and the effects are taken to be additive. Linearizing the thermodynamic equation about the zonal flow and using (16.67) gives the boundary condition at $z = 0$,

$$\frac{\partial}{\partial t} \left(\frac{\partial \psi'}{\partial z} \right) + \bar{u} \frac{\partial}{\partial x} \frac{\partial \psi'}{\partial z} - v' \frac{\partial \bar{u}}{\partial z} = - \frac{N^2}{f_0} \left(\bar{u} \frac{\partial h_b}{\partial x} + r \nabla^2 \psi' \right), \quad (16.68)$$

Solution

We look for solutions of (16.56) and (16.68) in the form

$$\psi' = \text{Re } \tilde{\psi}(z) \sin l y e^{ik(x-ct)} \quad \text{with} \quad h_b = \text{Re } \tilde{h}_b \sin l y e^{ikx} \quad (16.69)$$

with $\tilde{h}_b$ being purely real. Solutions must then satisfy

$$\left[\frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \tilde{\psi}}{\partial z} \right) \right] = \tilde{\psi} \left(K^2 - \frac{\partial \bar{q} / \partial y}{\bar{u} - c} \right) \quad (16.70)$$

in the interior, and the boundary condition

$$(\bar{u} - c) \frac{\partial \tilde{\psi}}{\partial z} - \tilde{\psi} \frac{\partial \bar{u}}{\partial z} + \frac{irN^2 K^2}{kf_0} \tilde{\psi} = -\frac{N^2 \bar{u} \tilde{h}_b}{f_0}, \quad \text{at } z = 0, \quad (16.71)$$

as well as a radiation condition at plus infinity (and we must have that $\rho_0 \tilde{\psi}^2$ be finite). Let us simplify by considering the case of constant $\bar{u}$ and N^2 and with $r = 0$. As before we let $\Phi(z) = \tilde{\psi}(z) \exp(-z/2H)$ and obtain the interior equation

$$\frac{d^2 \Phi}{dz^2} + m^2 \Phi = 0, \quad \text{where } m^2 = \frac{N^2}{f_0^2} \left(\frac{\beta}{\bar{u} - c} - K^2 - \gamma^2 \right), \quad (16.72a,b)$$

and $\gamma^2 = f_0^2 / (4N^2 H^2) = 1 / (2L_d)^2$, where L_d is the deformation radius, which is essentially the same as (16.60). The conditions for wave propagation, and in particular the Charney–Drazin condition, are just as previously derived. The surface boundary condition is now

$$(\bar{u} - c) \left(\frac{d\Phi}{dz} + \frac{\Phi}{2H} \right) = -\frac{N^2 \bar{u} \tilde{h}_b}{f_0}, \quad \text{at } z = 0. \quad (16.73)$$

This leads to an expression for the streamfunction amplitude, namely

$$\Phi_0 = \frac{N^2 \tilde{h}_b / f_0}{(\alpha - im) - (2H)^{-1}}, \quad \text{where } \alpha = +\frac{N}{f_0} \left(K^2 + \gamma^2 - \frac{\beta}{\bar{u}} \right)^{1/2}, \quad (16.74a,b)$$

and $(\alpha, -im)$ refers to the (trapped, oscillatory) case. A little algebra gives the solutions in the form

$$\psi'(x, y, z) = \text{Re} \exp[i(kx + mz) + z/2H] \sin ly \frac{f_0 \tilde{h}_b [im - (2H)^{-1}]}{K_s^2 - K^2}, \quad m^2 > 0 \quad (16.75a)$$

$$\psi'(x, y, z) = \text{Re} \exp[(2H)^{-1} - \alpha z + ikx] \sin ly \frac{N^2 \tilde{h}_b}{f_0 [\alpha - (2H)^{-1}]}, \quad m^2 < 0 \quad (16.75b)$$

where $K_s^2 = \beta / \bar{u}$.

Evidently, as resonance is possible when $\alpha = 1/(2H)$ or $K^2 = K_s^2$, that is when barotropic Rossby waves are stationary. The wave resonates because the wave is a solution of the unforced, inviscid equations for the barotropic wave. If $K > K_s$ then $\alpha > 1/(2H)$ and the forced wave (i.e., the amplitude of ψ) decays with height with no phase variation. If $\alpha < 1/(2H)$ then $\tilde{\psi}$ increases with height (although $\rho_R |\tilde{\psi}|^2$ decreases with height), and this occurs when $(K_s^2 - \gamma^2)^{1/2} < K < K_s$. If $(K_s^2 - \gamma^2)^{1/2} > K$ then the amplitude of ϕ , (i.e., $\rho_R |\tilde{\psi}|^2$) is independent of height; their vertical structure is oscillatory, like $\exp(imz)$.

16.5.3 Properties of the solutions

Upward propagation and the Charney–Drazin condition

Let us return to the criterion for upward propagation given in (16.63) and illustrated in Fig. 16.6. One way to interpret this condition is note that in a resting medium the Rossby wave frequency has a minimum value (and maximum absolute value), when $m = 0$, of

$$\omega = -\frac{\beta k}{K^2 + \gamma^2}. \quad (16.76)$$

Suppose that the waves are generated by bottom topography, and that $\bar{u}$ is uniform. In a frame moving with speed $\bar{u}$ our Rossby waves (stationary in the Earth's frame) have frequency $-\bar{u}k$, and this is the forcing frequency arising from the now-moving bottom topography. Thus, (16.63) is equivalent to saying that for oscillatory waves to exist *the forcing frequency must lie within the frequency range of vertically propagating Rossby waves*.

For westward flow, or for sufficiently strong eastward flow, the waves decay exponentially as $\Phi = \Phi_0 \exp(-\alpha z)$ where α is given by (16.74b). The critical velocity $u_c = (\beta/K^2 + \gamma^2)$ is a function of wavenumber, increasing with horizontal wavelength. Thus, for a given eastward flow long waves may penetrate vertically when short waves are trapped, an effect sometimes referred to as 'Charney–Drazin filtering'. There are three important consequences of this.

- (i) Stratospheric motion is typically of larger horizontal scale than that of the troposphere, because Rossby waves tend to be excited first in the troposphere (by both baroclinic instability and flow over topography), but the shorter waves are trapped and only the longer ones reach the stratosphere.
- (ii) The Rossby waves more commonly reach the stratosphere in the Northern Hemisphere than in the Southern Hemisphere. In both hemispheres the shorter, baroclinically forced Rossby waves are filtered, but the Northern Hemisphere also generates long Rossby waves through topographic interactions. The overturning circulation in the stratosphere (the Brewer–Dobson circulation) is therefore stronger in the Northern Hemisphere.
- (iii) Rossby waves find it hard to reach the stratosphere in summer, for then the stratospheric winds are often westwards (because the pole is warmer than the equator) and all waves are trapped in the troposphere. The eastward stratospheric winds that favour vertical penetration occur in the other three seasons, although very strong eastward winds can suppress penetration in mid-winter.

Other properties of the solution

Various other properties of the solution are described below.

Amplitudes and phases. The decaying solutions have no vertical phase variations (a property known as 'equivalent barotropic') and the streamfunction is exactly in phase or out of phase with the topography according as $K > K_s$ and $\alpha > (2H)^{-1}$, or $K < K_s$ and $\alpha < (2H)^{-1}$. In the latter case the amplitude of the streamfunction actually increases with height, but the energy, proportional to $\rho_R |\psi'^2|$ falls. The oscillatory solutions have (if there is no shear) constant energy with height but a shifting phase. The phase of the streamfunction at the surface may be in or out of phase with the topography, depending on m , but the potential temperature, $\partial\psi/\partial z$ is always out of phase with the topography. That is, positive values of h_b are associated with cool fluid parcels.

Vertical energy propagation. As noted, the energy propagates upwards for the oscillatory waves.

This may be verified by calculating $\overline{p'w'}$ (the vertical component of the energy flux), where p' is the pressure perturbation, proportional to ψ' , and w' is the vertical velocity perturbation. To this end, linearize the thermodynamic equation (16.66) to give

$$\frac{\partial}{\partial t} \left(\frac{\partial \psi'}{\partial z} \right) + \bar{u} \frac{\partial}{\partial x} \frac{\partial \psi'}{\partial z} - \frac{\partial \bar{u}}{\partial z} \frac{\partial \psi'}{\partial x} + \frac{N^2}{f_0} w' = 0. \quad (16.77)$$

Then, multiplying by ψ' and integrating by parts gives a balance between the second and fourth terms,

$$N^2 \overline{\psi' w'} = \overline{u b' v'}, \quad (16.78)$$

Upward Propagating Rossby Waves

In general Rossby waves satisfy

$$\psi' = \text{Re } \Phi_0 \exp [i(kx + ly + mz - kct) + z/2H], \quad (\text{T.1})$$

where Φ_0 is a constant determined by the lower boundary conditions and

$$m = \pm \frac{N}{f_0} \left(\frac{\beta}{\bar{u}} - K^2 - \gamma^2 \right)^{1/2}, \quad \text{with} \quad \gamma = \frac{f_0}{2NH}. \quad (\text{T.2})$$

If $m^2 > 0$ the solutions are propagating, or radiating, waves in the vertical direction. If $m^2 < 0$ the energy of the solution, $|\rho_R \psi'^2|$, is vertically evanescent. The condition $m^2 > 0$ is equivalent to

$$0 < \bar{u} < \frac{\beta}{k^2 + l^2 + (f_0/2NH)^2}, \quad (\text{T.3})$$

which is known as the Charney–Drazin condition. Vertical penetration is favoured when the winds are weakly eastwards, and the range of $\bar{u}$ -values that allows this is larger for longer waves. Some other properties of the solution are:

- In order that the energy propagate upwards the vertical component of the group velocity must be positive, and hence k and m must have the same sign.
- The meridional heat flux is proportional to km , and thus upward propagation of waves is associated with a poleward heat transport.
- In an atmosphere in which density falls exponentially with height the amplitude of the streamfunction grows exponentially, so eventually nonlinear terms will become important. The waves may break, even in the absence of a critical layer.

where $b' = f_0 \partial \psi' / \partial z$ and $v' = \partial \psi' / \partial x$. Thus, the upward transfer of energy is proportional to the poleward heat flux. Evidently, the transfer of energy is upward when $km > 0$, and from (16.65), this corresponds to the condition that the vertical component of group velocity is positive, which has to be the case from general arguments. For Rossby waves $k < 0$ so that upward energy propagation requires $m < 0$ and therefore downward phase propagation.

Meridional heat transport. The meridional heat transport associated with a wave is

$$\rho_R \overline{v' b'} = \rho_R f_0 \overline{\frac{\partial \psi'}{\partial x} \frac{\partial \psi'}{\partial z}}. \quad (16.79)$$

For an oscillatory wave the heat flux is proportional to km . Now, the condition that energy is directed upward is that km is positive, for then c_g^z is positive. Thus, upward propagation is associated with a polewards heat flux. The meridional transport associated with a trapped solution is identically zero.

Form drag. If the waves propagate energy upwards, there must be a surface interaction to supply that energy. There is a force due to *form drag* (see section 3.6) associated with this interaction

given by

$$\text{form drag} = \overline{p' \frac{\partial h_b}{\partial x}}. \quad (16.80)$$

In the trapped case, the streamfunction is either exactly in or out of phase with the topography, so this interaction is zero. In the oscillatory case

$$\overline{\psi' \frac{\partial h_b}{\partial x}} = \frac{f_0 \tilde{h}_b^2 k m}{4(K_s^2 - K^2)}, \quad (16.81)$$

where the factor of 4 arises from the x and y averages of the squares of sines and cosines. The rate of doing work is $\bar{u}$ times (16.81).

16.6 ♦ VERTICAL PROPAGATION OF ROSSBY WAVES IN SHEAR

In the real atmosphere the zonal wind and the stratification change with height and there may be regions in which propagation occurs and regions where it does not, and in this section we illustrate that phenomena with two examples. In one example the zonal wind increases sufficiently with height that wave propagation ceases because the wind is too strong, and in the other the zonal wind decreases aloft and becomes negative (westward), again causing wave propagation to cease. If the zonal wind and the stratification both vary sufficiently slowly with height — meaning that the scale of the variation is much greater than a vertical wavelength — then *locally* the solution will look like a plane wave and the analysis is straightforward, very similar to that performed in sections 7.5 (where we looked at internal waves with varying stratification) and section 16.2.2 (where we looked at horizontally propagating Rossby waves).

For simplicity we consider Rossby waves in a flow with vertical shear but no horizontal shear, with constant stratification and constant density. With reference to section 16.1.2, the equation of motion is

$$\left(\frac{\partial}{\partial t} + \bar{u}(z) \frac{\partial}{\partial x} \right) q' + \beta v' = 0. \quad (16.82)$$

We seek solutions of the form

$$\psi' = \tilde{\psi}(z) e^{ik(x-ct)+ly}, \quad (16.83)$$

obtaining

$$\frac{\partial^2 \tilde{\psi}}{\partial z^2} + m^2(z) \tilde{\psi} = 0, \quad \text{where} \quad m^2(z) = \frac{N^2}{f_0^2} \left[\frac{\beta}{\bar{u} - c} - (k^2 + l^2) \right]. \quad (16.84a,b)$$

The WKB solution to this equation (see appendix on page 247) is

$$\tilde{\psi}(z) = A m^{-1/2} e^{i \int m dz}. \quad (16.85)$$

where A is a constant. The local vertical wavenumber is just m itself (for this is the derivative of the phase), and the amplitude varies like $m^{-1/2}$. This variation of amplitude is consistent with the conservation of wave activity, which in this case means that the Eliassen–Palm flux is constant. As there is no horizontal divergence in this problem, the constancy of $\mathcal{F}$ in (16.19) implies $\partial_z v' b' = 0$ and therefore

$$k m |\tilde{\psi}|^2 = \text{constant}. \quad (16.86)$$

Since the horizontal wavenumber is constant the dependence of the amplitude on $m^{-1/2}$ immediately follows. The energy of the wave is not constant unless there is no shear, since it may be extracted or given up to the mean flow.

As discussed in earlier sections wave propagation requires that m^2 be positive. For stationary waves ($c = 0$) this gives the condition that $0 < \bar{u} < \beta/(k^2 + l^2)$. At the lower bound there is a critical line and $m^2 \rightarrow \infty$. At the upper bound $m^2 = 0$ and this is a turning line. Let us illustrate the behaviour approaching these regions with two examples.

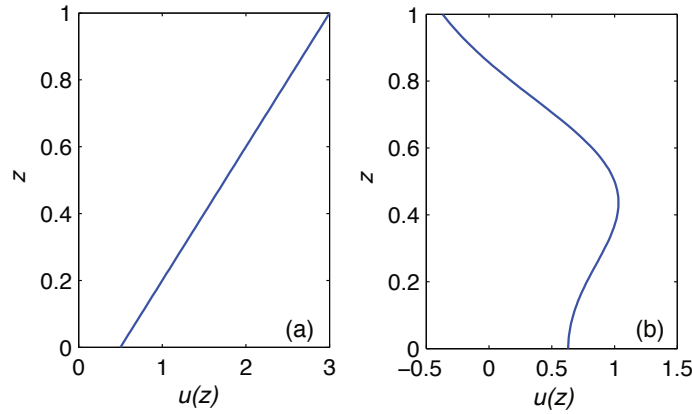


Fig. 16.8 Two profiles of nondimensional zonal wind used in the calculations illustrated in Fig. 16.9 and Fig. 16.10. (a) is a uniform shear that gives rise to a turning latitude, and (b) shows a profile in which the zonal wind diminishes to zero aloft, giving rise to a critical line.

16.6.1 Two examples

In much the same way as we illustrated horizontal propagation in section 16.2 we'll calculate the group velocity and wave amplitudes in two cases, one with a turning line and the other with a critical line. The same caveats apply — that the slowly-varying assumption fails at these lines, and we cannot use WKB or ray theory to properly calculate reflection or absorption.

Waves with a turning line

Consider Rossby waves propagating in a background state in which the zonal wind increases uniformly with height, as in Fig. 16.8a, but in which all other parameters are constant. Specifically, we choose (nondimensional) values of $\beta = 5$, $k = l = 1$ and $c = 0$ (the reader may re-dimensionalize). We also scale the vertical coordinate so that $Pr = 1$. For the profile chosen m^2 is positive for $\bar{u} < 2.5$ and so for $0 < z < 0.8$, as shown in Fig. 16.9. For l fixed and m given by (16.84b) we calculate the group velocity using (16.7) and these are displayed in Fig. 16.9. We choose upwardly propagating waves (i.e. $m > 0$); in any physical situation the group velocity will be directed away from the source, and we are assuming this occurs at the surface. We also show equatorward moving waves for $y < 0.5$ and poleward moving waves for $y > 0.5$, but this is for illustrative purposes. The right-hand panel of the figure shows the value of m^2 diminishing with height, along with the vertical profiles of the amplitude (which goes like $m^{-1/2}$) and the energy (which goes like $(k^2 + l^2 + m^2)m$).

We see from Fig. 16.9 that the group velocity turns away from the turning line, and we can understand this from the ratio of the group velocities given in (16.7), namely

$$\frac{c_g^z}{c_g^y} = \frac{Pr^2 m}{l}, \quad (16.87)$$

where $Pr = f/N$. The group velocity is horizontal at the turning line. The amplitude of the waves is infinite, but the waves do not necessarily break because the vertical wavelength is very large. In fact, for the reasons given in the horizontally propagating case, we expect the waves to reflect.

Waves with a critical line

Now consider waves in a zonal wind that initially increases with height and then decreases and becomes negative, as illustrated in Fig. 16.8b. There is a critical line where $\bar{u}$ passes through zero, but all the other parameters are the same as in the previous example. The value of m^2 now generally increases with height, as illustrated in the right-hand panel of Fig. 16.10, becoming infinite at the critical line and negative above it. The amplitude of the wave, being proportional to $m^{-1/2}$ actually goes to zero at the critical line but the energy increases without bound (in the linear, inviscid approximation).

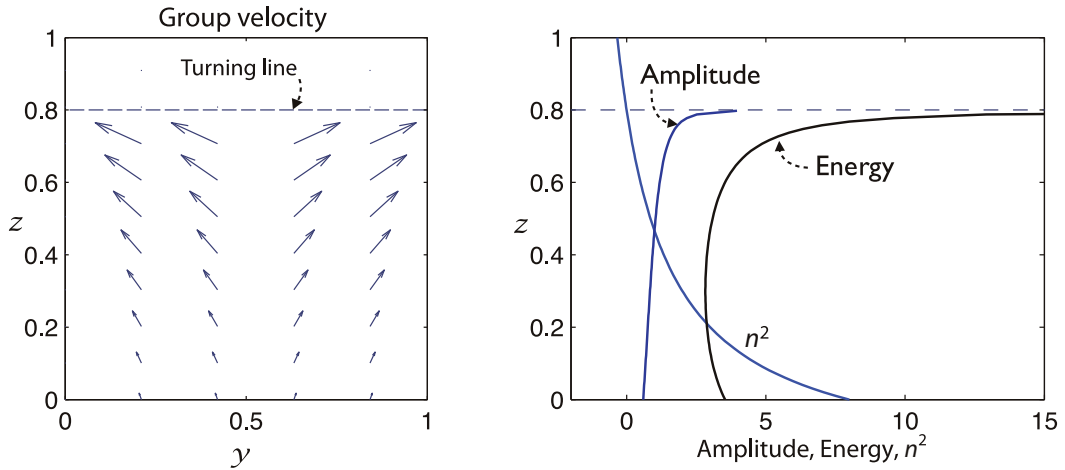


Fig. 16.9 Vertically propagating Rossby waves approaching a turning line. Left panel: group velocity vectors calculated using (16.7) for the parameters shown in Fig. 16.8a and with a source at $y = 0.5$. Right: profiles of m^2 , wave amplitude and energy. The horizontal line at $z = 0.8$ marks a turning line, and the group velocity turns away from it.

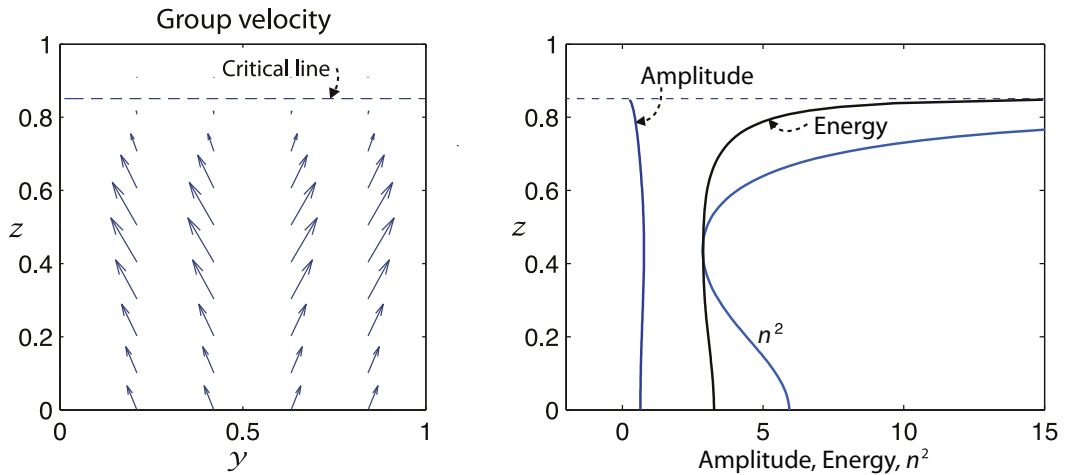


Fig. 16.10 Vertically propagating Rossby waves approaching a critical line. Left panel: group velocity vectors calculated using (16.7) for the parameters shown in Fig. 16.8b. Right: profiles of m^2 , wave amplitude and energy. The horizontal line at $z \approx 0.85$ marks a critical line; the group velocity turns toward it but its amplitude diminishes as the critical line is approached.

The group velocity, shown in the left panel of Fig. 16.10, turns upward and toward the critical line and, from (16.87), is purely vertical at the critical line. The amplitude of the group velocity also diminishes, frictional and nonlinear effects become more important, and the notion of group velocity itself ceases to be meaningful close to the critical line where the WKB approximation breaks down.

As a final remark, we note that the condition that there be neither a critical line or a turning line is essentially a version of the Charney–Drazin condition.

16.7 FORCED AND STATIONARY ROSSBY WAVES

We now turn our attention to understanding the large-scale zonally asymmetric circulation of the atmosphere, much of which is determined by the presence of stationary Rossby waves forced by topographic and thermal anomalies at the surface.⁶

16.7.1 A simple one-layer case

Many of the essential ideas can be illustrated by a one-layer quasi-geostrophic model, with potential vorticity equation

$$\frac{Dq}{Dt} = 0, \quad q = \zeta + \beta y - \frac{f_0}{H}(\eta - h_b), \quad (16.88)$$

where H is the mean thickness of the layer, η is the height of the free surface, h_b is the bottom topography, and the velocity and vorticity are given by $\mathbf{u} = (g/f_0)\nabla^\perp \eta \equiv (g/f_0)\mathbf{k} \times \nabla \eta$ and $\zeta = (\partial v/\partial x - \partial u/\partial y) = (g/f_0)\nabla^2 \eta$. Linearizing (16.88) about a flat-bottomed state with zonal flow $\bar{u}(y) = -(g/f_0)\partial \bar{\eta}/\partial y$ gives

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad (16.89)$$

where $q' = \zeta' - (f_0/H)(\eta' - h_b)$ and $\partial \bar{q}/\partial y = \beta + \bar{u}/L_d^2$ with $L_d = \sqrt{gH}/f_0$, the radius of deformation. Equation (16.89) may be written, after the cancellation of a term proportional to $\bar{u}\partial \eta'/\partial x$, as

$$\frac{\partial}{\partial t} \left(\zeta' - \frac{\psi'}{L_d^2} \right) + \bar{u} \frac{\partial \zeta'}{\partial x} + \beta v' = -\bar{u} \frac{\partial \hat{h}}{\partial x}, \quad (16.90)$$

where $\psi' = (g/f_0)\eta'$ and $\hat{h} = h_b f_0/H = h_b g/(L_d^2 f_0)$.

The solution of this equation consists of the solution to the homogeneous problem (with the right-hand side equal to zero, as considered in section 6.4 on Rossby waves) and the particular solution. We proceed by decomposing the variables into their Fourier components

$$(\zeta', \psi', \hat{h}) = \text{Re}(\tilde{\zeta}, \tilde{\psi}, \tilde{h}_b) \sin ly e^{ikx}. \quad (16.91)$$

where such decomposition is appropriate for a channel, periodic in the x -direction and with no variation at the meridional boundaries, $y = (0, L)$. The full solution will be a superposition of such Fourier modes and, because the problem is linear, these modes do not interact. The free Rossby waves, the solution to the homogeneous problem, evolve according to

$$\psi = \text{Re} \tilde{\psi} \sin ly e^{i(kx - \omega t)}, \quad (16.92)$$

where ω is given by the dispersion relation [cf. (6.65)]

$$\omega = k\bar{u} - \frac{k\partial_y \bar{q}}{K^2 + k_d^2} = \frac{k(\bar{u}K^2 - \beta)}{K^2 + k_d^2}, \quad (16.93a,b)$$

where $K^2 = k^2 + l^2$ and $k_d = 1/L_d$. Stationary waves occur at the wavenumbers for which $K = K_s \equiv \sqrt{\beta/\bar{u}}$. To the free waves we add the solution to the steady problem,

$$\bar{u} \frac{\partial \zeta'}{\partial x} + \beta v' = -\bar{u} \frac{\partial \hat{h}}{\partial x}, \quad (16.94)$$

which is, using the notation of (16.91)

$$\tilde{\psi} = \frac{\tilde{h}_b}{(K^2 - K_s^2)}. \quad (16.95)$$

Now, $\tilde{h}_b$ is a complex amplitude; thus, for $K > K_s$ the streamfunction response is *in phase* with the topography. For $K^2 \gg K_s^2$ the steady equation of motion is

$$\bar{u} \frac{\partial \zeta'}{\partial x} \approx -\bar{u} \frac{\partial \hat{h}}{\partial x}, \quad (16.96)$$

and the topographic vorticity source is balanced by zonal advection of relative vorticity. For $K^2 < K_s^2$ the streamfunction response is *out of phase* with the topography, and the dominant balance for very large scales is between the meridional advection of planetary vorticity, $v \partial f / \partial y$ or βv , and the topographic source. For $K = K_s$ the response is infinite, with the stationary wave resonating with the topography. Now, any realistic topography can be expected to have contributions from *all* Fourier components. Thus, for *any* given zonal wind there will be a resonant wavenumber and an infinite response. This, of course, is not observed, and one reason is that the real system contains friction. The simplest way to include this is by adding a linear damping to the right-hand side of (16.90), giving

$$\frac{\partial}{\partial t} \left(\zeta' - \frac{\psi'}{L_d^2} \right) + \bar{u} \frac{\partial \zeta'}{\partial x} + \beta v' = -r \zeta' - \bar{u} \frac{\partial \hat{h}}{\partial x}. \quad (16.97)$$

The free Rossby waves all decay monotonically to zero. However, the steady problem, obtained by omitting the first term on the left-hand side, has solutions

$$\tilde{\psi} = \frac{\tilde{h}_b}{(K^2 - K_s^2 - iR)}, \quad (16.98)$$

where $R = (rK^2 / \bar{u}k)$, and the singularity has been removed. The amplitude of the response is still a maximum for the stationary wave, and for this wave the phase of the response is shifted by $\pi/2$ with respect to the topography. The solution is shown in Fig. 16.11.⁷ It is typical that for a mountain range whose Fourier composition contains all wavenumbers, there is a minimum in the streamfunction a little downstream of the mountain ridge.

16.7.2 Application to Earth's atmosphere

With three parameters, I can fit an elephant.

William Thomson, Lord Kelvin (1824–1907).

Rather surprisingly, given the complexity of the real system and the simplicity of the model, when used with realistic topography a one-layer model gives reasonably realistic answers for the Earth's atmosphere. Thus, we calculate the stationary response to the Earth's topography using (16.97), using a reasonably realistic representation of the Earth's topography and, with qualification, the zonal wind. The zonal wind on the left-hand side of (16.97) is interpreted as the wind in the mid-troposphere, whereas the wind on the right-hand side is better interpreted as the surface wind, and so perhaps is about 0.4 times the mid-troposphere wind. Since the problem is linear, this amounts to tuning the amplitude of the response. The results, obtained using a rather crude representation of the Earth's topography, are plotted in Fig. 16.12. Also plotted is the observed time averaged response of the real atmosphere (the 500 hPa height field at 45° N). The agreement between model and observation is quite good, but this must be regarded as somewhat fortuitous if only because the other main source of the stationary wave field — thermal forcing — has been completely omitted from the calculation. Quantitative agreement is thus a consequence of the aforesaid tuning. Nevertheless, the calculation does suggest that the stationary, zonally asymmetric, features of the Earth's atmosphere arise via the interaction of the zonally symmetric wind field and the zonally asymmetric lower boundary, and that these may be calculated to a reasonable approximation with a linear model.

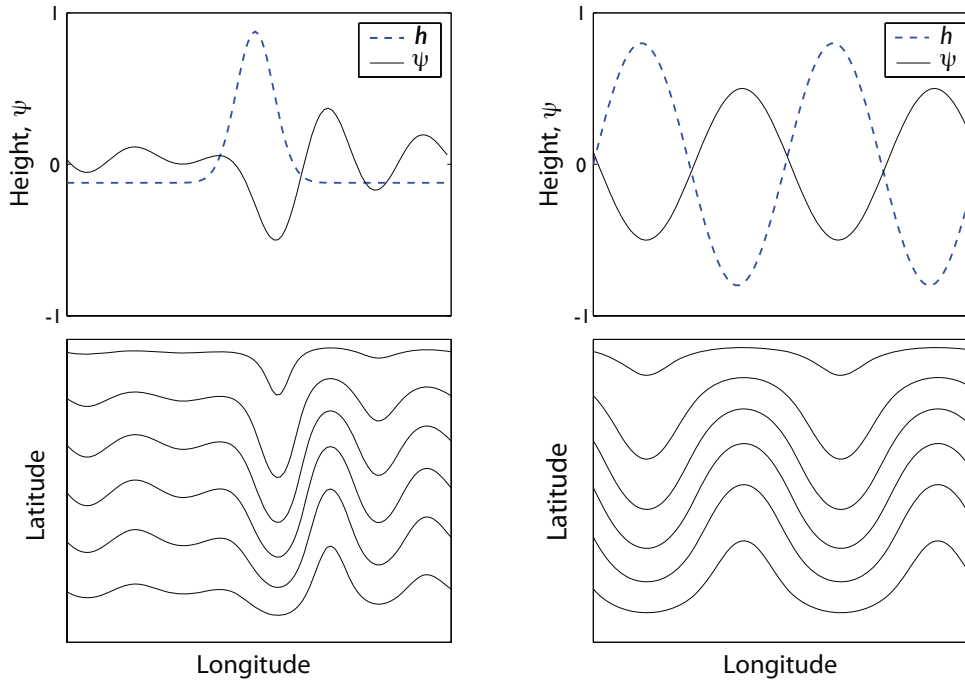


Fig. 16.11 The response to topographic forcing, i.e., the solution to the steady version of (16.97), for topography consisting of an isolated Gaussian ridge (left panels) and a pure sinusoid (right panels). The wavenumber of the stationary wave is about 4 and $r/(\bar{u}k) = 1$. The upper panels show the amplitude of the topography (dashed curve) and the perturbation streamfunction response (solid curve). The lower panels are contour plots of the streamfunction, including the mean flow. With the ridge, the response is dominated by the resonant wave and there is a streamfunction minimum, a ‘trough’, just downstream of the ridge. In the case on the right, the flow cannot resonate with the topography, which consists only of wavenumber 2, and the response is exactly out of phase with the topography.

16.7.3 ♦ One-dimensional Rossby wave trains

Although the Fourier analysis above gives exact (linear) results, it is not particularly revealing of the underlying dynamics. We see from Fig. 16.11 that the response to the Gaussian ridge is largely downstream of the ridge, and this suggests that it will be useful to consider the response as being due to Rossby *wavetrains* being excited by local features. This is also suggested by Fig. 16.13, which shows that the response to realistic topography is relatively local, and may be considered to arise from two relatively well-defined wavetrains each of finite extent one coming from the Rockies and the other from the Himalayas.

One way to analyse these wavetrains, and one which also brings up the concept of group velocity in a natural way, is to exploit (as in section 15.1.2) a connection between changes in wavenumber and changes in frequency. Consider the linear barotropic vorticity equation in the form

$$\frac{\partial}{\partial t}(\zeta - k_d^2 \psi) + \bar{u} \frac{\partial \zeta}{\partial x} + \beta \frac{\partial \psi}{\partial x} = -r\zeta, \quad (16.99)$$

where r is a frictional coefficient, which we presume to be small. Setting $k_d = 0$ for simplicity, the linear dispersion relation is

$$\omega = \bar{u}k - \frac{\beta k}{K^2} - ir \equiv \omega_R(k, l) - ir, \quad (16.100)$$

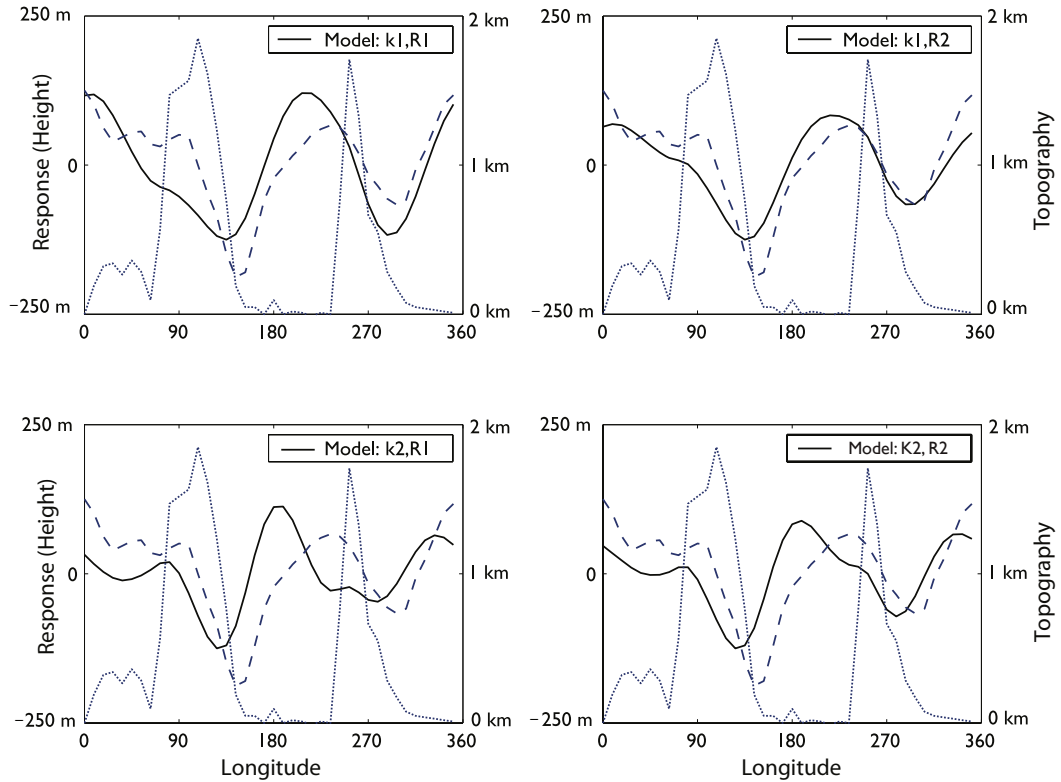


Fig. 16.12 Solutions of the Charney–Eliassen model. The solid lines are the steady solution of (16.97) using the Earth’s topography at 45° N with two values of friction ($R1 \approx 6$ days, $R2 \approx 3$ days) and two values of resonant zonal wavenumber (2.5 for $k1$, 3.5 for $k2$), corresponding to zonal winds of approximately 17 and 13 m s^{-1} . The solutions are given in terms of height, η' , where $\eta' = f_0 \psi' / g$, with the scale on the left of each panel. The dashed line in each panel is the observed average height field at 500 hPa at 45° N in January. The dotted line is the topography used in the calculations, with the scale on the right of each panel.

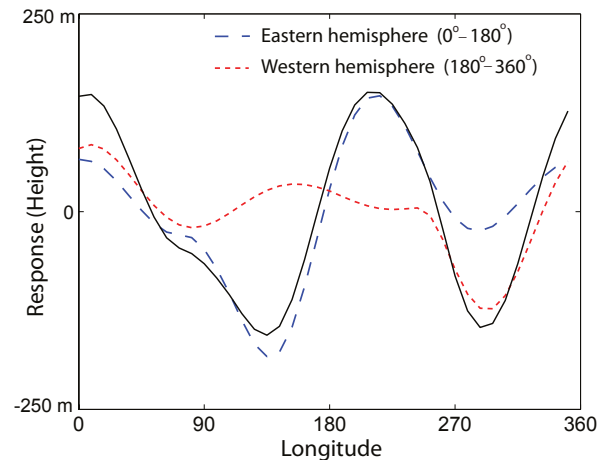


Fig. 16.13 The solution of the upper left-hand panel of Fig. 16.12 (solid line), and the solution divided into two contributions (dashed lines), one due to the topography only of the western hemisphere (i.e., with the topography in the east set to zero) and the other due to the topography only of the eastern hemisphere.

where $K^2 = k^2 + l^2$ and $\omega_R(k, l)$ is the inviscid dispersion relation for Rossby waves. Now, if there is a local source of the waves, for example an isolated mountain, we may expect to see a *spatial* attenuation of the wave as it moves away from the source. We may then regard the system as having a fixed, real frequency, but a changing, possibly complex, wavenumber. To determine this wavenumber for stationary waves (and so with $\omega = 0$), for small friction we expand the dispersion relation in a Taylor series about the inviscid value of ω_R at the real stationary wavenumber k_s , where $k_s = (K_s^2 - l^2)^{1/2}$ and $K_s = \sqrt{\beta/\bar{u}}$. This gives

$$\omega + ir = \omega_R(k, l) \approx \omega_R(k_s, l) + \left. \frac{\partial \omega_R}{\partial k} \right|_{k=k_s} k' + \dots \quad (16.101)$$

Thus, $k' \approx ir/c_g^x$, where c_g^x is the zonal component of the group velocity evaluated at a fixed position and at the stationary wavenumber; using (6.61b) this is given by

$$c_g^x = \left. \frac{\partial \omega_R}{\partial k} \right|_{k=k_s} = \frac{2\bar{u}k_s^2}{k_s^2 + l^2}. \quad (16.102)$$

The solution therefore decays away from a source at $x = 0$ according to

$$\psi \sim \exp(ikx) = \exp[i(k_s + k')x] \approx \exp(ik_s x - rx/c_g^x) \quad (16.103)$$

and, because $c_g^x > 0$, the response is east of the source. The approximate solution for the streamfunction (denoted ψ_δ) of (16.97) in an infinite channel, with the topography being a δ -function mountain ridge at $x = x'$, and with all fields varying meridionally like $\sin ly$, is thus

$$\psi_\delta(x - x', y) \sim \begin{cases} 0 & x \leq x' \\ -\frac{1}{k_s} \sin ly \sin[k_s(x - x')] \exp[-r(x - x')/c_g^x] & x \geq x'. \end{cases} \quad (16.104)$$

In the more general problem in which the topography is a general function of space, every location constitutes a separate source of wavetrains, and the complete (approximate) solution is given by the integral

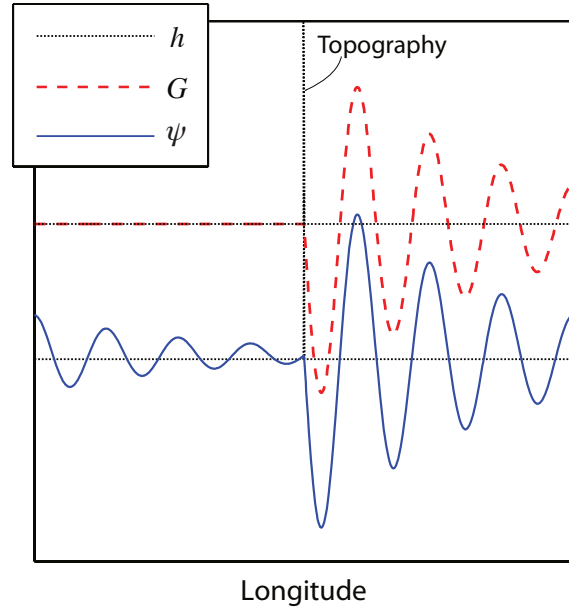
$$\psi'(x, y) = \int_{-\infty}^{\infty} \hat{h}(x) \psi_\delta(x - x', y) dx'. \quad (16.105)$$

The field $\psi_\delta(x - x', y)$, is the *Green function* for the problem, sometimes denoted $G(x - x', y)$.

Example solutions calculated using both the Fourier and Green function methods are illustrated in Fig. 16.14. As in Fig. 16.11 there is a trough immediately downstream of the mountain, a result that holds for a broad range of parameters. In these solutions, the streamfunction decays almost completely in one circumnavigation of the channel, and thus, downstream of the mountain, both methods give virtually identical results. Such a correspondence will not hold if the wave can circumnavigate the globe with little attenuation, for then resonance will occur and the Green function method will be inaccurate; thus, whether the resonant picture or the wavetrain picture is more appropriate depends largely on the frictional parameter. A frictional time scale of about 10 days is often considered to approximately represent the Earth's atmosphere, in which case waves are only slightly damped on a global circumnavigation, and the Fourier picture is natural with the possibility of resonance. However, the smaller (more frictional) value of 5 days seems to give quantitatively better results in the barotropic problem, and the solution is more evocative of wavetrains. The larger friction may perform better because it is crudely parameterizing the meridional propagation and dispersion of Rossby waves that is neglected in the one-dimensional model.⁸

Fig. 16.14 A one-dimensional Rossby wave train excited by uniform eastward flow over a δ -function mountain ridge (h) in the centre of the domain. The upper curve, G , shows the Green's function (16.104), whereas the lower curve shows the exact (linear) response, ψ , in a re-entrant channel calculated numerically using the Fourier method.

The two solutions are both centred around zero and offset for clarity; the only noticeable difference is upstream of the ridge, where there is a finite response in the Fourier case because of the progression of the wave-train around the channel. The stationary wavenumber is 7.5.



16.7.4 The adequacy of linear theory

Having calculated some solutions, we are in a position to estimate, *post facto*, the adequacy of the linear theory by calculating the magnitude of the omitted nonlinear terms. The linear problem here differs in kind from that which arises when using linear theory to evaluate the stability of a flow, as in chapter 9. In that case, we assume a small initial perturbation and the initial evolution of that perturbation is then accurately described using linear equations. In this case the amplitude of the perturbation is arbitrary, for it may grow exponentially and its size at any given time is proportional to the magnitude of the initial perturbation, which is assumed small but which is otherwise unconstrained. In contrast, when we are calculating the stationary linear response to flow over topography or to a thermal source, the amplitude of the solution is *not* arbitrary; rather, it is determined by the parameters of the problem, including the size of the topography, and represents a real quantity that might be compared to observations. Of course, because the problem is linear, the amplitude of the solution is directly proportional to the magnitude of the topography or thermal perturbation.

From (16.98), and recalling that the amplitude of $\tilde{h}$ is scaled relative to the real topography by the factor $(g/L_d^2 f_0)$, we crudely estimate the amplitude of the response to topography to be

$$|\psi'| \sim \frac{\alpha g h_b}{f_0} \approx \alpha \times 10^8 \text{ m}^2 \text{ s}^{-1} = 2 \times 10^7 \text{ m}^2 \text{ s}^{-1}, \quad |\eta'| = \frac{f_0 |\psi'|}{g} \sim \alpha h_b \approx 0.2 \text{ km}. \quad (16.106)$$

where the non-dimensional parameter α accounts for the distance of the response from resonance and the ratio of the length scale to the deformation scale. Choosing $\alpha = 0.2$ and $h_b = 1 \text{ km}$ gives the numerical values above, which are similar to those calculated more carefully, or observed (Fig. 16.12).

If linear theory is to be accurate, we must demand that the self advection of the response is much smaller than the advection by the basic state, and so that

$$|J(\psi', \nabla^2 \psi')| \ll |\bar{u} \frac{\partial}{\partial x} \nabla^2 \psi'|, \quad (16.107)$$

or, again rather crudely, that $|\psi'|/L \ll \bar{u}$. For $L = 5000 \text{ km}$ we have $\psi'/L = 4 \text{ m s}^{-1}$, which is a

few times smaller than a typical mid-troposphere zonal flow of 20 m s^{-1} , suggesting that the linear approximation may hold water. However, the inequality is not a large one, especially as a different choice of numerical factors would give a different answer, and the use of a simple barotropic model also implies inaccuracies. Rather, we conclude that we must carefully calculate the linear response, and compare it with the observations and the implied nonlinear terms, before concluding that linear theory is appropriate, although it certainly does give qualitative insight.

16.8 ♦ EFFECTS OF THERMAL FORCING

How does thermal forcing influence the stationary waves? To give an accurate answer for the real atmosphere is a little more difficult than for the orographic case where the forcing can be included reasonably accurately in a quasi-geostrophic model with a term $\bar{u} \cdot \nabla h_b$ at the lower boundary. Anomalous (i.e., variations from a zonal or temporal mean) thermodynamic forcing typically also arises initially at the lower boundary through, for example, variations in the surface temperature. However, such anomalies may be felt throughout the lower troposphere on a relatively short time scale by way of such non-geostrophic phenomena as convection, so that the effective thermodynamic source that should be applied in a quasi-geostrophic calculation has a finite vertical extent. However, an accurate parameterization of this may depend on the structure of the atmospheric boundary layer and this cannot always be represented in a simple way.⁹ Because of such uncertainties our treatment concentrates on the fundamental and qualitative aspects of thermal forcing.

The quasi-geostrophic potential vorticity equation, linearized around a uniform zonal flow, is [cf. (16.57)]

$$\left(\frac{\partial}{\partial t} + \bar{u} \frac{\partial}{\partial x} \right) \left[\nabla^2 \psi' + \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \psi'}{\partial z} \right) \right] + \frac{\partial \psi'}{\partial x} \left[\beta - \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \bar{u}}{\partial z} \right) \right] = \frac{f_0}{N^2} \frac{\partial Q}{\partial z} \equiv T, \quad (16.108)$$

where Q is the source term in the (linear) thermodynamic equation,

$$\frac{\partial}{\partial t} \left(\frac{\partial \psi'}{\partial z} \right) + \bar{u} \frac{\partial}{\partial x} \frac{\partial \psi'}{\partial z} - v' \frac{\partial \bar{u}}{\partial z} + \frac{N^2}{f_0} w' = \frac{Q}{f_0}. \quad (16.109)$$

A particular solution to (16.108) may be constructed if $\bar{u}$ and N^2 are constant, and if Q has a simple vertical structure. If we again write $\psi' = \text{Re } \tilde{\psi}(z) \sin ly \exp(ikx)$ and let $\Phi(z) = \tilde{\psi}(z) \exp(-z/2H)$ we obtain

$$\frac{d^2 \Phi}{dz^2} + m^2 \Phi = \frac{T}{ik\bar{u}} e^{-z/2H}, \quad \text{where} \quad m^2 = \frac{N^2}{f_0^2} \left(\frac{\beta}{\bar{u}} - K^2 - \gamma^2 \right). \quad (16.110)$$

If we let $T = T_0 \exp(-z/H_Q)$, so that the heating decays exponentially away from the Earth's surface, then the particular solution to the stationary problem is found to be

$$\tilde{\psi} = \text{Re} \frac{i\hat{T} e^{-z/H_Q}}{k\bar{u} \left[(N/f_0)^2 (K_s^2 - K^2) + H_Q^{-2} (1 + H_Q/H) \right]}, \quad (16.111)$$

where $\hat{T}$ is proportional to T . This solution does not satisfy the boundary condition at $z = 0$, which in the absence of topography and friction is

$$\bar{u} \frac{\partial}{\partial x} \frac{\partial \psi'}{\partial z} - v' \frac{\partial \bar{u}}{\partial z} = \frac{Q(0)}{f_0}. \quad (16.112)$$

A homogeneous solution must therefore be added, and just as in the topographic case this leads to a vertically radiating or a surface trapped response, depending on the sign of m^2 . One way to

calculate the homogeneous solution is to first use the linearized thermodynamic equation (16.109), or the linearized vorticity equation (16.114), to calculate the vertical velocity at the surface implied by (16.111), $w_p(0)$ say. We then notice that the homogeneous solution is effectively forced by an equivalent topography given by $h_e = -w_p(0)/(iku(0))$, and so proceed as in the topographic case. The complete solution is rather hard to interpret, and is in any case available only in special cases, so it is useful to take a more qualitative approach.

16.8.1 Thermodynamic balances

It is the properties of the particular solution that mainly distinguish the response to thermodynamic forcing from that due to topography, because the homogeneous solutions of the two cases are similar. Far from the source region the homogeneous solution will dominate, giving rise to wavetrains as discussed previously.

We can determine many of the properties of the response to thermodynamic forcing by considering the balance of terms in the steady linear thermodynamic equation, which we write as

$$\bar{u} \frac{\partial}{\partial x} \frac{\partial \psi'}{\partial z} - \frac{\partial \psi'}{\partial x} \frac{\partial \bar{u}}{\partial z} + \frac{N^2}{f_0} w' = \frac{Q}{f_0} \equiv R \quad (16.113a)$$

or

$$f_0 \bar{u} \frac{\partial v'}{\partial z} - f_0 v' \frac{\partial \bar{u}}{\partial z} + N^2 w' = Q. \quad (16.113b)$$

The vorticity equation is

$$\bar{u} \frac{\partial \zeta'}{\partial x} + \beta v' = \frac{f_0}{\rho_R} \frac{\partial \rho_R w'}{\partial z}. \quad (16.114)$$

Assuming that the diabatic forcing is significant, we may imagine three possible simple balances in the thermodynamic equation:

- (i) zonal advection dominates, and $v' = \partial \psi' / \partial x \sim Q H_Q / (f_0 \bar{u})$;
- (ii) meridional advection dominates, and $v' \sim Q H_u / (f_0 \bar{u})$;
- (iii) vertical advection dominates, and $w' \sim Q / N^2$. Then, for large enough horizontal scales the balance in the vorticity equation is $\beta v' \sim f_0 w'_z$ and $v' \sim f_0 Q / (\beta N^2 H_Q)$. For smaller horizontal scales advection of relative vorticity may dominate that of planetary vorticity, and β is replaced by $\bar{u} K^2$.

Here, H_Q is the vertical scale of the source (so that $\partial Q / \partial z \sim Q / H_Q$) and H_u is the vertical scale of the zonal flow (so that $\partial \bar{u} / \partial z \sim \bar{u} / H_u$). We also assume that the vertical scale of the solution is H_Q , so that $\partial v' / \partial z \sim v' / H_Q$. Which of the above three balances is likely to hold? Heuristically, we might suppose that the balance with the smallest v' will dominate, if only because meridional motion is suppressed on the β -plane. Then, zonal advection dominates meridional advection if $H_u > H_Q$, and vice versa. Defining $\hat{H} = \min(H_u, H_Q)$ then horizontal advection will dominate vertical advection if

$$\mu_1 = \frac{\beta N^2 H_Q \hat{H}}{\bar{u} f_0^2} \ll 1. \quad (16.115)$$

More systematically, we can proceed in *reductio ad absurdum* fashion by first neglecting the vertical advection term in (16.113), and seeing if we can construct a self-consistent solution. If $\psi' = \text{Re } \tilde{\psi}_p(z) e^{ikx}$, and noting that $\bar{u} \partial \tilde{\psi}_p / \partial z - \tilde{\psi}_p \partial \bar{u} / \partial z = \bar{u}^2 (\partial / \partial z) (\tilde{\psi}_p / \bar{u})$ we obtain

$$\tilde{\psi}_p = \frac{i \bar{u}}{k f_0} \int_z^\infty \frac{\tilde{Q}}{\bar{u}^2} dz, \quad (16.116)$$

where $\tilde{Q}$ denotes the Fourier amplitude of Q . Then, from the vorticity equation (16.114), we obtain the (Fourier amplitude of the) vertical velocity

$$\tilde{w}_p = \frac{-ik}{f_0 \rho_R} \int_z^\infty \rho_R \bar{u} (K_s^2 - K^2) \tilde{\psi}_p dz. \quad (16.117)$$

Using this one may, at least in principle, check whether the vertical advection in (16.113) is indeed negligible. If $\bar{u}$ is uniform (and so $H_u \gg H_Q$) then we find

$$\tilde{\psi}_p \propto \frac{iQH_Q}{kf_0\bar{u}} \quad \text{and} \quad \tilde{w}_p \propto \frac{QH_Q^2(K_s^2 - K^2)}{f_0^2}. \quad (16.118a,b)$$

Using this, vertical advection indeed makes a small contribution to the thermodynamic equation provided that

$$\mu_2 = \frac{N^2 H_Q^2 |K_s^2 - K^2|}{f_0^2} \ll 1 \quad (16.119)$$

If $K_s^2 \gg K^2$ and $\hat{H} = H_Q$ then (16.119) is equivalent to (16.115). If $\bar{u}$ is not constant and if $H_u \ll H_Q$ then H_u replaces H_Q and the criterion for the dominance of horizontal advection becomes

$$\mu = \frac{N^2 \hat{H} H_Q |K_s^2 - K^2|}{f_0^2} \ll 1. \quad (16.120)$$

This is the condition that the first term in the denominator of (16.111) is negligible compared with the second. For a typical tropospheric value of $N^2 = 10^{-4} \text{ s}^{-2}$ and for $K > K_s$ we find that $\mu \approx (H_Q/7 \text{ km})^2$, and so we can expect $\mu < 1$ in extra-equatorial regions where the heating is shallow. At low latitudes f_0 is smaller, β is bigger and $\mu \approx (H_Q/1 \text{ km})^2$, and we can expect $\mu > 1$. However, there is both uncertainty and variation in these values.

Equivalent topography

In the case in which zonal advection dominates, the equivalent topography is given by

$$h_e = \frac{-\tilde{w}_p(0)}{iku(0)} = \frac{1}{u(0)f_0\rho_R(0)} \int_0^\infty \rho_R \bar{u} (K_s^2 - K^2) \tilde{\psi}_p dz, \quad (16.121)$$

where $\tilde{\psi}_p$ is given by (16.116). The point to notice here is that if $K < K_s$ the equivalent topography is in phase with ψ_p .

16.8.2 Properties of the solution

In the tropics μ may be large for H_Q greater than a kilometre or so. Heating close to the surface cannot produce a large vertical velocity and will therefore produce a meridional velocity. However, away from the surface the heat source will be balanced by vertical advection. For scales such that $K < K_s$, a criterion that might apply at low latitudes for wavelengths longer than a few thousand kilometres, the associated vortex stretching $f\partial w/\partial z > 0$ is balanced by βv and a poleward meridional motion occurs. This implies a trough west of the heating and/or a ridge east of the heating, although the use of quasi-geostrophic theory to draw tropical inferences may be a little suspect.

In mid-latitudes μ is typically small and horizontal advection locally balances diabatic heating. In this case there is a trough a quarter-wavelength downstream from the heating, and equatorward motion at the longitude of the source. [To see this, note that if the heating has a structure like $\cos kx$ then from either (16.111) or (16.116) the solution goes like $\psi_p \propto -\sin kx$.] The trough may be warm or cold, but is often warm. If $H_Q \ll H_u$, as is assumed in obtaining (16.111), then θ is

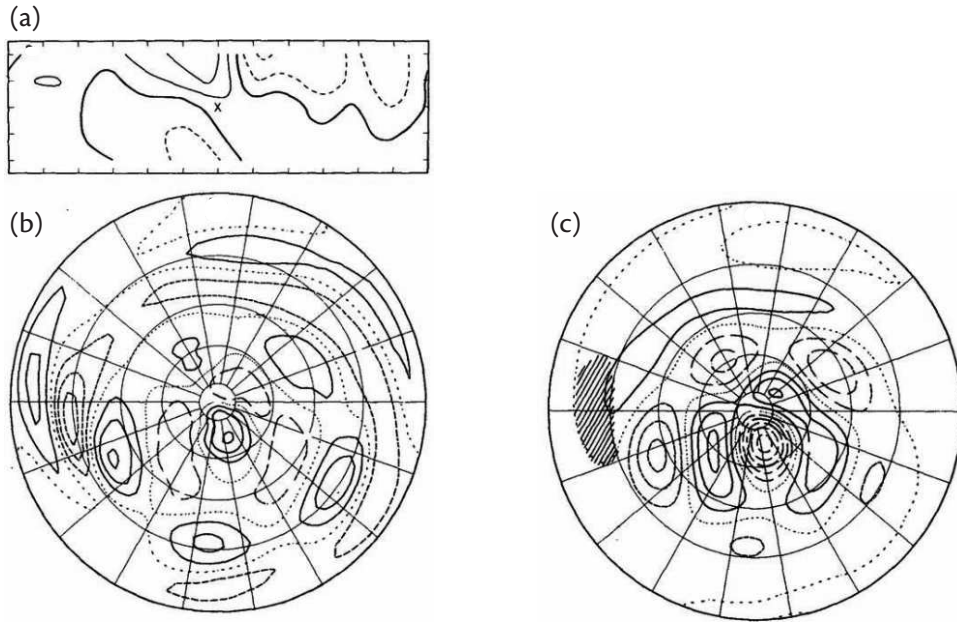


Fig. 16.15 Numerical solution of a baroclinic primitive equation model with a deep heat source at 15° N and a zonal flow similar to that of northern hemisphere winter. (a) Height field in a longitude height at 18° N (vertical tick marks at 100, 300, 500, 700 and 900 hPa); (b) 300 hPa vorticity field; (c) 300 hPa height field. The cross in (a) and the hatched region in (c) indicate the location of the heating.¹⁰

positive and warm. This is because zonal advection dominates and so the effect of the heating is advected downstream. If $H_Q \gg H_u$ and meridional advection is dominant, then the trough is still warm provided Q decreases with height. The vertical velocity can be inferred from the vorticity balance. If $f_0 \partial w / \partial z \approx \beta v$ and if $w = 0$ at the surface (in the absence of Ekman pumping and any topographic effects) there is *descent* in the neighbourhood of a heat source. This counter-intuitive result arises because it is the horizontal advection that is balancing the diabatic heating. (This result cannot be inferred from the particular solution alone.) If the advection of relative vorticity balances vortex stretching, the opposite may hold.

The homogeneous solution can be inferred from (16.121) and (16.75). Consider, for example, waves that are trapped ($m^2 < 0$) but still have $K < K_S$; that is $K^2 < K_S^2 < K^2 + \gamma^2$. The homogeneous solution forced by the equivalent topography is out of phase with that topography, and so out of phase with ψ_p , using (16.121). For still shorter waves, $K > K_S$, the homogeneous solution is in phase with the equivalent topography, and so again out of phase with ψ_p . Thermal sources produced by large-scale continental land masses may have $K^2 < K_S^2$ and, if $K^2 + \gamma^2 < K_S^2$ they will produce waves that penetrate up into the stratosphere and typically these solutions will dominate far from the source. Evidently though, the precise relationship between the particular and homogeneous solution is best dealt with on a case-by-case basis. A few more general points are summarized in the box on page 619.

16.8.3 Numerical solutions

The numerically calculated response to an isolated heat source is illustrated in Figs. 16.15 and 16.16. The first figure shows the response to a ‘deep’ heating at 15° N. As the reasoning above would suggest, the vertical velocity field (not shown) is upwards in the vicinity of the source. Away from

Thermal Forcing of Stationary Waves

- (i) The solution is composed of a particular solution and a homogeneous solution.
- (ii) The homogeneous solution may be thought of as being forced by an 'equivalent topography', chosen so that the complete solution satisfies the boundary condition on vertical velocity at the surface.
- (iii) For a localized source, the far field is dominated by the homogeneous solution. This solution has the same properties as a solution forced by real topography. Thus, it may include waves that penetrate vertically into the stratosphere as well as wavetrains propagating around the globe with an equivalent barotropic structure.
- (iv) In the extratropics, a diabatic heating is typically balanced by horizontal advection, producing a trough a quarter wavelength east (downstream) of a localized heat source. The heat source is balanced by advection of cooler air from higher latitudes, and there may be sinking air over the heat source. This can occur when $\mu \ll 1$ [see (16.120)].
- (v) In the tropics, a heat source may be locally balanced by vertical advection, that is adiabatic cooling as air ascends. This can occur when $\mu \gg 1$.
- (vi) In the real atmosphere, the stationary solutions must coexist with the chaos of time-dependent, nonlinear flows. Thus, they are likely to manifest themselves only in time averaged fields and in a modified form.

the source, the solution is dominated by the homogeneous solutions in the form of wavetrains, as described in section 16.7.3, with a simple vertical structure. (In fact, the pattern is quite similar to that obtained with a suitably forced barotropic model, as was found earlier in the topographically forced case.)

Figure 16.16 shows the response to a perturbation at 45°N , and again the solutions are qualitatively in agreement with the reasoning above. The local heating is balanced by an equatorward wind, and there is a surface trough about 20° east of the source, and an upper-level pressure maximum, or ridge, about 60° east. The scale height of the wind field, H_u is about 8 km, greater than that of the source, and the balance in the thermodynamic equation is between the zonal advection of the temperature anomaly $\bar{u}\theta'_x$ and the heat source, so producing a temperature maximum downstream. Again, the far field is dominated by the wavetrain of the homogeneous solution.

Finally, we show a calculation (Fig. 16.17) that, although linear, includes realistic forcing from topography, heat sources and observed transient eddy flux convergences, and uses a realistic zonally averaged zonal flow, although some physical parameters (e.g., representing frictional and diffusion) in the calculation must be changed in order that a steady solution can be achieved. Such a calculation is likely to be the most accurate achievable by a linear model, and discrepancies from observations indicate the presence of nonlinearities that are neglected in the calculation. In fact, a generally good agreement with the observed fields is found, and provides some *post facto* justification for the use of linear, stationary wave models.¹³

In such realistic calculations it is virtually impossible to see the wavetrains emerging from iso-

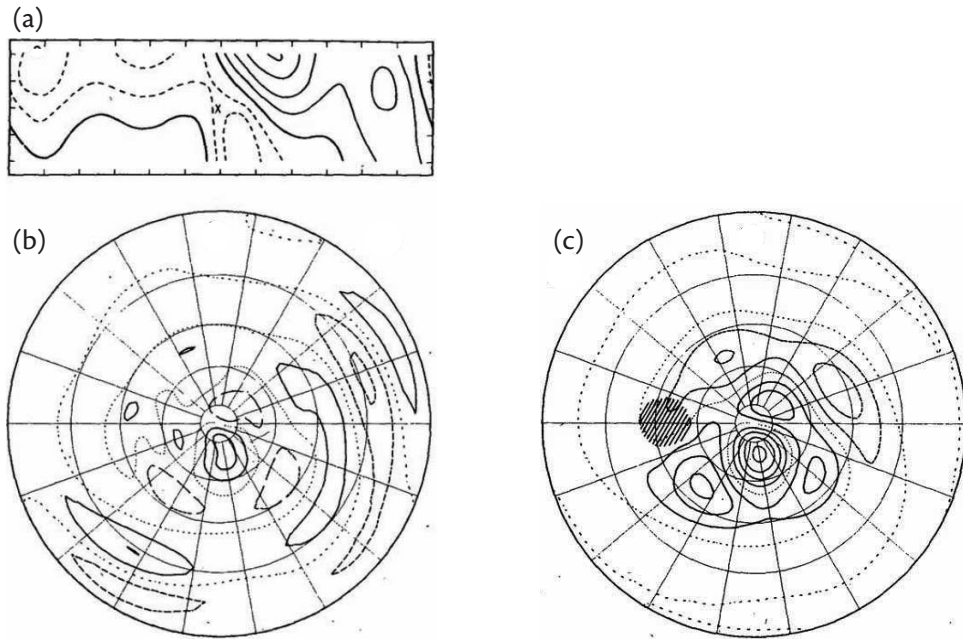


Fig. 16.16 As for Fig. 16.15, but now the solution of a baroclinic primitive equation model with a deep heat source at 45° N. (a) Height field in a longitude height at 18° N; (b) 300 hPa vorticity field; (c) 300 hPa height field. The cross in (a) and the hatched region in (c) indicate the location of the heating.¹¹

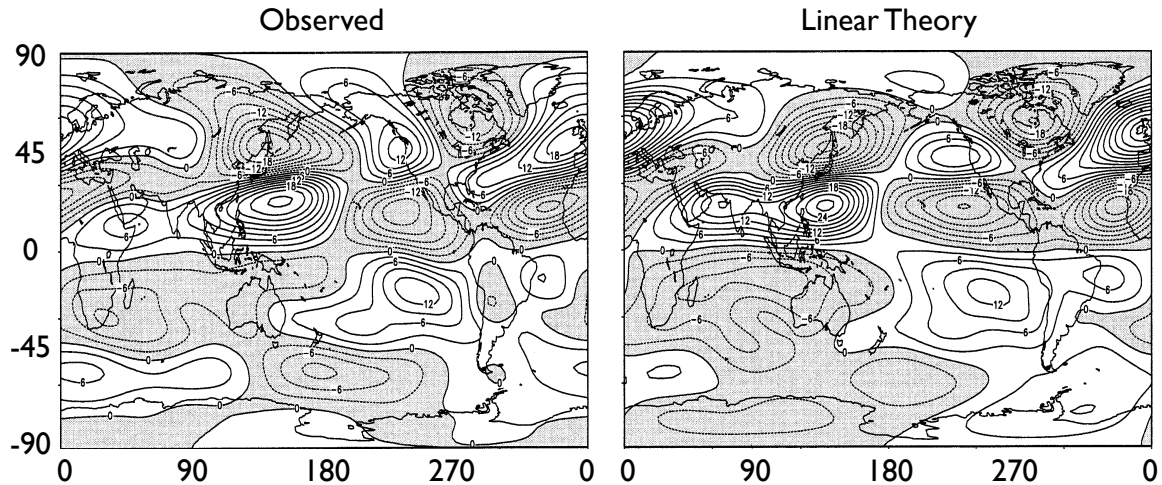


Fig. 16.17 Left: the observed stationary (i.e., time-averaged) streamfunction at 300 hPa (about 8.5 km altitude) in northern hemisphere winter. Right: the steady, linear response to forcing by orography, heat sources and transient eddy flux convergences, calculated using a linear model with the observed height-varying zonally averaged zonal wind. Contour interval is $3 \times 10^6 \text{ m}^2 \text{ s}^{-1}$, and negative values are shaded. Note the generally good agreement, and also the much weaker zonal asymmetries in the southern hemisphere.¹²

lated features like the Rockies or Himalayas, because they are combined with the responses from all the other sources included in the calculation. Breaking up the forcing into separate contributions from orographic forcing, heating, and the time averaged momentum and heat fluxes from transient eddies reveals that all of these separate contributions have a non-negligible influence. We should also remember that the effects of the fluxes from the transient eddies are not explained by such a calculation, merely included in a diagnostic sense. Nevertheless, the agreement does reveal the extent to which we might understand the steady zonally asymmetric circulation of the real atmosphere as the response due to the interaction of a zonally uniform zonal wind with the asymmetric features of the Earth's geography and transient eddy field. The quasi-stationary response of the planetary waves to surface anomalies, and the interaction of transient eddies with the large-scale planetary wave field, are important factors in the natural variability of climate, and their understanding remains a challenge for dynamical meteorologists.

16.9 ♦ WAVE PROPAGATION USING RAY THEORY

Rossby waves propagate meridionally as well as zonally, and we can expect the major mountain ranges on Earth, as well as thermal anomalies, to generate Rossby waves that propagate both zonally and meridionally. The coefficients of the linear equations that determine this propagation will vary with space: on the sphere β is a function of latitude and in general topography is a function of both latitude and longitude; thus calculating the trajectories of the waves will be difficult, and we cannot expect to solve the full problem except numerically, but a few ideas from wave tracing illustrate many of the features of the response, and indeed of the stationary wave pattern in the Earth's atmosphere.¹⁴

Let us first recall a few results about rays and ray tracing that we encountered in section 6.3. Most of the important properties of a wave, such as the energy (if conserved) and the wave activity, propagate along rays at the group velocity. Rays themselves are lines that are parallel to the group velocity, generally emanating from some wave source. A ray is perpendicular to the local wave front, and in a homogeneous medium a wave propagates in a straight line. In non-homogeneous media the group velocity varies with position; however, if the medium varies only slowly, on a scale much larger than that of the wavelength of the waves, the wave activity still propagates along rays at the group velocity.

The variation of wavenumber and frequency vary according to, respectively,

$$\frac{\partial k_i}{\partial t} + c_{gj} \frac{\partial k_i}{\partial x_j} = -\frac{\partial \Omega}{\partial x_i}, \quad \frac{\partial \omega}{\partial t} + c_{gj} \frac{\partial \omega}{\partial x_j} = \frac{\partial \Omega}{\partial t}. \quad (16.122)$$

If the frequency is not an explicit function of space then the wavenumber is constant along a ray, and similarly for time and frequency. Thus, in problems of the form

$$\frac{\partial}{\partial t} \nabla^2 \psi + \beta(y) \frac{\partial \psi}{\partial x} = 0, \quad (16.123)$$

then the frequency and the x -wavenumber, but not the y -wavenumber, are constant along a ray. The wavenumber is not constant in the y -direction because the frequency is a function of y .

16.9.1 Rossby waves and Rossby rays

If the wave source is localized, then ray theory provides a useful way of calculating and interpreting the response. On the β -plane and away from the orographic source the steady linear response to a zonally uniform but meridionally varying zonal wind will obey

$$\bar{u}(y) \frac{\partial}{\partial x} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \psi + \beta \frac{\partial \psi}{\partial x} = 0. \quad (16.124)$$

In fact, an equation of this form applies on the sphere. To see this, we transform the spherical coordinates (λ, ϑ) into Mercator coordinates with the mapping¹⁵

$$x' = a\lambda, \quad \frac{1}{a} \frac{\partial}{\partial \lambda} = \frac{\partial}{\partial x'}, \quad y' = \frac{a}{2} \ln \left(\frac{1 + \sin \vartheta}{1 - \sin \vartheta} \right), \quad \frac{1}{a} \frac{\partial}{\partial \vartheta} = \frac{1}{\cos \vartheta} \frac{\partial}{\partial y'}. \quad (16.125)$$

The spherical-coordinate vorticity equation then becomes

$$\bar{u}_M \frac{\partial}{\partial x} \left(\frac{\partial^2}{\partial x'^2} + \frac{\partial^2}{\partial y'^2} \right) \psi + \beta_M \frac{\partial \psi}{\partial x'} = 0, \quad (16.126)$$

where $\bar{u}_M = \bar{u} / \cos \vartheta$ and

$$\beta_M = \frac{2\Omega}{a} \cos^2 \vartheta - \frac{d}{dy'} \left[\frac{1}{\cos^2 \vartheta} \frac{d}{dy} (\bar{u}_M \cos^2 \vartheta) \right] = \cos \vartheta \left(\beta_s + \frac{1}{a} \frac{\partial \bar{\zeta}}{\partial \vartheta} \right), \quad (16.127)$$

where $\beta_s = 2a^{-1}\Omega \cos \vartheta$. Thus, β_M is the meridional gradient of the absolute vorticity, multiplied by the cosine of latitude. An advantage of Mercator coordinates over their spherical counterparts is that (16.126) has a Cartesian flavour to it, with the metric coefficients being absorbed into the parameters $\bar{u}_M$ and β_M . Of course, unlike the case on the true β -plane, the parameter β_M is not a constant, but this is not a particular disadvantage if $\bar{u}_{yy}$ is also varying with y .

Having noted the spherical relevance we revert to the Cartesian β -plane and seek solutions of (16.124) with the form $\psi' = \tilde{\psi}(y) \exp(ikx)$, whence

$$\frac{d^2 \tilde{\psi}}{dy^2} = \left(k^2 - \frac{\beta}{\bar{u}} \right) \tilde{\psi} = (k^2 - K_s^2) \tilde{\psi}, \quad (16.128)$$

where $K_s = (\beta/\bar{u})^{1/2}$. From this equation it is apparent that if $k < K_s$ the solution is harmonic in y and Rossby waves may propagate away from their source. On the other hand, wavenumbers $k > K_s$ are trapped near their source; that is, short waves are meridionally trapped by eastward flow. Without solving (16.128), we can expect an isolated mountain to produce two wavetrains, one for each meridional wavenumber $l = \pm(K_s^2 - k^2)^{1/2}$. These wavetrains will then propagate along a ray, and given the dispersion relation this trajectory can be calculated (usually numerically) using the expressions of the previous section. The local dispersion relation of Rossby waves is

$$\omega = \bar{u}k - \frac{\beta k}{k^2 + l^2}, \quad (16.129)$$

so that their group velocity is

$$c_g^x = \frac{\partial \omega}{\partial k} = \bar{u} - \frac{\beta(l^2 - k^2)}{(k^2 + l^2)^2} = \frac{\omega}{k} + \frac{2\beta k^2}{(k^2 + l^2)^2}, \quad c_g^y = \frac{\partial \omega}{\partial l} = \frac{2\beta kl}{(k^2 + l^2)^2}. \quad (16.130a,b)$$

The sign of the meridional wavenumber thus determines whether the waves propagate polewards (positive l) or equatorwards (negative l). Also, because the dispersion relation (16.130) is independent of x and t , the zonal wavenumber and frequency in the wave group are constant along the ray, and the *meridional wavenumber* must adjust to satisfy the local dispersion relation (16.129). Thus, from (16.128), the meridional scale becomes larger as K_s approaches k from above and an incident wavetrain is reflected, its meridional wavenumber changes sign, and it continues to propagate eastwards.

Stationary waves have $\omega = 0$, and the trajectory of a ray is parameterized by

$$\frac{dy}{dx} = \frac{c_g^y}{c_g^x} = \frac{l}{k}. \quad (16.131)$$

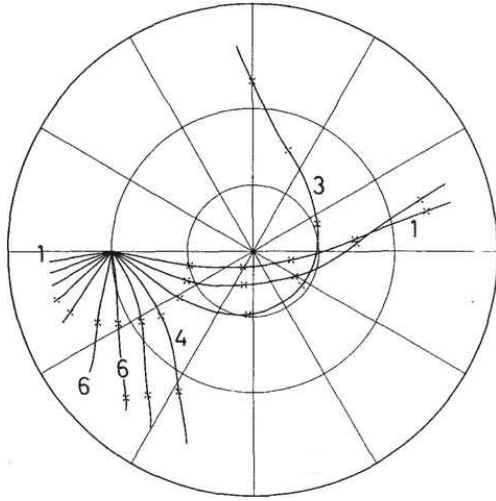


Fig. 16.18 The rays emanating from a point source at 30° N and 180° (nine o'clock), calculated using the observed value of the wind at 300 hPa.¹⁶ The crosses mark every 180° of phase, and mark the positions of successive positive and negative extrema of the ray. The numbers indicate the zonal wavenumber of the ray. The ray paths may be compared with the full linear calculation shown in Fig. 16.19.

For a given zonal wavenumber the trajectory is then fully determined by this condition and that for the local meridional wavenumber which from (16.129) is

$$l^2 = K_s^2 - k^2. \quad (16.132)$$

Finally, from (16.130) the magnitude of the group velocity is

$$|c_g| = [(c_g^x)^2 + (c_g^y)^2]^{1/2} = 2 \frac{k}{K_s} \bar{u}, \quad (16.133)$$

which is double the speed of the projection of the basic flow, $\bar{u}$, onto the wave direction. Given the above relations, and the zonal wind field, we can compute rays emanating from a given source, although the calculation must still be done numerically. One example is given in Fig. 16.18.

♦ A WKB solution

The wave amplitudes along a ray can be obtained using a WKB approach. We write (16.128) as

$$\frac{d^2 \tilde{\psi}}{dy^2} + l^2(y) \tilde{\psi} = 0, \quad \text{where} \quad l^2(y) = K_s^2 - k^2, \quad (16.134a,b)$$

and if $l(y)$ is varying sufficiently slowly in y then the WKB solution for the stationary streamfunction is

$$\psi(x, y) = A l^{-1/2} \exp \left[i \left(kx + \int^y l(y) dy \right) \right], \quad (16.135)$$

where A is a constant. Consider, for example, the disturbance excited by an isolated low-latitude peak, with $\bar{u}$ increasing, and so K_s decreasing, polewards of the source. Assuming that initially there exists a zonal wavenumber k less than K_s then two eastward propagating wavetrains are excited. The meridional wavenumber of the poleward wavetrain diminishes according to (16.134b), so that, using (16.131), the ray becomes more zonal. At the latitude where $k = K_s$, the 'turning latitude' the wave is reflected but continues propagating eastwards. The southward propagating wavetrain is propagating into a medium with smaller $\bar{u}$ and larger K_s . At the critical latitude, where $\bar{u} = 0$, $l \rightarrow \infty$ but c_g^x and c_g^y both tend to zero, but [using (16.130)] in such a way that $c_g^x/c_g^y \rightarrow 0$. That is, the rays become meridionally oriented and their speed tends to zero, and the waves may be absorbed.¹⁷ Finally, we mention without derivation that for zonal flows with constant angular velocity the trajectories are great circles.

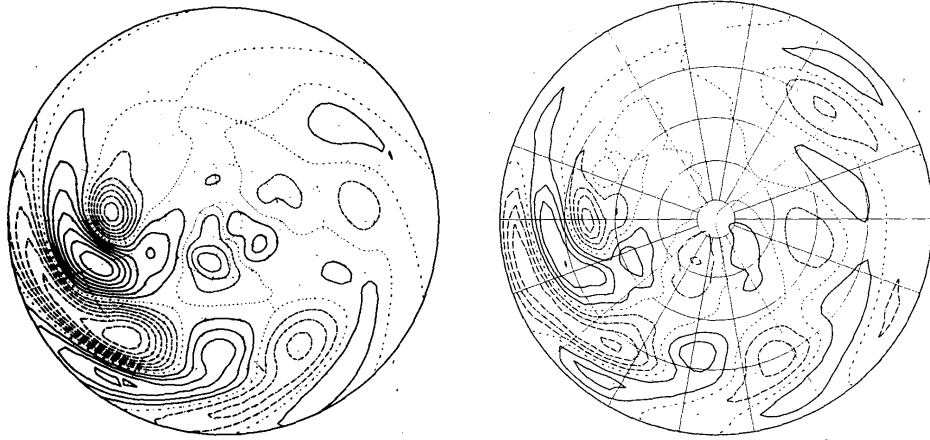


Fig. 16.19 The linear stationary response induced by circular mountain at 30° N and at 180° longitude (nine o'clock). The figure on the left uses a barotropic model, whereas the figure on the right uses a multi-layer baroclinic model.¹⁸ In both cases the mountain excites a low-wavenumber polar wavetrain and a higher-wavenumber subtropical train.

16.9.2 Application to an idealized atmosphere

Given the complexity of the real atmosphere, and the availability of computers, it is probably best to think of the remarks above as helping us interpret more complete numerical, but still linear, calculations of stationary Rossby waves — for example, numerical solutions of the stationary barotropic vorticity equation in spherical coordinates,

$$\frac{\bar{u}}{a \cos \vartheta} \frac{\partial \zeta'}{\partial \lambda} + v' \left(\frac{1}{a} \frac{\partial \bar{\zeta}}{\partial \vartheta} + \beta \right) = -\frac{\bar{u} f_0}{aH \cos \vartheta} \frac{\partial h_b}{\partial \lambda} - r \zeta', \quad (16.136)$$

where $[u, v] = a^{-1}[-\partial\psi/\partial\vartheta, (\partial\psi/\partial\lambda)/\cos\vartheta]$, $\beta = 2\Omega a^{-1} \cos\vartheta$ and $\zeta = \nabla^2\psi$. The last term in (16.136) crudely represents the effects of friction and generally reduces the sensitivity of the solutions to resonances. Solutions to (16.136) may be obtained first by discretizing and then numerically inverting a matrix, and although the actual procedure is quite involved it is analogous to the Fourier methods used earlier for the simpler one-dimensional problem. Such linear calculations, in turn, help us interpret the stationary wave pattern from more comprehensive models and in the Earth's atmosphere.

Figure 16.19 shows the stationary solution to the problem with a realistic northern hemisphere zonal flow and an isolated, circular mountain at 30° N. The topography excites two wavetrains, both of which slowly decay downstream because of frictional effects, rather like the one-dimensional wavetrain in Fig. 16.14. The polewards propagating wavetrain develops a more meridional orientation, corresponding to a smaller meridional wavenumber l , before moving southwards again, developing a much more zonal orientation eventually to decay completely as it meets the equatorial westward flow. The equatorially propagating train decays a little more rapidly than its polewards moving counterpart because of its proximity to the critical latitude. More complicated patterns naturally result if a realistic distribution of topography is used, as we see in Fig. 16.17. We can see wavetrains emanating from both the Rockies and the Himalayas, but distinct poleward and equatorward wavetrains are hard to discern.

Notes

- 1 Many thanks to Isaac Held and Peter Haynes for various comments on Rossby waves and critical lines.
- 2 Early discussions of Rossby wave critical layers include those of Warn & Warn (1976), Stewartson (1977) and Killworth & McIntyre (1985). Booker & Bretherton (1967) consider gravity wave critical layers. Haynes (2015) provides an accessible review of both.
- 3 Adapted from Haynes (2015).
- 4 After Charney & Drazin (1961).
- 5 A quite extensive discussion of the thermal (and the topographically forced) problem is given by Pedlosky (1987a).
- 6 Much of our basic understanding in this area stems from conceptual and numerical work on forced Rossby waves by Charney & Eliassen (1949), who looked at the response to orography using a barotropic model. This was followed by a study by Smagorinsky (1953) on the response to thermodynamic forcing using a baroclinic, quasi-geostrophic model. Seeking more realism later studies have employed the primitive equations and spherical coordinates in studies that are at least partly numerical (e.g., Egger 1976, and a host of others), although most theoretical studies perforce still use the quasi-geostrophic equations. We also draw from various review articles, among them Smith (1979), Dickinson (1980), Held (1983), particularly for sections 16.7.3 and 16.8, and Wallace (1983). See also the collection in the *Journal of Climate*, vol. 15, no. 16, 2002.
- 7 To obtain the solutions shown in Fig. 16.11 and Fig. 16.12, the topography is first specified in physical space. Its Fourier transform is taken and the streamfunction in wavenumber space is calculated using (16.98). The inverse Fourier transform of this gives the streamfunction in physical space.
- 8 The difference between wavetrains emanating from an isolated topographic feature and a global resonant response is relevant for intra-seasonal variability, which might be considered a quasi-stationary response to slowly changing boundary conditions like the sea-surface temperature. If resonance is important, we might expect to see global-scale anomalies, whereas the viewpoint of damped wave-trains is more local. This whole area is one of continuing, active, research with deep roots going back to Namias (1959) and Bjerknes (1959) and beyond.

A different point of view, one that we do not explore in this book, is that the zonally asymmetric features of the Earth's atmosphere are predominantly due to *nonlinear* effects. One possibility is that eddies might significantly modify (and perhaps amplify and sustain) stationary patterns through their large-scale turbulent transfers; see, for example, Green (1977) and Shutts (1983). We could incorporate such effects into a linear model by including the eddy effects as forcing term on the right-hand side of a linear equation such as (16.90), or its two- or three-dimensional analogue, although the forcing term would have to be calculated using a nonlinear theory or taken from observations. Different again is the notion, inspired by models of low-order dynamical systems, that the atmosphere might have *regimes* of behaviour, and that the zonally asymmetric patterns are manifestations of the time spent in a particular regime before transiting to another. See for example Kimoto & Ghil (1993) and Palmer (1997).
- 9 Because of such difficulties, understanding the effects of sea-surface temperature anomalies on the atmosphere has become largely the subject of GCM experiments, and one plagued with ambiguous results that depend in part on the particular configuration of the GCM. Some of the modelling issues are reviewed by Kushnir *et al.* (2002).
- 10 From Hoskins & Karoly (1981).
- 11 From Hoskins & Karoly (1981).
- 12 Adapted from Held *et al.* (2002).
- 13 Such solutions are nearly always most easily obtained numerically. One way is to use a Fourier method described earlier. A related method is to write the equations in finite difference form, schematically as $\mathbf{AX} = \mathbf{F}$, where $\mathbf{X}$ is the vector of all the model fields, $\mathbf{F}$ represents the known forcing and $\mathbf{A}$ is a matrix obtained from the equations of motion and boundary conditions, and

solve for X . A quite different method is to use a nonlinear time-dependent model, such as a GCM: prescribe or hold steady the zonally averaged zonal flow as well as all the zonally asymmetric forcing terms, but multiply the asymmetric terms by a small number (e.g., 0.01) to ensure the response is linear; then calculate the steady response by forward time integration, and then divide that solution by the small number to obtain the final solution.

- 14 The description of the stationary waves in terms of wavetrains comes from Hoskins & Karoly (1981), with some earlier theoretical results having been derived by Longuet-Higgins (1964).
- 15 Steers (1962) and Phillips (1973).
- 16 From Hoskins & Karoly (1981).
- 17 At the critical latitude the wkb analysis fails and both dissipative and nonlinear effects are likely to play a role, as discussed by Dickinson (1968), Tung (1979) and others.
- 18 From Grose & Hoskins (1979) and Hoskins & Karoly (1981).

*And beyond it, the deep blue air, that shows
Nothing, and is nowhere, and is endless.*
Philip Larkin, *High Windows*, 1974.

CHAPTER 17

The Stratosphere

THE STRATOSPHERE IS THE REGION OF THE ATMOSPHERE above the troposphere and below the mesosphere; thus, it extends from the tropopause at a height of about 8–15 km, or a pressure of around 200–300 hPa, to the stratopause at about 50 km or about 1 hPa (see Fig. 15.23 on page 573). In this chapter our goal is to provide an introductory explanation of the dynamics governing the structure and variability of the stratosphere. The *middle atmosphere* is the somewhat larger region that also includes the mesosphere, and so that extends up to the mesopause at about 90 km or 2×10^{-3} hPa, but we won't consider the mesosphere here.¹

The outline of this chapter is roughly as follows. We begin with a rather descriptive overview of the stratosphere as a whole. Then, starting in section 17.2, we discuss the Rossby and gravity waves that in many ways serve to drive the circulation. We come back to the circulation itself in section 17.4, focusing mainly on the generation of zonal flows and the meridional residual overturning circulation. We round out the chapter with discussions of two striking examples of stratospheric variability, namely the quasi-biennial oscillation in section 17.6, and extratropical variability and sudden warmings in section 17.7, with these terms to be defined in the sections ahead.

17.1 A DESCRIPTIVE OVERVIEW

In the troposphere the stratification is determined by dynamical processes — largely by convection at low latitudes and additionally by baroclinic instability at high latitudes — and the tropopause is the height to which the dynamical activity reaches, as discussed in chapter 15. In contrast, in the stratosphere the temperature is determined to a much greater degree by radiative processes and the dynamics are, compared to those in the tropopause, slow. Over much of the stratosphere the temperature actually increases with height, and this is due to a layer of ozone that absorbs solar radiation in the mid-stratosphere between about 20 and 30 km. If there were no ozone we would certainly have a tropopause and a stratosphere, but the temperature in the stratosphere would increase much less with height than it in fact does.

The resulting radiative-equilibrium temperature for January is illustrated in Fig. 17.1. The calculation to produce this takes as given the downward solar radiation at the top of the atmosphere and the upward solar and infra-red radiative flux from the troposphere, as well as the physical properties of the fluid, and then calculates the temperature that would ensue without any stratospheric fluid motion. There is quite a strong lateral gradient in the winter hemisphere and a weaker

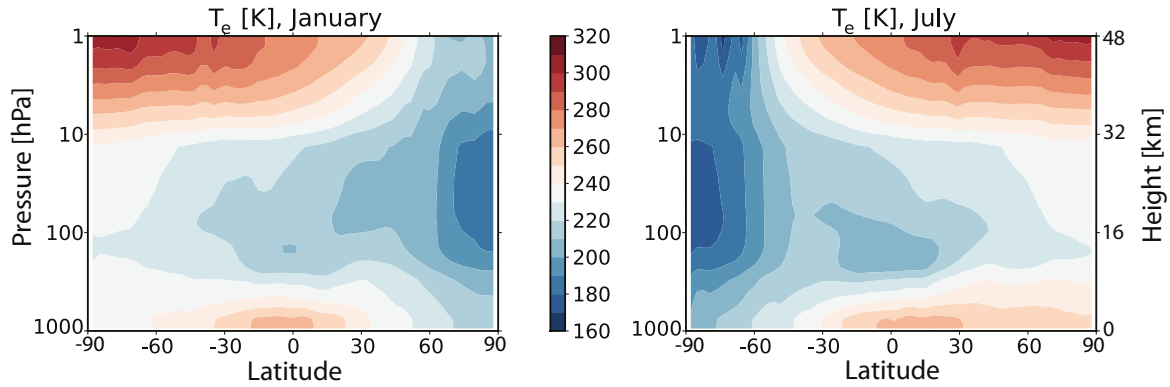


Fig. 17.1 The zonally averaged radiative-equilibrium temperature in January and July; that is, the temperature that would nominally arise in the absence of fluid motion in the stratosphere, but with the actual distribution of radiative absorbers.² Above about 50 km the equilibrium temperature generally diminishes with height. The ordinate is pressure, and the height values are approximate.

reversed temperature in the summer hemisphere, and in fact the part of the stratosphere with the highest radiative equilibrium temperature is the upper-stratosphere summer pole, at around 1 hPa. The observed zonally averaged temperature and zonal-wind structure are plotted in Fig. 17.2. From these we infer the following.

- The stratosphere is very stably stratified, with a typical lapse rate corresponding to $N \approx 2 \times 10^{-2} \text{ s}$, about twice that of the troposphere on average. This is in part due to the absorption of solar radiation by ozone between 20 and 50 km.
- In the summer the solar absorption at high latitudes leads to a reversed temperature gradient (warmer pole than equator) and, by thermal wind balance, a negative vertical shear of the zonal wind. The temperature distribution is, in fact, not far from the radiative equilibrium distribution, and over much of the summer stratosphere the mean winds are indeed negative (westwards).
- In winter high latitudes receive very little solar radiation and there is a strong meridional temperature gradient and consequently a strong vertical shear in the zonal wind. Nevertheless, this temperature gradient is significantly weaker than the radiative equilibrium temperature gradient, implying a poleward heat transfer by the fluid motions.

The observed fields are a consequence of the dynamics as well as the radiative forcing and we may ask how and why does the dynamics of the stratosphere differ from the tropopause? A proximate reason is that the higher stratification of the stratosphere tends to weaken the baroclinic instability, and the instability that does occur is at a much larger scale. A typical value of the static stability in the stratosphere is $N \approx 2 \times 10^{-2} \text{ s}^{-1}$, and using a height scale of 20 km gives a value of the deformation radius NH/f of about 4000 km, as opposed to a canonical 1000 km in the troposphere. Thus, even with the same horizontal temperature gradient as the troposphere, a typical instability scale (of the stratosphere in isolation) is large, perhaps at wavenumber 2 rather than wavenumber 8. The growth rate is also much less than in the troposphere: the Eady growth rate is given by $\sigma_E \equiv 0.31\Lambda H/L_d = 0.31U/L_d$, where Λ is the shear, so even when a horizontal temperature gradient is present the growth rate will typically be several times smaller than its tropospheric value. Furthermore, there is no real opportunity for counter-propagating edge waves or Rossby waves to interact in the stratosphere, so taken in isolation the stratosphere may be baroclinically stable. Of course, if baroclinic instability has a modal form then the instability grows at the same

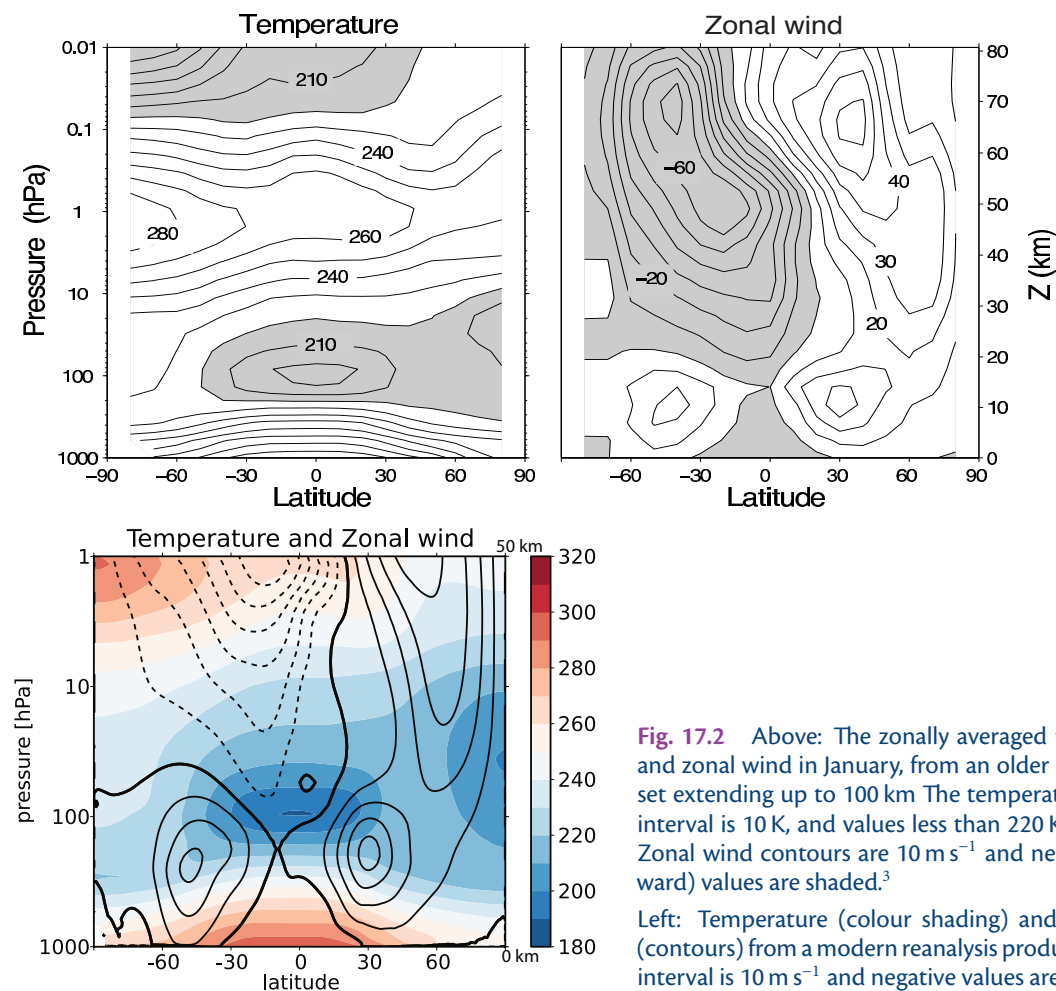


Fig. 17.2 Above: The zonally averaged temperature and zonal wind in January, from an older (1980s) data set extending up to 100 km. The temperature contour interval is 10 K, and values less than 220 K are shaded. Zonal wind contours are 10 m s⁻¹ and negative (westward) values are shaded.³

Left: Temperature (colour shading) and zonal wind (contours) from a modern reanalysis product. Contour interval is 10 m s⁻¹ and negative values are dashed.

rate in the stratosphere as the tropospheric one (it is the same mode!), but in this case the higher lapse rate suppresses the amplitude of the stratospheric instability, as shown in Fig. 9.21.

For all these reasons, baroclinic instability is not the main process leading to a circulation in the stratosphere — the main process is the propagation and subsequent breaking and dissipation of gravity and Rossby waves from the troposphere to the stratosphere. This breaking will produce an acceleration of the zonal flow, and/or a meridional overturning circulation, and a good fraction of this chapter will be devoted to describing that process. But first we'll provide a little more description about the circulation itself, and it is convenient to divide that into two parts:

- (i) A quasi-horizontal circulation.
- (ii) A meridional overturning circulation that is most usefully described as residual circulation using the TEM formalism.

We now describe these.

17.1.1 The quasi-horizontal circulation

In the extra-tropics the stratification is high and the Rossby number small and, at least to the extent that the scales of motion are not truly hemispheric the circulation is well described by the quasi-geostrophic equations. Now, not only does any stratospheric baroclinic instability tend to occur

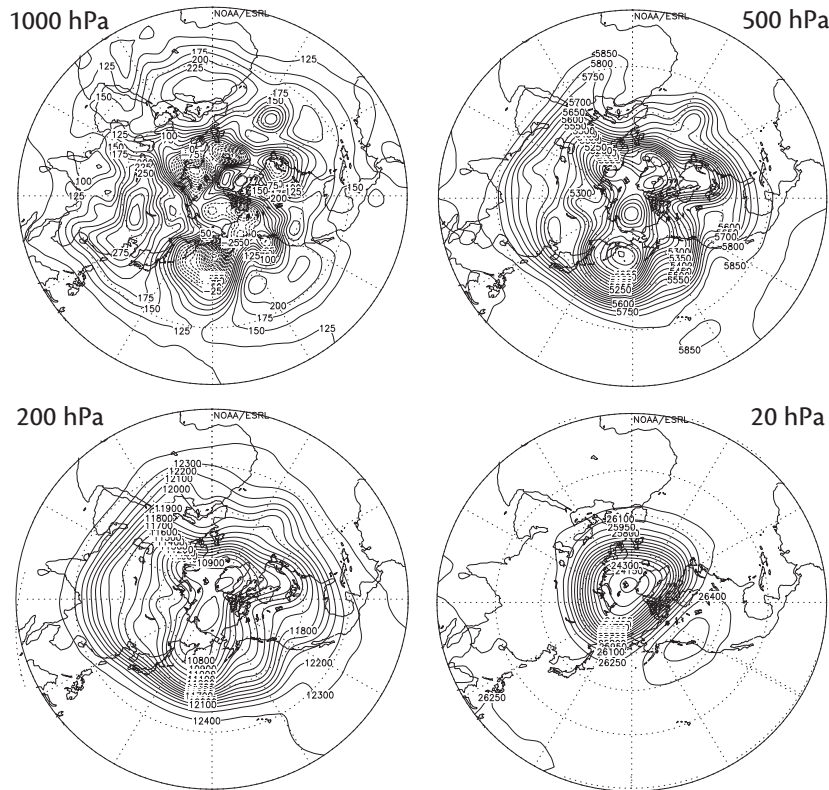


Fig. 17.3 The geopotential height on February 1, 2000, at various levels in the atmosphere — 1000 and 500 hPa are in the troposphere, 200 hPa is around the tropopause and 20 hPa is in the mid stratosphere, at about 30 km. Note the general increase in scale of the variations with height.

on a large scale, but so does any wave activity that arises from the propagation of Rossby waves up from the troposphere. This is because of Charney–Drazin filtering, summarized in Fig. 16.6: the smaller the wavelength the smaller is the range of zonal winds through which the waves can propagate. If the wind is too high the waves encounter a turning surface, whereas if the wind is too low they encounter a critical layer. Thus, we would expect that the general horizontal scale of motion is larger in the stratosphere than in the troposphere, and this is borne out by inspection of Fig. 17.3, which shows geopotential height at various levels. The complex patterns of the lower and mid troposphere are well filtered, and in mid-stratosphere the pattern is dominated by wave-numbers 1 and 2. Indeed it seems from the figure (which is typical) that much of the motion is concentrated around a *polar vortex*.

Looking at geopotential (which roughly corresponds to a streamfunction) gives a somewhat misleading impression of the lack of activity away from the poles. Here, because diabatic effects occur on a somewhat longer time scale than advective processes, the flow may be characterized by the advection of potential vorticity on more slowly evolving isentropic surfaces, as illustrated in Figs. 17.4 and 17.5. Both the potential vorticity and the tracer are evocative of two-dimensional turbulence. We see Rossby waves breaking and vortices stretched into filaments and tendrils, the features of an enstrophy cascade. We also perceive some idea of the spectral non-locality of the enstrophy transfer — a single large vortex overturns and breaks and there is little sense of a spectrally-local cascade of enstrophy to dissipative scales. For this reason, the mid-latitude region is sometimes known as the *surf zone*. It is precisely this wave breaking that gives rise to the enstrophy flux to

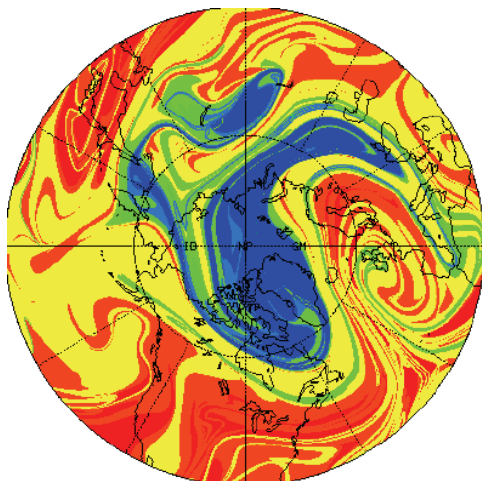


Fig. 17.4 The tracer distribution in the northern hemisphere lower stratosphere on 28 January 1992. The tracer was initialized on 16 January by setting it equal to the potential vorticity field calculated from an observational analysis, and then advected for 12 days by the observed wind fields.⁴

small scales and its dissipation, and which in turn gives rise to the overturning circulation that we discuss below.

The surf-zone does not usually extend to the pole, and in winter dense cold air over the pole forms itself into a cyclonic vortex, apparent in both Fig. 17.3 and 17.5. Although the vortex is ultimately the result of diabatic forcing, and has a preferred location, the tendency of quasi-two-dimensional flow to organize itself into vortices (as we see in Figs. 9.6 and 11.8) contributes to its coherence and isolation from the rest of the hemisphere. The boundary of the vortex, as measured by the value of the potential vorticity or of the tracer, is quite sharp with the value of PV often jumping by a factor of 2 or so, and the vortex is quite persistent — in fact it is a near-permanent feature of the winter hemisphere. Within the vortex potential vorticity tends to homogenize, and once formed the main communication that the vortex has with the surf zone is via occasional wave breaking at its boundary. It is interesting that, although the potential vorticity gradient is strong at the edge of the vortex, the exchange of properties is weak, implying a failure of notions of diffusion; the edge of the vortex is a *mixing barrier*. We saw this property before, in our discussion of potential vorticity staircases in section 12.1.3.

Stable as it is, the polar vortex is nevertheless sometimes disrupted by wave activity from below; this tends to occur when the wave activity itself is quite strong, and when the mean conditions are such as to steer that wave activity polewards. Occasionally, this activity is sufficiently strong so as to cause the vortex to break down, or to split into two smaller vortices, and so allow warm mid-latitude air to reach polar latitudes — an event known as a *stratospheric sudden warming*, and one such is illustrated in Fig. 17.22. We come back to the mechanism of such warmings later in the chapter.

17.1.2 The overturning circulation

That there is a meridional overturning circulation in the stratosphere was inferred by A. Brewer and G. Dobson based on observations of water vapour and chemical transport, and is often called the *Brewer–Dobson circulation*.⁷ Brewer and Dobson both inferred the circulation on the basis of tracer transports, rather than performing an Eulerian average of velocity measurements (which would have been impossible then, and is still difficult now). Thus, the circulation they inferred was, in modern parlance, a residual circulation and some modern observations of this circulation are shown in Fig. 17.6. The figure actually shows the observed thickness-weighted circulation, which is almost equivalent to the residual circulation (section 10.3.3), and which represents both the Eulerian mean and eddy-contributed components. We see a single, equator-to-pole cell in each

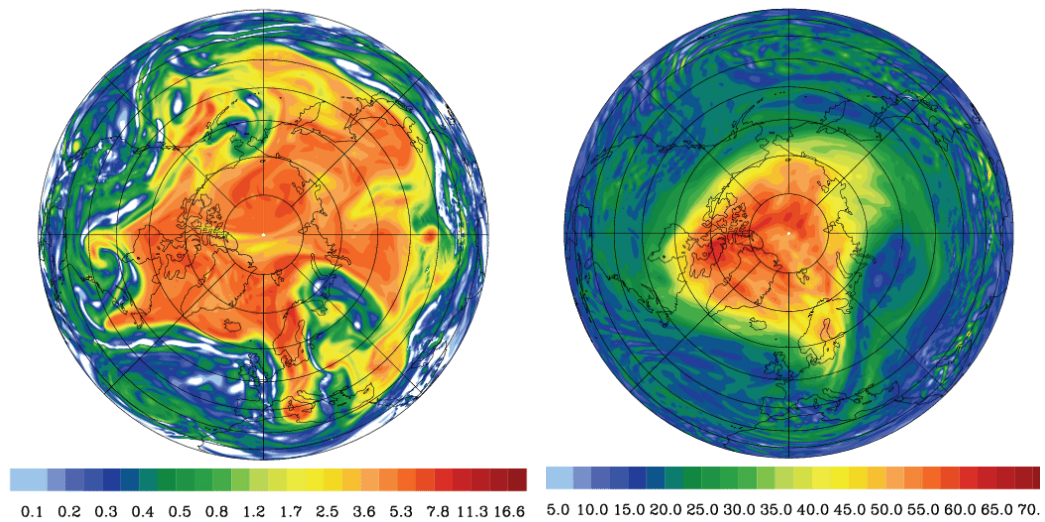


Fig. 17.5 The potential vorticity on two isentropic surfaces, the 310 K surface (left) and the 475 K surface (right), on January 19, 2005. The shaded bar is in PV units. The 310 K surface is mainly in the troposphere (see Fig. 15.16) where baroclinic instability is abundant. The 475 K surface is at about 20 km altitude, and on it we see a polar stratospheric vortex with a fairly sharp boundary where the PV gradient is high, and a mid-latitude region of smaller-scale features and wave breaking.⁵

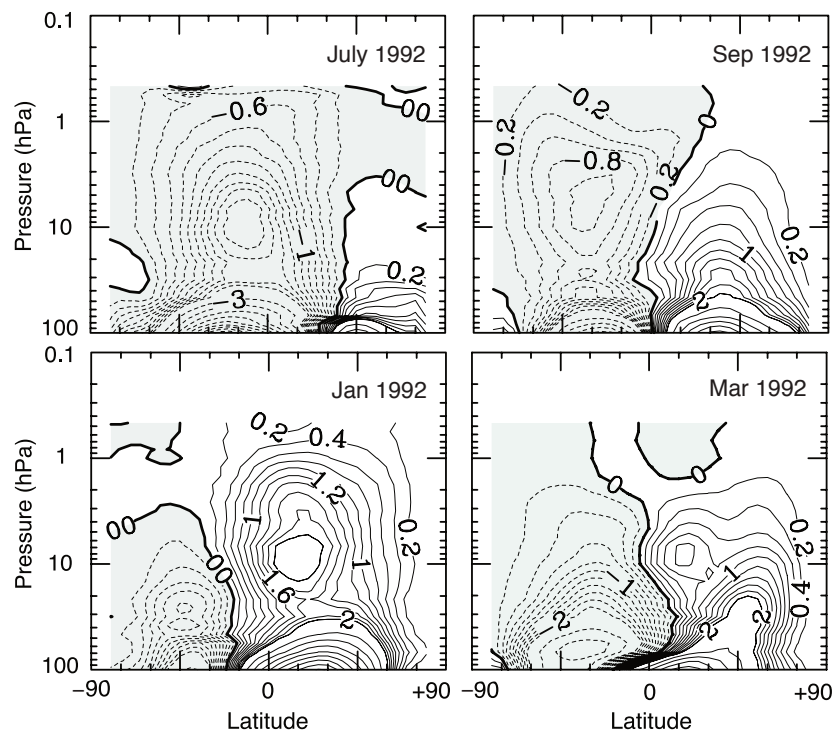


Fig. 17.6 The observed thickness-weighted (residual) streamfunction in the stratosphere, in Sverdrups (10^9 kg s^{-1}). The circulation is clockwise where the contours are solid.

The circulation is stronger in the winter hemispheres, whereas the equinoctial circulations (September, March) are more inter-hemispherically symmetric.⁶

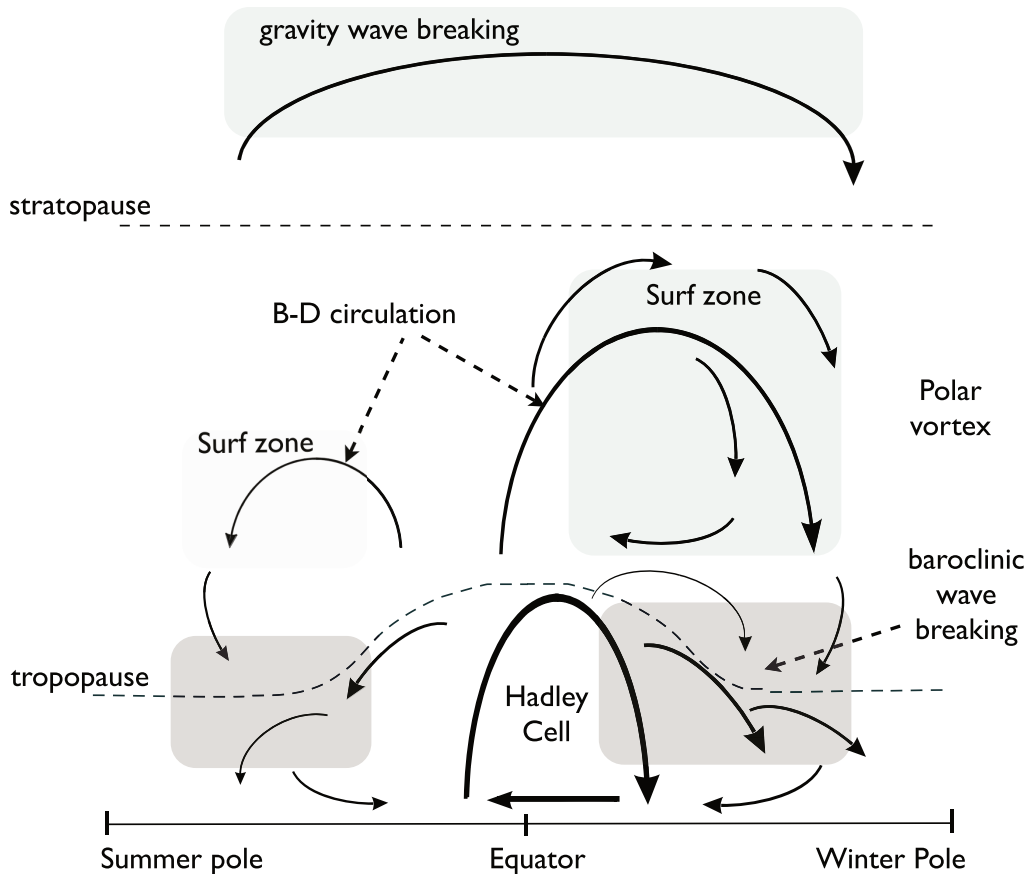


Fig. 17.7 A sketch of the residual mean meridional circulation of the atmosphere. The solid arrows indicate the residual circulation (B-D for Brewer–Dobson) and the shaded areas the main regions of wave breaking (i.e., enstrophy dissipation) associated with the circulation. In the surf zone the breaking is mainly that of planetary Rossby waves, and in the troposphere and lower stratosphere the breaking is that of baroclinic eddies. The surf zone and residual flow are much weaker in the summer hemisphere. Only in the Hadley Cell is the residual circulation comprised mainly of the Eulerian mean; elsewhere the eddy component dominates.

hemisphere, stronger in the winter hemisphere where it goes high into the stratosphere. There is also a distinct lower branch to the circulation, present in all seasons although strongest in winter, that is confined to the lower stratosphere and is in some ways a vertical extension of (the residual circulation of) the tropospheric Ferrel Cell. Not all the upper circulation is ventilated by the troposphere — some of it recirculates within the stratosphere. This circulation and some of the associated dynamics is illustrated schematically in Fig. 17.7,⁸ and three regions may usefully be delineated: (i) A tropical region; (ii) a mid-latitude region; (iii) the polar vortex. The tropical region is relatively quiescent, an area of upward motion where air is drawn up from the troposphere. In mid-latitudes the residual flow is generally polewards before sinking at high latitudes. In winter the extreme cold leads to the formation of the polar vortex, a strong cyclonic vortex that appears quite isolated from mid-latitudes although, especially in the Northern Hemisphere, it is not always centred over the pole.

Let us now turn to the dynamics of the circulation, and since this in large measure dependent on the waves that exist in the stratosphere we first discuss them.

17.2 WAVES IN THE STRATOSPHERE

Both gravity waves and Rossby waves are important in the stratosphere and we have already discussed aspects of both. Our goal now is to see how they affect the stratosphere, first in midlatitudes and then equatorial region, where Rossby waves and gravity waves are intertwined.

17.2.1 Linear equations of motion

Because we are dealing explicitly with compressible atmosphere we will use the ideal gas equations with log-pressure coordinates, as discussed in section 2.6.3. (Nevertheless we will often find that, to a decent approximation, the equations reduce to the Boussinesq form, with compressibility having only a small effect.) We will restrict attention to the hydrostatic case, thereby limiting ourselves to relatively large scales. On a β -plane the equations of motion in log-pressure coordinates, linearized about a resting state, may be written as

$$\frac{\partial u}{\partial t} - fv = -\frac{\partial \Phi}{\partial x}, \quad \frac{\partial v}{\partial t} + fu = -\frac{\partial \Phi}{\partial y}, \quad (17.1a,b)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{1}{\rho_R} \frac{\partial(\rho_R w)}{\partial z} = 0, \quad \frac{\partial}{\partial t} \frac{\partial \Phi}{\partial z} + w N_*^2 = 0. \quad (17.1c,d)$$

The equations are, respectively, the two horizontal momentum equations, the mass continuity equation and the thermodynamic equation. The notation is as in section 2.6.3 except we use lower case z and w , not ω ; thus, $z = -H \ln(p/p_0)$ where p_0 is a constant reference pressure and H is a reference height, and $w = Dz/Dt$. As usual u and v are the horizontal velocities and Φ is the geopotential. The density profile ρ_R is an exponential, $\rho_R(z) = \rho_0 \exp(-z/H)$, where we may take $\rho_0 = 1$, and N_*^2 is a reference stratification similar to but not exactly the same as the buoyancy frequency; we will take it to be constant and drop the subscript *. The Coriolis parameter f varies as $f = f_0 + \beta y$; when we consider equatorial waves we will take $f_0 = 0$, and when we consider gravity waves in mid-latitudes we will take $\beta = 0$.

As in section 16.5 it is convenient to extract that part of the solution that grows exponentially with height, and so seek wave solutions of the form

$$[u, v, w, \Phi] = [\tilde{u}(y), \tilde{v}(y), \tilde{w}(y), \tilde{\Phi}(y)] e^{z/2H} e^{i(kx + mz - \omega t)}. \quad (17.2)$$

We cannot assume a simple harmonic form in the y -direction because the equations of motion have coefficients (i.e., f) that depend on y . Substituting (17.2) into (17.1) yields

$$-i\omega \tilde{u} - f \tilde{v} = -ik \tilde{\Phi}, \quad -i\omega \tilde{v} + f \tilde{u} = -\frac{\partial \tilde{\Phi}}{\partial y}, \quad (17.3a)$$

$$ik \tilde{u} + \frac{\partial \tilde{v}}{\partial y} + i \left(m + \frac{i}{2H} \right) \tilde{w} = 0, \quad -i\omega \left(\frac{1}{2H} + im \right) \tilde{\Phi} + \tilde{w} N^2 = 0. \quad (17.3b)$$

Perhaps surprisingly, in many situations we can ignore the factor $1/2H$ in this system. Many observed stratospheric waves have a vertical wavelength, λ , is of order a few kilometers and usually less than 10 km. Also, $T_0 = 240$ K then $H \approx 7$ km. The $1/2H$ factor is small when $m \gg 1/2H$ or $4\pi H/\lambda \gg 1$. In fact, it is the square of this ratio that needs to be large, and this is true for all but the deepest stratospheric waves. Compressibility remains in the system because, using (17.2), all the perturbation variables grow exponentially with height, albeit slowly.

In the sections that follow we look some of the waves supported by this system. The analysis is more complicated for equatorial region, where gravity and Rossby waves are intertwined, so we begin with the mid-latitudes.

17.2.2 Waves in mid-latitudes

In mid-latitudes there is a good frequency separation between Rossby waves and gravity waves so they can be treated separately, and since we have already treated both in chapters 7 and 16 our discussion focuses on their stratospheric relevance.

Rossby waves

If we neglect the factor of $1/H^2$ the x - and z -components of the group velocity are

$$c_g^x = \frac{(k^2 - l^2 - f_0^2 m^2 / N^2) \beta}{(k^2 + l^2 + f_0^2 m^2 / N^2)^2}, \quad c_g^z = \frac{2km\beta f_0^2 / N^2}{(k^2 + l^2 + f_0^2 m^2 / N^2)^2}. \quad (17.4a,b)$$

Since $k < 0$ for Rossby waves then, in order for the waves to be upwardly propagating, (17.4b) requires that $m < 0$. Thus, the lines of constant phase tilt westward with height. The ratio of the vertical to the horizontal components of group velocity is not, unlike the case with gravity waves, a simple function of the wavenumbers and it is not possible to determine whether c_g^x is positive or negative without knowing the value of l , the meridional wavenumber. To obtain a typical value of the vertical group velocity in the atmosphere we may take $k^{-1} = 1000 \text{ km}$, $m^{-1} = 10 \text{ km}$, $f_0/N = 10^{-2}$, $\beta = 10^{-11} \text{ m}^{-1} \text{ s}^{-1}$, giving $c_g^z \sim 0.1 \text{ m s}^{-1} \approx 10 \text{ km/day}$.

Because Rossby waves grow in amplitude as they ascend the linearity assumption will eventually fail and the waves may break and dissipate, and in doing so they will deposit momentum. They may also break and/or dissipate if they encounter a critical line. This deposition is responsible for the production of the stratospheric meridional overturning circulation that we discuss later.

Gravity waves

Gravity waves may also propagate up into the stratosphere from the troposphere. If we take f to be a constant, f_0 , then, except for the factor of ρ_R the set (17.1) is the same as the f -plane hydrostatic Boussinesq equations, namely (7.144) on page 280. It is a straightforward matter to show that the dispersion relation is

$$\omega^2 = \frac{f^2 m'^2 + (k^2 + l^2) N^2}{m'^2} = f^2 + \frac{N^2(k^2 + l^2)}{m'^2}, \quad (17.5)$$

where $m'^2 = m^2 + 1/4H^2$. As noted above the factor of $1/4H^2$ is often small and we shall ignore it. If we suppose that the horizontal component of the wave vector is aligned with the x -axis (i.e. $l = 0$) then the group velocity components are

$$c_g^x = \frac{N^2 k}{\omega m^2} = \frac{N^2}{\omega m} \cos \vartheta, \quad c_g^z = -\frac{N^2(k^2 + l^2)m}{\omega m^4} = \frac{-N^2}{\omega m} \cos^2 \vartheta. \quad (17.6a,b)$$

where $\cos^2 \vartheta = k^2/(k^2 + m^2) \approx k^2/m^2 \ll 1$. The above expressions are most easily derived directly from (17.5) but are also the hydrostatic limit of the full expression (7.140). The directional aspects of these expressions are the same as those given for the non-rotating case in (7.74) with $\sin \vartheta = 1$, consistent with the hydrostatic limit — indeed we can obtain (17.6) from (7.74) by setting $\cos \lambda = 1$, $\sin \vartheta = 1$, $\omega = N \cos \vartheta$ and $\kappa = m$. Thus, the relation of the group velocity to the phase speed is much the same as for the gravity waves considered in chapter 7, and in particular we have $c_g^z/c_g^x = -k/m$. If the waves are generated in the troposphere then c_g^z must be positive and so m must be negative. In mid-latitudes there is however no requirement that the horizontal propagation be in any particular direction. The distribution of velocity, pressure and temperature are illustrated in the two panels of Fig. 17.8 for waves with a positive and negative horizontal wavenumber and a negative vertical wavenumber.

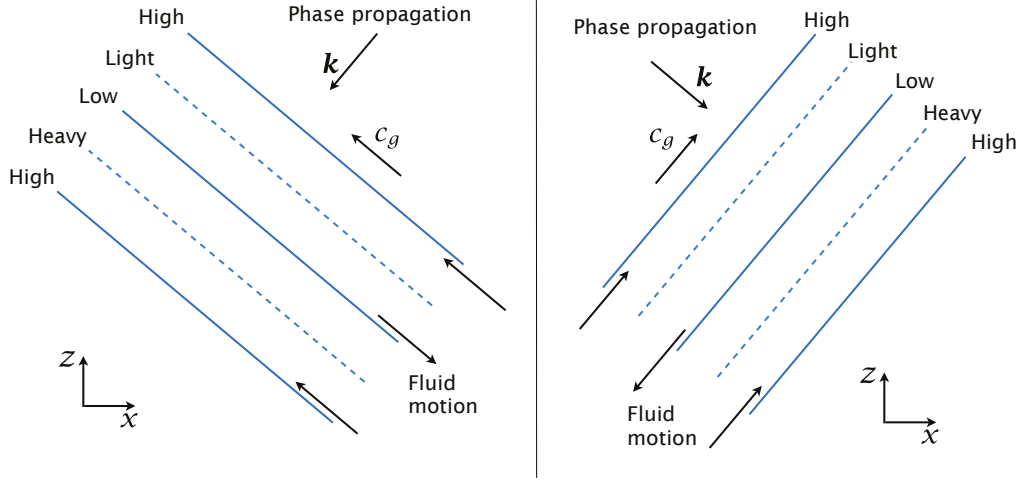


Fig. 17.8 Phase relationships for two examples of upwardly propagating gravity waves. The sketch on the left shows waves propagating to the left, with $k < 0$, and the one on the right sketch shows waves with $k > 0$. Lines of constant phase are solid and dotted lines, with 'high' and 'low' referring to pressure and 'light' and 'heavy' referring to density and corresponding to warm and cold, respectively. In both sketches m is negative and the group velocity is directed upwards and phase propagates downwards. For hydrostatic flow the phase lines would be nearly horizontal. The figure may be compared with Fig. 7.3.

17.2.3 Waves in the Equatorial Stratosphere

Equatorial waves in the stratosphere — Kelvin waves and Rossby waves — turn out to be particularly important in generating stratospheric variability. It transpires that there can be vertically propagating waves with both eastward and westward phase speeds, even at relatively low frequencies, and this difference in phase speed has some important consequences. We first look at Kelvin waves, for these provide a gentle introduction via a simple special treatment, and follow this by a more general treatment that includes Rossby and gravity waves.

Kelvin waves

We obtain the Kelvin wave solution by setting $\bar{v} = 0$ everywhere in (17.3), whence, after eliminating $\bar{w}$, (17.3) straightforwardly becomes

$$\omega \bar{u} = k \bar{\Phi}, \quad f \bar{u} = -\frac{\partial \bar{\Phi}}{\partial y}, \quad \omega \left(m^2 + \frac{1}{4H^2} \right) \bar{\Phi} - N^2 k \bar{u} = 0 \quad (17.7a,b,c)$$

Eqs. (17.7a,b) give $\omega \partial \bar{u} / \partial y + k f \bar{u} = 0$, which upon integration and with $f = \beta y$ yields

$$\bar{u}(y) = \bar{u}_0 e^{-\beta y^2 / 2c_p}, \quad (17.8)$$

where $c_p = \omega/k$ and $\bar{u}_0$ is the value of $\bar{u}$ at the equator. The exponential fall-off is familiar from our earlier studies of Kelvin waves in chapter 8 and requires that $c_p > 0$, meaning that the phase speed of the waves is eastward. Also, since $\omega > 0$ by convention, the x -wavenumber is positive (i.e., $k > 0$). The dispersion relation for Kelvin waves follows easily from (17.7a,c) and is

$$\omega^2 = \frac{N^2 k^2}{m^2 + 1/4H^2}. \quad (17.9)$$

Aside from the factor of $1/4H^2$, which in any case is often small compared to m^2 , (17.9) is essentially the same as the dispersion relation for hydrostatic gravity waves, namely (7.60) on page 261. The zonal and vertical components of the group velocity are

$$c_g^x = \frac{N}{(m^2 + 1/4H^2)^{1/2}}, \quad c_g^z = \frac{\partial \omega}{\partial m} = \frac{-Nkm}{(m^2 + 1/4H^2)^{3/2}}. \quad (17.10)$$

Now, for upwardly propagating waves (and so for waves that emanate from the troposphere) we require $c_g^z > 0$ and therefore (because $k > 0$) $m < 0$. The combined conditions of $k > 0$ and $m < 0$ means that the phase lines tilt eastward with height, as in the right panel of Fig. 17.8. Finally, note that the frequency of Kelvin waves, unlike inertia-gravity waves, is uninfluenced by rotation and thus, as seen in Fig. 8.6, can extend over a broad range.

A more general treatment of equatorial waves

For simplicity let us assume that the scale height H is very large compared to the vertical wavelengths of interest; that is, $m^2 \gg 1/H^2$ and $\exp(-z/H) = 1$. Eqs. (17.1b) and (17.1c) then combine to give

$$\frac{\partial}{\partial t} \frac{\partial^2 \Phi}{\partial z^2} - N^2 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0. \quad (17.11)$$

If we assume a vertical structure of the form $\Phi(x, y, z, t) = \tilde{\Phi}(x, y, t) \exp(imz)$, and similarly for u and v , then we obtain

$$\frac{\partial \tilde{\Phi}}{\partial t} + \frac{N^2}{m^2} \left(\frac{\partial \tilde{u}}{\partial x} + \frac{\partial \tilde{v}}{\partial y} \right) = 0, \quad (17.12a)$$

with corresponding momentum equations

$$\frac{\partial \tilde{u}}{\partial t} - f\tilde{v} = -\frac{\partial \tilde{\Phi}}{\partial x}, \quad \frac{\partial \tilde{v}}{\partial t} + f\tilde{u} = -\frac{\partial \tilde{\Phi}}{\partial y}. \quad (17.12b,c)$$

Evidently, (17.12) are isomorphic to the linear shallow water equations (8.20) on page 304 with the replacement $c^2 = N^2/m^2$, where in (8.20) $c^2 \equiv gH_e$ where H_e is the equivalent depth. (Note that c is not necessarily the phase speed in this problem; we will denote that as c_p .) The correspondence between the continuously-stratified and shallow water equations is a general one, as we found in section 3.4. All of the machinery following (8.20) can now be applied to (17.12), with Rossby-gravity waves and Kelvin waves emerging in the same way (and the Kelvin waves identified above emerge as a special case); thus, in what follows we draw directly from section 8.2. We will use m to denote the vertical wavenumber and n to denote the order of the Hermite function, akin to a meridional wavenumber.

Rossby-gravity waves

The dispersion relation that emerges from (17.12) is, by analogy with (8.37b), (8.53) and (8.63),

$$\omega^2 - c^2 k^2 - \beta \frac{kc^2}{\omega} = (2n+1)\beta c, \quad n > 0, \quad (17.13a)$$

$$\omega^2 - \omega kc - \beta c = 0, \quad n = 0, \quad (17.13b)$$

$$\omega = ck, \quad n = -1. \quad (17.13c)$$

where $c = N/m$. The case with $n = 0$ is the Yanai wave (a Rossby-gravity wave) and the cases with $n \geq 1$ are planetary waves or gravity waves. The ' $n = -1$ ' case is the Kelvin wave, and all cases are illustrated in Fig. 8.6.

We now note that for waves whose origin is in the troposphere we may think of the frequency as being given and (17.13) then provides a condition on the vertical wavenumber. This in turn suggests that we nondimensionalize the wavenumbers by defining

$$\hat{m} = \frac{\omega^2}{\beta N} m = \frac{\omega^2}{\beta c}, \quad \hat{k} = \frac{\omega}{\beta} k, \quad (17.14)$$

with the hats denoting nondimensional variables. Equations (17.13) become

$$\hat{m}^2 - (2n + 1)\hat{m} - \hat{k}^2 - \hat{k} = 0, \quad n > 0, \quad (17.15a)$$

$$\hat{m} - \hat{k} - 1 = 0, \quad n = 0 \quad (17.15b)$$

$$\hat{m} = \hat{k}, \quad n = -1. \quad (17.15c)$$

These equations define a set of curves in $\hat{m}$ - $\hat{k}$ space that are similar to the curves in ω - k space defined by (17.13), although (17.15) are lower order. Equation (17.15a) has the solution

$$\hat{m} = \left(n + \frac{1}{2}\right) \pm \left[\left(\hat{k} + \frac{1}{2}\right)^2 + n(n + 1)\right]^{1/2}, \quad (17.16)$$

and the complete set of curves is plotted in Fig. 17.9. The curves at the top and bottom are gravity waves, corresponding to the positive sign in (17.16), and the planetary waves are the curves just to the left of the origin, corresponding to the negative sign. The $n = 0$ curve (the Yanai wave) and the $n = -1$ curve (the Kelvin wave) are labelled.

We can infer the group velocity from the figure by noting that, since β and N are constant and using (17.14) the curves are contours of constant frequency. The group velocity is the gradient of frequency in wavenumber space and so is at right angles to these curves and directed toward a higher frequency, and is marked by short arrows. For waves propagating up from the troposphere the group velocity must be upward, and therefore have negative m , as can be seen from (17.6) and (17.10).

17.3 WAVE MOMENTUM TRANSPORT AND DEPOSITION

Steady waves by themselves don't affect the mean flow. Rather, they affect the mean flow when they are generated or dissipated, and typically they are generated in the troposphere and dissipated in the stratosphere. Let's look into that, beginning with Rossby waves.

17.3.1 Rossby waves

The vertical momentum transport of zonal momentum by eddies is given by $\overline{w'u'}$, where an overbar denotes a zonal average. However, directly evaluating this expression from the quasi-geostrophic equations is not particularly simple because w is not a first order variable — it results from the divergence of the ageostrophic horizontal velocities.⁹ In fact, under quasi-geostrophic scaling we neglect the vertical eddy flux divergences, but nevertheless the eddy fluxes may certainly make themselves felt aloft, among other things by generating a form stress that acts to transfer momentum vertically and generating a meridional overturning circulation.

To proceed we will use the transformed Eulerian mean (TEM) framework of section 10.3, for which the inviscid and adiabatic zonally-averaged momentum and thermodynamic equations are

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \overline{v'q'}, \quad \frac{\partial \bar{b}}{\partial t} + N^2 \bar{w}^* = 0, \quad (17.17a,b)$$

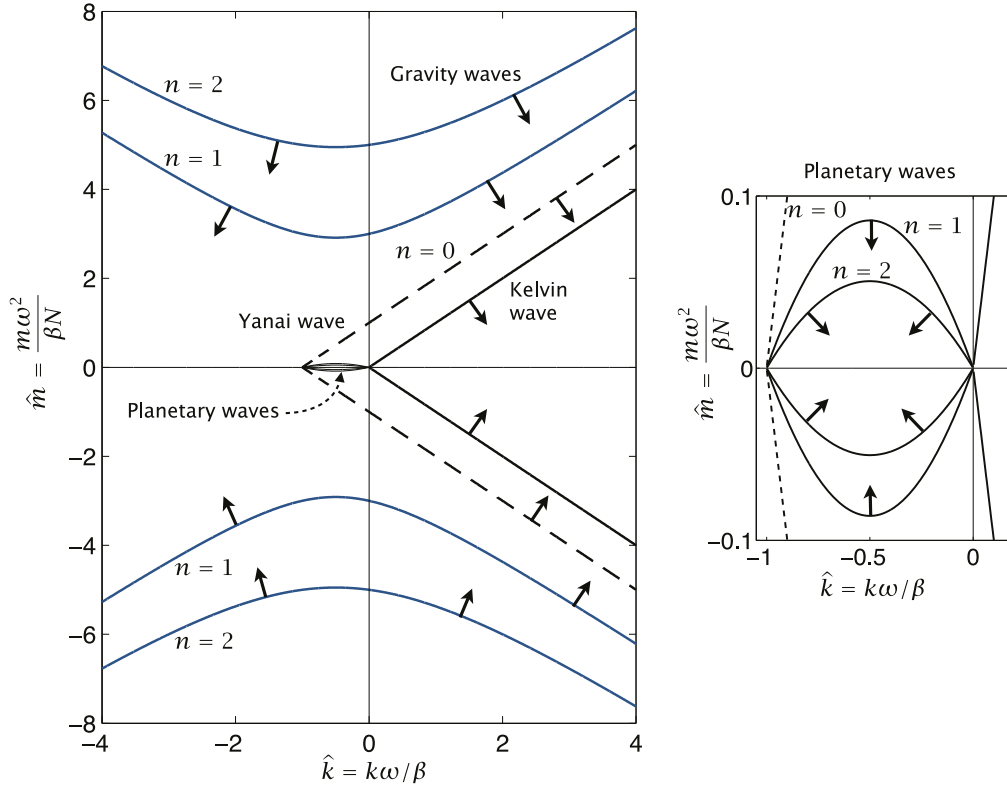


Fig. 17.9 Dispersion curves plotted in $\hat{m}$ – $\hat{k}$ space, using (17.15). Shown are gravity waves, the Yanai wave and the Kelvin wave for positive and negative $\hat{m}$, with the plot at the right showing a magnification of the region near the origin. The arrows in the figure indicate the group velocity, which, being the gradient of the frequency, is perpendicular to the curves. Upward propagating waves occur for negative m . Compare with Fig. 8.6.

where potential vorticity flux is related to the EP flux, $\mathcal{F}$, by

$$\overline{v'q'} = \nabla \cdot \mathcal{F}, \quad \mathcal{F} = -\overline{u'v'} \mathbf{j} + \frac{f_0}{N^2} \overline{v'b'} \mathbf{k}, \quad (17.18)$$

and now for simplicity we use the Boussinesq equations with b as buoyancy. If the eddy fluxes are due to the presence of waves that satisfy a dispersion relation then, as shown in section 10.2.2, the EP flux is related to the group velocity by

$$\mathcal{F} = (\mathcal{F}^y, \mathcal{F}^z) = \mathbf{c}_g \mathcal{A}. \quad (17.19)$$

Combining the above equations we obtain

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \frac{\partial}{\partial y} (c_g^y \mathcal{A}) + \frac{\partial}{\partial z} (c_g^z \mathcal{A}) = \nabla \cdot \mathcal{F}. \quad (17.20)$$

Now, reprising (10.29a), the wave activity itself satisfies a conservation law of the form

$$\frac{\partial \mathcal{A}}{\partial t} + \nabla \cdot \mathcal{F} = \mathcal{D}, \quad (17.21)$$

where $\mathcal{A} = \overline{q'^2}/2\beta$, which is a positive quantity and $\mathcal{D}$ represents dissipation. Evidently if $\mathcal{D} = 0$ and the waves are steady then $\nabla \cdot \mathcal{F} = 0$ so that the left hand side of (17.20) is zero. (This is the

non-acceleration result of section 10.4.2.) Evidently in order to get a zonal flow acceleration or an MOC we need to invoke some dissipative or time-dependent processes.

Consider the case in which Rossby waves propagate up from the troposphere, with $c_g^z > 0$. Suppose that there is some dissipation in the system (and/or that Rossby waves break as they ascend) and that wave activity $\mathcal{A}$ falls with height, and suppose further that we are in a statistically steady state. In this case $\nabla \cdot \mathcal{F} < 0$ and from (17.17a) this will produce a mean flow deceleration (i.e., a westward acceleration) and/or a polewards residual flow. The balance between these two possibilities is discussed later, but one may intuit close to the equator the more likely outcome is a zonal acceleration rather than a meridional circulation. Why is there a preferred sense of acceleration when the waves break? Ultimately it is because of the beta effect which distinguishes east from west, and the wave activity (or pseudomomentum) $\mathcal{A}$ is proportional to beta. The beta effect leads to a particular orientation of the phase of Rossby waves and this carries westward momentum away from the source region. When the waves break, be it in the tropospheric subtropics or in the stratosphere, a westward momentum is deposited.

17.3.2 Gravity and Kelvin waves

Now consider the vertical momentum transport in gravity waves. If these are uninfluenced by rotation, or if they reside on the f -plane, then there is no preferred horizontal direction of propagation. Kelvin waves, on the other hand, propagate their phase eastwards only. The upwards transport of momentum from waves originating in the troposphere can occur only for waves with a positive group velocity, and thus for either of the examples illustrated in Fig. 17.8. Those waves that propagate phase westward (i.e., have $k < 0$) have $\overline{u'w'} < 0$, and those that propagate phase eastward, such as equatorial Kelvin waves, have $\overline{u'w'} > 0$. The contribution to the zonal flow acceleration by the wave transport is given by

$$\frac{\partial \bar{u}}{\partial t} = -\frac{\partial}{\partial z} \overline{u'w'} + \text{other terms.} \quad (17.22)$$

and if the amplitude of the waves stays constant with height then no mean flow acceleration is induced. However, if the amplitude diminishes with height, because of dissipative processes, then the waves that have a westward (eastward) phase propagation will cause the zonal flow to accelerate westward (eastward). Thus *the dissipation of Kelvin waves as they propagate vertically will cause an eastward acceleration of the zonal flow.*

17.3.3 ♦ The processes of wave attenuation

We now explicitly consider the dissipation of waves and the associated momentum deposition as gravity waves propagate vertically. (The reader may wish to skim sections 16.3.2 and 16.3 before proceeding, for there we consider similar but algebraically simpler problems.) To keep the algebra manageable we will consider the propagation of two-dimensional (x - z) gravity waves in a Boussinesq fluid uninfluenced by rotation. The momentum and buoyancy equations, linearized about a zonal flow $U(z)$ and constant stratification N^2 , are

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} = -\frac{\partial \phi}{\partial x}, \quad \frac{\partial w}{\partial t} + U \frac{\partial w}{\partial x} = -\frac{\partial \phi}{\partial z} + b, \quad (17.23a)$$

$$\frac{\partial b}{\partial t} + U \frac{\partial b}{\partial x} + wN^2 = -\alpha b. \quad (17.23b)$$

We include a damping term, $-\alpha b$, where α is a constant, in the buoyancy equation but neglect viscous effects in the momentum equation. If we cross-differentiate the momentum equation and use the mass continuity equation ($\partial u / \partial x + \partial w / \partial z = 0$) we obtain the linear vorticity equation

$$\left(\frac{\partial}{\partial t} + \frac{\partial}{\partial x} \right) \nabla^2 \psi + \frac{\partial \psi}{\partial x} \frac{d^2 U}{dz^2} = \frac{\partial b}{\partial x}, \quad (17.23c)$$

where ψ is such that $w = \partial\psi/\partial x$ and $u = -\partial\psi/\partial z$. We seek solutions of (17.23) in the form

$$[\psi, b] = [\tilde{\psi}(z), \tilde{b}(z)] e^{ik(x-ct)}, \quad (17.24)$$

and a little algebra reveals

$$i(-kc + Uk) \left(-k^2 \tilde{\psi} + \frac{d^2 \tilde{\psi}}{dz^2} \right) + ik\tilde{\psi} \frac{d^2 U}{dz^2} = ik\tilde{b}, \quad (17.25a)$$

$$i(-kc + Uk)\tilde{b} + ikN^2 \tilde{\psi} = -\alpha \tilde{b}. \quad (17.25b)$$

These last two equations combine to give

$$\frac{d^2 \tilde{\psi}}{dz^2} + m^2(z) \tilde{\psi} = 0, \quad m^2(z) = \left[\frac{N^2 [1 + i\alpha/k(U-c)]}{(U-c)^2 + \alpha^2/k^2} - k^2 - \frac{d^2 U/dz^2}{(U-c)} \right]. \quad (17.26)$$

This is an equation for the vertical structure of the streamfunction. If U and N^2 were constant in (17.26) then m would be constant and its real part would be the vertical wavenumber. There is an imaginary component to m which can be expected to cause the solution to decay in the vertical. In most circumstances the decay is slow but in the neighbourhood of a critical layer where $c = U$ then the decay will be rapid. The wave can be expected to deposit momentum and accelerate or decelerate the mean flow, depending on the direction of the phase propagation of the wave. When $\alpha = 0$ there is no dissipation and no deposition (except possibly at the critical layer itself). Although the equation seems complicated, we can proceed to a solution if we make some reasonable simplifying assumptions.

- (i) We consider (as is realistic) low aspect ratio flows ($L_z/L_x \ll 1$) so that the factor of k^2 is small compared to m^2 and may be neglected.
- (ii) We will assume that the variation of the mean flow occurs on a long vertical scale compared to m , and so neglect the term in $d^2 U/dz^2$. This assumption will also allow us to use WKB methods.
- (iii) We assume dissipation is small, and in particular that $\alpha/k(U-c) \ll 1$

With these approximations (17.26b) becomes

$$m(z) \approx \left[\frac{N^2 [1 + i\alpha/k(U-c)]}{(U-c)^2 + \alpha^2/k^2} \right]^{1/2} \approx \frac{N}{U-c} \left[1 + \frac{i\alpha}{2k(U-c)} \right], \quad (17.27)$$

and we can proceed with a WKB solution.

WKB solution and momentum flux

The WKB solution to (17.26) is

$$\tilde{\psi}(z) = A m^{-1/2} \exp \left(\pm i \int^z m dz' \right), \quad (17.28)$$

where A is a constant. The wave momentum flux, F , associated with the wave is

$$F_k(z) = \overline{u'w'} = -ik \left(\tilde{\psi} \frac{\partial \tilde{\psi}^*}{\partial z} - \tilde{\psi}^* \frac{\partial \tilde{\psi}}{\partial z} \right), \quad (17.29)$$

where the overbar denotes a zonal average and the right-hand side is always real, and the subscript on F indicates we are considering the effects of a single wave of wavenumber k .

Now, the fact that m varies only slowly with z (specifically $m^2 \gg |dm/dz|$) means that when we take the vertical derivative of $\tilde{\psi}$ we can ignore the derivative of the vertical derivative of the amplitude, $Am^{-1/2}$. Given this, and using (17.28) and (17.27) in (17.29) we obtain

$$F_k(z) = F_0 \exp \left(i \int_0^z (m - m^*) dz' \right) = F_0 \exp \left(\int_0^z \frac{-N\alpha}{k(U - c)^2} dz' \right). \quad (17.30)$$

where F_0 is the value of the flux at $z = 0$ and we have chosen the sign in the exponent to be appropriate for upwardly propagating gravity waves. The integrand in the right-most expression is the attenuation rate of the wave and, referring to (7.80) on page 266, it can be written as

$$\text{Attenuation rate} = \frac{\alpha}{k(U - c)^2/N} = \frac{\text{Dissipation rate}}{\text{Vertical group velocity}}. \quad (17.31)$$

As $U - c$ diminishes the group velocity falls, giving the dissipative processes more time to act. The result of (17.31) is a general one; we found an almost identical result when looking at the absorption of Rossby waves in section 16.3 — see (16.49). (The dissipation rate in expressions like (16.49) and (17.31) is that of wave activity, which in the gravity wave case here equals the thermal dissipation rate.)

Effect on the mean flow

If a wave propagating upwards is attenuated there will be a divergence in the eddy momentum flux associated with that wave. In particular, if α in (17.31) is nonzero then momentum flux deposition will increase rapidly as a critical layer is approached, $\partial F_k / \partial z$ will be nonzero and the zonal mean flow will be accelerated or decelerated. For definiteness, consider a wave propagating upwards with a positive phase speed (so $m < 0$ and $k > 0$). From (17.30) F_k diminishes with height and the mean flow is accelerated eastward. Similarly, absorption of a wave of negative phase speed leads to a negative, or westward mean-flow acceleration. It is not inconceivable to imagine that the wave deposition will affect the mean flow to an extent that the position of the deposition is significantly altered, leading to interesting dynamical behaviour. Indeed this is precisely what happens in the quasi-biennial oscillation of the equatorial stratosphere. But before we discuss variability let us discuss the maintenance of the mean state.

17.4 PHENOMENOLOGY OF THE RESIDUAL OVERTURNING CIRCULATION

We now return to a discussion of the general circulation of the stratosphere and in particular the maintenance of the residual meridional overturning circulation (RMOC), or the Brewer–Dobson circulation. We expect the circulation to be a consequence of waves coming up from the troposphere and breaking, with both tropospheric baroclinic instability and flow over zonal asymmetries being sources of wave activity. We may naturally ask such questions as whether wavebreaking can give rise to a circulation of the right strength and the right sense, whether it will give the correct seasonal variability, and what determines vertical extent of the circulation. We begin with some elementary theory and phenomenology, for we will find that it will explain a number of the main features of the RMOC, and the advanced or confident reader may skip ahead to section 17.5.

17.4.1 Wave breaking and residual flow

The equations of motion governing the mean fields are the zonally averaged momentum and thermodynamic equations, which with quasi-geostrophic scaling and in residual form (see section 10.3.1) may be written as

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \nabla \cdot \bar{\mathcal{F}} + \bar{F}, \quad \frac{\partial \bar{\theta}}{\partial t} + \frac{\partial \bar{\theta}}{\partial z} \bar{w}^* = \bar{J}, \quad (17.32a,b)$$

where $\bar{F}$ represents frictional effects and $\bar{J}$ represents heating, and on the β -plane the residual velocities are related to the Eulerian means by

$$\bar{v}^* = \bar{v} - \frac{1}{\rho_R} \frac{\partial}{\partial z} \left(\rho_R \frac{\overline{v'\theta'}}{\partial_z \bar{\theta}} \right), \quad \bar{w}^* = \bar{w} + \frac{\partial}{\partial y} \left(\frac{\overline{v'\theta'}}{\partial_z \bar{\theta}} \right). \quad (17.33)$$

The vector $\mathcal{F}$ is the Eliassen–Palm flux, and this is related to the meridional flux of potential vorticity by $\nabla \cdot \mathcal{F} = \overline{v'q'}$. The wave activity itself obeys the Eliassen–Palm relation

$$\frac{\partial \mathcal{A}}{\partial t} + \nabla \cdot \mathcal{F} = \mathcal{D}, \quad (17.34)$$

where $\mathcal{A}$ is the wave activity, $\mathcal{F}$ is its flux and $\mathcal{D}$ is its dissipation.

From the autumn to the spring, the zonal wind in the stratosphere is generally receptive to planetary-scale Rossby waves propagating up from the troposphere (Fig. 16.6), although at high latitudes in winter there may be a period when the eastward zonal winds are too strong for waves to propagate. If these waves break in the stratosphere then there will be an enstrophy flux to small scales and dissipation. In a statistically-steady state and with small frictional effects the dominant balance in the zonal momentum equation (17.32a) is

$$-f_0 \bar{v}^* \approx \overline{v'q'}, \quad (17.35)$$

where $\bar{v}^*$ is the residual velocity and the potential vorticity flux on the right-hand side is induced by the Rossby wave breaking. In dissipative regions the zonally averaged potential vorticity flux will tend to be down its mean gradient and, if the potential vorticity gradient is polewards (largely because of the β -effect), the residual velocity will be positive if f_0 is positive. That is, the residual flow will be *polewards*, in both hemispheres, and the mechanism giving rise to this is called the ‘Rossby wave pump’. Put another way, Rossby waves propagating up from the troposphere break and deposit westward momentum in the stratosphere, and in the mean this wave drag is largely balanced by the Coriolis force on the polewards residual meridional circulation.

This meridional circulation is weakest in summer mainly because linear Rossby waves cannot propagate upwards through the westward mean winds, as illustrated in Fig. 17.10. It is quite striking how the EP vectors avoid the region of westward winds in the summer hemisphere, even though the level of wave activity at low elevations is relatively similar in the summer and winter hemispheres (look between 10 and 15 km in the figure). We can interpret this by noting that for nearly plane waves the EP flux obeys the group velocity property, meaning that $\mathcal{F} = c_g \mathcal{A}$; however, as discussed in section 16.5, if the mean winds are westward the waves evanesce instead of propagate, and thus almost the entire summer hemisphere is shielded from upwardly propagating waves, leaving it in a near-radiative equilibrium state. In the other seasons, the EP flux is able to propagate into the stratosphere and a circulation is generated. This acts to weaken the pole–equator temperature gradient, as we see by inspection of the thermodynamic equation: if the heating is represented by a simple relaxation to a radiative equilibrium state, θ_E , then in a steady state we have

$$N^2 \bar{w}^* = \frac{\theta_E - \theta}{\tau}. \quad (17.36)$$

Poleward flow in mid-latitudes must be supplied by rising air at low latitudes, and sinking air at high latitudes. Thus, from autumn to spring, at low latitudes we have $\theta < \theta_E$ and at high latitudes $\theta > \theta_E$.

Although cause and effect can be very difficult to disentangle in fluid dynamical problems, and the ultimate cause of nearly all fluid motions in the atmosphere is the differential heating from the Sun, it is important to realize that the meridional overturning in the stratosphere is not a direct response to differential solar heating: note that the most intense solar heating is over the summer

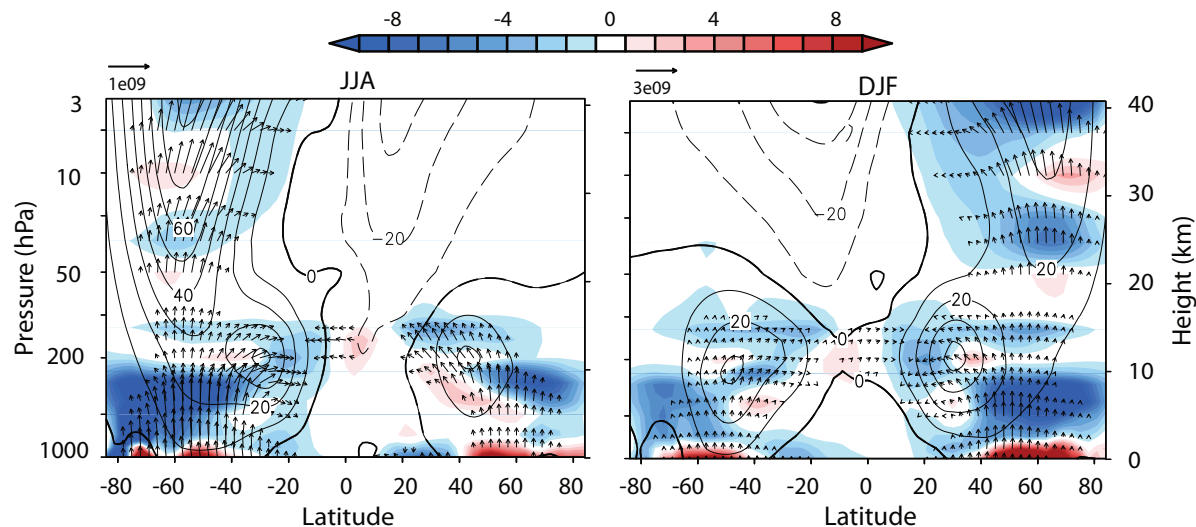


Fig. 17.10 The EP flux vectors (arrows), the EP flux divergence (shading) and the zonally averaged zonal wind (contours) for northern hemisphere summer and winter. The tropospheric EP fluxes are of similar magnitudes in the summer and winter hemispheres, but are almost zero in the summer in the stratosphere. Note also strong convergence at high latitudes in the stratospheric winter hemispheres, leading to poleward residual flow and/or zonal flow acceleration.¹⁰

pole, yet here there is little or no ascent. Rather, the circulation is more usefully thought of as a response to potential vorticity fluxes which in turn are determined by the upward propagation of Rossby waves from the troposphere combined with the poleward gradient of potential vorticity in the stratosphere. It is salutary to note that without motion we have $\theta = \theta_E$, so there is no net heating at all — the heating is a consequence of the wave forcing.

17.5 ♦ DYNAMICS OF THE RESIDUAL OVERTURNING CIRCULATION

We now discuss the dynamics of the residual meridional overturning circulation, the RMOC, in rather more detail than in the previous section, without shying away from the repetition of important matters.¹¹ The dynamics of the RMOC can be usefully couched as a quasi-linear problem. That is, the governing equations can be written with the linear terms on the left-hand side and the nonlinear terms as forcing terms on the right-hand side. Of course, this cannot be regarded as a full solution, but if the right-hand sides can be determined, if approximately, by independent means then the equations can be solved fairly straightforwardly and the structure of the RMOC so determined. Such a procedure is actually likely to be more successful in the stratosphere than in the troposphere; in the latter, the nonlinear terms are a truly essential part of the solution and cannot properly be separated from the linear dynamics of the RMOC (although we might choose to separate the terms to *diagnose* what forces the RMOC). In the stratosphere the nonlinear terms represent the effects of waves and wave breaking on the mean flow. These waves — both gravity waves and Rossby waves — often have their origins in the troposphere, and although the propagation and breaking of the waves does depend strongly on the background flow the basic features of the forcing of the RMOC can still usefully be considered independently of the RMOC itself.

17.5.1 Equations of motion

Away from the equator the Rossby number is small and the equations governing the large scale flow are in good geostrophic balance. The equations of motion governing the mean fields are the

zonally averaged momentum and thermodynamic equations, along with the thermal wind equation and the mass continuity equations. In what follows we will write the equations in their full form in spherical co-ordinates using the ideal gas equations in log-pressure co-ordinates, since both sphericity and compressibility are important effects, using the transformed Eulerian mean formalism. The equations of motion may be written as

$$\frac{\partial \bar{u}}{\partial t} - f \bar{v}^* = \mathcal{G} + \mathcal{D} = \nabla \cdot \mathcal{F} - \gamma \bar{u}, \quad f \frac{\partial \bar{u}}{\partial z} + \frac{R}{aH} \frac{\partial \bar{T}}{\partial \vartheta} = 0, \quad (17.37a,b)$$

$$\frac{\partial \bar{T}}{\partial t} + \bar{w}^* \mathcal{S} = Q_s + Q_l = \mu(T_R - \bar{T}), \quad \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (\bar{v}^* \cos \vartheta) + \frac{1}{\rho_R} \frac{\partial}{\partial z} (\rho_R \bar{w}^*) = 0. \quad (17.37c,d)$$

These equations are, respectively, the zonal momentum equation, the thermal wind equation, the thermodynamic equation and the mass continuity equation, with an overbar denoting a zonal average. The vertical co-ordinate, z is log pressure and $S = H_\rho N^2 / R$ where R is the gas constant and H_ρ is the scale height used to define z ; thus, z has dimensions of height and $\rho_R = \exp(-z/H_\rho)$. (We denote our vertical coordinate as z rather than Z for aesthetic reasons.) The equations have a very similar form if written in height co-ordinates using the anelastic approximation (section 2.5); in that case, ρ_R is a reference profile of density and the thermodynamic equation is written using potential temperature or buoyancy as the thermodynamic variable, the factor H_ρ/R no longer appears, and z really is physical height.

The other variables in (17.37) use our standard notation, with $\bar{v}^*$ and $\bar{w}^*$ being the residual, or transformed Eulerian mean, meridional and vertical velocities. The right-hand side of (17.37a) represents wave forcing and friction: $\mathcal{G} = \nabla \cdot \mathcal{F}$ is the divergence of the Eliassen–Palm flux and $\mathcal{D}$ is the frictional force, which we take to be a simple linear drag. Such a drag is a little arbitrary but its form greatly simplifies the ensuing analysis. On the right-hand side of the thermodynamic equation Q_s and Q_l represent the forcing due to solar and long wave radiation; we take $Q_l = -\mu \bar{T}$ where μ is a constant thermal damping rate, and we may write $Q_s = \mu T_r(\vartheta, z)$ where T_r is a radiative equilibrium temperature, assumed known. There are no fluid-dynamic wave-forcing terms in the TEM form of the thermodynamic equation. Typically, the momentum dissipation is small and $\gamma \ll \mu$, and indeed we may take $\gamma = 0$ without much loss of realism, except close to the ground.

17.5.2 An equation for the MOC

If the right-hand sides are known, (17.37) constitutes a closed set of equations for the response of the temperature and the three components of the velocity to an applied wave force $\mathcal{G}$ and solar heating Q_s . Although nominally the equations have two time derivatives, the zonal wind and temperature are related through the thermal wind relation and the equations are balanced and no gravity waves are present. Our interest here is in the meridional overturning circulation and hence $\bar{v}^*$ and $\bar{w}^*$, and it is possible to derive a single equation for either of these variables that provides considerable insight into that circulation; we will focus on $\bar{w}^*$.

The procedure is similar to that used in section 14.5.3 and the reader may wish to review that section before proceeding. Essentially, we differentiate (17.37a) with respect to z and (17.37c) with respect to ϑ and then use (17.37b) to eliminate the time derivatives. We then use the mass continuity equation to obtain a single equation in $\bar{w}^*$. We are particularly interested in the dependence of the RMOC on the spatial structure and time dependence of $\mathcal{G}$ and Q_s , and to this end it is instructive to consider the case in which the time dependence is harmonic; that is, $\mathcal{G} = \tilde{\mathcal{G}} e^{i\omega t}$, $Q_s = \tilde{Q}_s e^{i\omega t}$ and $\bar{w} = \bar{w}^* e^{i\omega t}$. After a little algebra, we obtain

$$\begin{aligned} & \frac{\partial}{\partial z} \left[\frac{1}{\rho_0} \frac{\partial(\rho_0 \bar{w})}{\partial z} \right] + \left(\frac{i\omega + \gamma}{i\omega + \mu} \right) \frac{N^2}{4\Omega^2 a^2 \cos \vartheta} \frac{\partial}{\partial \vartheta} \left[\frac{\cos \vartheta}{\sin^2 \vartheta} \frac{\partial \bar{w}}{\partial \vartheta} \right] \\ &= \frac{1}{2\Omega a \cos \vartheta} \frac{\partial}{\partial \vartheta} \left[\frac{\cos \vartheta}{\sin \vartheta} \frac{\partial \tilde{\mathcal{G}}}{\partial z} \right] + \left(\frac{i\omega + \gamma}{i\omega + \mu} \right) \frac{R}{4H\Omega^2 a^2 \cos \vartheta} \frac{\partial}{\partial \vartheta} \left[\frac{\cos \vartheta}{\sin^2 \vartheta} \frac{\partial \tilde{Q}_s}{\partial \vartheta} \right]. \end{aligned} \quad (17.38a)$$

The equation is quite a TeXful, but it is useful to realize that, schematically and without all the metric factors, it is of the form

$$\frac{\partial^2 \tilde{w}}{\partial z^2} + A \frac{N^2}{f^2} \frac{\partial^2 \tilde{w}}{\partial y^2} \sim \frac{1}{f} \frac{\partial}{\partial y} \frac{\partial \tilde{G}}{\partial z} + \frac{A}{f^2} \frac{\partial^2 \tilde{Q}_s}{\partial y^2}. \quad (17.38b)$$

where

$$A = \frac{i\omega + \gamma}{i\omega + \mu}.$$

Equation (17.38b) is similar to (10.63), since with the addition of diabatic terms and a slight change in notation (10.63) is

$$f_0^2 \frac{\partial^2 \tilde{\psi}}{\partial z^2} + AN^2 \frac{\partial^2 \tilde{\psi}}{\partial y^2} = f_0 \frac{\partial \tilde{G}}{\partial z} + A \frac{\partial \tilde{Q}_s}{\partial y}, \quad (17.38c)$$

where $\tilde{\psi} = \psi^* e^{i\omega t}$ is the amplitude of residual streamfunction of the overturning circulation. Since $\tilde{w}^* = \partial \psi^* / \partial y$, (17.38b) is almost the y -derivative of (17.38c). (In (10.63) we took $\gamma = \mu$ so that $A = 1$.) Much of the physical interpretation in what follows comes from (17.38b) and (17.38c), although we will allow f to vary spatially — that is, we use f and not f_0 .

With quasi-geostrophic scaling the wave forcing term in (17.38) is

$$\mathcal{G} = \overline{v'q'} = -\frac{\partial}{\partial y} \overline{u'v'} + \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} \overline{v'b'} \right) = \nabla \cdot \mathcal{F}, \quad \text{where } \mathcal{F} = -\overline{u'v'} \mathbf{j} + \frac{f_0}{N^2} \overline{v'b'} \mathbf{k}, \quad (17.39)$$

and $\mathcal{F}$ is the Eliassen–Palm flux.

17.5.3 The nature of the response

The operator on the left-hand side of (17.38) is elliptic, similar to a that of a Poisson equation. Thus, the response will be much less localized than the forcing itself, a concept that is familiar from potential vorticity inversion. To understand the equation better it is useful to take a heuristic look at some special cases, as follows. The cases are not all ‘orthogonal’ to each other — thus, for example, the low-latitude limit could be either low frequency or high frequency.

(i) The aspect ratio of the response

From (17.38c) the natural aspect ratio of the response, α_r say, is given by

$$\alpha_r = \frac{H_r}{L_r} = \frac{1}{A^{1/2}} \frac{f}{N} = \left(\frac{i\omega + \mu}{i\omega + \gamma} \right)^{1/2} \frac{f}{N}. \quad (17.40)$$

where H_R and L_R are the vertical and horizontal scales of the response. If the thermal and mechanical dissipation are zero, or have the same time scale, then $A = 1$, but more generally the presence of dissipation can alter the aspect ratio considerably. Also, the thermal dissipation can be expected to be much stronger than the mechanical dissipation, meaning that $\mu \gg \gamma$. The high- and low-frequency limits then have somewhat behaviour, as we see.

(ii) The high-frequency limit

In this case the thermal and mechanical damping are negligible and $A = (i\omega + \gamma)/(i\omega + \mu) \approx 1$. Since μ typically varies between 1/(20 days) in the lower stratosphere and 1/(5 days) in the upper stratosphere, and $1/\gamma$ is an even longer time, phenomena of order a few days fall into this category. Sudden stratospheric warmings are one example, although since the timescale of warmings is of order days thermal effects are not wholly negligible.

Using (17.40) we see that the aspect ratio of the response is simply of the order of Prandtl’s ratio. That is, $\alpha_r = H/L \sim f/N$ and since $f \sim 10^{-4} \text{ s}^{-1}$ and $N \sim 10^{-2} \text{ s}^{-1}$ or larger, the

response to rapid forcing is typically quite shallow. Of course, although the Prandtl ratio is small it is a natural scaling of vertical to horizontal scales in atmospheric dynamics and shallowness should be interpreted in that context. Still, as we approach the equator the response shallows still further, although at the equator itself (17.38) ceases to be valid. A quantitative analysis of the right-hand side of (17.38a) further suggests that both waves (the $\mathcal{G}$ term) and solar forcing act to drive the overturning circulation.

(iii) *The low-frequency limit*

In this case the frequency is less than the thermal damping rate; that is, $\omega \ll \mu$, with one obvious example being the annual cycle. If as is realistic, $\gamma \ll \mu$ then $|A|$ becomes small. The effect of the solar heating thus also becomes small in (17.38c). The response generally deepens with the ratio of the vertical to the horizontal scales being given by

$$\alpha_r = \frac{H_r}{L_r} \approx \left(\frac{1}{A^{1/2}} \right) \frac{f}{N} = \left(\frac{\mu}{i\omega + \gamma} \right)^{1/2} \frac{f}{N} \gg \frac{f}{N}, \quad (17.41)$$

if $\mu \gg \gamma$. In the steady-state limit the solar heating is balanced by the thermal relaxation and the right-hand side of (17.37c) nearly vanishes. We discuss this limit in more detail below, in the section on downward control.

(iv) *Deep and shallow forces*

A force may be regarded as deep or shallow depending on whether its aspect ratio (vertical to horizontal, α_F say) is greater or less than f/N , and it turns out that deep force tends to give rise to an acceleration of the zonal wind (a non-zero $\partial \bar{u}/\partial t$) whereas a shallow force tends to give rise to a meridional circulation. To see this we will consider the simplified forms of the momentum equation

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \mathcal{G}, \quad (17.42)$$

along with the MOC equation, (17.38c) which, if f_0 and N are both constant, is a Poisson equation with a right-hand side equal to $\partial \mathcal{G}/\partial z$. Solutions can be obtained by Fourier series methods with terms having the form

$$\tilde{\psi} = \Psi \cosh k_z z \sin k_y y. \quad (17.43)$$

Let us suppose the forcing is also of this form, so that by a deep forcing we mean that $k_y/k_z \gg f_0/N$ and shallow means $k_y/k_z \ll f_0/N$. We'll also suppose that $A = \mathcal{O}(1)$. From (17.38c) we see that

$$(f_0^2 k_z^2 - N^2 k_y^2) \tilde{\psi} \sim f_0 k_z G, \quad (17.44)$$

where G is the forcing amplitude. From this we can informally infer the form of the solution for deep and shallow forcing, as follows.

- *Deep forcing:* The dominant balance in (17.44) is between the second term on the left-hand side and the right-hand side and therefore $\tilde{\psi} \sim f_0 k_z G / (N^2 k_y^2)$ and

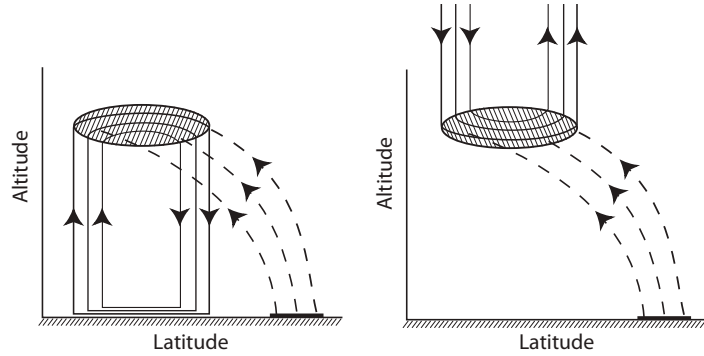
$$\tilde{w} \sim \frac{f_0 k_z G}{N^2 k_y}, \quad \tilde{v} \sim \frac{f_0 k_z^2 G}{N^2 k_y^2}. \quad (17.45)$$

Now look at the momentum equation (17.42). The ratio of the Coriolis force to the forcing on the right-hand side is given by

$$\frac{|f_0 v|}{|G|} \sim \frac{f_0^2 k_z^2}{N^2 k_y^2} \ll 1, \quad (17.46)$$

where the inequality follows by definition of what is deep. The dominant balance in the momentum equation must then be between the wave forcing and the acceleration.

Fig. 17.11 Downward control. Left panel: wave activity propagates upwards (dashed lines) from the troposphere, breaking and depositing zonal momentum in the shaded region. This induces an overturning circulation (solid lines) connecting the wave-breaking region with a bottom frictional boundary layer. Right panel: putative ‘upward control’, requiring friction above the wave breaking region for a steady response.



- *Shallow forcing*: The dominant balance in (17.44) is between the first term on the left-hand side and the right-hand side and therefore $\tilde{\psi} \sim G/(f_0 k_z)$ and

$$\tilde{w} \sim \frac{k_y G}{f_0 k_z}, \quad \tilde{v} \sim \frac{G}{f_0}. \quad (17.47)$$

The ratio of the Coriolis term to the forcing term in the momentum equation is now $\mathcal{O}(1)$, and therefore the response to a shallow forcing appears in the meridional circulation rather than as an acceleration.

The underlying reason for the two different responses arises from the need to satisfy the thermal wind equation. A shallow force means that a shear is generated, and hence via thermal wind the temperature must change, but this can only be accomplished if there is an RMOC to affect temperature. If the force is deep then the response can and will be in the form of an acceleration. At the equator all forces are deep because f is so small.

(v) *Deep and shallow heating*

A similar analysis may be applied to a heating field, but given the intuition we have just developed about the response to a mechanical forcing there is no need to go through the mathematical details (although these are straightforward). A shallow heating (perhaps more usefully thought of as a broad heating) can and will produce a direct response in the temperature field itself. However, if the heating is deep (or, more usefully, latitudinally confined) then any direct response in the temperature field would have to be associated with a response in the zonal wind. Instead of this, a latitudinally confined heating will tend to produce a response in the meridional circulation.

(vi) *The low-latitude limit*

At low-latitudes most forces become deep because f is small. More precisely, the criterion for deepness, that $H_f/L_f \gg N/f$ where H_f and L_f are the vertical and horizontal scales of the forcing, becomes easier to satisfy. Thus, wavebreaking at low-latitudes is more likely to induce an acceleration than a similar wavebreaking in midlatitudes, which will tend to induce an overturning circulation. By the same token, a heating source at low-latitudes has a greater tendency to induce an overturning circulation than a similar heat source in midlatitudes. The above statements are rules of thumb and not necessarily quantitative.

17.5.4 The steady-state limit and downward control

Let us now consider in a little more detail the steady-state response in which $\omega/\mu \rightarrow 0$, and we also take $\gamma = 0$, so that there is no momentum forcing. Although (17.38) of course still holds, it is also

useful to look directly at the momentum equation and thermodynamic equations. The momentum equation, (17.37a) reduces to a balance between the Coriolis force and the wave driving, namely

$$-f\bar{v}^* = \mathcal{G}, \quad (17.48)$$

and the thermodynamic equation becomes

$$\bar{w}^* \mathcal{S} = Q_s + Q_l = \mu(T_R - \bar{T}). \quad (17.49)$$

The thermodynamic equation gives us very little information about the vertical velocity because the right-hand side contains the unknown temperature, $\bar{T}$; rather, we can glean nearly all the information we want from (17.48).

If we differentiate (17.48) with respect to y (or ϑ) and use the mass continuity equation we obtain

$$\frac{1}{\rho_0} \frac{\partial \rho_0 w}{\partial z} = \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} \left(\frac{\mathcal{G} \cos \vartheta}{f} \right). \quad (17.50)$$

This is a first-order partial differential equation for the vertical velocity, and we can obtain the vertical velocity itself by a vertical integration, using a single boundary condition. If we require that vertical velocity stays finite at $z \rightarrow \infty$, and that $\rho_0 w = 0$ then we obtain

$$\bar{w}^* = \frac{1}{a \rho_0(z) \cos \vartheta} \frac{\partial}{\partial \vartheta} \int_z^\infty \left(\frac{\rho_0(z') \mathcal{G}(\vartheta, z') \cos \vartheta}{f} \right) dz'. \quad (17.51a)$$

The Cartesian quasi-geostrophic version of this equation is just

$$\bar{w}^*(z) = -\frac{1}{\rho_R} \frac{\partial}{\partial y} \int_z^\infty \rho_R(z') \frac{\mathcal{G}(\vartheta, z')}{f_0} dz'. \quad (17.51b)$$

Equation (17.51) implies that, in the steady-state limit, the vertical velocity at a given height is determined by the wave forcing *above* that height. The physical situation is illustrated in the left panel of Fig. 17.11. Here, a wave source in the troposphere propagates upwards and breaks in the middle atmosphere depositing momentum. This induces a meridional circulation as illustrated in the left panel, with a response *below* the momentum source. The numerically-computed response to an imposed force is illustrated in Fig. 17.12, showing how the response changes depending on the time-scales of the forcing and damping, with the steady-state response illustrated in the bottom panel.

The above derivation may seem a little disingenuous, for surely we might just as well have assumed $w = 0$ at $z = 0$, leading (in the quasi-geostrophic case) to

$$\bar{w}^*(z) = \frac{1}{\rho_R} \frac{\partial}{\partial y} \int_0^z \rho_R \frac{\mathcal{G} + \mathcal{D}}{f_0} dz'. \quad (17.52)$$

This might appear to give ‘upward control’, as illustrated in the right panel of Fig. 17.11. However, if we are integrating from the ground up then frictional effects are important near the surface and must be included, as represented by the frictional term $\mathcal{D}$ in (17.52). Furthermore, mass conservation demands that

$$\int_0^\infty \rho_R \bar{v}^* dz = 0, \quad \text{implying} \quad \int_0^\infty \rho_R (\mathcal{G} + \mathcal{D}) dz = 0, \quad (17.53a,b)$$

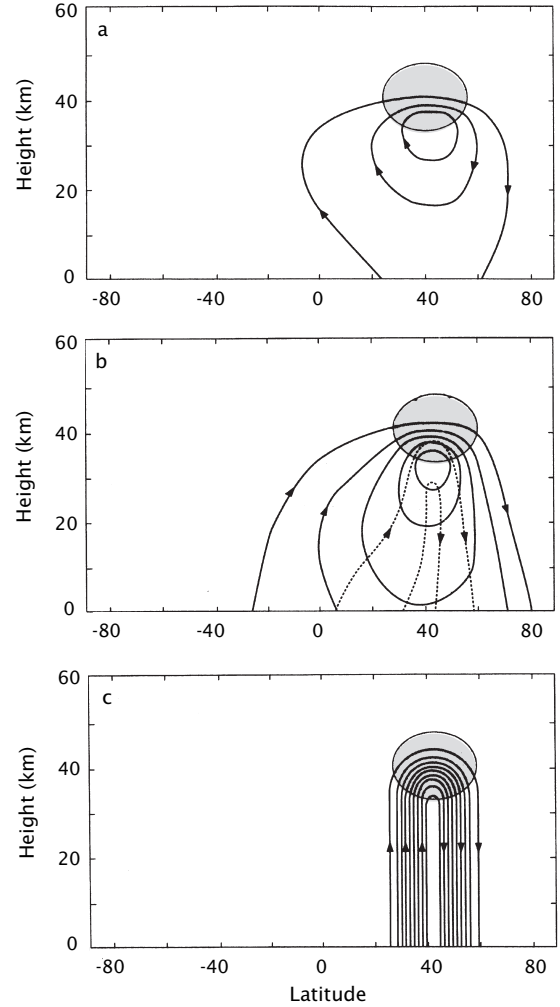
using (17.37a) for steady state conditions. Thus, above the level of the momentum source $\mathcal{G}$, (17.52) also in fact implies that the vertical velocity is zero, because $\mathcal{G}$ and $\mathcal{D}$ have cancelling effects. Thus,

Fig. 17.12 The numerically-computed response of the meridional overturning circulation to a longitudinally symmetric westward force, with the frequency of the forcing decreasing from top to bottom.¹² Contours are streamlines of the residual circulation, with the same uniform interval in all panels, and the shading denotes the forcing region.

(a) Response to high-frequency forcing, $\omega/\mu \gg 1$, $\omega \gg \gamma$. The response is adiabatic and weakly spreads into the opposite hemisphere.

(b) A lower frequency case with $\omega/\mu = 0.34$, corresponding to an annual cycle and a 20-day thermal relaxation timescale. The solid and dashed lines show the response that is in phase and out of phase with the forcing, respectively.

(c) Steady state response, $\omega/\mu \ll 1$. The circulation increases in magnitude and narrows as the frequency decreases, and in panel (c) it is given using the downward control expression (17.51a).



the location of the frictional boundary layer where the momentum is removed distinguishes up from down. Equation (17.53b) tells us that the frictional boundary layer at the bottom must adjust itself to remove the same amount of momentum that is deposited by wave breaking higher up, if there is to be a steady state. If there were a momentum sink above the momentum deposition region there would be no justification for downward control, for we would have to include that frictional term in (17.51). However, it is hard to envision how such a sink could exist without violating angular momentum conservation. If there were no frictional sink at the ground the disturbance would initially propagate down, but on reaching the ground would propagate up again.

From the point of view of the diagnostic equation for the meridional overturning circulation, in the steady state limit we have $A = 0$, and the solar forcing on the right-hand side and the y -derivative on the left-hand side of (17.38) both vanish, and the equation for the RMOC becomes

$$\frac{\partial}{\partial z} \left[\frac{1}{\rho_0} \frac{\partial(\rho_0 \tilde{w})}{\partial z} \right] = \frac{1}{2\Omega a \cos \vartheta} \frac{\partial}{\partial \vartheta} \left[\frac{\cos \vartheta}{\sin \vartheta} \frac{\partial \tilde{G}}{\partial z} \right] \quad (17.54a)$$

or, in the quasi-geostrophic limit,

$$f_0 \frac{\partial^2 \tilde{\psi}}{\partial z^2} = \frac{\partial \tilde{G}}{\partial z}. \quad (17.54b)$$

That is to say, in a steady state the solar forcing provides no input to the meridional circulation! This may seem a little counter-intuitive, but if there is no wave forcing and $\mathcal{G} = 0$ then the vertical velocity is zero and the temperature adjusts to the radiative equilibrium temperature, so that the diabatic forcing is zero.

The temperature field

Given that the in the steady state vertical velocity field is determined by the wave forcing, the temperature field can be determined diagnostically from the thermodynamic equation. Thus, using (17.37c) with no time-dependence and a vertical velocity given by (17.51), we obtain in the quasi-geostrophic case

$$\mu(\bar{T} - T_r) = \frac{\mathcal{S}}{\rho_R} \frac{\partial}{\partial y} \int_z^\infty \rho_R \frac{\mathcal{G}(\vartheta, z')}{f_0} dz', \quad (17.55)$$

with a similar but more complicated expression in the full case. The temperature field at a given height is determined purely by the momentum forcing, being given by the meridional gradient of the zonal force *above* that height.

An oceanic comparison

It is instructive to compare downward control with the Stommel problem in oceanography (section 19.1.1). In TEM (residual) form, the approximate zonally averaged zonal momentum equation may be written as [cf. (17.32a)]

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \mathcal{F} + \mathcal{D}, \quad (17.56)$$

The steady version of (17.56) and the equation for the streamfunction for the *horizontal* flow, ψ , in the ocean, (19.6), may thus respectively be written

$$f_0 \frac{\partial \psi^*}{\partial z} = \mathcal{F} + \mathcal{D}, \quad \beta \frac{\partial \psi}{\partial x} = \mathcal{F}_w + r \nabla^2 \psi, \quad (17.57a,b)$$

where $\mathcal{F}_w$ represents the wind forcing at the ocean surface, and the second term on the right-hand side of (17.57b) represents friction. In (17.57a), $\bar{v}^* = -\partial \psi^* / \partial z$ and in (17.57b), $v = \partial \psi / \partial x$. In the ocean interior, the frictional term is negligible, and in solving the resulting first-order equation ($\beta \partial \psi / \partial x = \mathcal{F}_w$) we may apply the boundary condition of $\psi = 0$ only at one meridional boundary. The natural choice is to choose the eastern boundary for this, and then invoke frictional processes to bring ψ to zero on the west. It is a natural choice because Rossby waves propagate westwards ('westward control', as in Fig. 19.14); thus, the boundary current (e.g., the Gulf Stream) is on the west *because* the wind's influence is carried westwards by Rossby waves, not vice versa. Westward control is an enormously robust effect that pervades almost every aspect of large-scale physical oceanography. In the atmospheric case there is no similar mechanism that demands that the influence of the momentum source be propagated downwards. Rather, in the steady state downward control results *because* the frictional boundary layer is at the bottom, not vice versa.

As we mentioned, the mechanism of downward control is related to that which gives rise to the Ferrel Cell in the troposphere, and that is certainly a strong and robust effect. Whether the downward control effect following wavebreaking *in the stratosphere* is strong enough to influence circulation in the troposphere, or the structure of the tropopause, remains an open question.

17.6 THE QUASI-BIENNIAL OSCILLATION

17.6.1 A brief review of the observations

The *quasi-biennial oscillation*, or QBO, is a quasi-periodic reversal of the zonal wind in the equatorial stratosphere, as illustrated in Fig. 17.13 and Fig. 17.14. It is the most dominant variability of that region, and the following lists some of the main features of the phenomenon.¹⁵

Parameter	Background	Rossby-gravity waves	Kelvin waves
Static stability, N	$2.2 \times 10^{-2} \text{ s}^{-1}$		
Beta at equator, β	$2.3 \times 10^{-11} \text{ m}^{-1} \text{ s}^{-1}$		
Coriolis parameter, f , at 5°	$1.27 \times 10^{-5} \text{ s}^{-1}$		
Wave period		4–5 days	10–20 days
Zonal wavelength		10,000 km	20,000–40,000 km
Zonal wavenumber		4	1–2
dimensionally		$6.3 \times 10^{-7} \text{ m}^{-1}$	$1.6\text{--}3.2 \times 10^{-7} \text{ m}^{-1}$
Meridional scale		1,200 km	1,500 km
Vertical wavelength		4–8 km	6–10 km
Phase speed,		20–25 m s^{-1}	25 m s^{-1}
relative to ground		(westward)	(eastward)
Amplitudes:			
zonal velocity		2–3 m s^{-1}	4–8 m s^{-1}
meridional velocity		2–3 m s^{-1}	0
vertical velocity		1–2 mm s^{-1}	1–2 mm s^{-1}
temperature		1 K	2–3 K
geopotential height		4 m	30 m
F_0 , wave forcing at		$3\text{--}6 \times 10^{-3} \text{ m}^2 \text{ s}^{-2}$	$4\text{--}10 \times 10^{-3} \text{ m}^2 \text{ s}^{-2}$
17 km, see (17.59).			
Thermal damping rate, α		$0.5\text{--}1.5 \times 10^{-6} \text{ s}^{-1}$	$0.5\text{--}1.5 \times 10^{-6} \text{ s}^{-1}$

Table 17.1 Typical, approximate, values of parameters appropriate for waves and background flow in the equatorial lower stratosphere.¹³

- The zonal winds in the equatorial region between about 5 and 100 hPa (about 40 and 18 km) alternate between being eastward and westward with an average period of about 28 months, with the period varying between 22 and 34 months.
- The QBO is almost latitudinally symmetric about the equator. The amplitude is approximately Gaussian with a half width of about 12° .
- The phenomenon is approximately zonal symmetric; that is, the longitudinal variation is small.
- The maximum amplitude of the oscillation is about $30 \text{ m s}^{-1} (\pm 15 \text{ m s}^{-1})$ at about 20 hPa on the equator. The westward winds are slightly stronger than the eastward winds, after removing the annual cycle.
- The wind pattern descends at about 1 km per month with little loss of amplitude until it reaches 100 hPa, and the cycle begins again. (This does not mean that information propagates downward, as we discuss later.)
- The QBO is mildly synchronized to the annual cycle, with transitions between eastward and westward flow having a tendency to occur more commonly in March–June than in the other months.

Although we will not discuss it here, another oscillatory phenomena occurs above the QBO known as the semi-annual oscillation, or SAO. The SAO is an oscillation in the zonal wind with an approximate period of six months (it should perhaps be called the quasi semi-annual oscillation) and it occurs between 1 hPa and 0.1 hPa and extends from about 30° N to 30° S (Fig. 17.14).

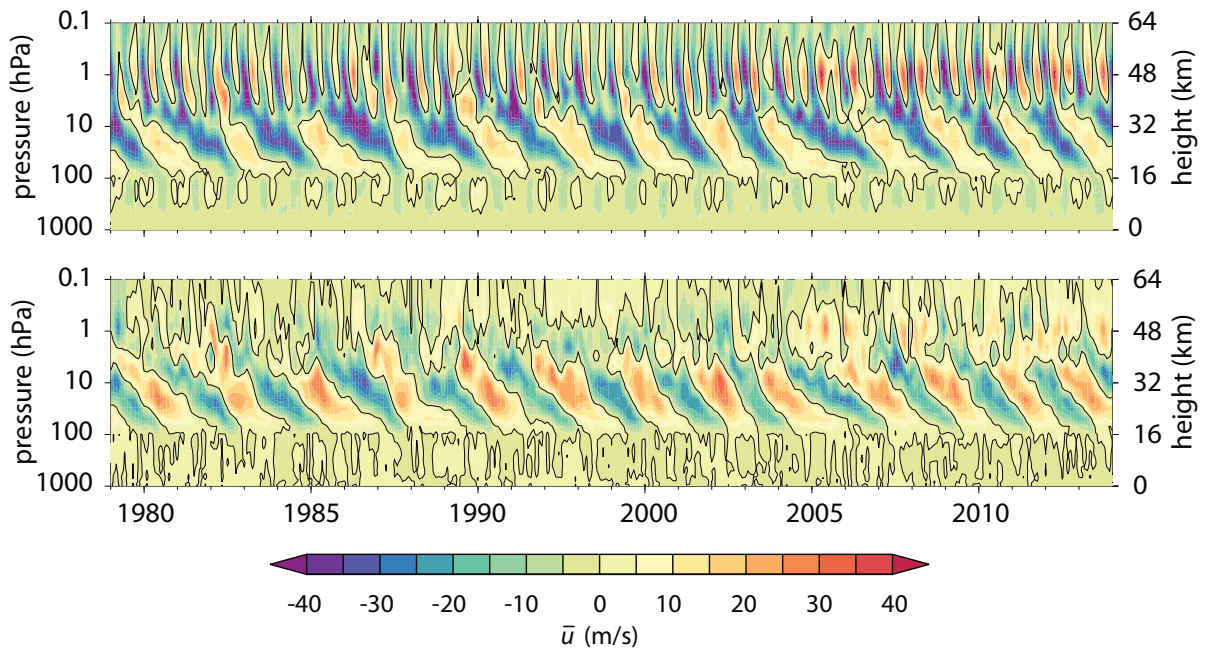


Fig. 17.13 Time height sections of the observed zonal-mean zonal wind for 1980–2014, averaged from 5° S from 5° N. The contour is the zero wind line, and in the bottom panel the seasonal cycle is removed. An oscillation, the ‘quasi-biennial oscillation’ or QBO, is quite visible between 20 and 40 km, or about 60 and 3 hPa.¹⁴

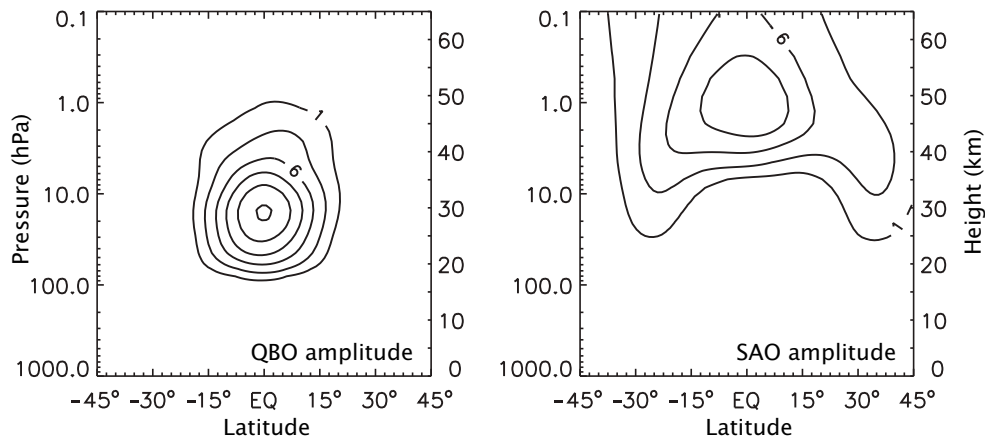


Fig. 17.14 The amplitudes (the root-mean square of the zonally-averaged wind of the oscillations after temporal filtering, with contours at 1, 3, 6, 9, 12 and 15 m s^{-1}) of the QBO and the SAO. The QBO evidently extends roughly from 15° S to 15° N and from 100 hPa to 1 hPa (about 17 km to 50 km). The SAO is broader, higher, weaker and faster.¹⁶

17.6.2 A qualitative discussion of mechanisms

Candidate mechanisms

We first note that the QBO *must* involve zonally asymmetric motions. Without such asymmetric or eddying motions there can be no maximum of angular momentum within the fluid interior, and therefore no eastward winds at the equator, as explained in sections 13.5.1 and 14.2.8. Given this, let us ponder for a moment what *might* be the mechanism of the QBO. One might suppose that horizontally propagating planetary waves would be a likely mechanism, for the transport of momentum by Rossby waves is known to be an important mechanism for the maintenance of jets in mid-latitudes. However, the descent of the wind pattern with no loss of amplitude cannot be easily explained by such a mechanism.¹⁷ Other proposed mechanisms have involved interactions with the annual cycle or its harmonics (natural enough given the period of the QBO) or have invoked external forcing or some nonlinear feedback. However, no candidate mechanism was able to explain all the features noted above, until a mechanism involving the vertical propagation and absorption of gravity waves was proposed, as we now describe.

A gravity wave mechanism

The mechanism for the QBO that is now generally accepted involves the upward propagation and absorption of gravity waves and their effect on zonal flow.¹⁸ We first describe the basic mechanism rather roughly and qualitatively.

A broad spectrum of gravity waves, with phase speeds in both eastward and westward directions, is generated in the upper equatorial troposphere by deep convection and various other instabilities. The waves will in general have a component of the group velocity that is directed upward, and if these waves are dissipated via mechanical or thermal damping then they will force mean flow accelerations (steady, non-dissipative waves cannot force a mean flow acceleration). A critical level, where the phase speed of the waves equals the speed of the mean flow (i.e., where $c = \bar{u}$) is one place where wave absorption and mean-flow acceleration will be particularly effective, because as a wave approaches a critical level it slows, giving more time for dissipation to act. However, it is not necessary for there to be an actual critical level; indeed, waves approaching a critical level will often be largely dissipated before reaching it.

Let us suppose that initially there is a westward shear (that is $\partial\bar{u}/\partial z > 0$) and that there are upward propagating gravity waves with positive phase speed c . These will be very efficiently absorbed as they approach the critical level, depositing momentum and causing the mean flow to accelerate. As pictured in Fig. 17.15, this causes the critical level to descend and hence the subsequent absorption of gravity waves and acceleration of the zonal flow will be at a lower level. The wind anomaly thus descends, and so on. A similar effect will still occur even if there is no critical level, provided there is enough dissipation for the gravity wave to be absorbed somewhere, causing the mean flow will accelerate. Even if this acceleration insufficient to induce a critical level, the difference between the wave speed and the zonal fluid speed will be reduced (i.e., $c - \bar{u}$ diminishes) and gravity wave absorption is enhanced. Gravity waves are thus absorbed at a lower level than previously and the anomaly in the zonal wind descends, as before.

Eventually, in the above models, the wind anomaly descends to the level of the gravity wave source. Depending on the strength of the dissipation one might imagine that dissipative processes could then wipe out the wind anomaly completely and the whole process would start all over again, or perhaps a low level westward wind anomaly would persist, so redefining the mean flow. However, in either case the zonal wind anomaly would not change sign (i.e., become westward), as is observed in the real QBO. For that to occur we may invoke a second wave in conjunction with an instability as we now explain.

Essentials of the QBO

What is the Quasi-Biennial Oscillation?

- The QBO is a quasi-periodic reversal of the zonal-mean zonal winds between about 20 km and 45 km and -15° S to 15° N, with a period of about 28 months. It is the dominant pattern of variability in the equatorial stratosphere and it is the clearest manifestation of a non-directly forced nearly periodic phenomenon in the atmosphere.
- The eastward and westward zonal winds appear to propagate downward at about 1 km per month, reversing at the end of each half cycle.
- The half-amplitude of the zonal wind cycle is about 15 m s^{-1} , with the westward winds being slightly stronger.

What is the mechanism?

- The oscillation is caused by the upward propagation and absorption of Kelvin waves and Rossby-gravity waves at the equator. If a wave has an eastward phase speed then, on absorption, it will cause the mean zonal flow to accelerate eastwards. Furthermore, the absorption is strongest near a critical layer, where the mean zonal wind speed equals the phase speed. An upward propagating Kelvin wave thus causes the mean flow aloft to accelerate eastwards, and then the maximum in eastward winds to move downwards and eventually to be dissipated. An upward propagating Rossby wave then generate a westward zonal wind anomaly aloft, which similarly propagates down. In this way the zonal wind oscillates between positive and negative values, as illustrated in Fig. 17.16.
- The waves are generally considered to be primarily excited by moist convection in the upper tropical troposphere.
- The period is determined by a combination of parameters involving the wave and mean flow. In the simplest model of two upwardly propagating gravity waves the period is given by

$$P = \frac{Akc^3}{\alpha N_0 F_0} \quad (\text{QBO.1})$$

where A is a nondimensional number weakly dependent on viscosity, and the other parameters, properties of the waves and mean flow, are defined in the text. The period is not proportional to the period of the waves; rather it is inversely proportional to their strength, F_0 , because stronger waves cause more mean flow acceleration and a faster descent of the pattern.

Why is the phenomenon equatorially confined?

- The mechanism requires there to be upwardly propagating waves with very different phase speeds in order that the mean flow can oscillate between the two values. In equatorial regions such a forcing can be provided by Rossby waves and Kelvin waves.
- In midlatitudes the tropospheric flow is largely balanced and it is primarily long Rossby waves that reach the stratosphere with a spectrum of phase speeds. If and when they break they would provide only a westward acceleration. Furthermore, in mid- and high latitudes an imposed force tends to induce a mean meridional circulation, not a mean flow acceleration (section 17.5).

A two-wave model

Suppose now there are two upward propagating gravity waves with speeds $+c$ and $-c$ (where c itself is positive), each of which will slowly be dissipated as it propagates, with the dissipation enhanced for smaller values of $|\bar{u} - c|$ or $|\bar{u} + c|$, respectively. (There will of course be very large dissipation if there is a true critical level.) Suppose that the mean flow has no shear, then simply by symmetry that state can persist, with the eastward and westward waves being dissipated equally as they ascend with no zonal flow generation. However, that symmetric state is unstable; to see this suppose that there is a small eastward perturbation to the zonal wind, as illustrated in the left panel Fig. 17.15. The eastward propagating wave will then be preferentially dissipated, because $\bar{u} - c$ is smaller for it than for the westward wave. The eastward anomaly in the zonal wind will therefore grow and descend, just as described above. The upward propagation of the eastward wave is then limited, but the westward wave is unconstrained at so it reaches higher levels before eventually being absorbed, providing a westward acceleration to the zonal flow, as illustrated in the right panel of Fig. 17.15.

As the the westward anomaly descends it squeezes the eastward anomaly which becomes thinner and thinner. Dissipative processes then become more efficient and can erode the eastward anomaly completely, with the flow becomes entirely westward, as illustrated in panel (b) of Fig. 17.16. A high level eastward anomaly is then created (panel (c) of Fig. 17.16), descending and squeezing the westward anomaly, and a mirror image of the first stage takes place. The entire cycle repeats itself and an oscillation is born, with the period of the oscillation being determined by the strength of the gravity waves and the rate of dissipation: stronger gravity waves lead to a faster acceleration of the mean flow and so a greater rate of descent and so a shorter period. Finally, note that the waves need not have speeds symmetric on either side of zero, $+c$ and $-c$. Suppose, for example, the wave speeds were both positive, a and b say. The mean flow could accelerate to an average value of $(a + b)/2$, with the flow then oscillating between a and b in a fashion similar to the symmetric case.

17.6.3 A quantitative model of the QBO

We now consider the above wave–mean-flow interaction model a little more quantitatively, and our first goal will be to obtain equations of motion for the interaction. To this end we will parameterize the vertical propagation and absorption of gravity waves by simple expressions resulting from gravity wave theory described in section 17.2.3. The absorption leads to a zonal flow acceleration, which in turn affects the wave absorption, and so on.

Let us consider a semi-infinite (no top), non-rotating, stratified fluid subject to a standing wave forcing at the lower boundary. Specifically, the waves are of the form

$$w = \text{Re } \bar{w}_1(z) e^{ik(x-ct)} + \bar{w}_2(z) e^{ik(x+ct)}. \quad (17.58)$$

The waves have a dispersion relation as discussed in section 17.2.3, a positive (upward) group velocity, and we will take $\bar{w}_1 = \bar{w}_2$. If there is a source of gravity waves such as convection there is no difficulty in exciting waves with either an eastward or westward phase speed: a Kelvin wave has a purely eastward phase speed ($c_p > 0$), a Rossby-gravity wave has a westward phase speed, and gravity waves completely uninfluenced by rotation can have a phase speed in either direction. The Kelvin and Rossby-gravity waves, probably the most important waves for the QBO, typically have zonal wavenumbers 1–4, and so zonal wavelengths greater than 10,000 km, and periods of 3 days or longer.

As the waves propagate up they are dissipated, primarily by thermal rather than viscous dissipation, and their amplitude diminishes in the vertical and consequently they deposit momentum into the mean flow. From the WKB calculation of section 17.3.3 the wave momentum flux, $F_k(z) = \bar{u}' w'$, of a given upwards-propagating wave is of the form

$$\bar{F}_k(z) = \bar{F}_k(0) \exp \left[- \int_0^z g_k(z') dz' \right] \quad (17.59)$$

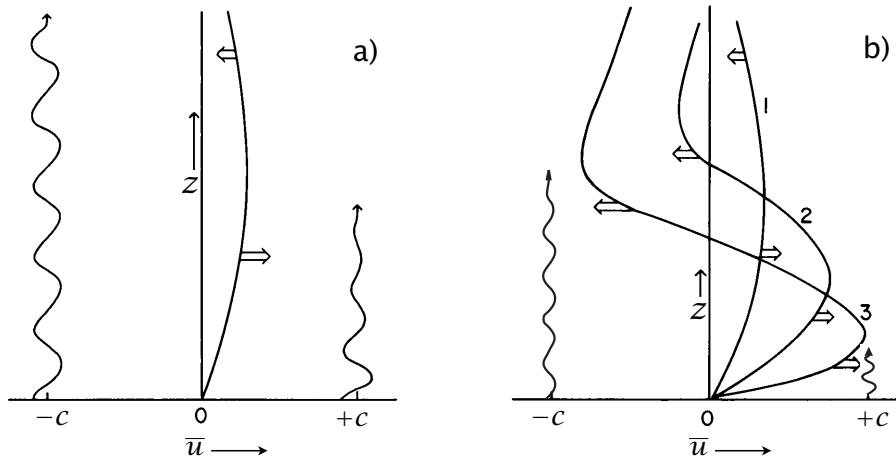


Fig. 17.15 Schema of the initial instability leading to the QBO.¹⁹ The solid lines show the zonal flow, the wavy arrows indicate the gravity-wave penetration from below and the double arrows indicate wave-induced acceleration. Initially, as in the left panel, a small eastward perturbation is added to a stationary mean flow. The eastward moving wave is preferentially absorbed and the perturbation is amplified and then descends, with the right panel showing the zonal flow at the successive times indicated, with the gravity wave penetration illustrated at $t = 3$. After the flow develops an eastward component the westward wave penetrates higher before being absorbed, inducing a westward flow aloft that then itself descends, making the eastward anomaly thinner. Subsequent stages and the development of a periodic oscillation are illustrated in Fig. 17.16.

where the subscript k indicates the zonal wave number and the attenuation rate, $g_k(z)$, for a given upward-propagating internal wave is given by

$$g_k(z) = \frac{\text{damping rate}}{\text{vertical group velocity}} = \frac{\alpha}{k(\bar{u} - c)^2/N}. \quad (17.60)$$

where c is the phase speed of the waves. The mean flow, $\bar{u}(z, t)$, is influenced by many such waves and so evolves according to

$$\frac{\partial \bar{u}}{\partial t} = -\sum_k \frac{\partial \bar{F}_k}{\partial z} + \nu \frac{\partial^2 \bar{u}}{\partial z^2}. \quad (17.61)$$

In writing (17.61) we include a dissipative term but neglect terms representing advection by the mean flow (such as $w\partial\bar{u}/\partial z$) and the Coriolis force $f v$. Equation (17.61), with (17.59) and (17.60), is a closed partial differential equation in a single unknown for the mean flow. If the forcing consists of two waves, one with a phase speed c that is positive and one with a negative phase speed, the model produces behaviour that is quite similar to that of the QBO, as we see shortly.

Direction of influence

Although both the observations and the schematic solutions illustrated in Fig. 17.16 suggest that influence is somehow propagating downwards, this is in fact not the case when the gravity waves are propagating upward. From (17.59) and (17.61) the wave-driven acceleration of the mean flow, A_w say, is given by

$$A_w = -\frac{\partial \bar{F}}{\partial z} = +\bar{F}(0)g(z) \exp \left[\int_0^z g(z') dz' \right] = +g(z)\bar{F}(z). \quad (17.62)$$

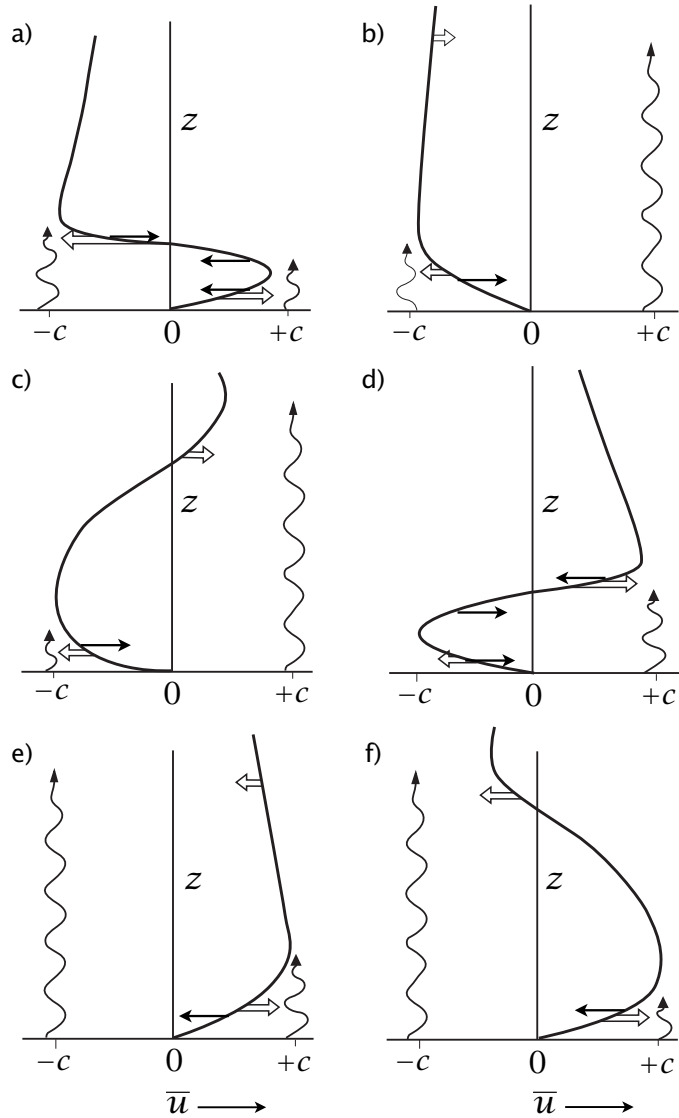


Fig. 17.16 Schema of the evolution of the QBO with gravity-wave forcing from below, following an initial perturbation illustrated in Fig. 17.15.²⁰ The solid lines show the mean flow and the wavy lines indicate the propagation of gravity waves. Horizontal double arrows indicate wave forcing and single arrows indicate viscous relaxation.

The panels are at successive times, with the top four panels showing a half cycle, and panels (d), (e) and (f) are mirror images of (a), (b) and (c). Wave-induced acceleration of the mean flow occurs preferentially near critical levels where $\bar{u} = c$.

That is, the acceleration is a function only of the profile of g in the region from 0 to z , that is the region through which the wave has propagated. Furthermore, attenuation rate $g(z)$ is itself, from (17.60), a function only of the local value of $\bar{u}(z)$ and not of the derivatives of $\bar{u}$. Thus, the wave forcing at some level z is a function only of the profile of $\bar{u}(z')$ for $z' < z$ and independent of the profile at higher altitudes. In other words, and in so far as the diffusivity term in (17.61) is negligible, there is no downward propagation of influence of the mean flow and the mean flow evolution is independent of what takes place above. The physical origin of this result is simply that waves are propagating upward and are absorbed by the mean profile as they ascend. If there were a *source* of waves at very high altitude, or if waves were reflected within the fluid (in which case the first-order WKB approximation is incomplete) then there could be a downward propagation of influence.

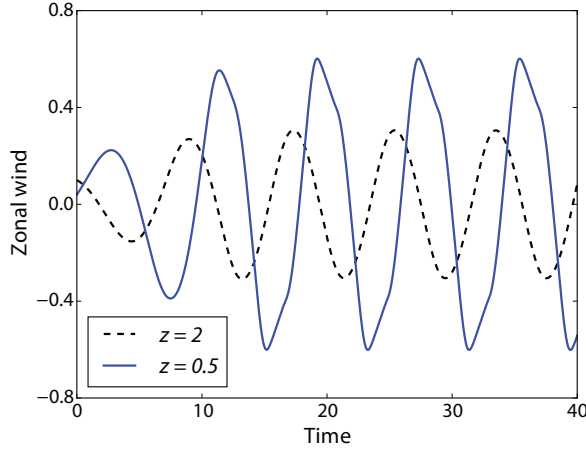


Fig. 17.17 The evolution of the mean zonal wind at two different levels in a numerical solution of (17.67). All the variables are nondimensional, with the only parameter in the problem being viscosity, and here $\hat{\nu} = 0.15$. A small perturbation is added to $\hat{u}$, and relatively quickly the solution becomes periodic.

17.6.4 Scaling and numerical solutions

Scaling the equations — nondimensionalizing in an intelligent way — not only makes numerical integration easier but also indicates what the natural height and time scales are for the problem. Important external parameters that determine the problem are the stratification N (which has units of inverse time, T^{-1}), the damping rate α (also units of inverse time) and the strength of the wave forcing, $\bar{F}$ (units of $(L/T)^2$). A natural horizontal scale is the inverse of the wavenumber k .

Denoting nondimensional quantities with a hat, let

$$\hat{F} = \bar{F}/F_0, \quad \hat{N} = N/N_0, \quad (17.63)$$

where $F_0 = \bar{F}(0)$ and N_0 is a typical value of N . If N were uniform then we would simply choose $N_0 = N$ whence $\hat{N} = 1$. To obtain sensible nondimensional quantities we note that the attenuation rate, g , has dimensions of inverse height so using (17.60) we choose a scaling height H as

$$H = \frac{kc^2}{\alpha N_0}. \quad (17.64)$$

This might suggest a time scaling of $T = kc/\alpha N$. However, because we have a forced problem, the form of (17.61) indicates that we choose

$$T = \frac{cH}{F_0} = \frac{kc^3}{\alpha N_0 F_0} \quad (17.65)$$

with a velocity scaling of $U = F_0 T/H = c$ (note that this is not an advective scaling). The non-dimensional coefficient of viscosity and thermal damping coefficients are then

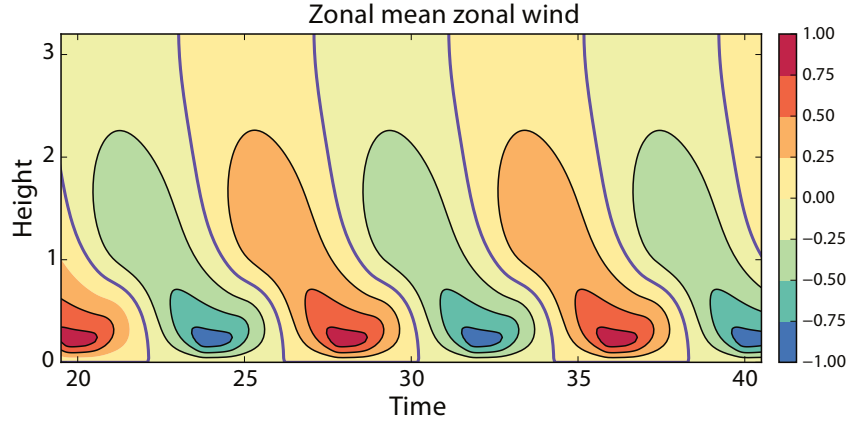
$$\hat{\nu} = \nu \frac{T}{H^2} = \nu \frac{\alpha N_0}{F_0 kc}, \quad \hat{\alpha} = \alpha T = \frac{kc^3}{N_0 F_0}. \quad (17.66)$$

The nondimensional equation for the mean flow evolution is then, for a single wave,

$$\frac{\partial \hat{u}}{\partial \hat{t}} = -\frac{\partial \hat{F}}{\partial \hat{z}} + \hat{\nu} \frac{\partial^2 \hat{u}}{\partial \hat{z}^2}, \quad \text{where} \quad \hat{F}(z) = \exp \left[-\int_0^{\hat{z}} \frac{1}{(\hat{u} - 1)^2} d\hat{z}' \right] \quad (17.67a,b)$$

and the hats indicate nondimensional quantities. The great simplification that (17.67) offers over (17.59)–(17.61) is that in (17.67) there are no parameters, save for the viscosity, and so the time and vertical scales of the problem are laid bare. In particular, if viscosity is small the only significant

Fig. 17.18 Time height section of nondimensional zonal wind in a numerical solution of (17.67), showing the last 20 time units of the same integration as Fig. 17.17. The zero contour is thicker.



timescale in the system is (17.65) and the period of the oscillation must be proportional to that, and the vertical scale of the oscillation must be given by (17.64). Evidently the period of the oscillation is inversely proportion to the strength of the waves, but the vertical extent and the amplitude of the oscillation are both independent of the wave strength.

A numerical solution

Equation (17.67) may readily be numerically integrated²¹ and solutions are illustrated in figures 17.17, 17.18 and 17.19. The simulations show many of the qualitative features of the observed QBO, including the decay of the pattern with height and its apparent downward propagation. The simulations are dependent on the viscosity $\hat{\nu}$ in that, if $\hat{\nu} = 0$, the jet at the bottom of the domain cannot be dissipated and the system in fact evolves to a steady state. On the other hand, if the viscosity is large then the jets become too broad, and the boundary layer near $z = 0$ that is evident in Fig. 17.18 is thicker. Still, if the viscosity is small but nonzero then over the bulk of the cycle it plays little role and the oscillation period is only weakly dependent on its value. Thus, for example, in Fig. 17.16 viscosity is needed to wipe out the low level westward jet between panels (d) and (e), but has little role in the rest of the half cycle and so only a small effect on the period. If viscosity is unimportant then the only timescale in the problem is that given by (17.65) and the period is proportional to it, and from numerical integrations we find that it is given by

$$P = A \frac{kc^3}{\alpha N_0 F_0} \quad (17.68)$$

where $A \approx 8$. Note that the period of the QBO is not directly dependent on the period of oscillation of the waves themselves, but is dependent on their strength.

17.6.5 The role of Rossby and Kelvin waves

The waves that propagate into the stratosphere in equatorial regions are of two main types, Kelvin waves and Rossby waves. Kelvin waves are a form of gravity wave but have only an eastward phase propagation, whereas Rossby waves are balanced waves with a westward phase propagation. The theoretical development paralleling section 17.6.3 is naturally more complex, in part because the problem is now, in principle, a three-dimensional one. However, it is much simplified if we consider motions at the equator and if we take note that, in general, the attenuation rate of a wave is equal to its damping rate divided by its group velocity, as in (17.60). The corresponding attenuation rates for Kelvin and Rossby waves are then given by

$$\text{Kelvin wave:} \quad g_K(z) = \frac{\alpha}{k_K(\bar{u} - c_K)^2/N}, \quad (17.69a)$$

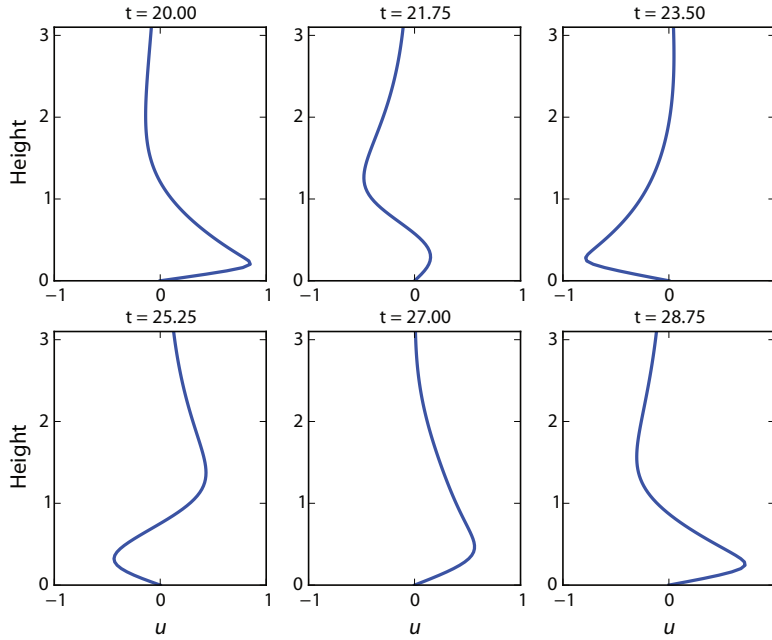


Fig. 17.19 Profiles of zonal wind at the times indicated from the numerical solution of (17.67), shown in Fig. 17.18.

Rossby wave:

$$g_R(z) = \frac{\alpha}{k_R(\bar{u} - c_R)^2/N} \left(\frac{\beta}{k_R^2(\bar{u} - c_R)} - 1 \right). \quad (17.69b)$$

The Kelvin wave attenuation rate is just the same as that for a non-rotating gravity wave, although the wave speed, c_K , is strictly positive. The Rossby wave attenuation rate (whose derivation requires a little work) involves the equatorial beta parameter and a negative phase speed, c_R . The full problem is now defined by (17.61) and (17.59), now with $g(z)$ given by (17.69). It is evident that the problem is no longer east-west symmetric, but the essential structure of the problem remains. Rossby wave absorption is enhanced near a critical layer where $\bar{u} = c_R$, and Kelvin wave absorption still occurs near $\bar{u} = c_K$, so we expect to see an oscillation contained between these two values. Further, just as in the gravity wave problem, influence propagates upward with the waves.

The equation set (17.61), (17.59) and (17.69) may be numerically integrated and solutions are illustrated in Fig. 17.20. It is the factor g_R that is responsible for the westward acceleration of the mean flow, for ‘dragging’ $\bar{u}$ toward the value c_R , which is negative. But g_R is zero when $\bar{u} - c_R = \beta/k_R^2$ which, for our numerical simulation, occurs when $\bar{u} = 1$. Thus, when $\bar{u}$ is close to its eastward (Kelvin-wave induced) peak the westward acceleration is small. Another source of east-west asymmetry is the likelihood that Rossby waves and Kelvin waves may have different amplitudes. If Kelvin waves were stronger, for example, then the eastward acceleration would be stronger than the westward and that part of the cycle would be faster.

17.6.6 General discussion

The above sections have described a couple of relatively simple models that seem to capture the essence of the QBO. The model using both Rossby waves and Kelvin waves is not noticeably more realistic in its predictions; rather, it is attractive because Rossby waves and Kelvin waves are observed in the equatorial stratosphere and may be more realistic in its assumptions. The observed east–west asymmetry in the observed QBO is not obviously caused by the differences between Rossby waves and Kelvin waves; other possibilities include the effects of a mean circulation and possible differences in the strength of the eastward and westward forcing.

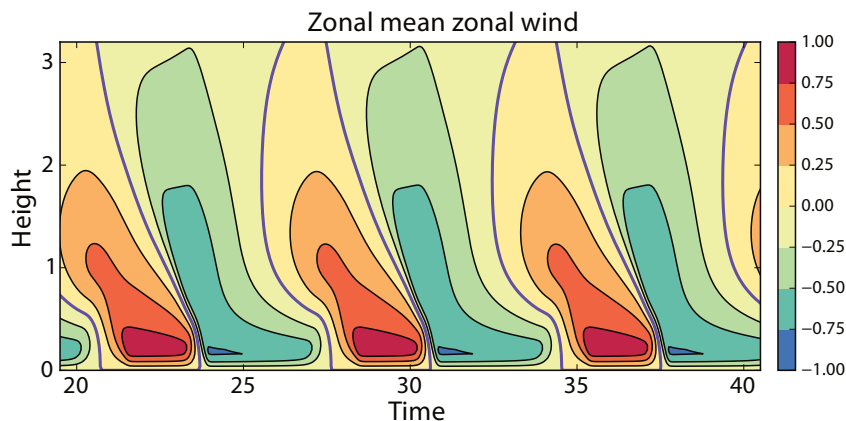


Fig. 17.20 Time height section of nondimensional zonal wind in a numerical solution using (17.69), with both Rossby and Kelvin waves. The zero contour is thicker.

The model with two non-rotating gravity waves is attractive because it allows a more complete analysis of its properties. In particular, the upward propagation of waves leading to a downward propagation of the zonal wind pattern, and the factors determining the period of the oscillation, are made transparent. The period of the problem is given by (17.68). Using table 17.1 as a guide, let us take the following dimensional values of the parameters: $k = 2 \times 10^{-7} \text{ m}^{-1}$, $\alpha = 1 \times 10^{-6} \text{ s}^{-1}$, $c = 25 \text{ m s}^{-1}$, $N_0 = 2.2 \times 10^{-2} \text{ s}^{-1}$, $F_0 = \times 10^{-2} \text{ m}^2 \text{ s}^{-2}$. We obtain a timescale of $T = (kc^3/\alpha N_0 F_0) \approx 160$ days or about 5 months and so, using (17.68), a period of 40 months. Obviously there is considerable uncertainty in the parameters chosen and it would not be difficult to choose a set of parameters giving the observed value of about 26 months — or for that matter, to choose a set that gave a still longer period.

The vertical scale of the oscillation is given by (17.64) and with the above set of parameters we obtain $H = (kc^2/\alpha N_0) \approx 6 \text{ km}$. From the numerical simulations we see that the vertical penetration of the phenomena is 2 or 3 times this, so about 15 km. This value again is reasonably close to the observed value of, from Fig. 17.13, about 20 km, but again we should be wary of too close an agreement, especially using a one-dimensional quasi-Boussinesq model. It is interesting that the period and vertical extent of the observed oscillation vary only a little, implying only a little interannual variability in the forcing strength and other parameters of the problem.

The actual waves themselves are primarily generated by convection in the tropical troposphere, then propagating up into the stratosphere. It is difficult to numerically simulate a QBO with an explicit representation of gravity waves because of the large range of scales involved in the problem, although three-dimensional simulations with parameterized gravity waves have been quite successful. Still, a striking demonstration of the mechanism above came from laboratory experiments, using an annulus of stratified water subject to a standing wave forced by a flexible lower boundary. Given a strong enough forcing an oscillating mean flow was generated whose structure was found to be in very good agreement with the two-wave theory. Thus, at the very least, the mechanism does describe a real physical phenomenon.²²

There are a great many aspects of the QBO that we have not discussed, including its latitudinal structure, the effects of a mean circulation, and three dimensional numerical simulations, and a few references that may serve as an introduction to these topics are given here.²³ The QBO is also not completely regular, as we see even from Fig. 17.13, and a somewhat unusual example occurred in early in 2016. The QBO normally shows a fairly steady downward propagation of the westerly phase, but in January 2016 the westerly winds in the lower stratosphere switched back to easterlies after only about 6 months (Fig. 17.21) the shortest period of westerlies above 20 km in a record going back to 1953. This behaviour may have been caused by anomalous horizontal propagation of Eliassen–Palm fluxes from midlatitudes because of the absence of a subtropical critical line, although we cannot be definitive. Finally, to make a personal remark, the QBO is both a curiosity

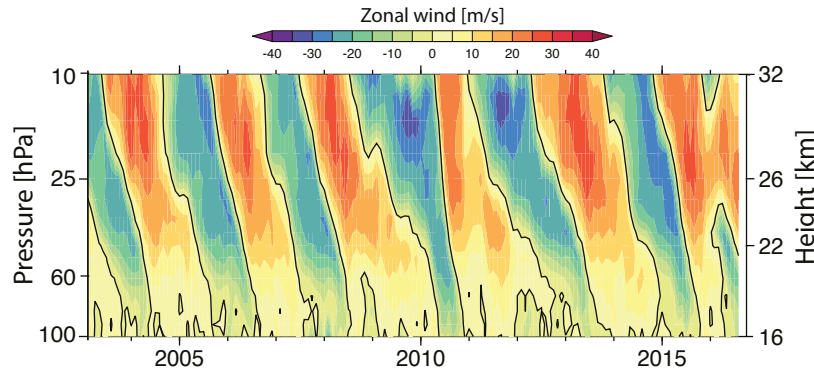


Fig. 17.21 Equatorial zonal wind, similar to Fig. 17.13, but now with cycles of the QBO from 2003 to 2016 and showing the interruption of the eastward winds below about 25 km in early 2016.²⁴

and a triumph. The former because its relationship to and influence on tropospheric circulation, and the climate and weather that affect humankind, is not obvious to the casual or even expert observer; it does not have the impact of an El Niño event or a cold winter, for example. Yet, excepting directly forced oscillations like the diurnal and seasonal cycle, it is the clearest example of a nearly periodic phenomenon in the atmosphere and its beautiful explanation must rank as a major achievement in geophysical fluid dynamics.

17.7 VARIABILITY AND EXTRA-TROPICAL WAVE–MEAN-FLOW INTERACTION

Compared to the troposphere the stratosphere has only mild baroclinic instability, both because of the lack of shear and because of the larger value of the deformation radius. The amplitude of a baroclinic mode reaching up from the troposphere thus tends to decay rapidly in the stratosphere, as in Fig. 9.21. If we take $N = 2.2 \times 10^{-2} \text{ s}^{-1}$, $H = 20 \text{ km}$ and $f = 10^{-4} \text{ s}^{-1}$ then $L_d = NH/f = 4,400 \text{ km}$, and given that for many problems in baroclinic instability the relevant scale is a few times L_d , sometimes there is simply no room for instability on a planet with a radius of 6,000 km. When there is an instability it will be at large scales, perhaps at wavenumbers 1, 2 or 3, as opposed to wavenumber 8 or so in the troposphere. None of this is to say there is no variability in the stratosphere, with the variability arising in two main ways.

- (i) From waves propagating up from the troposphere, with the stratospheric variability arising from the variability of the troposphere.
- (ii) From oscillatory or even chaotic flow arising from the interaction, within the stratosphere, of large-scale planetary waves with themselves and with the mean flow. The forcing may still come from the troposphere but, even when this is steady, intra-stratospheric interactions may give rise to unsteadiness.

In either case the variability tends to be relatively slow (compared to the troposphere) and at a large scale — the orographic forcing from the troposphere undergoes Charney–Drazin filtering and tends to occur at wavenumbers 1 and 2, and as noted any baroclinic instability is also at large scale. It is therefore useful to think of the variability as a wave–mean-flow problem rather than as a problem in geostrophic turbulence.

17.7.1 Upward propagating disturbances and sudden warmings

Consider planetary waves that are excited in the troposphere and propagate upward, as described in chapter 16, with this occurring predominantly in winter when the tropospheric forcing is strongest. The wave activity obeys

$$\frac{\partial \mathcal{A}}{\partial t} + \nabla \cdot \mathcal{F} = D \quad (17.70)$$

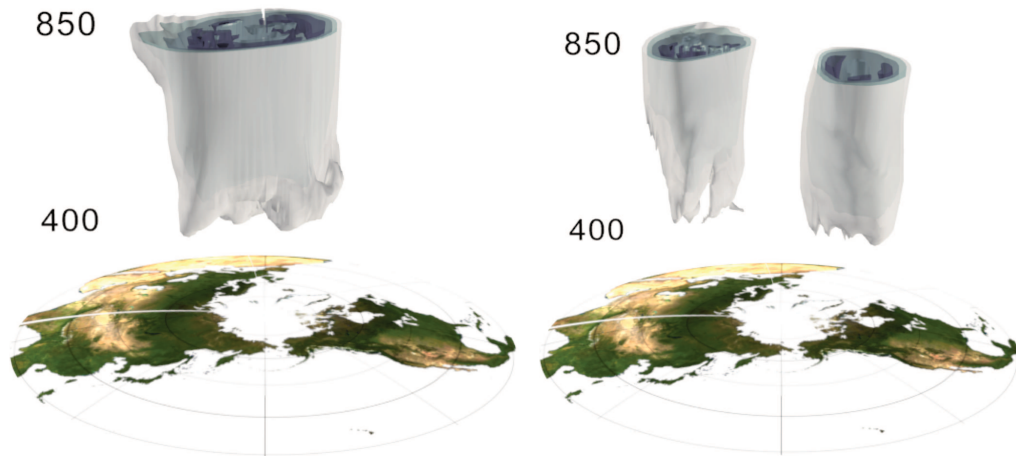


Fig. 17.22 The edge of the stratospheric polar vortex in December 1984. Plotted is the 35 PVU isosurface of $Q^* = Q(\theta/\theta_0)^{4.1}$, where Q is Ertel PV and $\theta_0 = 475\text{K}$. The vertical coordinate is potential temperature. Like Q , Q^* is materially conserved in adiabatic flow, and roughly compensates for the change in density with height that affects the Ertel PV. The left panel (14 December) shows the vortex in a fairly usual state, and the right panel (30 December) shows a split vortex following a stratospheric sudden warming.²⁵

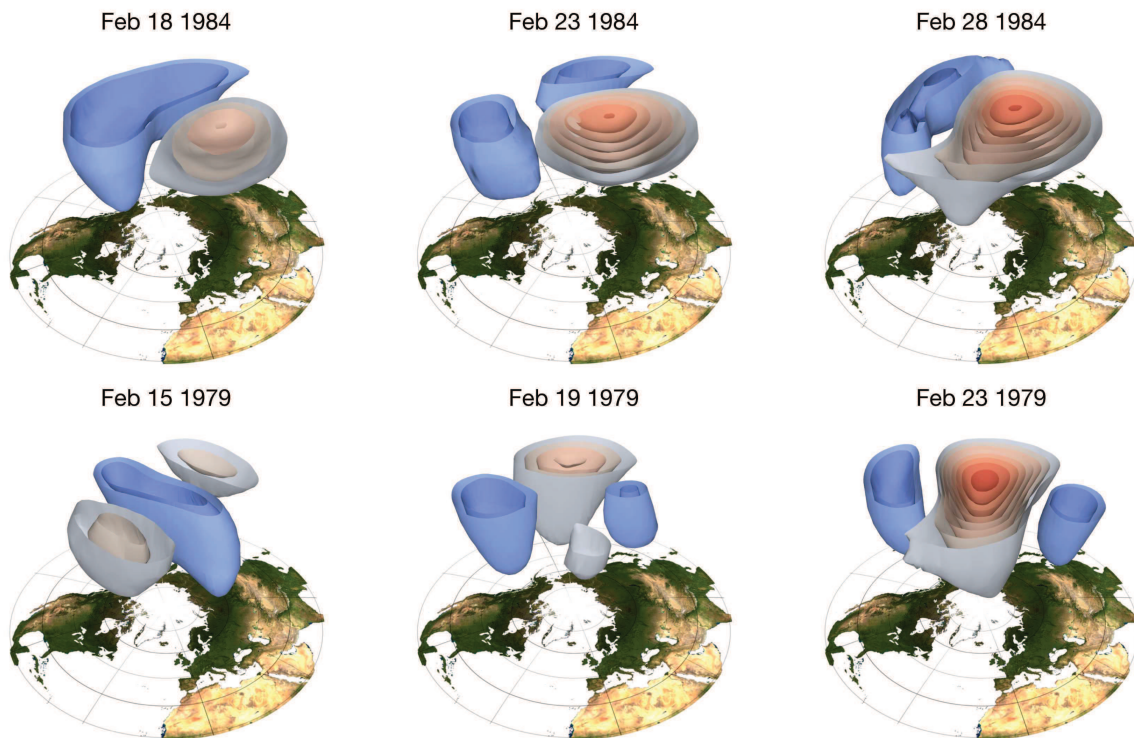


Fig. 17.23 Time sequence of two stratospheric warmings, with the top row showing a displacement of the initial (blue) polar vortex, and the bottom showing a split, with the dates marked. Contours are anomalous geopotential height larger (smaller) than 4km (-4km) in red (blue), between 200 and 10 hPa, spaced at 2km intervals. In both cases the initial polar vortex is cold (blue) with an anomalously warm end state (red).²⁶

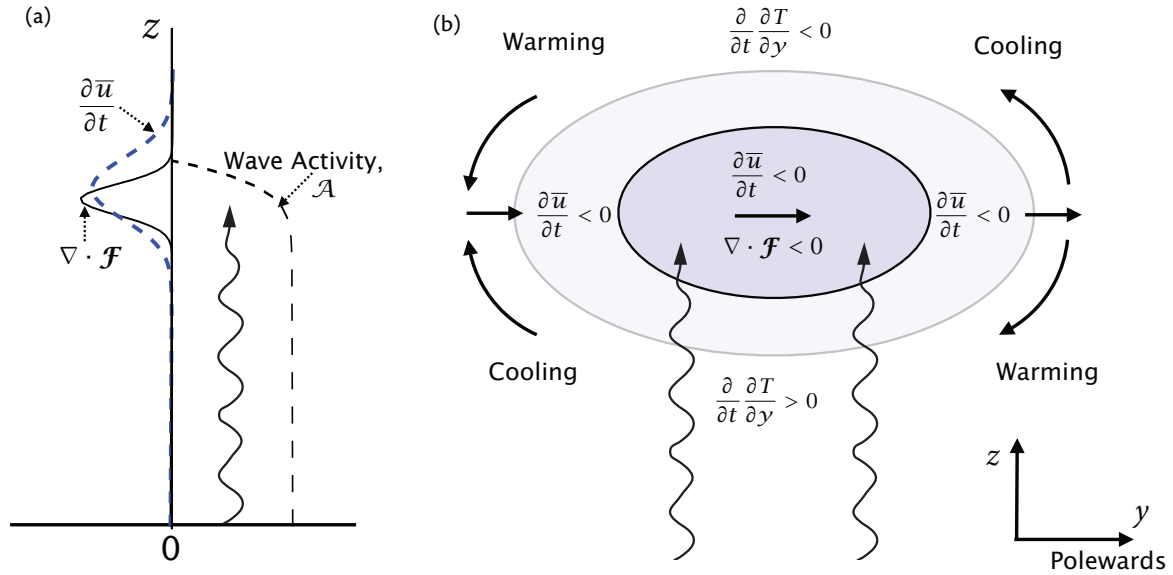


Fig. 17.24 The wave–mean-flow dynamics of a stratospheric warming. (a) Upward propagating Rossby waves (wavy lines) break in the stratosphere, and the wave activity (dashed line) diminishes. The EP flux divergence is negative ($\nabla \cdot \mathbf{F} < 0$), inducing a westward acceleration ($\partial \bar{u} / \partial t < 0$) over a broader region because of the ellipticity in (17.72). (b) The circulation and temperature response induced. The negative EP flux divergence is shaded dark, inducing westward acceleration over the broader region (light shading). Assuming there is no acceleration far away from the wavebreaking, the temperature response can be inferred from thermal wind, with a warming at the lower poleward end of the breaking and an induced residual circulation as shown by the arrows.

where $\mathcal{A} = \overline{q'^2} / 2\bar{q}_y$ and $\nabla \cdot \mathbf{F} = \overline{v'q'}$. Thus, if dissipation is small and $\partial \mathcal{A} / \partial t$ is positive, $\overline{v'q'}$ is negative. Now consider the zonal momentum equation in quasi-geostrophic TEM form, namely

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \overline{v'q'}. \quad (17.71)$$

Rossby waves propagating into the stratosphere or, by the same token, dissipating Rossby waves in a statistically steady state, thus induce a deceleration (i.e., a westward tendency) of the zonal mean flow and/or a poleward meridional flow, and only for a very deep forcing is the residual circulation response negligible. Also, the zonal-wind response to an wave forcing will tend to be of larger scale than the forcing itself, as can be seen from (10.87),

$$\left[\frac{\partial^2}{\partial y^2} + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \right] \frac{\partial \bar{u}}{\partial t} = \frac{\partial^2}{\partial y^2} \overline{v'q'}. \quad (17.72)$$

The elliptic nature of the operator acting on $\partial \bar{u} / \partial t$ produces a response on a large scale than the right-hand side.

Suppose, then, that Rossby waves propagate upwards from the troposphere and break in the stratosphere. The mean eastward flow (sometimes called the polar night jet) will be weakened, so allowing more waves to propagate up, since strong eastward flow inhibits propagation (Fig. 16.6). If the process continues the winds will eventually reverse, forming a critical layer (as described in section 16.6.1) where $\bar{u} = 0$. This completely inhibits further upward propagation and wave breaking is intensified, inducing a westward flow at the level to which the propagation reaches. There is a rapid changeover to westward flow and the critical layer descends. This sequence has an

obvious similarity with the westward acceleration phase of the QBO, but in the extra-tropics there is no eastward counterpart as there are no Kelvin wave, and thus not is no oscillation. Rather, the eastward winds of the polar night jet are gradually restored by radiative effects.

A reduced (or reversed) vertical shear is, by thermal wind, associated with a reduced (or reversed) meridional temperature gradient, so that the polar night jet is replaced by a warmer westward flow. Put simply, the deposition of westward wave momentum leads to a warming of the high-latitude stratosphere. Such an event can at times be strong enough to split asunder the cold polar vortex, as illustrated in Fig. 17.22 and Fig. 17.23, and when it does the event is known as a *sudden stratospheric warming*.²⁷

The interaction of the waves and mean flow is sketched in Fig. 17.24. In the left panel we see a wave propagating up from the troposphere and breaking, with wave activity then falling. The EP flux is negative in the breaking region causing a deceleration of the zonal flow over a somewhat broader region because of the elliptic operator in (17.72). The temperature response, shown in the right panel, can be inferred from thermal wind balance, noting that above the breaking region $\partial_t(\partial u/\partial z) > 0$ to that $\partial_t(\partial T/\partial y) < 0$, and oppositely for below, and the direction of the residual circulation follows by noting that adiabatic warming (cooling) results from descent. The residual circulation may also be inferred from (17.38b) or (17.38c).

Numerical simulations

To illustrate the above mechanism in a more realistic setting we shown some results from a primitive equation simulation that mimic the broad features the observations quite well. The advantage over showing the observations is that a great many events are simulated and full diagnostics can be obtained.²⁸ A primitive equation model with a resolved stratosphere and a representation of topography in the troposphere was integrated for many decades, during which time many sudden warmings occurred. Composites of these events were constructed, and these are shown in Fig. 17.25.

The main features are clear. An anomalous upward flux of wave activity (EP flux) that, on dissipating in the stratosphere, induces a westward acceleration of the zonal flow, a weakening of the polar vortex and a warming of the polar regions, extending equatorward as far as 60°. Vortex dynamics also play a role: a vortex, once formed, is rather stable and has a natural tendency to persist rather than break up. In two-dimensional turbulence vortices will merge, and not split. This property prevents warmings from occurring too frequently, since the wave activity must be strong enough to overcome the elastic properties of the vortex edge.

17.7.2 ♦ Wave–mean-flow interaction and stratospheric variability

Stratospheric variability need not arise solely from waves propagating up from the tropopause and we can illustrate this with a simple numerical model of wave–mean-flow interaction, similar to those discussed in section 10.1.3. Specifically the model consists of the following quasi-geostrophic ideal-gas equations.²⁹

The zonally-averaged fields obey

$$\frac{\partial \bar{q}}{\partial t} = \bar{F} - \frac{\partial}{\partial y} \overline{v'q'}, \quad (17.73a)$$

where

$$\bar{q}(y, z, t) - \beta y = \left[\frac{1}{\rho_R} \frac{\partial}{\partial z} \left(\rho_R \frac{f_0^2}{N^2} \frac{\partial \Psi}{\partial z} \right) + \frac{\partial^2 \Psi}{\partial y^2} \right], \quad \left(\bar{u}, \frac{R}{H f_0} \bar{T} \right) = \left(-\frac{\partial \Psi}{\partial y}, \frac{\partial \Psi}{\partial z} \right) \quad (17.73b,c)$$

The eddies obey

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = F', \quad (17.74a)$$

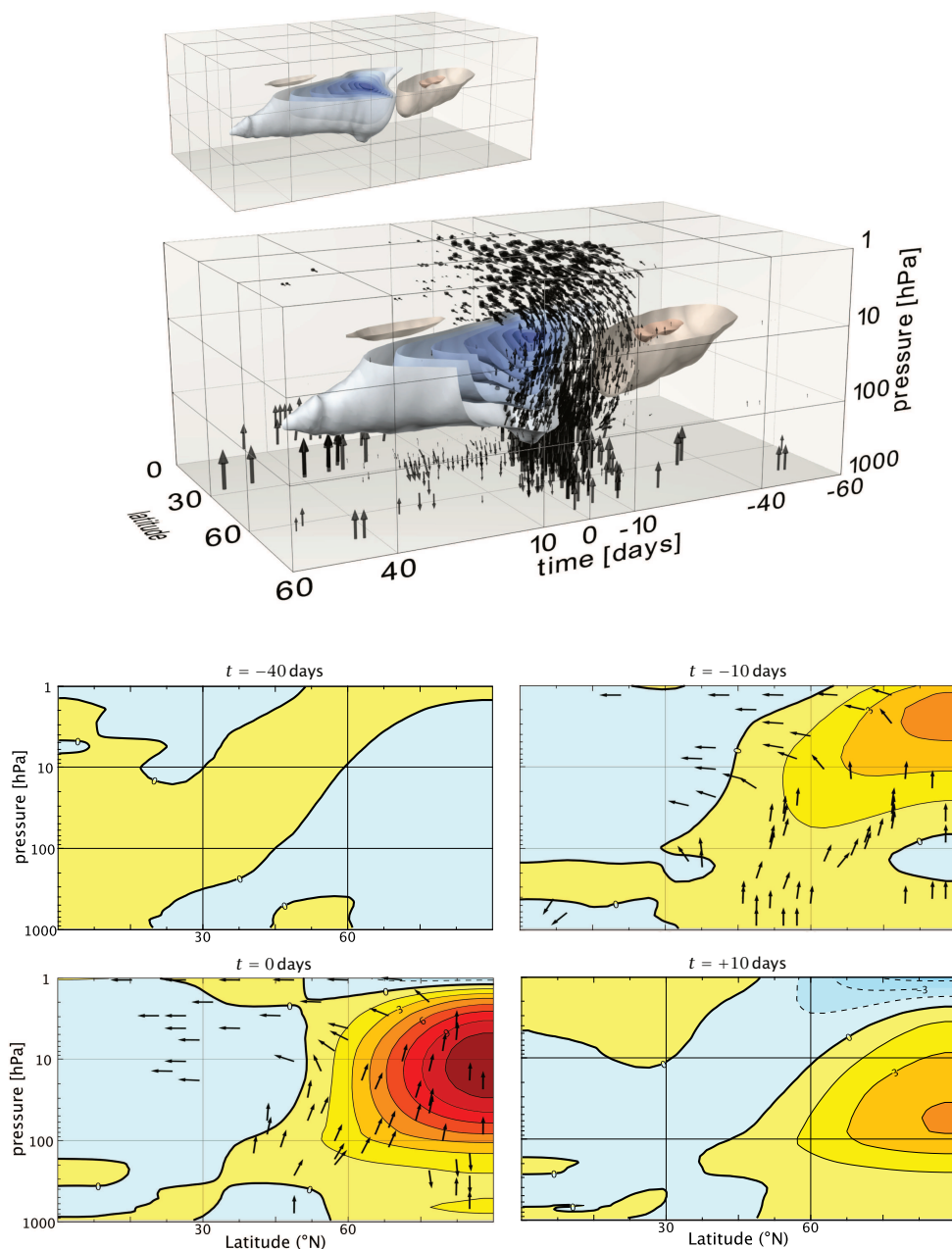


Fig. 17.25 The evolution of a stratospheric warming obtained with a general circulation model, compositing several events together. The top, three-dimensional plots show anomalous zonal mean zonal wind, with surface intervals of 2 m/s. The bottom plots show time slices of anomalous temperature in colour contours (interval 1.5 K, red is warm), with times relative to the peak warming. In both plots the black arrows show anomalous EP flux. Visible is a strengthening and northward propagation of the polar vortex prior to the onset, and a strong weakening during and after the onset. At forty days prior to the onset (upper left panel in lower set of plots), anomalies are very small. Ten days before onset (upper right), warming has appeared in the polar upper stratosphere, and anomalously strong EP fluxes appear throughout the atmosphere. At the onset (lower left), the warming and the EP flux anomalies are strongest. Ten days later (lower right), the temperature anomalies are weaker and confined to the lower stratosphere, and the EP fluxes (not shown) are very weak.³⁰

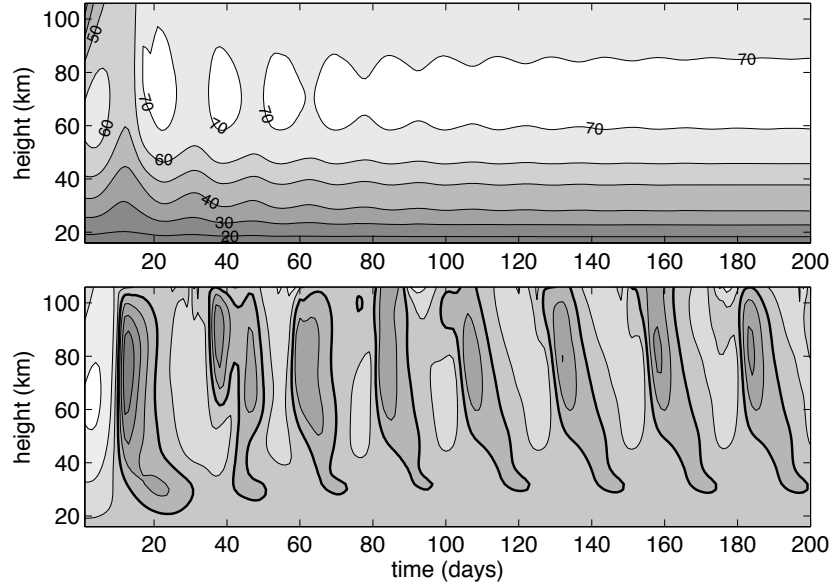


Fig. 17.26 Evolution of the zonal mean zonal wind in a case with steady wave forcing with a value of 200 in the top panel and 300 in the lower panel. In the bottom panel the contours are every 20 m s^{-1} , positive values have lighter shades and the zero contour is heavy.

where

$$q'(x, y, z, t) = \nabla^2 \psi' + \frac{1}{\rho_R} \frac{\partial}{\partial z} \left(\rho_R \frac{f_0^2}{N^2} \frac{\partial \psi'}{\partial x} \right), \quad (u', v') = \left(-\frac{\partial \psi'}{\partial y}, \frac{\partial \psi'}{\partial x} \right) \quad (17.74b)$$

and

$$\frac{\partial \bar{q}}{\partial y} = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} - \frac{1}{\rho_R} \frac{\partial}{\partial z} \left(\rho_R \frac{f_0^2}{N^2} \frac{\partial \bar{u}}{\partial z} \right). \quad (17.74c)$$

The notation follows our usual conventions, with $\rho_R(z)$ being a reference density profile and $\bar{F}$ and F' the forcing/dissipation terms for the mean flow and eddies, respectively. It is the domain, the boundary conditions and forcing that distinguish the model and make it representative of the stratosphere, as we now discuss.

The model domain is a channel nominally centred at 60° and of width 60° , extending upwards to about 100 km. The forcing on the zonal flow is a relaxation back to a specified radiative equilibrium temperature field (or equivalently a thermal wind field). In the results shown below this is independent of time and corresponds to a constant shear of 1 m s^{-1} per kilometre, or a temperature difference of about 15 K across the domain, with a relaxation timescale that varies from 20 days at 20 km to 4 days at 50 km. There is also a weak linear drag on the mean flow. The eddies are forced by imposing a constant perturbation at the lower boundary, with wavenumber 2 in the simulations shown. There is a radiative damping on the eddies ensuring that the eddies are mostly damped before reaching the top of the domain. The vertical variations are represented using finite differencing, whereas in the horizontal both the mean flow and the eddies are expanded in a Fourier series with only a very small number of terms retained. Thus, we write

$$[\bar{q}, \Psi] = [Q_0(z, t), \Psi_0(z, t)] \cos ly, \quad [q', \psi'] = \text{Re} [q_0(z, t), \psi_0(z, t)] \sin ly \exp(ikx). \quad (17.75)$$

and after some manipulation we can obtain evolution equations for Q_0 and q_0 with diagnostic equations for Ψ_0 and ψ_0 . The quadratic terms in the equations of motion create higher order terms that are projected back onto the retained terms. (Aside from the severe horizontal truncation the numerical method used to find results is not a key aspect of the model.)

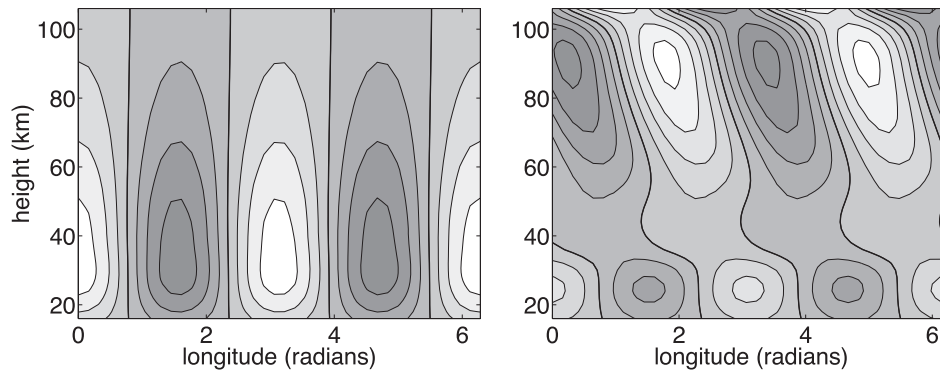


Fig. 17.27 Snapshots of the wave streamfunction in the steady case (forcing of 200) in the left panel and in the oscillatory case (forcing of 300) in the right panel. Zero contour is heavy.

Some numerical results and interpretation

Results of two numerical integrations are shown in Fig. 17.26 and Fig. 17.27. In one integration the geopotential forcing at the lower boundary has an amplitude of 200 m, whereas in the other it has an amplitude of 300 m. In the first case the flow evolves into an absolute steady state, whereas in the second the mean flow and the waves oscillate with a period of about 25 days, with the mean flow actually becoming negative over half the cycle. The streamfunction in the unsteady case is tilting into the mean shear (the right panel of Fig. 17.27), evocative of baroclinic instability. The oscillations are in some way redolent of stratospheric warmings. The climatological eastward winds transition rather quickly to westward winds (darker shading in Fig. 17.26), with the westward winds then descending and with a slower recovery back to climatology. There are two points to be made:

- (i) Interactions that are internal to the stratosphere can give rise to oscillatory motion.
- (ii) The source of energy for the waves may in part arise from a baroclinic instability and in part from tropospheric forcing.

To summarize, stratospheric warmings are not solely a response to tropospheric forcing; the internal dynamics of the stratosphere, and stratosphere-troposphere interactions, also play an important role.

Notes

- 1 To read more about the middle atmosphere see, for example, the review by Hamilton (1998), the collection of articles in *Journal of the Meteorological Society of Japan*, vol. 80, no. 4B, 2002, the book by Andrews *et al.* (1987) and the review by Haynes (2005). I am also grateful to Nili Harnik for many comments on this chapter.
- 2 Adapted from Jucker *et al.* (2013).
- 3 Upper figure courtesy of J. Wilson of GFDL, using data from Fleming *et al.* (1988). Lower figure uses data from the ECMWF ERA-interim reanalysis.
- 4 Courtesy of D. Waugh.
- 5 Courtesy of A. Dörnbrack.
- 6 Adapted from Eluszkiewicz *et al.* (1997).
- 7 Brewer (1949) and Dobson (1956). Brewer deduced upward motion into the stratosphere at low latitudes based on the water vapour distribution, while Dobson deduced a poleward transport within the stratosphere based on the ozone distribution — the circulation takes ozone from the low latitudes toward the poles. Although originally the Brewer–Dobson circulation was taken to

mean the chemical transport circulation, it is now usually taken to mean the residual (thickness or mass) circulation. The two may differ if there is mixing of chemical without mixing of mass, and the chemical transport may differ among chemicals.

- 8 See also Plumb (2002), which motivated this figure.
- 9 It turns out that $\overline{u'w'} > 0$ for upward propagating Rossby waves and thus, if the waves were to be dissipated, an eastward acceleration would seemingly be implied. In fact it is the form stress that is the most important aspect of vertical momentum transport in such waves, and when the waves are dissipated a westward acceleration ensues.
- 10 Blanca Ayarzagüena Porras kindly made this figure using an ERA-interim reanalysis.
- 11 Drawing from Garcia (1987) and Haynes (2005). I am also grateful for comments from Peter Haynes.
- 12 Adapted from Holton *et al.* (1995).
- 13 Values are taken from Wallace (1973), Plumb (1984) and Andrews *et al.* (1987).
- 14 I am grateful to Verena Schenzinger who kindly made this plot using winds from the ERA-interim reanalysis. See also Gray (2010).
- 15 A broad overview of QBO is provided by Baldwin *et al.* (2001), with updates and additions by Gray (2010), and a more theoretical review is given by Plumb (1984). The term quasi-biennial oscillation seems to have been coined by Angell & Korshover (1964), although the discovery of the QBO is generally credited to R. J. Reed and R. A. Ebdon, independently and at about the same time (Ebdon 1960, Veryard & Ebdon 1961, Reed 1960, Reed *et al.* 1961).
- 16 Figure adapted from Gray (2010), who used the method of Pascoe *et al.* (2005).
- 17 Wallace & Holton (1968), Baldwin *et al.* (2001).
- 18 The first theory of the QBO along these lines was put forward by Lindzen & Holton (1968) and Holton & Lindzen (1972), with clarifications and simplifications by Plumb (1977), and it is these models that we draw from. Prior this work, Booker & Bretherton (1967) had shown how the momentum deposition by gravity waves can be enhanced near critical lines and this was a key advance. A host of papers elaborating on the basic mechanism have since appeared, discussing such things as the particular type of gravity waves involved, the role of the Coriolis force and the meridional confinement of the QBO, the possible influence of the solar cycle and El Niño, the impact on tracer transport and so on.
- 19 Adapted from Plumb (1984).
- 20 Adapted from Plumb (1984).
- 21 Code is available to the reader.
- 22 Plumb & McEwan (1978). Regarding atmospheric relevance, and as with some other scientific theories of complex phenomena, it is hard to be absolutely certain that the theory is correct, for the sceptic can always point to observational disagreements or say that another theory might be the correct one. Yet the analogue of deciphering a complex communication seems apt: if an encrypted signal is deciphered to reveal a meaningful message, it seems perverse to ask whether the deciphering is unique, and whether some other message might have emerged from a different decryption.
- 23 Examples of additional theoretical development are Dunkerton (1982, 1997), Boyd (1976a) and Plumb & Bell (1982). Simulations of a QBO in an atmospheric GCM have been achieved by Takahashi (1996), Hamilton *et al.* (2001), Scaife *et al.* (2002), Giorgetta *et al.* (2002) and others. The possible effects of the QBO on the extra-tropical circulation are discussed by Holton & Tan (1980, 1982), Jones *et al.* (1998), Kushner (2010), Labitzke *et al.* (2006), Randel *et al.* (1999), Scott & Haynes (1998), Dunkerton *et al.* (1988) and others. The anomalous QBO of 2015–2016 is documented by Newman *et al.* (2016) and Osprey *et al.* (2016).
- 24 This plot was kindly made by Verena Schenzinger using data from the Singapore radiosonde. For a description of QBO datasets see <http://www.geo.fu-berlin.de/en/met/ag/strat/produkte/qbo/>.
- 25 Figure kindly prepared by M. Jucker using ERA-interim reanalysis. See Lait (1994) for a discussion

of the alternative PV.

- 26 Data from ECMWF ERA-Interim, visualization with the software 'pv-atmos', described in Jucker (2014).
- 27 The model described here was proposed by Matsuno (1971) and although nonlinear effects (and, to a lesser degree, non-geostrophic effects) play a quantitative role, Matsuno's model remains the foundation of our understanding. An early review is that of Schoeberl (1978) and there have been numerous studies since. To name but a few, Dunkerton *et al.* (1981) and Palmer (1981) explored the phenomenon from a TEM perspective, Limpasuvan *et al.* (2000) and Charlton & Polvani (2007) provide a comprehensive view of the observations of warmings using reanalysis datasets, Charlton *et al.* (2007) look at various simulations with GCMs, and Gray *et al.* (2001) look at external influences on the timing of warmings.
- 28 These simulations were kindly performed by Martin Jucker. See also Jucker *et al.* (2014).
- 29 This model was introduced by Holton & Mass (1976) and the numerical results we show use a code adapted from one by J. Holton. Plumb (1981), Yoden (1987, 1990) and others have explored the model further with a view to better understanding the parameters for which steady, oscillatory or chaotic motion was present. Christiansen (1999, 2000), Scaife & James (2000), Scott & Haynes (2000), Sjöberg & Birner (2012), Jucker *et al.* (2014) and others have explored related behaviour using various types of models and observations, for example with the primitive equations and/or including eddy-eddy interactions and with different boundary conditions.
- 30 Figure created using the software 'pv-atmos', described in Jucker (2014).

*I'm singing in the rain, Just singing in the rain,
What a glorious feelin', I'm happy again.
I'm laughing at clouds, So dark up above,
The sun's in my heart, And I'm ready for love.*
Lyrics by Arthur Freed, music by Nacio Herb Brown, 1929.

CHAPTER 18

Water Vapour and the Tropical Atmosphere

WATER IS AN ORDINARY SUBSTANCE WITH EXTRAORDINARY EFFECTS. The most obvious is that oceans themselves are made of water, and if our planet were dry this book would perforce be much shorter (if only). Leaving aside the dynamical effects of the oceans, water covers over two-thirds of Earth's surface and because it is warm in some places and cold in others, and because the atmosphere is in motion, water evaporates into the atmosphere in one place and condenses from it elsewhere. The condensation leads to rain, one of the most talked-about aspects of weather and climate. Water also freezes to form ice, so that at any given time water exists on Earth in all three phases. Radiatively, water vapour is a greenhouse gas, meaning that it absorbs infra-red radiation that might otherwise be lost to space and so maintains the surface of the planet at a temperature over 20 K higher than an equivalent dry planet. Dynamically, the condensation of water vapour in the atmosphere releases energy, warming the air and tending to make it more unstable than otherwise and leading to convection. Further, the net transport of water vapour from low to high latitudes is effectively a meridional transport of energy.

In this chapter we focus on a small number of these issues, mainly on the kinematics and dynamics of water vapour itself and on some aspects of the dynamics of the tropical atmosphere, where the effects of water vapour are most manifest. The tropics would certainly differ from the mid-latitudes even if the atmosphere were dry — its Coriolis parameter is small among other things — so our attention there is by no means confined to the effects of water vapour. Nevertheless, tropical convection and the attendant 'radiative-convective equilibrium' are greatly influenced by the presence of water. We begin with a discussion of the thermodynamic properties of water vapour itself. We then move onto an essentially kinematic description of the factors determining the large scale distribution of relative humidity, before finally looking at convection and at tropical dynamics more generally.¹

18.1 A MOIST IDEAL GAS

Water is the compound of hydrogen and oxygen with the chemical formula H_2O , although in informal conversation water is often understood to mean only the liquid form of the compound. *Water vapour* is a gas made up of molecules of H_2O , and *ice* is the solid form of water. *Steam*, in common

parlance, is a mixture of air, water vapour and suspended droplets of water, usually at a very high temperature. Steam is formed when water vapour at temperatures above boiling point cools under contact with air and some of the water vapour condenses, forming a fine mist. Cloud and fog are also mixtures of dry air, water vapour and water droplets, but need not be at high temperature. Our focus will be on water vapour which, as we will see, can exist over a wide range of temperatures; let us first say how we quantify it and how it affects the equation of state. A number of thermodynamic derivations are also given in Appendix A (page 720). Those derivations are more systematic but less pedagogical than those below, and may appeal to some.

18.1.1 Ideal gas equation of state

The thermal equation of state for an ideal gas is conventionally written in the form

$$pV = Nk_B T = nR^* T, \quad (18.1)$$

where N is the total number of molecules in the volume V , n is the number of moles in that volume, and $N = nN_A$ where N_A is Avogadro's number. A mole is the amount of a substance that contains the same number of elementary entities, usually atoms or molecules, as there are atoms in 12 grams of carbon-12, that number being Avogadro's number ($N_A \approx 6.02 \times 10^{23}$). Two moles of a substance contains two times Avogadro's number of elementary units. The constants in the above equation are Boltzmann's constant, k_B , and the universal gas constant, R^* , where $R^* \equiv N_A k_B = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$. As noted in chapter 1, for any particular gas it is convenient to define the specific gas constant by $R = R^*/\mu$ where μ is the molar mass (mean molecular weight in kg/mol). For a single component gas we then divide (18.1) by the total mass $M = n\mu$ to obtain

$$p = \rho RT. \quad (18.2)$$

Throughout this chapter we will be concerned only with 'simple ideal gases', or 'perfect gases', for which the gas constants at constant composition are, in fact, constant.

For a multi-component ideal gas the partial pressure of each component is independent of the presence of the other components (because the volume of the molecules is negligible) and so is equal to the hypothetical pressure of that gas if it alone occupied the volume of the mixture. The total pressure is therefore the sum of the partial pressures of each gas, a dictum known as Dalton's law of partial pressures. The partial pressure of each constituent in a mixture is proportional to the number of molecules of that constituent, and therefore also proportional to the number of moles. Because of Dalton's dictum we can obtain a simple expression for the equation of state of a mixture, as follows. Denoting the constituents by subscript i , the total pressure is given by

$$p = \sum_i p_i = \sum_i \frac{1}{V} n_i R^* T = \sum_i \left(\frac{M}{V} \right) \frac{n_i \mu_i}{M \mu_i} R^* T. \quad (18.3)$$

Let us define the effective molar mass, μ_e , by

$$\frac{1}{\mu_e} = \sum_i \frac{n_i \mu_i}{M \mu_i} = \sum_i \frac{\varphi_i}{\mu_i}, \quad (18.4)$$

where $\varphi_i = (n_i \mu_i)/M$ is the mass fraction of the i 'th constituent. We then have

$$p = \rho RT, \quad \text{where} \quad R = \frac{R^*}{\mu_e} = \sum_i \varphi_i R_i, \quad (18.5a,b)$$

and $R_i = R^*/\mu_i$. The effective gas constant of the mixture is thus the mass-weighted mean of the specific gas constants of its constituents. Any given gas has a specific constant that is inversely

proportional to its molecular weight. Thus, for a given fluid density and temperature a gas with a higher molecular weight will exert a lower pressure than one that smaller molecular weight, because it will have fewer molecules per unit mass. Similar expressions apply to the heat capacities c_p and c_v , so that a heavier gas (higher molecular weight) has a smaller specific heat capacity.

18.1.2 Application to moist air

Dry air has virtually constant composition and its mean molar mass is $\mu^d = 29.0 \times 10^{-3} \text{ kg mol}^{-1}$, giving $R^d = R^*/\mu^d = 287 \text{ J kg}^{-1} \text{ K}^{-1}$. Water vapour has a molar mass of $\mu^v = 18.014 \times 10^{-3} \text{ kg mol}^{-1}$ giving $R^v = 461.5 \text{ J kg}^{-1} \text{ K}^{-1}$. The two gas constants are related by

$$\frac{R^v}{R^d} = \frac{\mu^d}{\mu^v} \equiv \frac{1}{\epsilon} \approx 1.608 \quad (18.6)$$

Now consider a mixture of dry air and water vapour.

Measures of moisture

When mixtures are present we superscripts d , v and l to denote thermodynamic quantities associated with dry air, water vapour and liquid water. The *absolute humidity* is the amount of water vapour per unit volume, with units of kg m^{-3} , or informally g m^3 . The *mixing ratio*, w , is ratio of the mass of water vapour, m^v , to that of dry air, m^d , in some volume of air and is thus

$$w \equiv \frac{m^v}{m^d} = \frac{\rho^v}{\rho^d}. \quad (18.7)$$

It is a nondimensional measure but it is often expressed in terms of grams per kilogram. In the atmosphere values range from close to zero to about 20 g kg^{-1} (2×10^{-2}) in the tropics on a humid day.

The *specific humidity*, q , is the ratio of the mass of water vapour to the total mass of air — dry air plus water vapour — and so is

$$q \equiv \frac{m^v}{m^d + m^v} = \frac{w}{1 + w} \quad \text{and} \quad w = \frac{q}{1 - q}. \quad (18.8a,b)$$

The specific humidity is just the mass concentration of water vapour in air. In most circumstances in Earth's atmosphere $m^v \ll m^d$ so that $q \approx w$, usually to an accuracy of about one percent. In much of this chapter we will ignore the differences between w and q , but this is not appropriate for all planetary atmospheres.

The partial pressure of water vapour in air, e , is the pressure exerted by water molecules and is proportional to the number of moles of water vapour in the volume. It is given by

$$e = \frac{n^v}{n^d + n^v} P = \frac{m^v/\mu^v}{m^d/\mu^d + m^v/\mu^v} P. \quad (18.9)$$

where n^v and n^d are the number of moles of water vapour and dry air in the mixture and P is the total pressure. Using (18.7) we can write (18.9) as

$$e = \frac{wP}{w + \epsilon} \quad \text{or} \quad w = \frac{\epsilon e}{P - e}. \quad (18.10)$$

In terms of q instead of w these expressions are

$$e = \frac{qP}{q + \epsilon(1 - q)} \quad \text{and} \quad q = \frac{\epsilon e}{P - e(1 - \epsilon)}, \quad (18.11)$$

In Earth's atmosphere $w \ll 1$ so that

$$e \approx w \frac{p}{\epsilon} = 1.61wp \quad \text{and} \quad q \approx w \approx \epsilon \frac{e}{p}. \quad (18.12)$$

If the mixing ratio of water vapour is 10 g kg^{-1} (a typical tropical value) and $p = 1000 \text{ hPa}$ then $e \approx 16 \text{ hPa}$.

The *relative humidity*, $\mathcal{H}$, is the ratio of the actual vapour pressure to the saturation vapour pressure, e_s , which is the maximum vapour pressure that can occur at a given temperature before condensation occurs, as will be discussed in section 18.1.4. Thus, $\mathcal{H} = e/e_s \approx q/q_s$ where q_s is the specific humidity at saturation.

18.1.3 Equation of state and virtual temperature

Using (18.5b) the effective gas constant of moist air all varies with humidity according to

$$R = \frac{m^d R^d + m^v R^v}{m^d + m^v} = (1 - q)R^d + qR^v = R^d \left[1 + q \left(\frac{1}{\epsilon} - 1 \right) \right] \quad (18.13)$$

with similar expressions for c_p and c_v . In humid air, with $q = 0.02$ say, we have $R = R^d(1 + 0.02 \times 0.61) = 467.1 \text{ J kg}^{-1} \text{ K}^{-1}$.

The heat capacity of water vapour can be estimated from its molecular properties. Water vapour is triatomic molecule with three translational and three rotational degrees of freedom. If these were the only degrees of freedom then the internal energy would be given by $I = 6R^v T/2$, whence $c_v^v \approx 3R^v = 1384 \text{ J kg}^{-1} \text{ K}^{-1}$ and $c_p^v = R^v + c_v^v = 1846 \text{ J kg}^{-1} \text{ K}^{-1}$, where c_v^v and c_p^v are the specific heat capacities for water vapour at constant volume and pressure, respectively. In fact vibrational degrees of freedom can sometimes be excited and the measured values are a little higher, namely $c_v^v = 1397 \text{ J kg}^{-1} \text{ K}^{-1}$ and $c_p^v = 1859 \text{ J kg}^{-1} \text{ K}^{-1}$ (at 273 K , increasing very slightly with temperature). The heat capacity of moist air is thus slightly higher than that of dry air, but since values of q are small the difference is only about 1%.

The variation of gas constant with humidity can be inconvenient in numerical models. A workaround is to define a so-called virtual temperature, T_v , which is the temperature that dry would need to be in order to have the same density and pressure as moist air. That is, by definition,

$$p = \rho R T = \rho R^d T_v \quad (18.14)$$

where R is given by (18.13). Using (18.13) we obtain

$$p = \rho R^d T_v, \quad \text{where} \quad T_v = T \left[1 + q \left(\frac{1}{\epsilon} - 1 \right) \right] \approx T(1 + 0.61q). \quad (18.15)$$

The virtual temperature, T_v , increases with specific humidity and if $q = 20 \text{ g kg}^{-1}$ then T_v is about 12%, or 3 K, larger than the actual temperature. Such a temperature is often used in numerical models of the atmosphere because it enables various thermodynamic equations to keep their original form, with gas constants that actually are constant.

Because the concentration of water vapour in Earth's atmosphere is so small, the variations of the heat capacities are small and constant values are often used to calculate such quantities as the potential temperature and the adiabatic lapse rate. This is not always appropriate, and the appendix on page 720 indicates how, in principle, more accurate calculations could be made.

18.1.4 Saturation vapour pressure

Vapour pressure is the partial pressure of water vapour in the atmosphere. At any given temperature, there is a maximum value of that vapour pressure beyond which condensation normally

occurs and this is known as the *saturation vapour pressure*. Why should this be so, and why don't other atmospheric gases, such as oxygen or carbon dioxide, also condense? To understand this, we have to understand the thermodynamic equilibrium between a liquid and a gas.

Equilibration of the Gibbs function

Consider a system that consists of an enclosed, insulated container partially filled with liquid, and with vapour above it. The two subsystems can exchange mass and energy, with liquid potentially evaporating into vapour and vapour condensing into the liquid. Energy is required to evaporate the liquid into a vapour to overcome the molecular forces in the liquid and this is given by $M(h^v - h^l)$, where M is the mass that has evaporated, h^v is the specific enthalpy of the vapour and h^l is the specific enthalpy of the liquid. The enthalpy of vapourization, more commonly called the *latent heat of evaporation*, is defined by the difference between the two enthalpies at a temperature T , namely,

$$L(T) \equiv h^v - h^l, \quad (18.16)$$

with L having units of J kg^{-1} . It is a function of temperature, because the enthalpy of liquid water and water vapour are both functions of temperature, and a very weak function of pressure — see appendix A (page 720) for details. For water, L diminishes almost linearly by about 10% going from 0°C to 100°C , from 2.5×10^6 to $2.26 \times 10^6 \text{ J kg}^{-1}$.

Now suppose we leave the container alone for a long time so that the liquid and vapour come into equilibrium at a temperature T . If a mass M is to evaporate from liquid into vapour then the energy required, E , can be related to the entropy difference between the liquid and vapour phases,

$$E = ML = M(h^v - h^l) = MT(\eta^v - \eta^l), \quad (18.17)$$

where η^v and η^l are the specific entropies of the vapour and liquid, and the temperature is fixed because all the energy put into the liquid is used for evaporation. Re-arranging we find

$$h^v - T\eta^v = h^l - T\eta^l \quad \text{or} \quad g^v = g^l, \quad (18.18a,b)$$

where $g^l \equiv h^l - T\eta^l$ and $g^v \equiv h^v - T\eta^v$ are the specific Gibbs functions for the liquid and vapour (section 1.5.2). That is, *the specific Gibbs functions for the liquid and water phases of a substance are the same at equilibrium*. The result follows directly from (18.17): the energy required to evaporate a mass of liquid, which is equal to the mass times the specific enthalpy difference between the vapour and the liquid, is also equal to the mass times the specific entropy difference between the vapour and liquid. The equality is true only at equilibrium, when temperature remains fixed, so that (18.18b) is an equation, not an identity.

Another way to derive the above result is to begin with the fact that the total Gibbs function, for the entire system, must remain constant. That is, if M_l and M^v are the masses of liquid and water, then

$$G = M_l g^l + M^v g^v \quad (18.19)$$

and

$$\delta G = M_l \delta g^l + M^v \delta g^v + (g^v - g^l) \delta M = 0 \quad (18.20)$$

where δM is the mass exchanged between liquid and vapour arising from a small fluctuation. Now, from (1.79) changes in Gibbs functions arise because of changes in temperature and pressure; that is, in general,

$$\delta g = -\eta \delta T + \alpha \delta p, \quad (18.21)$$

and given this (18.20) becomes

$$\delta G = M_l(-\eta^l \delta T + \alpha^l \delta p) + M^v(-\eta^v \delta T + \alpha^v \delta p) + (g^v - g^l) \delta M = 0 \quad (18.22)$$

where α_l and α^v are the specific volumes (the inverse density) of the liquid and vapour, respectively. But the temperature and pressure are fixed, and thus, in order that $\delta G = 0$ we must have that $g^v = g^l$. The reason for equality of the two Gibbs functions — as opposed to the equality of some other thermodynamic potential — stems from the fact that the Gibbs function is the only potential for which the natural variables are intensive, namely temperature and pressure. The derivation we have given exploits this directly, for we kept p and T fixed.

18.1.5 Clausius–Clapeyron equation

Now suppose that the temperature of the liquid-vapour system changes by an amount δT , leading to a change in the vapour pressure and in the specific Gibbs functions for the liquid and water vapour. Using (18.21) the change in the two Gibbs functions is given by

$$\delta g^l = -\eta^l \delta T + \alpha^l \delta p, \quad \delta g^v = -\eta^v \delta T + \alpha^v \delta p \quad (18.23)$$

and, since the two changes must be the same, $\delta g^l = \delta g^v$. Re-arranging (18.23) and taking the limit of small changes then gives

$$\frac{dp}{dT} = \frac{\eta^l - \eta^v}{\alpha^v - \alpha^l}. \quad (18.24)$$

Using (18.16) and (18.17) this equation can be written

$$\frac{dp}{dT} = \frac{h^l - h^v}{T(\alpha^v - \alpha^l)} = \frac{L}{T(\alpha^v - \alpha^l)}, \quad (18.25)$$

where to obtain the rightmost expression we use the definition of L . The quantity p is the vapour pressure of the vapour above the liquid, which we are denoting e . Furthermore, it is the *saturation* vapour pressure, e_s , because the vapour is in equilibrium with the liquid: if more vapour were added it would immediately condense. Using this notation, the *saturation vapour pressure* of a condensable gas above a liquid is given by

$$\frac{de_s}{dT} = \frac{L}{T(\alpha^v - \alpha^l)}. \quad (18.26)$$

This is the *Clausius–Clapeyron* equation, and it tells us how the equilibrium pressure of a vapour, in thermodynamic equilibrium with an adjacent liquid, varies with temperature. If for some reason the pressure of the vapour pressure is higher than this value, and if there is an adjacent surface of liquid water, then the vapour will condense into a liquid — a common manifestation of which is the formation of clouds and rain. Evidently, since $L > 0$ and $\alpha^v > \alpha^l$, the saturation vapour pressure increases with temperature so that a falling of temperature can lead to saturation. At the temperature at which the saturation vapour pressure of a substance equals that of the ambient pressure then any liquid present will *boil*. For water at a pressure of 1000 hPa this occurs at about 100° C, with a lower temperature needed at a lower pressure, which is why it takes longer to properly boil an egg at high altitude.

The presence of a liquid surface is crucial to the derivation, and it means that the equation only applies to a condensable. For gases such as carbon dioxide or oxygen at temperatures encountered on Earth, the saturation vapour pressure is very much higher than the actual pressure at the Earth's surface and the gas never condenses; the partial pressure of the gas is then determined by the ideal gas relation and not by (18.26). On Mars, temperatures are sufficiently low that carbon dioxide (the main constituent of the Martian atmosphere) is a condensate and as much as 25% of the Martian atmosphere will condense in winter. On Titan temperatures are even lower and methane is a condensate, with methane lakes spread out over the dystopian surface.

On Earth, the partial pressure of water vapour, however, is constrained by (18.26), meaning that the partial pressure will often reach the saturated value and the vapour will then normally condense. Condensation is not, however, guaranteed, and to see this imagine a container that contains unsaturated water vapour and no liquid water, and suppose its temperature is then lowered; the pressure of the vapour will then fall following the ideal gas law. The saturation vapour pressure falls more quickly than this and so at some temperature the vapour pressure will exceed the saturation vapour pressure, but the vapour will not automatically condense since there is no liquid present and (18.26) does not apply. The vapour is then said to be *supersaturated*. A supersaturated state is unstable and condensation will eventually occur and liquid water will form, and subsequent changes in temperature induce pressure changes that satisfy the Clausius–Clapeyron equation. Supersaturated water vapour is fairly rare in Earth’s atmosphere because there is usually no shortage of condensation nuclei.

At the other end of the temperature scale, liquid water can exist at temperatures well below freezing when there is insufficient water for the molecules to become organized into a crystalline structure, and super-cooled water droplets rather than ice then result — a common situation in clouds. If the liquid that is present is in the form of small spherical droplets then the saturation vapour pressure will differ slightly from that when the vapour is over a flat surface, because surface tension will affect the energy required for a molecule to escape from the droplet and so the latent heat of vapourization will differ. Also, if the vapour is in contact with ice instead of water then its saturation vapour pressure will be different again, because the specific enthalpy of ice is different from that of liquid water.

Application to an ideal gas

In a mixture of ideal gases the partial pressure of one gas is unaffected by the presence of the other gases (because the volume of the gas molecules is assumed to be negligible) and in particular the saturation vapour pressure for a particular component is independent of the presence of other components. We can then, at least approximately, integrate (18.26) to see how the saturation vapour pressure of a particular component varies with temperature. Let us assume that density of the vapour is much less than that of the liquid, so that $\alpha^v \gg \alpha^l$. Given that partial pressure of the vapour satisfies the ideal gas law, namely $e^v \alpha^v = R^v T$, where R^v is the specific gas constant for the vapour, the Clausius–Clapeyron equation becomes

$$\frac{de_s}{dT} = \frac{Le_s}{R^v T^2}. \quad (18.27)$$

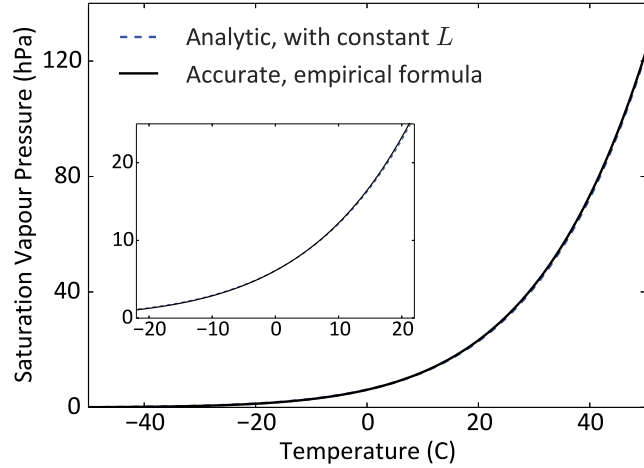
This is the form of Clausius–Clapeyron equation that is normally used in atmospheric applications. If we further assume that L is a constant then (18.27) can be integrated to give

$$e_s = e_0 \exp \left[\frac{L}{R^v} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right], \quad (18.28)$$

where e_0 and T_0 are constants, for example $T_0 = 273 \text{ K}$ and $e_0 = 6.12 \text{ hPa}$. Equation (18.28) is a good approximation if the temperature range is not too wide, and as seen in Fig. 18.1 the saturation vapour pressure of water increases approximately exponentially over commonly encountered terrestrial temperatures. Note that the expression for saturation vapour pressure does not depend on the presence or otherwise of dry air.

The fact that water vapour content cannot normally exceed the saturation value distinguishes the distribution of water from other tracers in the atmosphere, even without taking the heating effects of condensation into account. Note finally that if the atmosphere were motionless it would everywhere be in thermodynamic equilibrium with the moist surface and the surface layers would be saturated, and diffusion of water vapour upwards would then saturate the rest of the atmosphere. Thus, *the relative humidity of the atmosphere is determined by its circulation*, as we now discuss.

Fig. 18.1 The saturation vapour pressure of water vapour, calculated using the analytic formula (18.28), which assumes that L is constant, and using an accurate, semi-empirical formula.² The inset is the same plot over a smaller range, and for the analytic formula we use $T_0 = 273\text{ K}$, $e_0 = 6.12\text{ hPa}$, $L = 2.44 \times 10^6\text{ J kg}^{-1}$ and $R^v = 462\text{ J kg}^{-1}\text{ K}^{-1}$.



18.2 THE DISTRIBUTION OF RELATIVE HUMIDITY

To a first approximation, the distribution of water in the atmosphere is determined by the distribution of temperature. This is because of the near-exponential dependence of absolute humidity on temperature through the Clausius–Clapeyron equation, so that variations of relative humidity of even an order of magnitude, from 10% to 100% say, are barely noticeable in the specific humidity distribution, as seen in Fig. 18.2. It is, however, the relative humidity that determines such basic quantities as rainfall, and its distribution (Fig. 18.3 and Fig. 18.4) shows a quite different picture, with the following features evident.

- High, near-saturated values close to the ground.
- Low relative humidity in the subtropics at latitudes between 15° to 40° (depending on season) in both hemispheres.
- High values near the equator extending up to the tropopause.
- Vertically near-uniform values in mid- and high latitudes, increasing with latitude close to the pole in some cases.
- Very low values over much of the stratosphere.

The gross distribution of zonally-averaged temperature can be understood, at least in a rough way, using fairly simple arguments. The incoming solar radiation at the top of the atmosphere would lead, in the absence of atmospheric motion, to a strong meridional radiative equilibrium temperature gradient, as in Fig. 14.1. The meridional transport of heat by the Hadley Cell and by mid-latitude baroclinic eddies flattens that temperature gradient, and one might crudely model this transport as a diffusion. In the vertical heat is transported upwards by convection and baroclinic eddies, which might be modelled as a relaxation back to some specified neutrally stable profile or specified isentropic slope. However, arguments of this type cannot capture some basic features of the relative humidity distribution, and diffusive arguments in particular can be quite misleading. The effects of advection, either explicitly or as represented by a stochastic process, are crucial, and in this section we consider advection-diffusion-condensation models of the general form

$$\frac{\partial q}{\partial t} + \mathbf{v} \cdot \nabla q = \nabla \cdot \kappa \nabla q - S, \quad (18.29)$$

where q is the specific humidity, $\mathbf{v}$ is a specified velocity field, κ is a diffusion coefficient and S is the condensational sink.⁴ Condensation in the atmosphere involves complicated microphysical processes but the basic effect is to remove liquid water once the volume becomes saturated and to

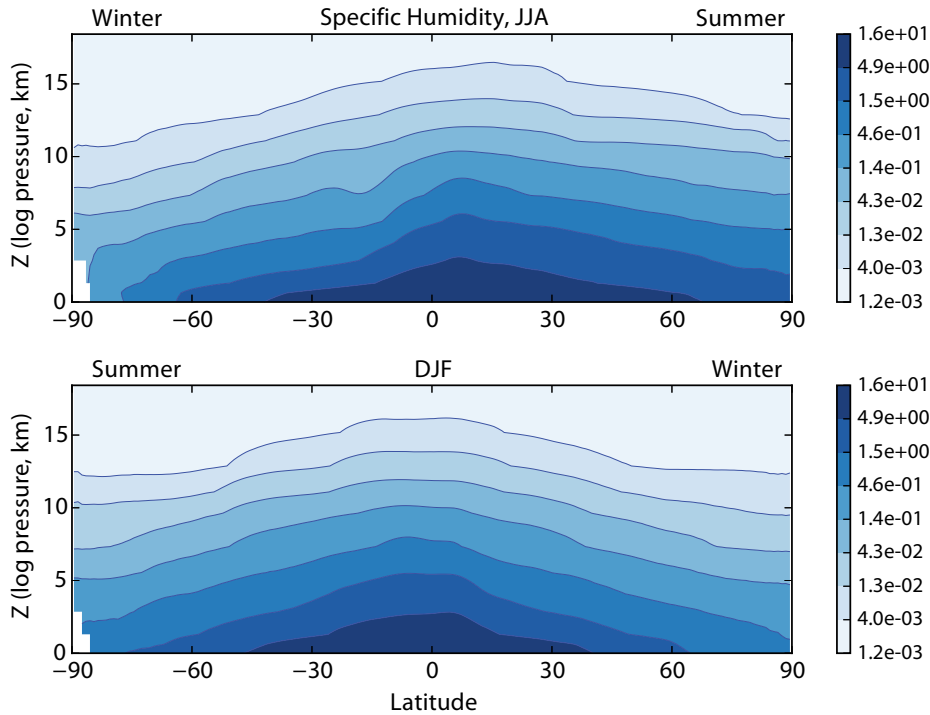


Fig. 18.2 Zonally-averaged specific humidity distribution (g/kg) in the atmosphere, as retrieved from a microwave satellite, for boreal summer in 2008 (top) and boreal winter 2008–2009 (bottom).³ Note the logarithmic scale.

largely prevent relative humidity from exceeding 100% (although supersaturation can occur locally if there are no particulates in the air onto which the water vapour may condense). A simple analytic way to represent such a condensation process is to let

$$S = \begin{cases} 0, & q \leq q_s, \\ (q - q_s)/\tau, & q > q_s, \end{cases} \quad (18.30)$$

where the time τ is much smaller than any large scale diffusion time, L^2/κ , where L is a characteristic length. Such a sink effectively prevents q from exceeding q_s except by a tiny amount. In (18.29) we may consider the advection and diffusion terms as representing larger scale processes and the sink as representing small-scale microphysical processes. We begin our exploration by omitting advection.

18.2.1 A diffusion-condensation model

If we omit advection in (18.29) we have a simple diffusion-condensation model

$$\frac{\partial q}{\partial t} = \nabla \cdot \kappa \nabla q - S. \quad (18.31)$$

Although superficially plausible, such a model is too-often unable to reproduce locally unsaturated regions. For simplicity consider the one-dimensional case satisfying

$$\frac{\partial q}{\partial t} = \kappa \frac{\partial^2 q}{\partial x^2} - S, \quad (18.32)$$

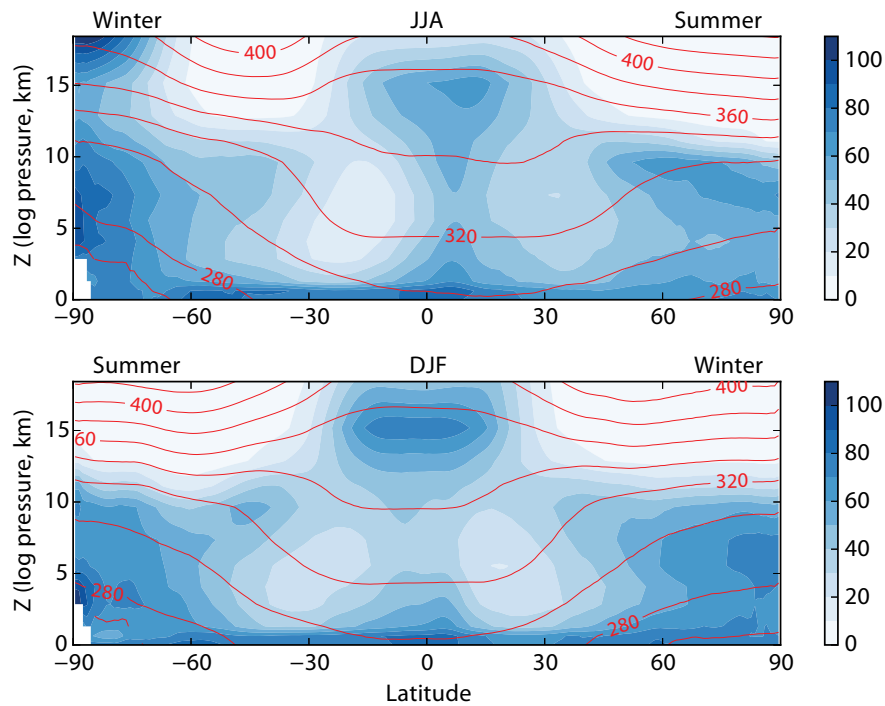


Fig. 18.3 Zonally-averaged relative humidity distribution in the atmosphere (in percent, shading), and isolines of equivalent potential temperature (contours), inferred from satellite as in Fig. 18.2. (Equivalent potential temperature is a modification of potential temperature to account for water vapour, and is approximately an adiabat; see section 18.3.2.)

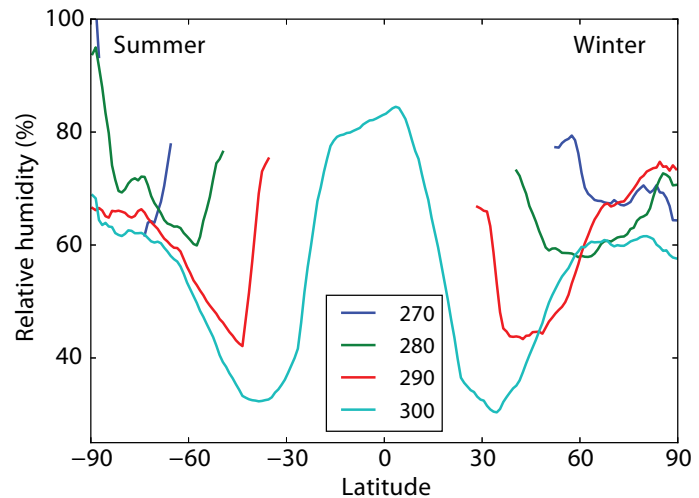


Fig. 18.4 Values of zonally-averaged relative humidity in boreal winter plotted along isolines of equivalent potential temperature (red contours in Fig. 18.3), with values of θ_{eq} as indicated in the legend.

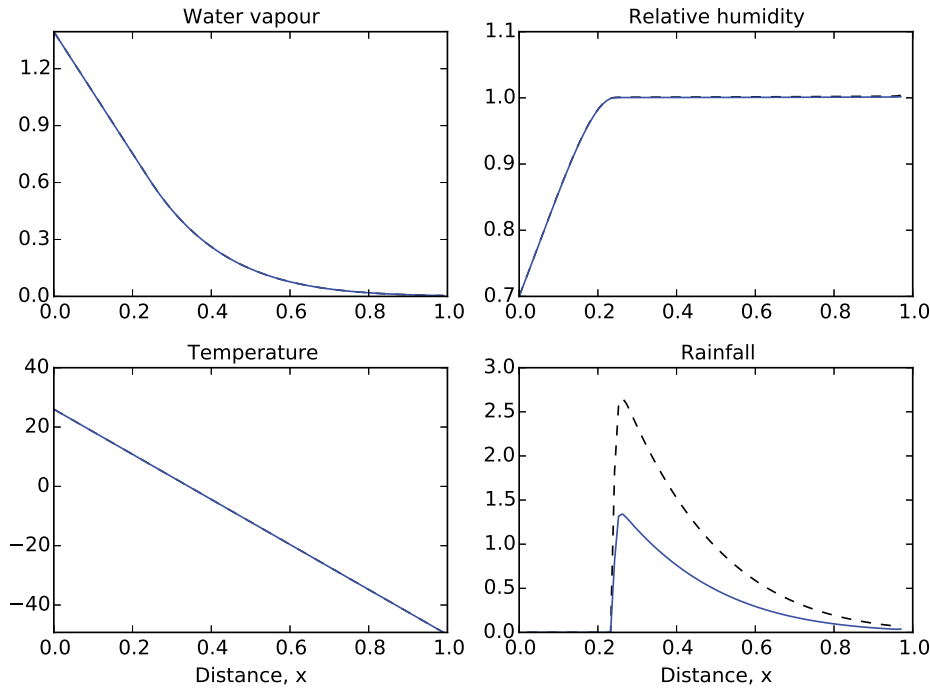


Fig. 18.5 Steady solution of the diffusion-condensation model (18.32) and (18.33), with $\mathcal{H}_b = 0.7$ at $x = 0$ and temperature ($^{\circ}\text{C}$) falling linearly from $x = 0$ as shown. Water vapour falls linearly away from the boundary at $x = 0$ until it becomes saturated at $x = 0.23$, after which the region is saturated. Two solutions are plotted, the one with dashed lines showing the solution with twice the diffusivity as that with solid lines. The two solutions are nearly identical except for the sink term, the rainfall.

with constant κ .

Regardless of the boundary conditions, any interior minimum of q_s will lead to saturation in that region. To see this, note that in any region with $\partial^2 q / \partial x^2 > 0$ and $q < q_s$ there will be a flux of water vapour into that region and that $S = 0$. Saturation must eventually occur, at which point q remains very close to the value of q_s . If $\partial^2 q_s / \partial x^2 > 0$ then the diffusive flux will maintain the saturated state. Given the monotonic dependence of q_s on temperature, this result means that, if water vapour is transported diffusively, the region of an interior minimum of temperature will become saturated.

A corollary of this result is that, unless there is a source of moisture at a boundary, a region with $\partial^2 q_s / \partial x^2 > 0$ everywhere will under many conditions lose *nearly all* its moisture. Suppose that $\partial q / \partial x = 0$ at $x = 0$ and that q_s has a maximum at $x = 0$, and that the region extends to infinity and is initially saturated. (Envision a semi-infinite domain with temperature decreasing linearly away from no-flux boundary at $x = 0$.) Moisture is transported to higher values of x , whence condensation occurs and moisture is removed. As time progresses the region of saturation moves to higher and higher values of x , but nevertheless water is continuously removed. In a finite domain, with no flux boundary conditions at either end, a small amount of moisture will remain in the system for all time because a finite amount of water vapour is needed for condensation to occur.

Now consider the more atmospherically relevant situation with a moisture source at $x = 0$, q_s decreasing monotonically away and with the other boundary either extending to infinity or being a no-flux boundary at finite x . Such a situation might represent an atmosphere sitting atop a moist

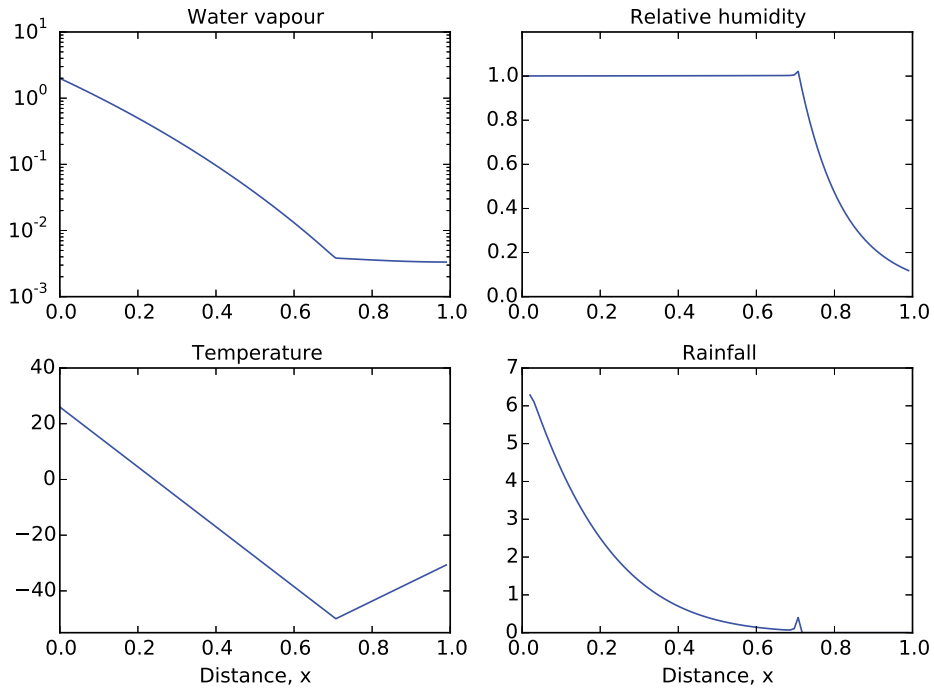


Fig. 18.6 As for Fig. 18.5, but now with $\mathcal{H}_b = 1$ and an interior temperature minimum, a ‘cold trap’, at $x = 0.7$ (see temperature panel), and water vapour has a log scale. Relative humidity falls rapidly beyond the cold trap and the rainfall there is zero.

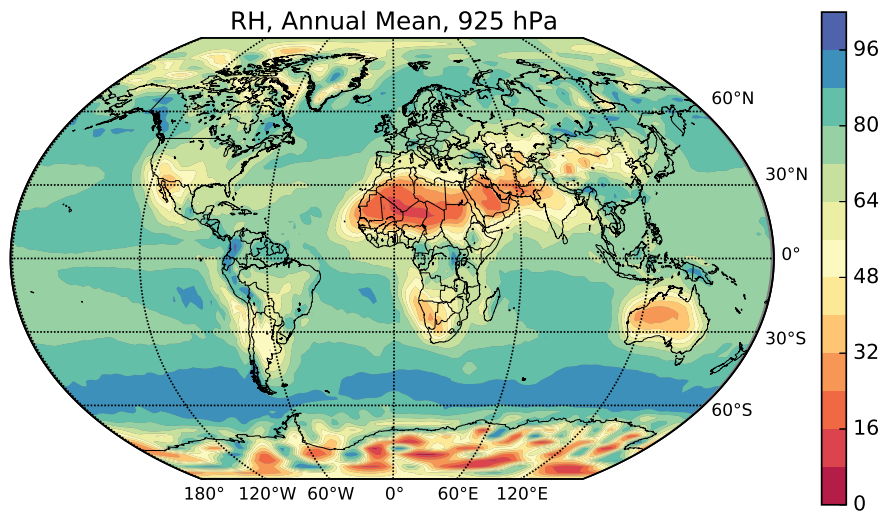


Fig. 18.7 Annually-averaged relative humidity (in percent) at 925 hPa, about 750 m above sea level. The contrast over land and ocean is apparent especially in the subtropics. (In regions of high topography values are interpolated.)

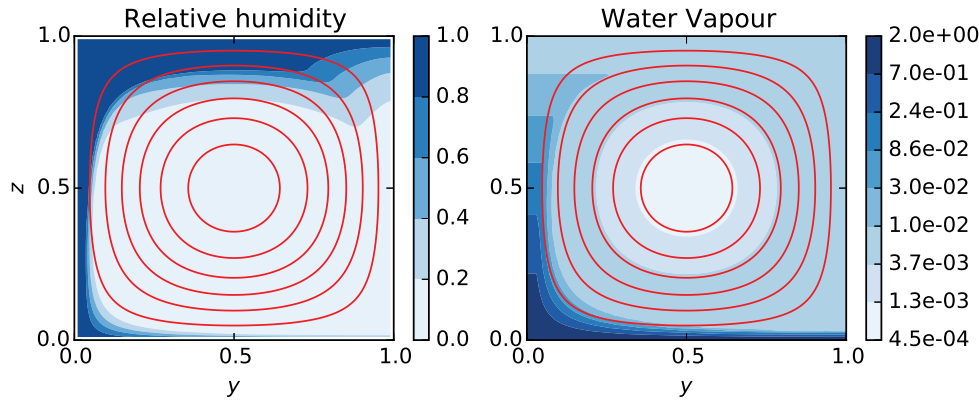


Fig. 18.8 Steady state distributions of relative humidity (filled contours, left) and water vapour (right) in a single cell as defined by the streamfunction (red contours, clockwise flow) in a closed domain. The domain boundary is saturated at the bottom, and temperature decreases linearly with height. The diffusivity $\kappa = 0.001$ and so the Peclet number, $Pe \equiv UL/\kappa \sim 1000$.

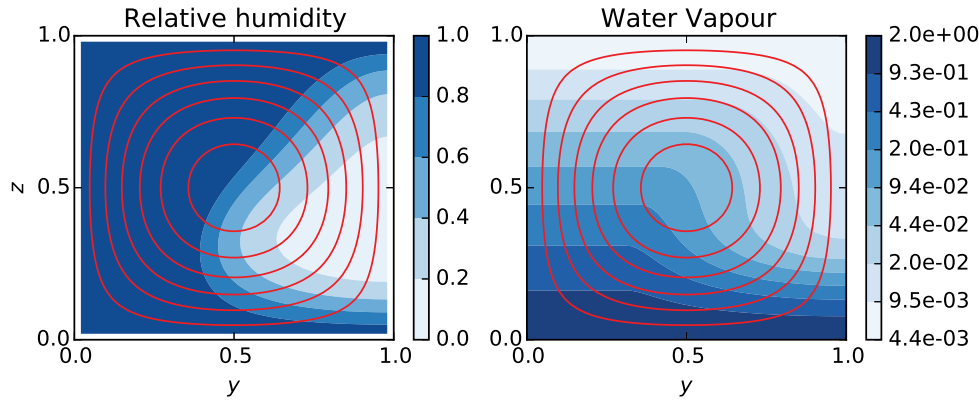


Fig. 18.9 As for Fig. 18.8, but with bigger diffusivity, $\kappa = 0.1$ and $Pe \sim 10$. Most of the domain is now much wetter.

surface with temperature decreasing with height. Consider the case

$$\begin{aligned} q &= \mathcal{H}_b q_s(T_0), & x &= 0, \\ \frac{\partial q}{\partial x} &= 0, & x &= 1, \end{aligned} \quad (18.33)$$

where $\mathcal{H}_b$ is a parameter such that if the boundary is effectively saturated then $\mathcal{H}_b = 1$, and $\mathcal{H}_b < 1$ otherwise, and we suppose that T decreases between T_0 at $x = 0$ and T_1 at $x = x_1$. Consider first the case with $\mathcal{H}_b = 1$. If T falls linearly then the value of q_s falls approximately exponentially between $x = 0$ and $x = 1$, with $\partial^2 q_s / \partial x^2 > 0$, and the steady solution of this problem is that the domain is saturated everywhere. To see this, suppose that $q < q_s$ is some region so that $S = 0$. Water vapour will then diffuse into that region until condensation begins, maintaining $q \approx q_s$ everywhere, with (if (18.30) applies, q in fact exceeding q_s by a very small amount so that the diffusion into the region is balanced by condensation. If $\mathcal{H}_b < 1$ then the region next to the surface will not be saturated and in steady state the water vapour content will decrease linearly, to satisfy $\partial^2 q / \partial x^2 = 0$, until at some value of x , x_s say, the atmosphere becomes saturated and remains so for $x > x_s$. The actual

solution is

$$q(x) = \begin{cases} \mathcal{H}_b q_s(0) + x \Delta q / x_s & x < x_s \\ q_s(x) & x \geq x_s, \end{cases} \quad (18.34)$$

where $\Delta q = (q_s(x_s) - \mathcal{H}_b q_s(0))$ and x_s is such that the flux is continuous there. A moment's thought reveals that $x_s = \Delta q / (\partial q_s / \partial x)_{x=x_s}$ and $x_s = 0$ if $\mathcal{H}_b = 1$. It is interesting that the water vapour levels in the solution (plotted in Fig. 18.5) do not depend upon κ and only very weakly on τ . In the fast condensation limit (in which τ is small compared to the diffusion time) variations of τ determine only the tiny amount by which q exceeds q_s . The amount of condensation is actually largely determined by the value of κ : large values of κ lead to a stronger diffusion of water vapour into dryer regions where it is almost immediately removed by condensation. This result illustrates the tenet that on large scales precipitation is at leading order determined by the motion of the fluid (here represented by diffusion), with variations in τ (crudely representing complex microphysical processes) being of less import. Microphysical processes are nevertheless important in many ways — a weather forecast model with poor microphysics would likely have little skill in forecasting the onset of precipitation, even if the climatology of the model were good.

Although the above model is over-simple in some respects, the dependence of the solution on $\mathcal{H}_b$ does capture the dependence of relative humidity in the lower atmosphere on the nature of the surface beneath, as seen in Fig. 18.7 showing relative humidity at 925 hPa. Over the desert regions the relative humidity is unsurprisingly low. Perhaps what is surprising is that the general dryness of the subtropics cannot be seen over the oceans — the surface moisture source simply overwhelms the drying effects of descending air (discussed more below).

A variation on the above theme introduces a temperature minimum, or ‘cold trap’, in the interior of the domain, as at $x = 0.7$ in Fig. 18.6. This configuration is a crude model of the tropopause, with temperatures increasing in the stratosphere beyond. Water vapour has to pass through the cold trap and so, since the specific humidity cannot be higher than the saturated value at the cold trap, the atmosphere will be unsaturated beyond it with relative humidity decreasing rapidly, as is seen in the real atmosphere in Fig. 18.3.

Although informative, diffusive-condensation models are fundamentally limited in what they can achieve, because of the deficiencies of diffusion in parameterizing the motion of a tracer in the presence of condensation. In particular, in the absence of a cold trap, diffusive models are prone to produce saturation everywhere. If the atmosphere obeyed (18.31) with a saturated surface, then the atmosphere would become saturated everywhere up to the tropopause, which from Fig. 18.3 is manifestly not the case. To remedy this we turn our attention to the effects of advection.

18.2.2 Advection-diffusion-condensation models

Consider now the effects of advection. Neglecting diffusion, the specific humidity of a parcel is conserved unless condensation occurs. If a moist parcel travels into a region of decreasing temperature it will eventually become saturated; however, if it passes into a region of increasing temperature then its relative humidity drops. Consider, for example, a cold trap with a temperature minimum at $x = x_{ct}$, as in the previous section. A parcel advected through the trap becomes saturated at x_{ct} , beyond which its specific humidity is constant, so that relative humidity is given by

$$\mathcal{H}(x) = \frac{q_s(T(x_{ct}))}{q_s(T(x))}. \quad (18.35)$$

That is to say, the relative humidity of a parcel is given by the value of the saturated vapour pressure at the point of last saturation, divided by the saturated vapour pressure at its current location.

Let us see the extent to which a simple advection-diffusion-condensation model can explain some of the features of the zonally-averaged relative humidity seen in Fig. 18.3. Consider a two-dimensional model in the y - z plane obeying (18.29), with a divergence-free advecting velocity, a

Water Vapour Transport and Relative Humidity

- Water vapour in the atmosphere is primarily transported by advection, much of this on large, near planetary scales but also by convection and smaller scale turbulence. Specific humidity is materially conserved in the absence of condensation and diffusion.
- If the vapour pressure exceeds the saturated value (as given by the Clausius–Clapeyron relation) condensation will occur provided condensation nuclei or liquid water are present. The condensation normally occurs much more quickly than large-scale advective processes, and this situation is known as the *fast condensation limit*.
- When dealing with large-scale flows in Earth's atmosphere the limit is a good approximation. Levels of relative humidity are then determined mainly by advective processes rather than the microphysical details of the condensation process. Specifically, the relative humidity of a parcel is determined by the temperature at the location of last saturation, as in (18.35).
- If the advection is not fully resolved — for example if there is some small-scale turbulence in the flow — then introducing some diffusion seems natural, as is commonly done for tracers, but a large diffusivity can give unrealistic results because of the irreversible nature of condensation. Diffusion is then not a good representation of small-scale quasi-random flow and is overly prone to produce saturation.
- In Earth's atmosphere some of the large-scale features of the relative humidity distribution may be explained as follows.
 - High levels of relative humidity close to the surface. These are due to transport from a saturated surface, especially over the ocean and moist ground.
 - High levels of relative humidity in the ascending branch of the Hadley Cell. These are due to upward advection from a nearly saturated surface. The branch is, however, not saturated on the zonal-average, because of the presence of smaller scale motion such as downdrafts that unsaturate the air.
 - A subtropical minimum of relative humidity. This largely arises because of the mean descending motion, advecting water vapour into a warmer region and decreasing its relative humidity.
 - Variable relative humidity in mid- and high latitudes, with locally strong gradients. Chaotic advection by baroclinic eddies takes moisture upwards and polewards into cooler regions where it becomes saturated, but also downwards and equatorwards so reducing relative humidity.
 - Very low levels of relative humidity in the stratosphere. The tropopause is a cold trap and so relative humidity is very low beyond it. The cold-trap effect occurs in both advective and diffusive models. In Earth's atmosphere, the little water vapour that is in the stratosphere mainly enters advectively through the tropical tropopause.

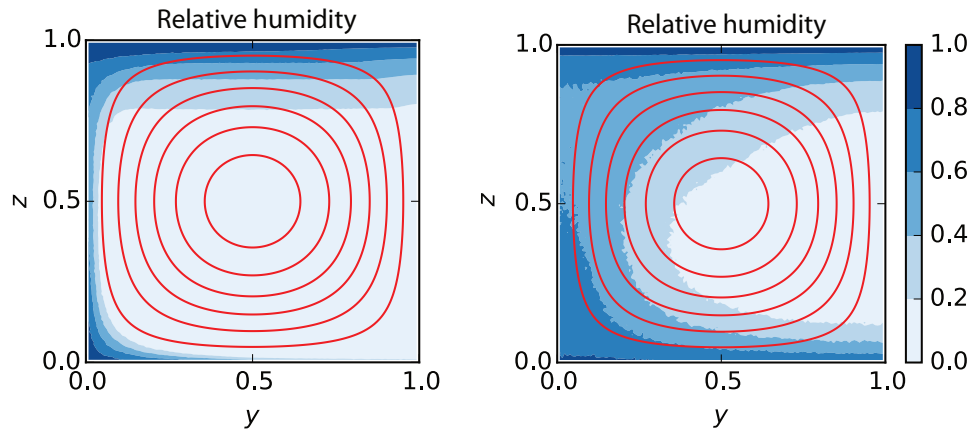


Fig. 18.10 The relative humidity produced using a stochastic advection-condensation model with the same imposed temperature, mean streamfunction (red contours) and boundary conditions as Fig. 18.8 and Fig. 18.9. The panel on the right has a larger stochastic component than the one on the left.

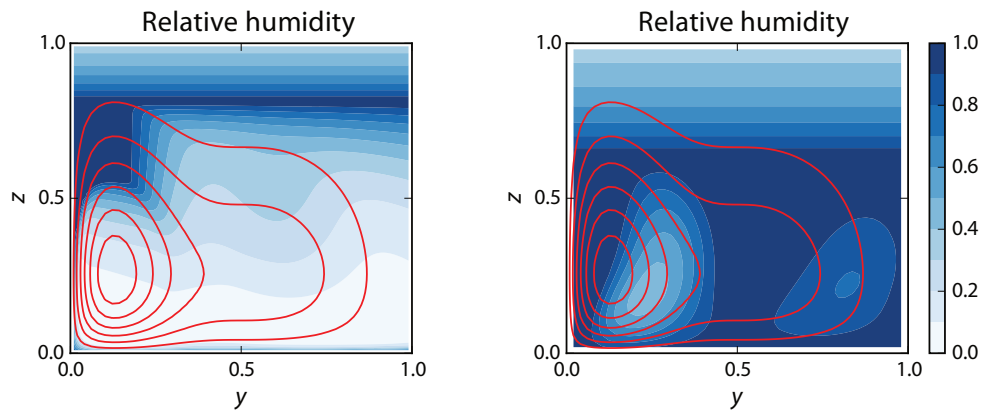


Fig. 18.11 Relative humidity (shading) for small diffusivity (left, $\kappa = 0.001$, $Pe = 1000$) and larger diffusivity (right, $\kappa = 0.05$, $Pe = 20$), and a clockwise overturning circulation as contoured. The imposed temperature falls linearly with height to a tropopause at $z = 0.8$ with $T = -50^\circ\text{C}$, then rises again to -40°C , and diminishes meridionally at the surface from 26°C at the equator ($y = 0$) to -15°C at the pole ($y = 1$).

saturated surface at $z = 0$ and no flux boundary conditions elsewhere. A simple model representing some of the features of the Hadley Cell is that of a single cell in a unit-sized square domain with a streamfunction of the form $\psi = \sin \pi y \sin \pi z$. We suppose the lower boundary is saturated, that there are no-flux boundary conditions on the other boundaries, and that temperature is uniform in latitude and falls linearly with height from 26°C to -50°C .

Solutions are illustrated in Fig. 18.8 and Fig. 18.9, the latter having diffusivity κ that is a hundred times larger. The upward branch in both cases is completely saturated. In the low diffusivity case the interior of the cell is quite dry, with a relative humidity that drops almost to zero in the centre. Increasing the diffusivity simply tends to moisten the interior, but does little to reduce the relative humidity in the upward branch of the cell. However, the upward branch of the real Hadley Cell is *not* saturated on the zonal mean (Fig. 18.3). Let us try to remedy this model failing.

Stochastic effects

The upwards branch of the Hadley Cell is not saturated because it is not steady — there is both ascending and descending motion, with the relative humidity of parcels falling as they descend so reducing the average value. We can mimic this by introducing a stochastic component into a Lagrangian model and performing ‘Monte Carlo’ simulations.⁵ The model advects infinitesimal parcels of water vapour — one million in the simulations shown in Fig. 18.10 — by a prescribed mean field plus a random component. Whenever the parcels touch the ground they become saturated and whenever their relative humidity exceeds one condensation occurs. We show results with two levels of stochastic motion, one nominally comparable to the diffusivity used in Fig. 18.8 and the other comparable to that of Fig. 18.9, all with the same temperature, boundary condition and mean velocity fields, and the results shown in Fig. 18.10 are coarse-grained time averages. Although the results with weak stochasticity do resemble those with small diffusion (compare the left-hand panel of Fig. 18.10 with Fig. 18.8), those with large stochasticity have less resemblance to those with large diffusivity. Most notably, the explicit randomness reduces the relative humidity in ascending branch without saturating the interior, a property that diffusion is unable to capture.

Relative humidity in mid-latitudes

The diffusive steady overturning model fails qualitatively when applied to an entire hemisphere, even with more realistic overturning circulation and temperature fields. To see this we construct a model in which the overturning streamfunction (representing a residual circulation) resembles that of the bottom panel of Fig. 15.22, as illustrated (red contours) in Fig. 18.11. To this we add a temperature distribution that decreases polewards and upwards until it reaches a tropopause at $z = 0.8$, then increases upwards again in the stratosphere, and we assume that the surface is saturated. With a small diffusivity (which is more realistic) the lower tropopause dries out too readily because of the sinking motion, and with larger diffusivity the atmosphere becomes more saturated but with little of the observed structure. Two features that the model *can* reproduce are the dry stratosphere, because of the tropospheric cold trap, and the subtropical minimum, a robust feature of the downwelling in the Hadley Cell.

The failings of this model arise from the fact that in mid-latitudes water vapour is not primarily advected by the overturning circulation, and diffusion is a poor model of the meridional transport. Rather, moisture is transported from the boundary layer into the free atmosphere by convection and polewards by larger-scale baroclinic eddies in a quasi-horizontal fashion, roughly along moist isentropic surfaces. Mid-latitude relative humidity is thus highly variable, as can be seen in Fig. 18.12, with sharp gradients between high and low values and with variations being strongly correlated with the parcel trajectory — note for example the swath of high relative humidity just west of the UK associated with moisture moving poleward and to a lower temperature. On the zonal average relative humidity is high in the boundary layer, diminishes as one moves upwards out of the boundary layer, then stays roughly constant as one moves polewards along a moist isentrope before increasing again at very high latitudes (Fig. 18.3 and Fig. 18.4).

We can understand some of these features using a simple advective-diffusive-condensation model, with the unsteadiness explicitly incorporated into the advection, and with no other stochastic component. Consider (18.29) in two horizontal dimensions in a channel of size (L_x, L_y) , periodic in x , with a saturated boundary at $y = 0$ and no flux at $y = L_y$, and with imposed meridional temperature gradient ($^{\circ}\text{C}$) and advecting streamfunction of the forms

$$T(y) = 20 - 30y/L_y, \quad \psi(x, y, t) = -Uy + \Psi_0 \sin(\pi y/L_y) e^{i(kx - \omega t)}. \quad (18.36)$$

We will choose $|\Psi_0| = 1$, $U = 10$, $k = 4\pi/L_x$, $\kappa = 0.005$ (so $Pe \sim 200$ on the domain scale), and a solution is shown in Fig. 18.13. The details of the solution depend on the parameters chosen, but there are couple of quite robust results.

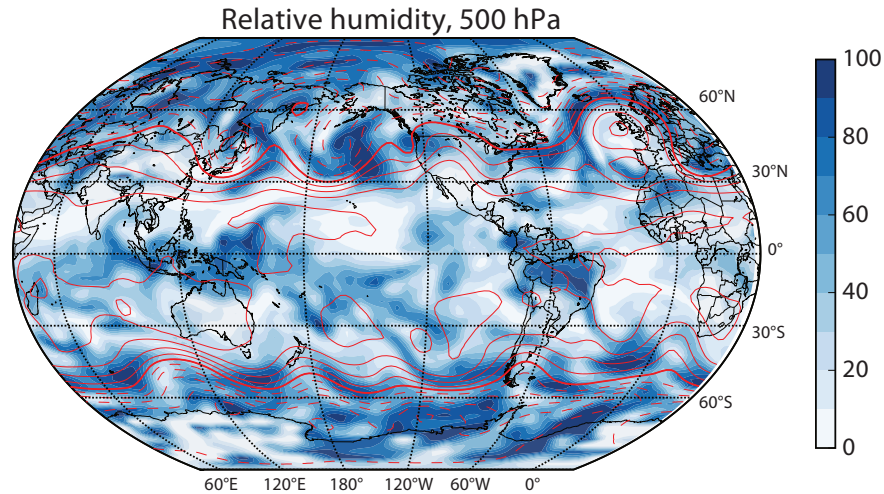


Fig. 18.12 Snapshot of relative humidity at 500 hPa on February 9, 2015 (shading, in percent), with red contours of geopotential height, from reanalysis.

- Relative humidity at a point depends on where a parcel has come from, and hence is correlated with the streamlines, and is generally higher for parcels moving to colder regions.
- Relative humidity is close to one (i.e., 100%) next to the saturated boundary, then decreases to a minimum in the interior, rising again at high latitudes (c.f., the observations shown in Fig. 18.4). The width of the saturated region near the saturated boundary diminishes as the Peclet number increases, and the mid-domain minimum arises because of the drying effects of advection, which diminish near either boundary.

There are many limitations to such a model, one being that in reality temperature itself is advected by the flow and another being that advection is three-dimensional, but exploring these effects is beyond the scope of our story.⁶ A summary of relative humidity transport is provided on page 687, but we now move on to the dynamical effects of water vapour, and in particular convection,

18.3 ATMOSPHERIC CONVECTION

As introduced in section 2.10, convection occurs when a parcel that is displaced upwards (downwards) finds itself lighter (heavier) than its surroundings, and hence becomes subject to a buoyancy force that amplifies the initial displacement. Convection is particularly prevalent in the tropics for reasons that are, at lowest order, independent of the presence of water vapour; rather, it is simply that the surface is warmer there, as sketched in Fig. 18.14. Consider, rather heuristically, a fluid that is heated by a warm surface from below and cools by radiation from above. If the surface is warmer at one end, as illustrated, a large-scale circulation will arise, transporting energy from warmer regions to cooler ones. In the absence of convection, there will be a larger vertical temperature gradient where the surface temperature is warm, and so these regions are more prone to convection. Having said this, moisture *does* affect convection in rather profound ways, one being that moisture is a *destabilizing* influence, as we now discover.

18.3.1 Generalized enthalpy and adiabatic lapse rates

A column of air that is convectively stable when dry can be made convectively unstable by the presence of water vapour, because when water vapour condenses heat is released, warming the

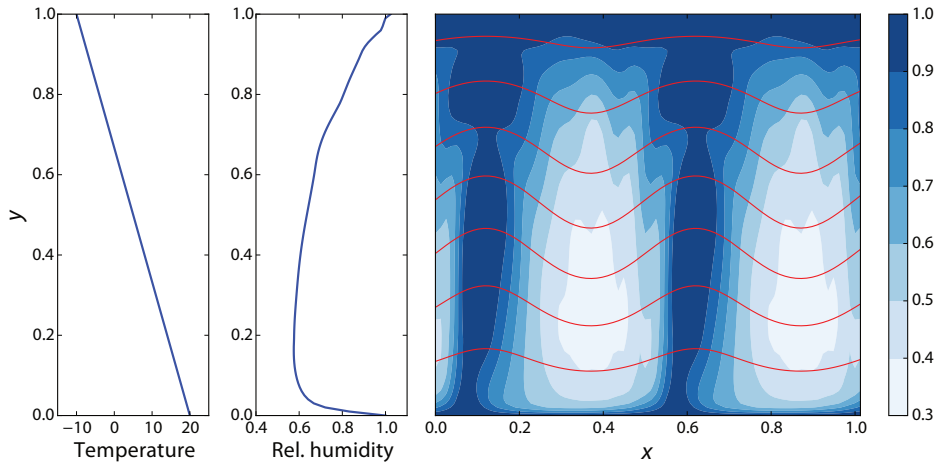


Fig. 18.13 A snapshot of relative humidity (right panel, shading) obtained by a numerical solution of the advective-diffusive-condensation model (18.29) and (18.36) in the horizontal (x - y) plane. The contours are the streamfunction. The middle panel is the zonally-averaged relative humidity and the left panel is the imposed temperature gradient ($^{\circ}\text{C}$).

parcel and potentially making it more buoyant than its surroundings. To explore this we begin with a derivation of the conditions under which a moist column of air may be unstable — essentially an extension of section 2.10 to include the effects of condensation.

An adiabatic (and convectively neutral) profile is one in which the entropy is constant. Recalling the arguments in sections 1.6 and 1.10.3, the entropy may be related to the enthalpy by writing the fundamental relation as

$$dh = Td\eta + \alpha dp, \quad (18.37)$$

and where, if the column is hydrostatically balanced, $\alpha dp = -g dz$. Thus, the condition that $d\eta = 0$ is equivalent to

$$d(h + gz) = 0. \quad (18.38)$$

That is, the generalized enthalpy or dry static stability, $h^* = h + gz$, is constant in an adiabatic,

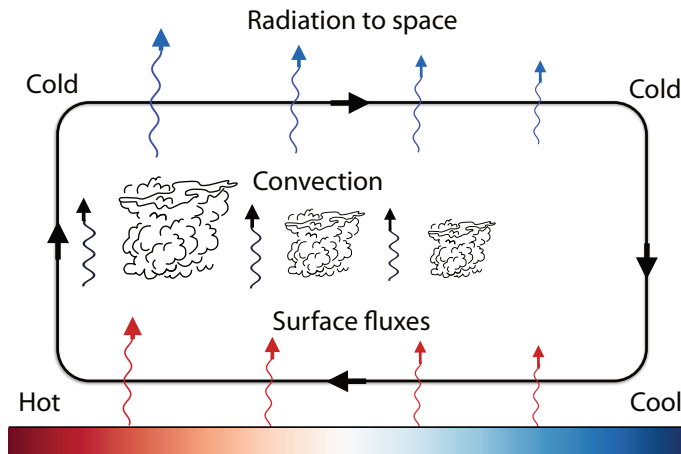


Fig. 18.14 Sketch of small scale convection embedded within a large scale circulation. The fluid is in contact with a warm surface below, and cools by radiation to space from above. The horizontal temperature gradient drives a large-scale meridional overturning circulation, embedded in which convection forms, especially at the warmer end.

hydrostatic, profile. The enthalpy of a dry parcel of ideal gas is given by $h^d = c_p T$ and thus,

$$h^* \equiv h^d + gz = c_p T + gz. \quad (18.39)$$

Since an adiabatic profile has $dh^* = 0$ we recover the dry adiabatic lapse rate (section 2.10.3), to wit

$$\Gamma^d = -\left.\frac{dT}{dz}\right|_{ad} = \frac{g}{c_p}. \quad (18.40)$$

The adiabatic lapse rate for a moist parcel is given using similar reasoning, but we need an appropriate expression for the enthalpy. A parcel of total mass M^t composed of dry air, water vapour and liquid water such that $M^t = M^v + M^l$ has an enthalpy given by

$$H = h^d M^d + h^v M^v + h^l M^l \quad (18.41)$$

where h denotes specific enthalpy and the superscripts denote dry air, water vapour and liquid water respectively. Dividing by the total mass, the specific enthalpy of the parcel is thus

$$h = (1 - q^w)h^d + q^v h^v + q^l h^l, \quad (18.42)$$

where

$$q^w = q^v + q^l, \quad q^v = \frac{M^v}{M^t}, \quad q^l = \frac{M^l}{M^t}. \quad (18.43)$$

As in (18.16) the latent heat of evaporation is defined to be $L = h^v - h^l$ so that (18.42) becomes

$$h = (1 - q^w)h^d + q^w h^l + Lq^v. \quad (18.44)$$

The generalized enthalpy is equal to this quantity plus a potential ϕ , and if that potential is equal to gz the quantity is the *moist static energy*,

$$h_m^* = (1 - q^w)h^d + q^w h^l + Lq^v + gz. \quad (18.45)$$

Since $h^d = c_p^d T$ and $h^l = c^l T$ (to a very good approximation and where c^l is the heat capacity of liquid water) we have

$$h_m^* = c_p^{dl} T + Lq^v + gz, \quad (18.46)$$

where $c_p^{dl} = (1 - q^w)c_p^d + q^w c^l$. This quantity varies with liquid water content but in Earth's atmosphere the variation is small and in the derivation below we will take c_p^{dl} to be a constant and denote it c_p . For adiabatic motion, the moist static energy is a constant, whether or not water is evaporating or condensing: the latent heat of evaporation or condensation is merely exchanged with the dry generalized enthalpy.

The saturated adiabatic lapse rate

In a moist atmosphere the moist static energy is conserved as a parcel ascends and $dh_m^*/dz = 0$. Using (18.46), an ascending parcel then has a lapse rate given by

$$c_p \frac{dT}{dz} = -L \frac{dq^v}{dz} - g. \quad (18.47)$$

If the air is moist but not saturated then an ascending parcel will follow the dry adiabatic lapse rate (because $dq^v/dz = 0$) but if it is saturated (and $q^v = q_s$) then as a parcel ascends it will cool

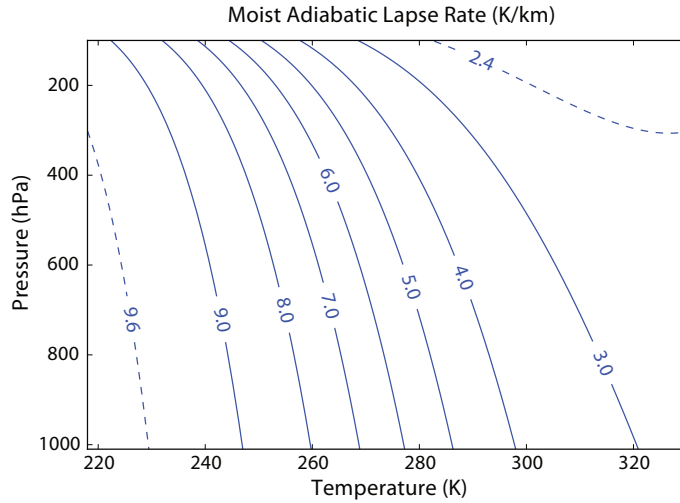


Fig. 18.15 Contours of saturated adiabatic lapse rate (K/km) as a function of pressure and temperature, calculated using (18.50) with $q_s = \epsilon e_s/p$.

and some water vapour will condense. Since $q_s \approx \epsilon e_s/p$, and so is a function of temperature and pressure, we have

$$\frac{dq_s}{dz} = \left(\frac{\partial q_s}{\partial T} \right)_p \frac{dT}{dz} + \left(\frac{\partial q_s}{\partial p} \right)_T \frac{dp}{dz} = \left(\frac{\partial q_s}{\partial T} \right)_p \frac{dT}{dz} + \left(\frac{q_s}{p} \right) \rho g, \quad (18.48)$$

using hydrostasy. Using the Clausius–Clapeyron equation and the ideal gas equation of state gives

$$\frac{dq_s}{dz} = \frac{Lq_s}{R^v T^2} \frac{dT}{dz} + \frac{gq_s}{R^d T}. \quad (18.49)$$

Using (18.47) and (18.49) we obtain an expression for the lapse rate of an adiabatically ascending saturated parcel,

$$\Gamma_s = - \left. \frac{dT}{dz} \right|_{ad} = \frac{g}{c_p} \frac{1 + Lq_s/(R^d T)}{1 + L^2 q_s/(c_p R^v T^2)}. \quad (18.50)$$

The quantity Γ_s is the *saturated adiabatic* lapse rate, plotted in Fig. 18.15. (This quantity is often called the moist adiabatic lapse rate, but that name is better given to the lapse rate of moist air that does not condense, and which differs slightly from the dry adiabatic lapse rate.) The lapse rate is a function of temperature and pressure because $q_s = \epsilon e_s/p$ and e_s is given by the solution of the Clausius–Clapeyron equation, (18.28). Values of Γ_s are typically around 6 K km^{-1} in the lower atmosphere although since dw_s/dT is an increasing function of T , Γ_s decreases with increasing temperature and can be as low as 3 K km^{-1} . The second term in the numerator of (18.50) is usually quite small (around 0.1, although it can become large at very high temperatures) but the second term in the denominator is positive and order unity. Thus, and since $\Gamma_s = g/c_p$ is the dry adiabatic lapse rate, the saturated adiabatic lapse rate is smaller than the dry adiabatic lapse rate (i.e., $\Gamma_s < \Gamma_d$), as can be seen directly from (18.47) since $dq^v/dz < 0$.

The saturated adiabatic lapse rate determines the stability of a saturated profile. Using parcel theory, just as in section 2.10, a profile will be stable or unstable depending on whether the lapse rate is less than or greater than (18.50); that is

$$\text{Stability : } - \frac{\partial \bar{T}}{\partial z} < \Gamma_s, \quad \text{Instability : } - \frac{\partial \bar{T}}{\partial z} > \Gamma_s. \quad (18.51a,b)$$

where $\tilde{T}$ is the environmental temperature. The lapse of the atmosphere rarely exceeds the saturated adiabatic lapse rate, as was seen in 15.24. In the tropics and subtropics the lapse is, on average, very close to the saturated adiabat up to about 300 hPa (about 9km), whereas in midlatitudes it is considerably less, and so more stable, because of the upward transport of heat by baroclinic eddies. Convection only directly determines the lapse rate over a small fraction of the tropics where convection actually occurs, but nevertheless the average lapse rate is close to moist neutral. This is because the gravity waves emanating from convective regions adjust the tropics to have weak horizontal temperature gradient, in a process akin to geostrophic adjustment, hence maintaining approximately the same vertical profile even away from regions of active convection, as discussed more in sections 18.8 and 18.9.

18.3.2 Equivalent potential temperature

In a dry atmosphere adiabatic motion is characterized by the material conservation of potential temperature, θ , a surrogate for entropy. Thus in adiabatic flow $D\theta/Dt = 0$, and the dry adiabatic lapse rate can be characterized equivalently either as $dT/dz = -g/c_p$ or $d\theta/dz = 0$. We can construct a similar quantity for moist air. Suppose that a parcel is lifted, and so cooled, until all its moisture condenses, and that all the latent heat released goes into heating the parcel. The *equivalent potential temperature*, θ_{eq} , is the potential temperature that the parcel then achieves.⁷ If the parcel is then moved along a dry adiabat to a reference pressure p_R (commonly 1000 hPa) the actual temperature it will then have is θ_{eq} . As a consequence of the near adiabatic nature of the process, θ_{eq} is an approximate measure of the entropy of the parcel, as we will show (see also Appendix A).

We may obtain an approximate analytic expression for θ_{eq} by noting that the first law of thermodynamics, $dQ = T d\eta$, implies, by definition of potential temperature,

$$-L dq = c_p T d \ln \theta, \quad (18.52)$$

during the condensation process, where dq is the change in water vapour content. Integrating gives, by definition of equivalent potential temperature,

$$-\int_q^0 \frac{L}{c_p T} dq = \int_\theta^{\theta_{eq}} d \ln \theta. \quad (18.53)$$

Here, q is the initial amount of water vapour contained in the parcel and T is the temperature at which condensation occurs. We might imagine lifting a parcel from its initial position to a temperature T at which saturation first occurs. To remove *all* the water vapour the parcel must be lifted to great height, because all of the water vapour will only condense if the final temperature is very low, with condensation occurring continuously and so with varying temperature along the way. But if we assume that temperature is constant during condensation, and that L and c_p are also constants, then (18.53) gives

$$\theta_{eq} = \theta \exp\left(\frac{Lq}{c_p T}\right) = T \left(\frac{p_R}{p}\right)^{R/c_p} \exp\left(\frac{Lq}{c_p T}\right). \quad (18.54)$$

The equivalent potential temperature so defined is approximately conserved during condensation, the approximation arising going from (18.53) to (18.54). It is a useful expression for diagnostic purposes and in constructing theories of convection, but it is not accurate enough to use as a primary prognostic variable in a numerical model that aims to be realistic. In saturated adiabatic flow a parcel will follow a ‘moist isentrope’ (an isoline of equivalent potential temperature) more closely than a dry isentrope.

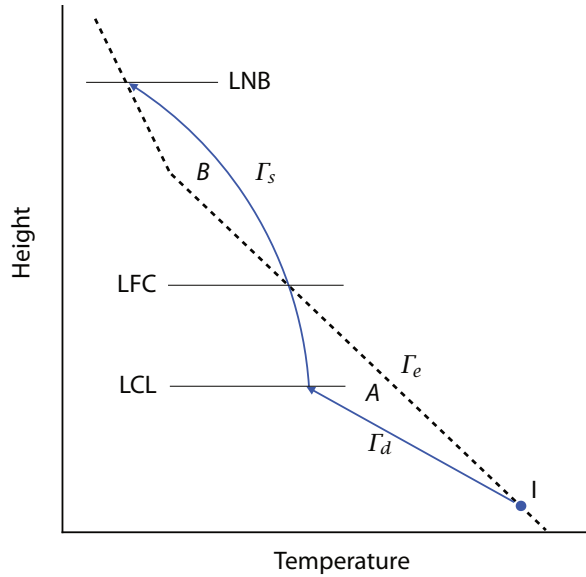


Fig. 18.16 Schematic of a conditional instability in an atmosphere with environmental lapse rate Γ_e (dashed line). A parcel at I is forced to rise, and it does so along the dry adiabat, Γ_d , until it is saturated at the lifting condensation level, LCL. It will then rise along the saturated adiabat, Γ_s , and after reaching the level of free convection (LFC) it is convectively unstable. The parcel continues to rise along the saturated adiabat, without any external forcing, until it reaches the level of neutral buoyancy (LNB). Not to scale.

To see that the equivalent potential temperature is an approximate measure of the entropy of a saturated parcel we begin with the entropy of dry air, which referring to (1.108) is

$$\eta^d = c_p \ln \theta = c_p \ln(T/T_0) - R \ln(p/p_0), \quad (18.55)$$

where T_0 and p_0 are constants. If we add to this a contribution from liquid water (see Appendix A on page 720 for a more complete treatment) then the saturation entropy, η_s , is approximately given by

$$\eta_s \approx \eta^d + \frac{Lq_s}{T} = c_p \ln(T/T_0) - R \ln(p/p_0) + \frac{Lq_s}{T}, \quad (18.56)$$

where q_s is the saturation specific humidity and final liquid water content. If we now define θ_{eq} by

$$c_p \ln(\theta_{eq}/T_0) \equiv \eta_s, \quad (18.57)$$

then using (18.56) and (18.57) we recover (18.54) with $q = q_s$ — meaning that logarithm of the equivalent potential temperature is, approximately, the entropy. To relate this result to the lapse rate, consider a parcel ascending into a colder region and condensing. The saturation moist entropy will stay the same: the last term on the right-hand side of (18.56) will fall, but the ensuing heating causes the temperature to increase. If we differentiate (18.56) with respect to z and set $\partial \eta_s / \partial z = 0$ then, using hydrostasy and a little algebra, we recover the saturated adiabatic lapse rate given by (18.50), which, therefore, has constant θ_{eq} and constant η_s .

18.4 CONVECTION IN A MOIST ATMOSPHERE

Over a wide range of temperatures and pressures the saturated adiabatic lapse rate is considerably less than the dry lapse rate (Fig. 18.15), and thus a moist atmosphere can be stable to dry convection but unstable to moist convection. If an environmental profile is unstable then convection ensues, transporting energy upward, until the atmosphere nearly stabilizes. But if the atmosphere is not saturated it is the dry adiabatic lapse rate is the relevant one (or, strictly, the adiabatic lapse rate of moist, noncondensing air, which is very slightly different). The observed environmental profile in convecting situations may thus be a combination of the dry adiabatic and saturated adiabatic profiles. An unsaturated parcel that is unstable by the dry criterion will rise and cool following a

dry adiabat, Γ_d , until it becomes saturated, above which it will rise following a saturated adiabat, Γ_s . The temperature then does not fall as rapidly as the dry case because the latent heat release warms the parcel as it rises.

These facts give moist convection a particular flavour and lead to the notion of *conditional stability*, whereby a parcel is stable to an infinitesimal perturbation but unstable to a finite perturbation. Consider an environment in which the lapse rate lies between the moist and dry rates, as in Fig. 18.16, and consider a parcel near the surface at position I, as in Fig. 18.16. Suppose the parcel is adiabatically lifted (perhaps mechanically) then its temperature profile follows the dry adiabat until it is saturated at, by definition, the *lifting condensation level* (LCL). However, the parcel is actually negatively buoyant and so would sink unless the parcel is forced to continue rising, but if it does continue to rise it will be along a saturated adiabat and eventually, at the *level of free convection* (LFC) it will become buoyant and convectively unstable. It will continue to rise until its buoyancy no longer exceed that of the environment, at the *level of neutral buoyancy* which may well be at or close to the tropopause where the temperature starts to increase again. (The height of the tropopause is not independent of the convection itself, a matter we discuss in section 18.6.) Evidently, to trigger such an instability a *finite* perturbation is needed (for example a large scale flow over a hill forcing a parcel upwards) and this is known as *conditional instability*. A parcel that is initially sitting at the LCL will only be unstable if it is saturated, which is by no means usually the case: the water has to get there somehow.⁸

Convection in the atmosphere need not be ‘conditional’, in the sense described above, nor is the atmosphere necessarily always in a conditionally unstable state. Indeed, on average the tropical atmosphere is closer to a moist profile than a dry profile. It is also important to remember that convection ultimately arises because the radiative forcing produces a vertical profile that is convectively unstable, with large-scale horizontal temperature gradients confining the convection to warmer regions, as in 18.14.

18.4.1 Energetics of convection

How much energy is available for a parcel in convection? One way is to calculate the work released by, or required for, a parcel as it moves through an environment that has a different density. The upward force per unit mass on a parcel of density ρ_p in an environment with density ρ_e is just the familiar buoyancy force,

$$F_b = g \frac{\rho_e - \rho_p}{\rho_p} = g \frac{\alpha_p - \alpha_e}{\alpha_e}. \quad (18.58)$$

Thus, the energy released as a parcel ascends from z_1 to z_2 is

$$\text{Energy} = \int_{z_1}^{z_2} g \frac{\alpha_p - \alpha_e}{\alpha_e} dz = - \int_{p_1}^{p_2} (\alpha_p - \alpha_e) dp, \quad (18.59)$$

if the environment is in hydrostatic balance. If we take the limits of integration to be the level of free convection and the level on neutral buoyancy, the energy is known as the *convective available potential energy*, or CAPE. Using the ideal gas relation, and assuming that the parcel has the same pressure as the environment, (18.59) becomes

$$\text{CAPE} = -R \int_{\text{LFC}}^{\text{LNB}} (T_p - T_e) \frac{dp}{p}. \quad (18.60)$$

If the height axis in Fig. 18.16 were log pressure then the CAPE would be proportional to the area *B*. If the limits were taken from the initial height to the LFC then the integral would be proportional

to the area A and would be negative, and is known as the ‘convective inhibition,’ or CIN. That is

$$\text{CIN} = -R \int_{p_{\text{bot}}}^{\text{LFC}} (T_p - T_e) \frac{dp}{p}, \quad (18.61)$$

and it is the amount of energy that must be supplied to initiate convection. In some accounts the definition of CAPE includes the CIN, in which case CAPE may be negative (but this is unusual). It is often the case that observed profiles of temperature in the tropics exhibit both CIN and CAPE, and a parcel in the lower atmosphere is stable and the instability is only conditional. That a finite perturbation is needed to initiate convection is a distinguishing feature of moist convection.

Given the initiation of convection, one may envision an unstable parcel of air experiencing a buoyant force and accelerating through the unstable region, gaining kinetic energy, and this energy can be quite significant. Suppose that $T_p - T_e = 1^\circ \text{C}$ and that the limits of integration are 250 hPa and 750 hPa. Then we estimate

$$\text{CAPE} \sim R(T_p - T_e) \frac{\Delta p}{p} = R \times 1 \times \frac{750 - 250}{500} = 286 \text{ J kg}^{-1}. \quad (18.62)$$

This is actually an underestimate, by a factor of a few, of the value of CAPE often found in tropical atmospheres, but nevertheless if we translate it to a vertical velocity using $w = 2\sqrt{\text{CAPE}}$ then we find $w \approx 24 \text{ m s}^{-1}$! In fact, CAPE is a significant overestimate of the kinetic energy that is imparted to a buoyant parcel: a parcel will not keep its identity because of mixing or entrainment with its surroundings and the environmental profile itself becomes altered, and even if the parcel did keep its identity it would give up some of its kinetic energy to the environment in pushing it out of the way. Nevertheless, the estimate gives the sense that atmospheric convection is a vigorous process, occurring on a fast timescale compared to large-scale dynamics. Let us suppose that the above estimate is too big by a factor of 10. The time taken for a parcel to travel half the height of the troposphere in the tropics is then approximately $7500 \text{ m} / 2.4 \text{ m s}^{-1} = 3000 \text{ s}$, or less than one hour. *Thus, the timescales of atmospheric convection are measured in hours, not days.*

18.4.2 Effects of convection

We have seen that, with even small differences in between an environmental profile and a stable profile, convection will be vigorous and act on rather fast timescales. What will the result of that convection be? Parcels will ascend until they are no longer unstable, at which point they can mix irreversibly their environment, warming the environment aloft. (Indeed the parcels will mix as they ascend, but they have more time to mix as the ascent slows.) The convection and the mixing will proceed until the environmental profile is no longer unstable, a process that, if it occurs on a short timescale, is called *convective adjustment* or, if on a longer timescale, *convective relaxation*. One might suppose that the profile to which the environment will adjust will be the saturated adiabatic or dry adiabatic lapse rates, depending on whether the atmosphere is saturated or not, although this is something of an oversimplification.⁹ The convection will also transfer moisture vertically, and if this transfer causes the profile to become saturated then condensation and precipitation may occur.

Let us denote the adjusted reference profiles of temperature and specific humidity as $T_r(z)$ and $q_r(z)$ (or equivalents in pressure). If the vertical structure of these profiles is given, the magnitude of the profile will be determined by the fact that convection will conserve enthalpy. If, for example, atmosphere is far from saturated then the reference profiles will satisfy

$$\int_{p_B}^{p_T} c_p (T_r - T_i) dp = 0 \quad \text{and} \quad \int_{p_B}^{p_T} (q_r - q_i) dp = 0, \quad (18.63)$$

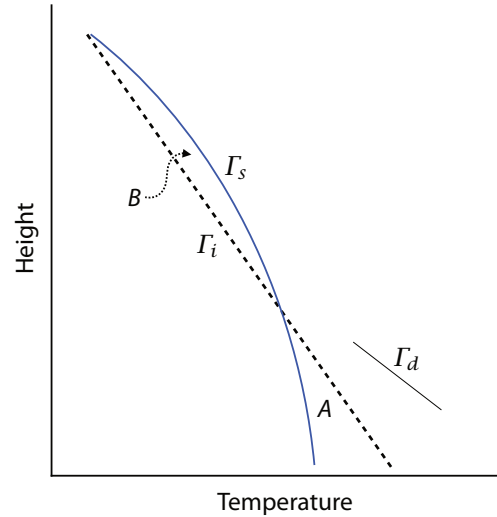


Fig. 18.17 Schematic of convective adjustment in a saturated atmosphere. The initial environment, with profile Γ_i , is unstable to moist convection and adjusts to the saturated adiabatic lapse rate Γ_s , with the dry adiabat Γ_d shown for reference. The average temperature of the final profile is such that the enthalpy is conserved, which provides a relationship between the areas A and B.

where the subscript i denotes the initial environmental profile and p_B and p_T denote the levels at the base and top of the convection. If precipitation does occur then only the total enthalpy is conserved and we have

$$\int_{p_B}^{p_T} (h_r - h_i) dp = 0. \quad (18.64)$$

where $h = c_p T + Lq$ is the (moist) specific enthalpy. One then needs a second constraint to determine the profiles of T_r and q_r separately.

The specification of these profiles, and the levels of the base and the top of the convection, is not an exact science since the notion of adjustment is an approximation. As a first estimate in a moist atmosphere one might suppose that T_r is the dry adiabat below saturation and the saturated adiabatic lapse rate above, and that the reference profile of q_r is such that the atmosphere remains saturated, or nearly so, after adjustment, and that p_B and p_T are the levels of free convection and of neutral buoyancy, respectively. Application of (18.64) then uniquely determines the final profiles of temperature and humidity, as schematically illustrated for a saturated atmosphere in Fig. 18.17. Details apart, two robust, key points should be emphasized.

- (i) To a first approximation, convection determines the *profiles* of temperature and humidity rather than the actual heating rate or the precipitation. The heating and precipitation released by convection is a function of the environmental profile that in turn is largely determined by the larger-scale processes, although convection feeds back onto this profile.
- (ii) After convection has occurred the local profile is, by construction, stable. However, this profile is not necessarily in equilibrium with the large-scale motion, or with the radiative processes that initially gave rise to the unstable profile. This disequilibrium may cause the profile to evolve further, possibly giving rise to more convection.

The above state of affairs — a putative initial profile set up by radiation and/or large-scale dynamics, modified quickly by convection, leading to a further slow evolution and convection — is called *quasi-equilibrium*.

18.4.3† Quasi-equilibrium

Convective quasi-equilibrium is a posited state in which the forcing of a convectively unstable profile by large-scale dynamics and/or radiation is statistically balanced by convection.¹⁰ The large-scale forcing may change, and the system may evolve, but in quasi-equilibrium the convection is

Convection, Quasi-Equilibrium and Radiative-Convective Equilibrium

- Atmospheric convection ultimately arises because the radiative forcing tends to produce temperature profiles, and hence buoyancy profiles, that are statically unstable. Large-scale advection, moving cold air over warm air, also leads to convection.
- Convection is most prevalent in the tropics because it is here that the surface is warmest, and the radiative-equilibrium lapse rate would be most unstable (section 18.5). This would be true even in the absence of moisture, but moisture is most important in the tropics because it is warmest, and because water is imported via the Hadley Cell.
- The presence of moisture may cause the critical lapse rate for convection to be lower than that for dry air, because of the release of heat when water vapour condenses. The effect only arises if a parcel is saturated. Thus, a column of air may be *conditionally unstable* if it requires a finite perturbation to raise a parcel to a level where it saturates (Fig. 18.16).
- Atmospheric convection tends to occur on much faster timescales than those associated with the large-scale circulation. Once convection occurs the lapse rate is closely constrained to the critical lapse rate in the regions of convection.
- In a process analogous to geostrophic adjustment, internal waves propagating away from convection maintain weak horizontal temperature gradients and ensure that the lapse rate does not deviate too far from saturated adiabatic anywhere in the tropics (Fig. 15.24). In midlatitudes, in contrast, energy is also transported upward by large-scale baroclinic eddies and the lapse rate is lower (more stable).
- When the large-scale dynamical or radiative forcing changes, the convective flux changes to constrain the lapse rate, and the production of CAPE by the large scale is *approximately* balanced by the relaxation of CAPE by convection. This state of affairs is called quasi-equilibrium, and its occurrence is really a hypothesis that appears to be fairly well-satisfied, if the larger timescales are sufficiently long, in Earth's tropical atmosphere.
- Quasi-equilibrium, if and where closely satisfied, leads to considerable simplifications regarding the dynamics of large-scale flow for it de-emphasizes the need to explicitly calculate vertical energy fluxes because the lapse rate is constrained. Equations resembling the shallow-water equations, with a small number of vertical modes, then become a decent approximation for large-scale baroclinic flow, with diabatic effects forcing the first baroclinic mode.
- A radiative-convective equilibrium state arises when the radiative-equilibrium state is convectively unstable state. Convection then balances the radiative forcing, producing at first approximation a convectively active state up to some finite height, with a radiative equilibrium state beyond (Fig. 18.18).
- The height to which the constrained lapse rate extends is determined by the need to maintain an overall radiative balance — the incoming solar radiation must equal the outgoing infrared radiation — and this is the leading-order determinant of the tropopause height (section 18.6.2). A similar calculation could be applied in mid-latitudes, with a lapse rate determined as much or more by baroclinic instability as by convection.
- Quasi-equilibrium is a useful idealization when convection occurs quickly and efficiently, but it does not mean that the large scale always controls the convection. There are many situations when there is no causal separation between convection and the large scale — indeed the convection may create its own large scale flow, the MJO being one example.

assumed to occur on a faster timescale than the other processes and the fluid quickly enters into a statistical equilibrium state. Even as the large-scale evolves, the vertical profile at each step is largely determined by convection. That such a state exists is an assumption, but in Earth's tropical atmosphere it seems a fairly good one when dealing with timescales of longer than a few days, because convective timescales are relatively short and so convection is able to significantly alter the stratification. The assumption of quasi-equilibrium is marginally satisfied over the diurnal cycle.

The most obvious consequence of quasi-equilibrium is that the temperature profile is constrained to be close to being neutrally stable, which generally means a saturated-adiabatic profile except possibly in some regions in the lower atmosphere where the appropriate profile may dry adiabatic, and in the boundary layer where properties are well mixed. A second important consequence concerns CAPE, convective available potential energy. Quasi-equilibrium implies that the CAPE is released by convection at about the same rate as it is generated by the forcing. Thus, at the same time as radiative forcing may be creating an unstable profile with a finite amount of CAPE, convection is destroying that potential energy, largely by converting it to kinetic energy where it is dissipated. The amount of CAPE in a profile may (and does) vary on slow timescales, the variation being produced by the weak imbalance between production and destruction; observations indicate it can vary by a factor of a few over the course of many days, and the environmental profile may also depart from the saturated adiabat away from convectively active regions. However, it turns out that the *average* stratification over the tropics will not deviate too much from moist neutral because internal waves extend the moist-neutral profile in the convective regions to the cloud-free regions, and horizontal pressure and temperature gradients are constrained to be relatively small. This 'weak temperature gradient' effect is described in section 18.8.

Finally, albeit a little less tangibly, quasi-equilibrium gives rise to a *point of view* of tropical dynamics, that convection does not act as a primary determinant of the heat source for the large-scale circulation. Rather, convection controls the temperature and water vapour profiles, with the heating associated with the convection being determined by the requirement that such a profile be maintained, and thus ultimately arising from the larger scale forcing. (This process is analogous to the production of condensation in the fast-condensation models of section 18.2, in which the condensation criteria determine the profile of water vapour but not the amount of condensation.) The resulting environmental profile then lies on the edge of stability, with a non-zero CAPE because of the finite time-scale and finite efficiency of convection and with the intensity of convection being just what is required to maintain that profile. The value of CAPE then evolves slowly, compared to convective timescales, if and as external factors change.

Having said this, quasi-equilibrium should not be thought of as being exact or even especially profound — it is a way to gain an understanding of the structure of the tropics but is not a universal recipe. There are many situations — the diurnal cycle, aspects of boundary layer development, weak convection — in which the quasi-equilibrium assumptions do not hold. Furthermore, the recipe may mislead if misapplied: even when quasi-equilibrium holds in so far as the buoyancy profile is constrained to be moist neutral, it should not be thought that convection and the associated condensation is always a quasi-passive process. That is, the scale separation between convective events and the large-scale does not mean that the large-scale always 'controls' the convection and the convection merely 'feeds back' on the large scale. For example, a localised source of convection near the equator may itself create a large-scale flow pattern similar to that of the Matsuno–Gill problem (section 8.5), and this effect may be at the heart of the Madden–Julian oscillation (section 18.10). Here one might say the convection controls the large-scale and the large-scale feeds back on the convection! Also, in the moist model of the Hadley Cell (section 14.2.7) the release of latent heat in the upward branch serves to change the horizontal distribution of heating and greatly intensify the overturning circulation. The convection should not be thought of as the primary driver of the Hadley Cell, but it has a lowest-order effect.

Caveats aside, quasi-equilibrium is a very useful concept and in following sections we look at two particularly important consequences. First (albeit after an introduction to radiative equilib-

rium) we discuss radiative-convective equilibrium, and we then use quasi-equilibrium to simplify the equations of motion for the larger-scale circulation.

18.5 RADIATIVE EQUILIBRIUM

In order to understand the effects of convection we must first determine what the profile of temperature would be in its absence — the *radiative-equilibrium* state. The electromagnetic radiation in the Earth's atmosphere may usefully be divided into two types, solar (or shortwave) radiation and infra-red (or longwave) radiation. We will assume the atmosphere is semi-grey (i.e., grey in the infra-red) and, in some instances, transparent to solar radiation. Neither assumption is quantitatively good, but they capture the essence. In Appendix B (page 724) we show that the upward, U , and downward, D , streams of longwave radiation then satisfy the radiative-transfer equations,

$$\frac{dD_L}{d\tau} = B - D_L, \quad \frac{dU_L}{d\tau} = U_L - B. \quad (18.65a,b)$$

Here, $B = \sigma T^4$ is the radiative flux emitted by a black body, τ is the optical depth and σ is Stefan's constant. The optical depth is related to the geometric height z by $d\tau = -e_L dz$ where e_L is the longwave emissivity. The net flux of longwave radiation is $N_L = U_L - D_L$ and the longwave heating is proportional to the net flux divergence, $-\partial N_L / \partial z$.

18.5.1 Solutions

Formal solution to radiative transfer equations

If the temperature profile is known then B is known and (18.65) is a pair of first order differential equations in the two unknowns U and D . The solution requires two boundary conditions and in atmospheric problems these might be provided at the top of the atmosphere, for example by requiring that the downward infrared radiation be zero (i.e., $D_L(\tau = 0)$), and by specifying the upward radiation (i.e., let $U_L(\tau = 0) = U_0$ where U_0 is given). Alternatively, the upward radiation at the bottom of the atmosphere could be specified if the ground temperature were known. The solution in physical space is found by specifying the form of $\tau(z)$; that is, by specifying the emissivity. To obtain the solution we multiply (18.65) by the integrating factors $\exp(\tau)$ and $\exp(-\tau)$ to give

$$\frac{d}{d\tau}(D_L e^\tau) = B e^\tau, \quad \frac{d}{d\tau}(U_L e^{-\tau}) = -B e^{-\tau}. \quad (18.66a,b)$$

Integrating between 0 and τ' straightforwardly gives

$$D_L(\tau') = e^{-\tau'} \left[D_L(0) - \int_0^{\tau'} B(\tau) e^\tau d\tau \right], \quad U_L(0) = U_L(\tau') e^{-\tau'} + \int_0^{\tau'} B(\tau) e^{-\tau} d\tau. \quad (18.67a,b)$$

There are other ways to write the solutions that may be appropriate depending on the boundary conditions, but in any case the solutions are in general *nonlocal*, for they depend on the temperature along the path. The terms in (18.67) represent the attenuation of radiation as it travels along its path as well as the cumulative emission. However, in some important special cases we can get a local solution, as we now see.

Radiative equilibrium solution

A *radiative equilibrium* state has, by definition, no radiative heating. If the atmosphere is transparent to solar radiation then the condition implies that the vertical divergence of the longwave radiation is zero:

$$\frac{\partial(U_L - D_L)}{\partial z} = 0 \quad \text{implying} \quad \frac{\partial(U_L - D_L)}{\partial \tau} = 0. \quad (18.68a,b)$$

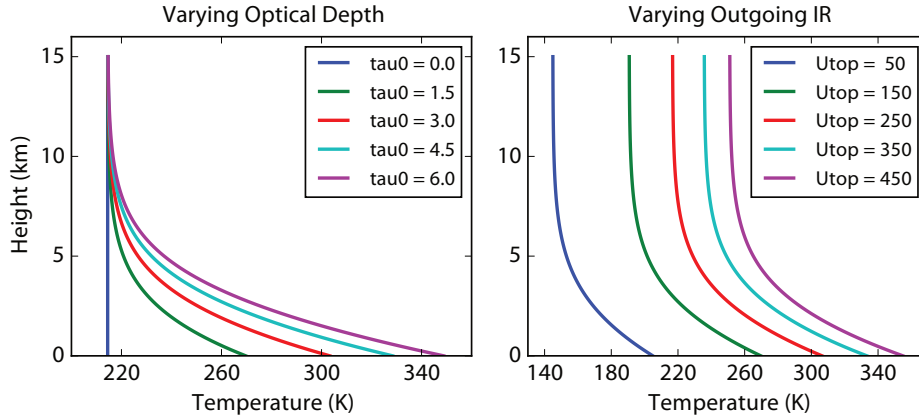


Fig. 18.18 Radiative equilibrium temperature calculated using (18.73), with $H_a = 2$ km. Left: incoming solar radiation is 240 W m^{-2} and surface optical depth varies from 0 to 6. Right: the optical depth is 3.0 and outgoing IR radiation varies from 50 to 450 W m^{-2} .

This condition is normally *not* satisfied in the atmosphere because the air is in motion. If it were satisfied then (18.65) and (18.68b) form three equations in three unknowns, U_L , D_L , and B , and a solution can be found as follows.

Consider an atmosphere with net incoming solar radiation S_{net} and suppose the planet is in radiative equilibrium with the incoming solar radiation balanced by outgoing infra-red radiation. That is, $U_{Lt} \equiv U_L(\tau = 0) = S_{net}$ where U_{Lt} is the net outgoing longwave radiation (OLR) at the top of the atmosphere. The downward infrared radiation at the top of the atmosphere is zero, so that the boundary conditions on the radiative transfer equations at the top of the atmosphere are

$$D_L = 0, \quad U_L = U_{Lt} \quad \text{at} \quad \tau = 0. \quad (18.69)$$

To obtain a solution we rewrite (18.65) as

$$\frac{\partial}{\partial \tau}(U_L - D_L) = U_L + D_L - 2B, \quad \frac{\partial}{\partial \tau}(U_L + D_L) = U_L - D_L. \quad (18.70a,b)$$

and a little algebra reveals that a solution of these equations that satisfies (18.68b) is

$$D_L = \frac{\tau}{2} U_{Lt}, \quad U_L = \left(1 + \frac{\tau}{2}\right) U_{Lt}, \quad B = \left(\frac{1 + \tau}{2}\right) U_{Lt}, \quad (18.71a,b,c)$$

as can be easily verified by substitution back into the equations.

It remains to explicitly relate τ to z , and one approximate recipe is to suppose that τ has an exponential profile,

$$\tau(z) = \tau_0 \exp(-z/H_a), \quad (18.72)$$

where τ_0 is the optical depth at $z = 0$ and H_a is the scale height of the absorber. In the Earth's atmosphere the optical depth is determined by the concentrations of water vapour (primarily) and carbon dioxide (secondarily) and τ_0 (the scaled optical depth) typically varies between 2 and 4, depending on the water vapour content of the atmosphere, and $H_a \approx 2$ km, this being a typical scale height for water vapour. From (18.71c) the temperature then varies as

$$T^4 = U_{Lt} \left(\frac{1 + \tau_0 e^{-z/H_a}}{2\sigma} \right), \quad (18.73)$$

which is illustrated in Fig. 18.18. The following aspects of the solution deserve mention.

- (i) Temperature increases rapidly with height near the ground.
- (ii) The upper atmosphere, where τ is small, is nearly isothermal.
- (iii) The temperature at the top of the atmosphere, T_t is given by

$$\sigma T_t^4 = \frac{U_{Lt}}{2} \quad (18.74)$$

Thus, if we define the emitting temperature, T_e , to be such that $\sigma T_e^4 = U_{Lt}$, then $T_t = T_e/2^{1/4} < T_e$; that is, the temperature at the top of the atmosphere is *lower* than the emitting temperature.

- (iv) Related to the previous point, $B_t/U_{Lt} = 1/2$. That is, the upwards long wave flux at the top of the atmosphere is twice that which would arise if there were a black surface at a temperature T_t . The reason is that there is radiation coming from all heights in the atmosphere.

Interestingly, in obtaining a solution we have not imposed a boundary condition at the ground — in fact there is no ground at all in this problem! What happens if we add one? That is, suppose that we declare that there is a black surface at some height, say $z = 0$, and we require that the atmosphere remain in radiative equilibrium with the same temperature profile. What temperature does the ground take? From (18.71) the upward irradiance and temperature at any height z are related by

$$U_L(z) = \left(\frac{2 + \tau(z)}{1 + \tau(z)} \right) \sigma T^4(z). \quad (18.75)$$

At $z = 0$ the ground will have to supply upwards radiation equal to that given by (18.75), and therefore its temperature, T_g is given by

$$\sigma T_g^4 = \left(\frac{2 + \tau_0}{1 + \tau_0} \right) \sigma T_s^4, \quad (18.76)$$

where T_s is the temperature of the fluid adjacent to the ground (the ‘surface temperature’). That is, $T_g > T_s$ and there is a temperature discontinuity at the ground, especially if optical depth is small. In fact, in very still and clear conditions a very rapid change of temperature near the ground can sometimes be observed, but usually the presence of conduction and convection, as we discuss below, ensure that T_g and T_s are equal.

We note that in the limit in which $\tau = 0$ in the upper atmosphere we have

$$D_L = 0, \quad U_L = U_{Lt}, \quad B = \frac{U_{Lt}}{2}. \quad (18.77)$$

That is, the upper atmosphere is isothermal, there is no downwelling irradiance and the upward flux is constant. The upper-atmosphere temperature, T_{ua} say, and the emitting temperature are related by $T_{ua} = T_e/2^{1/4}$.

The above results imply that at a given optical depth, the temperature difference between the surface and the ground increases with temperature. From (18.73) we have that

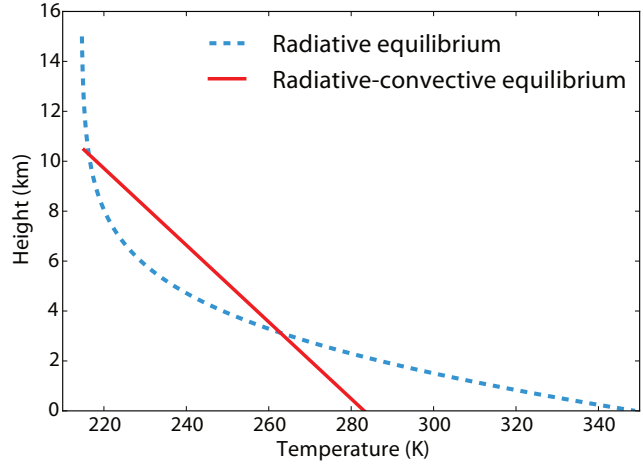
$$T_t^4 = \frac{U_{Lt}}{2\sigma}, \quad T_s^4 = \frac{U_{Lt}(1 + \tau_0)}{2\sigma} \quad (18.78)$$

whence

$$T_s^4 - T_t^4 = T_t^4 \tau_0 \quad \text{or} \quad \Delta T \approx \frac{\tau_0}{4} T_t, \quad (18.79)$$

where $\Delta T = T_s - T_t$ and we assume $\Delta T \ll T_s, T_t$. Thus, in the absence of other effects, higher temperatures lead to larger average lapse rates, and so are potentially more conducive to convection.

Fig. 18.19 Radiative and radiative-convective equilibrium profiles. The initial radiative equilibrium temperature adjusts to a specified profile (in this instance a constant lapse rate) that extends to a finite height, beyond which radiative equilibrium holds. This height (the tropopause) is determined by the requirement of radiative balance, and the overall adjustment process is not adiabatic. The curves are results of numerical calculations with $\tau_0 = 6$, $\Gamma = 6.5 \text{ K km}^{-1}$ and outgoing radiation of 240 W m^{-2} (see section 18.6.2).



18.6 RADIATIVE-CONVECTIVE EQUILIBRIUM

In the radiative equilibrium solution the temperature gradient near the ground varies so rapidly that $-\partial T/\partial z$ may exceed even the dry adiabatic lapse rate. If so, the radiative equilibrium solution is convectively unstable and convection will be triggered and energy (and moisture) will be redistributed throughout the column. What is the resulting temperature profile?

18.6.1 ♦ The general case

We first approach the problem without making any simplifying quasi-equilibrium assumptions, supposing only that we have a balance between longwave and solar radiation and the effects of convection. Let us assume that upward and downward shortwave fluxes, U_S and D_S , obey the two-stream radiative equations with no thermal emission and no scattering, namely

$$\frac{dD_S}{d\tau_S} = -D_S, \quad \frac{dU_S}{d\tau_S} = U_S. \quad (18.80a,b)$$

Here, τ_S is the shortwave optical depth, which is related to the physical height by $d\tau_S = -e_S dz$, where e_S is the shortwave emissivity. and we can then write the radiative equations as

$$\frac{dD_S}{d\tau_L} = -\frac{e_S}{e_L} D_S, \quad \frac{dU_S}{d\tau} = \frac{e_S}{e_L} U_S, \quad (18.81a,b)$$

The solar heating is proportional to the convergence of the net solar fluxes, $-\partial N_S/\partial z$ where $N_S = U_S - D_S$. The convection also provides a local heating, and this is proportional to the convergence of the upwards enthalpy flux, $-\partial \mathcal{H}/\partial z$. Thus, in a steady state,

$$\frac{\partial}{\partial z}(N_S + N_L + \mathcal{H}) = 0 \quad \text{implying} \quad \frac{\partial}{\partial \tau}(N_S + N_L + \mathcal{H}) = 0. \quad (18.82a,b)$$

Using (18.70a) we write (18.82b) as

$$U_L + D_L - 2B + \frac{1}{e_L} Q_S + \frac{\partial \mathcal{H}}{\partial \tau} = 0, \quad (18.83)$$

where Q_S is the solar heating given by $Q_S = -\partial N_S/\partial z = e_L^{-1} \partial N_S/\partial \tau$. If we had a theory for the enthalpy fluxes $\mathcal{H}$ then (18.83), in conjunction with the Schwarzschild equations and knowledge of the solar radiation and atmospheric emissivity, would determine the temperature profile with height. We have no such theory, but we can still make progress.

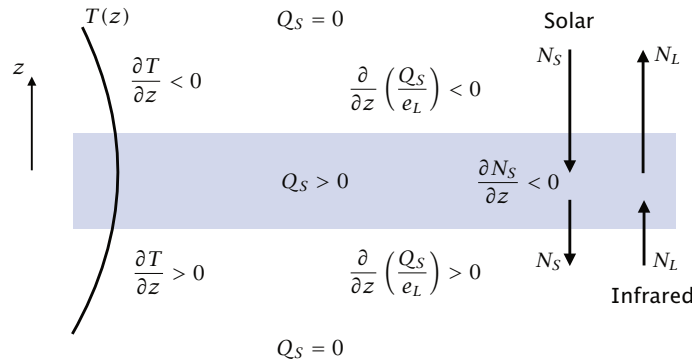


Fig. 18.20 Effects of a band of solar radiation absorption. Shading marks the absorbing region where $\partial N_s/\partial z < 0$, with consequences for the heating, Q_s , and temperature profile, $T(z)$, determined by (18.86b), as indicated.

Effects of solar fluxes

Suppose first (and taking a diversion from convection) that the enthalpy fluxes are zero, much as in the stratosphere. We then have a radiative equilibrium profile then satisfies

$$\frac{\partial}{\partial z}(N_L + N_s) = 0, \quad (18.84)$$

implying

$$\frac{\partial N_L}{\partial z} = Q_s \quad \text{or} \quad \frac{\partial N_L}{\partial \tau} = -\frac{Q_s}{e_L}. \quad (18.85a,b)$$

If we differentiate (18.85b) with respect to τ and use (18.70a,b) we obtain

$$2\frac{\partial B}{\partial \tau} = N_L + \frac{\partial}{\partial \tau}\left(\frac{Q_s}{e_L}\right) \quad \text{or} \quad 8\sigma T^3 \frac{\partial T}{\partial z} = -e_L N_L + \frac{\partial}{\partial z}\left(\frac{Q_s}{e_L}\right). \quad (18.86a,b)$$

The middle of the stratosphere in Earth's atmosphere contains a layer of ozone that absorbs solar radiation. Thus, $\partial N_s/\partial z < 0$ and Q_s is positive. The net longwave radiation N_L is positive in this region, as there is much more upwelling radiation than downwelling, and this tends to produce a negative vertical temperature gradient, but the effect is weak because e_L is small. However, the solar radiation can have a strong effect because e_L appears in the *denominator* of the term involving Q_s . Just below the heating region $\partial Q_s/\partial z$ is positive and this produces a positive value of $\partial T/\partial z$, with the converse above the ozone layer. Thus, region of solar heating corresponds to a maximum of temperature, or at least a region of small vertical temperature gradient, with negative values of $\partial^2 T/\partial z^2$, as schematically illustrated in Fig. 18.20. The result is intuitively reasonable, but (18.86) provides the explicit solution. A non-intuitive result that follows from (18.86b) concerns the effects of an increasing infra-red emissivity, such as happens with global warming. Suppose that $\partial T/\partial z > 0$, as in the stratosphere because of the presence of a layer of ozone, and suppose further that e_L then increases, because of an increase concentration of greenhouse gases. Changes in the longwave term are small, because e_L is small, but changes in the solar term are large, and these cause $\partial T/\partial z$ to diminish. That is, an increase in the concentration of greenhouse gases will cause stratospheric temperatures to *fall*.¹¹ The result arises because of the balance, in this case, between solar heating and infrared cooling, as in (18.85a) and Fig. 18.20. If the emissivity increases there will still be the same infrared loss to space, to achieve radiative balance, and if the emissivity is higher the loss can be achieved at a lower temperature.

Effects of enthalpy fluxes

Now consider the effects of an enthalpy flux, supposing that the atmosphere is transparent to solar radiation. The equilibrium condition, (18.83), becomes

$$U_L + D_L - 2B + \frac{\partial \mathcal{H}}{\partial \tau} = 0. \quad (18.87)$$

Differentiating with respect to τ and using (18.70b) gives

$$2 \frac{\partial B}{\partial \tau} = N_L + \frac{\partial^2 \mathcal{H}}{\partial \tau^2}. \quad (18.88)$$

Now, from (18.82a) we must have $N_S + N_L + \mathcal{H} = 0$ for all z , if the atmosphere is in an overall radiative balance and if $\mathcal{H} = 0$ at the atmosphere's top. If the solar flux is constant with height then, also for all z ,

$$N_L = S_{net} - \mathcal{H}, \quad (18.89)$$

where S_{net} is the net incoming solar radiation, a positive quantity. Using this expression (18.88) becomes

$$2 \frac{\partial B}{\partial \tau} = S_{net} - \mathcal{H} + \frac{\partial^2 \mathcal{H}}{\partial \tau^2} \quad \text{or} \quad \frac{8\sigma T^3}{e_L} \frac{\partial T}{\partial z} = -S_{net} + \mathcal{H} - \frac{1}{e_L} \frac{\partial}{\partial z} \left(\frac{1}{e_L} \frac{\partial \mathcal{H}}{\partial z} \right). \quad (18.90a,b)$$

These expression explicitly tell us how the lapse rate is affected by enthalpy fluxes. If we knew $\mathcal{H}$ we could integrate (18.90) to give us the temperature at every level, and if $\mathcal{H} = 0$ the equations are equivalent to the derivative of (18.71c). Enthalpy fluxes are usually positive (i.e., upwards) in the atmosphere and so they tend increase $\partial T/\partial z$ and reduce the lapse rate; that is, they tend to warm the upper atmosphere and cool the lower atmosphere and the surface. Since we do not have a quantitative theory for these fluxes let us invoke some ideas of quasi-equilibrium and specify the resulting lapse rate rather than the fluxes themselves.

18.6.2 Convective adjustment and the height of the tropopause

Let us assume that convection occurs with sufficient efficiency so that it will establish a convectively neutral lapse rate, Γ say, which for simplicity we here assume is constant. We may also assume that the convection equalizes the ground temperature and the surface temperature (the temperature layer of air immediately above the ground). Such a model simplifies the problem enormously, for we avoid completely the problem of solving the radiative transfer equations. However, radiation is still present, and we must still satisfy an overall radiative balance, and this determines the height to which the convection extends — which is, effectively, the height of the tropopause. In the first instance we can suppose that the convection occurs quickly and adiabatically, so that the re-arrangement of the temperature profile conserves energy. However, such a profile will not necessarily be a solution of the radiative transfer equations (18.67) and the profile must adjust until a solution is found. If the lapse rate is fixed, the only degree of freedom is the height to which convection reaches — that is, the height of the tropopause — above which radiative equilibrium holds.

The computation of this height follows a straightforward algorithm. If the top of the convecting layer occurs at a height, H_T , at which the optical depth is small, then outgoing long-wave radiation there will be approximately equal to the outgoing radiation at the top of the atmosphere, $U_L(\tau = 0)$, which is equal to the incoming solar radiation and therefore known. This gives the tropopause temperature (from (18.76), namely $T_T^4 = U_L(\tau = 0)/2\sigma$) and therefore, if we know H_T and the lapse rate, the temperature at all heights. We can then calculate the upwards radiative flux using a variant of (18.67b), which we may write as

$$U_L(z = 0) = U_L(z = H_T) e^{\tau_s} + \int_0^{H_T} B(\tau) e^{\tau_s - \tau} \frac{d\tau}{dz} dz, \quad (18.91)$$

where τ_s is the optical depth at the surface. However, for an arbitrary tropopause height the upwelling radiation at the bottom, $U_L(z = 0)$, as given by (18.91), will *not* equal σT_s^4 , which it must do if radiative balance is to be satisfied. The height of the tropopause must therefore adjust, and an iterative algorithm for finding the equilibrium solution goes as follows.

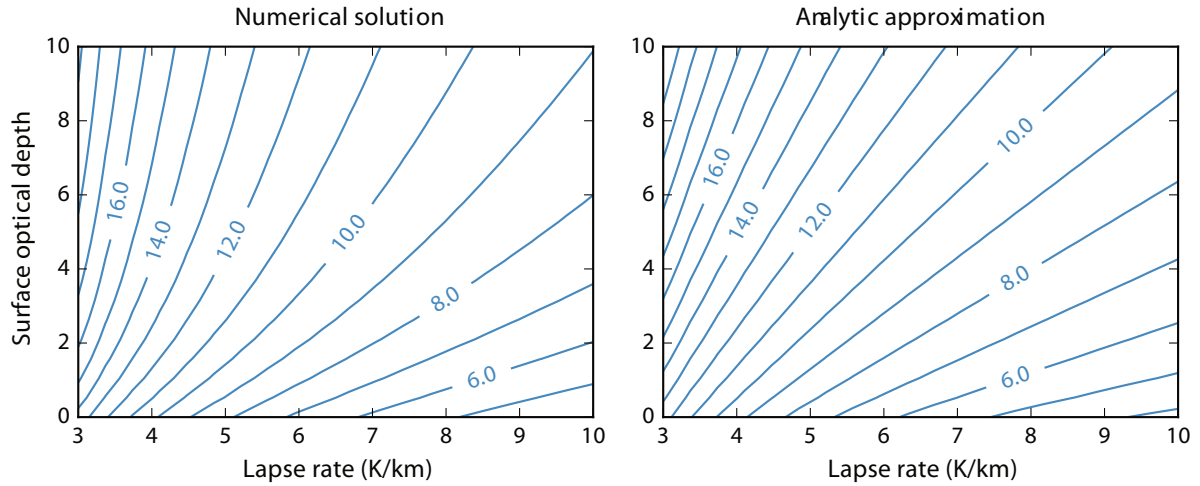


Fig. 18.21 Contours of tropopause height (km) as a function of lapse rate and surface optical depth in a grey atmosphere, calculated numerically by iterating the Schwarzschild equations (left) and using an analytic approximation (right), with $H_a = 2\text{ km}$ and outgoing longwave radiation of 242 W m^{-2} , or tropopause temperature of 215 K .

- (i) Obtain the radiative equilibrium temperature profile.
- (ii) Make a guess for the height of the tropopause, and using the given lapse rate obtain the temperature all the way down to the ground.
- (iii) Integrate the radiative transfer equations, (18.91), down from the top. The outgoing radiative balance is achieved this way but there is no balance at the surface if temperature is continuous. That is, $\sigma T_s^4 \neq U_L(z=0)$.
- (iv) Change the height of the tropopause, find another solution, and iterate until the surface radiative balance is properly achieved.

A profile of temperature so calculated, along with the radiative-equilibrium temperature, are shown in Fig. 18.19, and solutions as a function of lapse rate and temperature are given in Fig. 18.21.

18.6.3 Approximate analytic solution

In Appendix C (page 725) we show that an approximate analytic solution of the above algorithm for the height of the tropopause, H_T , is

$$8\Gamma H_T^2 - C H_T T_T - \tau_s H_a T_T = 0. \quad (18.92)$$

or

$$H_T = \frac{1}{16\Gamma} \left(C T_T + \sqrt{C^2 T_T^2 + 32\Gamma \tau_s H_a T_T} \right), \quad (18.93)$$

where $C = \log 4 \approx 1.4$, Γ is the lapse rate, T_T is the temperature at the tropopause, τ_s is the surface optical depth and H_a is the scale height of the main infra-red absorber. For Earth's atmosphere, $H_a \approx 2\text{ km}$, $\tau_s \approx 5$ and $\Gamma \approx 6.5\text{ K km}^{-1}$. All three terms in (18.92) are then approximately the same size and (18.93) gives $H_T \approx 11\text{ km}$. We can now be a little more precise about what it means for an atmosphere to be optically thin or thick. Using (18.93) and approximating $C^2 = 2$ we see that the optically thick limit arises when

$$\tau_s H_a \gg \frac{T_T}{16\Gamma} \quad \text{whence} \quad H_T \approx \sqrt{\frac{T_T \tau_s H_a}{8\Gamma}}. \quad (18.94)$$

The optically thin case has

$$\tau_s H_a \ll \frac{T_T}{16\Gamma} \quad \text{whence} \quad H_T \approx \frac{CT_T}{8\Gamma}. \quad (18.95)$$

With parameters appropriate for Earth's atmosphere both of the above limits give estimates in the range 6–15 km, and they are additive effects. Plots of the tropopause height as a function of lapse rate and optical depth, calculated using (18.93), as well as numerical solutions using the algorithm of the previous subsection, are given in Fig. 18.21, and the agreement is fairly good in Earth's parameter regime. A robust result is that as the lapse rate diminishes ($-\partial T/\partial z$ becomes smaller) the tropopause height increases, essentially to maintain the same tropopause temperature.

The main quantitative deficiency of this argument is that the atmosphere is not grey — the absorption of radiation is a function of wavelength. A less severe approximation is to suppose that the infra-red radiation occurs in two bands — a 'window' region and the remainder. In the window region the atmosphere is fairly transparent, meaning that a fraction ($\sim 1/4$ or $1/3$) of the radiation emitted by the surface goes straight to space. This has the effect of slightly decoupling the tropopause temperature from the effective emitting temperature. Nonetheless, the general arguments leading to (18.93) remain valid and the way in which the tropopause height varies with optical depth and stratification will carry through in the more realistic case.¹²

Effects of lateral energy transport and application to mid-latitudes.

Although we have set the above calculation in the context of convection, it applies in any situation where the lapse rate can be specified; the argument would equally well apply if the lapse rate were set by baroclinic instability. We can also apply a modified argument even when there is a horizontal transport of energy, for such a transport effectively just acts to change the radiative emitting temperature. If the lateral convergence of energy and the incoming solar radiation are known then the amount of radiation that the column must emit to space in order to maintain energy balance is easily calculated. For example, if there is a convergence of energy at high latitudes because of the transport of energy by baroclinic eddies, then the column needs to emit less infra-red radiation to space that it would otherwise in order to maintain radiative equilibrium. In actuality, in mid-latitudes the lapse rate cannot be regarded as being specified independently of the lateral heat transport, and the determination of the lapse rate and the tropopause height become intertwined. The reader is referred back to section 15.5 to follow this argument.

18.7 VERTICALLY-CONSTRAINED EQUATIONS OF MOTION FOR LARGE SCALES

Let us now segue toward the large scale dynamics of the tropics. We first show how having a nearly constant lapse rate constrains the vertical degrees of freedom, so reducing the equations of motion to something akin to the shallow water equations, and the readers may wish to skim 3.4 before continuing.

18.7.1 Reduction of vertical degrees of freedom

The assumption that convection maintains a moist-adiabatic profile everywhere constrains the vertical structure of the horizontal temperature gradient and, at least in so far as the momentum dynamics are linear, reduces the equations of motion governing the large-scale to a set similar to the shallow-water equations.¹³ To see this we begin with hydrostasy in pressure coordinates applied to the fluctuating fields,

$$\frac{\partial \phi'}{\partial p} = -\alpha', \quad (18.96)$$

where ϕ' and α' are variations in the geopotential and the specific volume. The latter be taken to be a function of the state variables pressure and saturation entropy, η (dropping the subscript s). Variations in α at constant pressure thus obey

$$\alpha' = \left(\frac{\partial \alpha}{\partial \eta} \right)_p \eta' = \left(\frac{\partial T}{\partial p} \right)_\eta \eta', \quad (18.97)$$

using one of Maxwell's equations (page 19). If we integrate (18.96) in the vertical and use (18.97) we obtain

$$\phi'(x, y, p, t) = - \int \alpha \, dp = \int \left(\frac{\partial T}{\partial p} \right)_\eta \eta' \, dp = \eta' \int dT, \quad (18.98)$$

where we can take η' out of the integral *because it is constant if the atmospheric has a saturated adiabatic profile*. Integrating gives

$$\phi(x, y, p, t) - \bar{\phi}(x, y, t) = (\bar{T}(x, y, t) - T(x, y, p, t)) \eta(x, y, t), \quad (18.99)$$

where we have written the constant of integration such that $\bar{\phi}$ and $\bar{T}$ are the vertical means, in pressure coordinates, of ϕ and T . Following conventional usage, we call the vertically integrated components 'barotropic' and the deviations 'baroclinic'.

The temperature difference $(\bar{T}(x, y, t) - T(x, y, p, t))$ in (18.99) is determined by the saturated adiabatic lapse rate and in the horizontal this varies quite weakly with the temperature variations found in the tropics. Thus, the horizontal variation of the terms on the right-hand side are dominated by variations in entropy and to a good approximation we can write

$$\nabla \phi(x, y, p, t) = \nabla \bar{\phi}(x, y, t) + M(p) \nabla \eta(x, y, t), \quad (18.100)$$

where $M(p) = \bar{T}(x, y, t) - T(x, y, p, t)$ and which *we take to be a function of pressure alone*. The profile of $M(p)$ has a single node in the vertical, and this constrains the horizontal velocity to have a similar variation in the vertical, and the vertical integral of $M(p)$ vanishes.

Velocity decomposition

First consider the vertical velocity. In pressure coordinates the mass continuity equation is

$$\frac{\partial \omega}{\partial p} = -\nabla_p \cdot \mathbf{u} \quad (18.101)$$

where $\bar{\mathbf{u}} = (u, v)$. Let $\mathbf{u} = \bar{\mathbf{u}} + \mathbf{u}^*$, where $\bar{\mathbf{u}}$ and $\mathbf{u}^*$ are the barotropic and baroclinic components of $\mathbf{u}$, and ω is the vertical (pressure) velocity. We suppose that the fluid is confined between the ground at $p = p_g$ and a rigid tropopause at p_t , with $\omega = 0$ at both. The baroclinic components of the velocity will, as a consequence of their definition, vanish when integrated over the troposphere. An integration of (18.101) over the troposphere implies the barotropic flow is divergence-free:

$$\nabla_p \cdot \bar{\mathbf{u}} = 0. \quad (18.102)$$

The vertical velocity is thus related to the baroclinic flow by,

$$\frac{\partial \omega}{\partial p} = -\nabla_p \cdot \mathbf{u}^*. \quad (18.103)$$

The horizontal flow itself is given using the momentum equation which, using (18.100), we write as

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{u} + \mathbf{f} \times \mathbf{u} = -\nabla \bar{\phi} + M(p) \nabla_p \eta. \quad (18.104)$$

where $\mathbf{v}$ is the three-dimensional velocity and we omit forcing and viscous terms, and ∇ without a subscript denotes horizontal derivative. The above equation suggests that it will be a good approximation to suppose that the horizontal velocity has the same vertical structure as $M(p)$ and we let

$$\mathbf{u}(x, y, p, t) = \mathbf{u}_0(x, y, t) + m(p)\mathbf{u}_1(x, y, t), \quad (18.105)$$

where $\mathbf{u}_0 (= \bar{\mathbf{u}})$ and $\mathbf{u}_1$ are the ‘barotropic’ and ‘first baroclinic’ components of the flow, and $m(p) = M(p)/N$ where N is a normalizing coefficient, say $[\int M^2 dp / (p_t - p_g)]^{1/2}$, so that $m(p)$ is a dimensionless basis function. The reader will appreciate the similarity of this approach to that of section 3.4, now with the quasi-equilibrium constraints determining the form of the vertical eigenfunctions for us.

Momentum equations

Neglecting the advective term the horizontal momentum equation decomposes exactly into separate barotropic and baroclinic equations which, omitting frictional terms, are

$$\frac{\partial \mathbf{u}_0}{\partial t} + \mathbf{f} \times \mathbf{u}_0 = -\nabla \phi_0, \quad \frac{\partial \mathbf{u}_1}{\partial t} + \mathbf{f} \times \mathbf{u}_1 = -\nabla \phi_1, \quad (18.106a,b)$$

where $\phi_0 = \bar{\phi}$ and $\phi_1 = N\eta$. These have the same form as the linearized shallow water equations. We may obtain a diagnostic equation for the barotropic pressure by taking the divergence of (18.106a), whence the time derivatives disappear because $\nabla \cdot \mathbf{u}_0 = 0$. Alternatively, take the curl of (18.106a) to give the linear barotropic vorticity equation,

$$\frac{\partial \zeta_0}{\partial t} + \beta v_0 = 0. \quad (18.107)$$

This equation is closed because, since the flow is divergence-free, there exists a streamfunction ψ such that $(u_0, v_0) = (-\partial\psi/\partial y, \partial\psi/\partial x)$ and $\zeta_0 = \nabla^2\psi$.

If we add nonlinearity back in then the baroclinic and barotropic modes interact. Thus, the terms $\mathbf{u}_1 \cdot \nabla \mathbf{u}_1$ and $\mathbf{u}_0 \cdot \nabla \mathbf{u}_0$ both project onto the barotropic flow, and the terms $\mathbf{u}_1 \cdot \nabla \mathbf{u}_0$ and $\mathbf{u}_0 \cdot \nabla \mathbf{u}_1$ project onto the baroclinic flow, in a way that is analogous to that occurring in two-layer quasi-geostrophic flow (section 12.2.2). A baroclinic-baroclinic interaction also arises because of vertical advection and we obtain

$$\frac{\partial \mathbf{u}_0}{\partial t} + \mathbf{u}_0 \cdot \nabla \mathbf{u}_0 + D_0(\mathbf{u}_1, \mathbf{u}_1) + \mathbf{f} \times \mathbf{u}_0 = -\nabla \phi_0, \quad (18.108a)$$

$$\frac{\partial \mathbf{u}_1}{\partial t} + \mathbf{u}_1 \cdot \nabla \mathbf{u}_0 + \mathbf{u}_0 \cdot \nabla \mathbf{u}_1 + D_1(\mathbf{u}_1, \mathbf{u}_1) + \mathbf{f} \times \mathbf{u}_1 = -\nabla \phi_1, \quad (18.108b)$$

where D_0 and D_1 are nonlinear (largely advective) operators whose exact form is not of concern here. The barotropic flow is divergence free and this determines the pressure ϕ_0 , but ϕ_1 is as yet undetermined.

Thermodynamics

To close the equation for baroclinic flow we use an equation for the entropy or for the equivalent potential temperature in conjunction with (18.100). Since entropy is assumed invariant with height there is no vertical dependence and the equation might be written in the general form,

$$\frac{\partial \eta}{\partial t} + \mathbf{u} \cdot \nabla \eta = S, \quad (18.109)$$

where S are various source and sink terms of both heat and moisture. This term hides a multitude of sins, for it potentially includes radiative, condensational, evaporative and turbulent flux terms

with complex flow-dependent specifications, possibly needing separate equations for dry air and moisture. However, since entropy is (it is assumed) constant with height, we need a thermodynamic equation only at one level, which we might take to be near the surface, or we may use a vertical average.

In summary, the constraints on the vertical temperature structure imposed by convection lead to a relatively simple vertical structure of all the dynamical fields, and as a consequence a heat source will primarily force the first baroclinic mode. In the linear approximation the baroclinic flow is uncoupled from the barotropic flow. The procedure provides some justification for using shallow water-like equations in tropical dynamics, as in chapter 8, with the velocity fields to be interpreted as being those of the first baroclinic mode, an approach that is especially useful if the dynamics are predominantly linear. More generally, it is perhaps useful to think of this type of quasi-equilibrium model in a similar light as quasi-geostrophy: neither are quantitatively accurate for flow evolution (we would use neither model for an accurate weather forecast) but they may be able to provide insight where the full equations are too complex.

The equations derived above are still unbalanced, and are simpler than the primitive equations only in their constrained vertical structure. Let us try to go a little further and see if we can incorporate any balances in the horizontal that might simplify matters.

18.8 SCALING AND BALANCED DYNAMICS FOR LARGE-SCALE FLOW IN THE TROPICS

In earlier chapters we looked at flow with small Rossby number and, by performing a scale analysis, determined what the dominant balance of terms was in the equations of motion. This allowed us to derive the quasi-geostrophic equations, which are the basis for much of the theory of mid-latitude motion. Is such a program possible for the tropics? The answer is, ‘well, in part’. It is possible to some balanced sets of equations for the tropics, and that will be the main topic of this section. However, these reduced sets of equations have not proven nearly as useful for the tropics as quasi-geostrophy has been for the mid-latitudes, in part because it is harder to make relevant equations that are simple, or simple equations that are relevant. In any case, let us begin with the stratified primitive equations, without explicitly invoking a constrained vertical structure.

18.8.1 Balanced, adiabatic flow

We will present a scaling for tropical flow side-by-side with the corresponding scaling for mid-latitude flow.¹⁴ We begin with the hydrostatic primitive equations for adiabatic, frictionless flow, which, reprising (5.15b), may be written as,

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \phi, \quad \frac{\partial \phi}{\partial z} = b, \quad (18.110a,b)$$

$$\frac{Db}{Dt} + N^2 w = 0, \quad \nabla \cdot (\bar{\rho} \mathbf{v}) = 0. \quad (18.110c,d)$$

These are nominally the anelastic equations in height co-ordinates in our standard notion, but an entirely equivalent derivation could use pressure co-ordinates, as these have the same form. The reader will recall that $b = g \delta\theta/\theta_0$ is the buoyancy and $N^2 = d\tilde{b}/dz$ where $\tilde{b}(z)$ is a reference stratification. We will suppose that the basic variables scale according to

$$(x, y) \sim L, \quad z \sim H, \quad (u, v) \sim U, \quad w \sim W, \quad t \sim \frac{L}{U}, \quad \phi \sim \Phi, \quad b \sim B, \quad f \sim f_0. \quad (18.111)$$

The quantity B is representative of horizontal variations in buoyancy. Vertical variations scale differently, hence their separate representation in (18.110c). By choosing the time t to scale advectively we are implicitly eliminating gravity waves. The nondimensional numbers that will arise are

the Rossby, Burger and Richardson numbers,

$$Ro = \frac{U}{f_0 L}, \quad Bu = \left(\frac{L_d}{L} \right)^2 = \left(\frac{NH}{f_0 L} \right)^2, \quad Ri = \left(\frac{NH}{U} \right)^2, \quad (18.112)$$

and evidently

$$Bu = Ri \times Ro^2. \quad (18.113)$$

The Rossby number Ro is generally small in mid-latitudes for large-scale flow, but in the tropics it is $\mathcal{O}(1)$ or larger. The Richardson number (which is the inverse square of a Froude number) is usually large in both midlatitudes and tropics, *except* in regions of active convection where N is very small. If we take $N = 10^{-2} \text{ s}^{-1}$, $H = 10^4 \text{ m}$ and $U = 10 \text{ m s}^{-1}$ then $Ri = 100$. In fact, for large-scale flow the Richardson number is usually sufficiently large that $1/(Ro \ Ri)$ is small in both mid-latitudes and tropics.

The difference between the tropics and mid-latitudes is apparent from the dominant balance in the momentum equation. At small Rossby number we have the familiar geostrophic balance and with hydrostatic balance we obtain the scaling:

$$\mathbf{f} \times \mathbf{u} \approx -\nabla_z \phi, \quad \frac{\partial \phi}{\partial z} = b, \quad \implies \quad \Phi = f_0 UL, \quad B = \frac{f_0 UL}{H}. \quad (18.114)$$

In the tropics the advective term, or the advectively-scaled time derivative, balances the pressure gradient meaning that $D\mathbf{u}/Dt \sim \nabla_z \phi$ and, since we still have $\partial \phi / \partial z = b$ we find

$$\Phi = U^2, \quad B = \frac{U^2}{H}. \quad (18.115)$$

If U is of similar magnitude in the tropics and mid-latitudes (and in the absence of a dynamical analysis this is an assumption), then *variations of pressure and temperature are smaller in the tropics than in mid-latitudes*. This is an important and not-quite obvious result and it is the foundation of the *weak temperature gradient approximation*, discussed further in the next section. If we were to carry through the derivation in pressure co-ordinates (see the shaded box on page 81) with geopotential and temperature as the variables we would find

$$\text{Mid-latitudes: } \Phi = f_0 UL, \quad T_s = \frac{f_0 UL}{R}, \quad (18.116a)$$

$$\text{Tropics: } \Phi = U^2, \quad T_s = \frac{U^2}{R}, \quad (18.116b)$$

where Φ is now the scaling for geopotential, T_s is the scaling for temperature and R is the ideal gas constant. For $U = 10 \text{ m s}^{-1}$, $L = 10^6 \text{ m}$ and, for mid-latitudes only, $f_0 = 10^{-4} \text{ s}^{-1}$, we obtain

$$\text{Mid-latitudes: } \Phi \sim 1000 \text{ m}^2 \text{ s}^{-2}, \quad T_s \sim 3 \text{ K} \quad (18.117a)$$

$$\text{Tropics: } \Phi \sim 100 \text{ m}^2 \text{ s}^{-2}, \quad T_s \sim 0.3 \text{ K}. \quad (18.117b)$$

The implications of these results are illustrated in Fig. 18.22, which shows a snapshot of observed contours of geopotential height, temperature and zonal wind; the large-scale variability of geopotential and temperature is evidently much smaller in the tropics than mid-latitudes.

Vertical velocity

The obvious scaling for the vertical velocity is that suggested by the mass continuity equation, namely $W = UH/L$. However, as we know from chapter 5, the vertical velocity may be much less

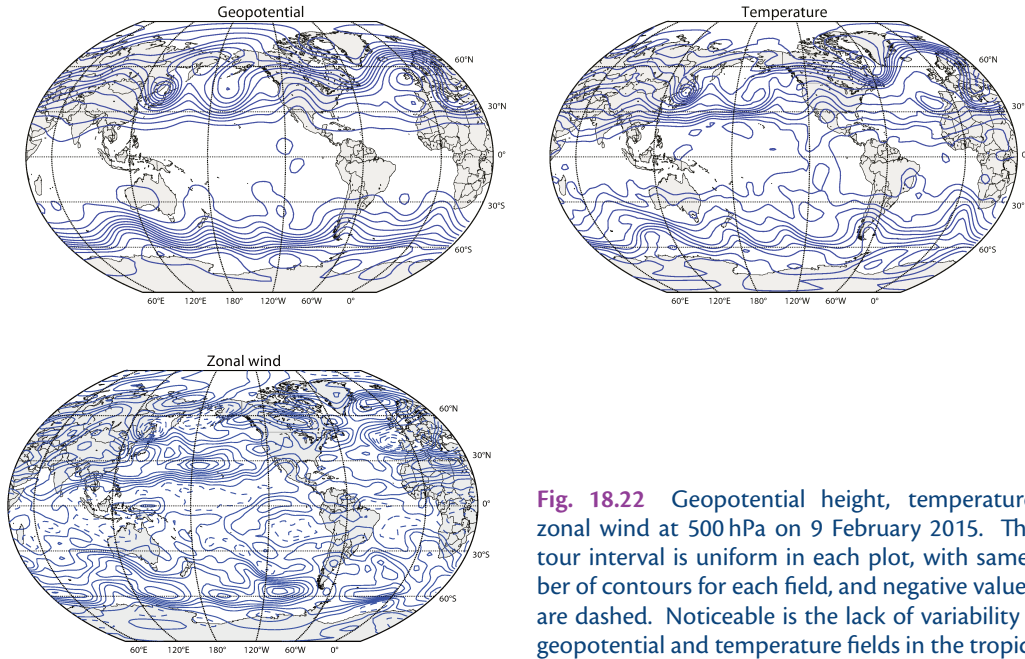


Fig. 18.22 Geopotential height, temperature and zonal wind at 500 hPa on 9 February 2015. The contour interval is uniform in each plot, with same number of contours for each field, and negative values for u are dashed. Noticeable is the lack of variability of the geopotential and temperature fields in the tropics.

than this estimate in a stratified, rotating fluid, and here we start with the thermodynamic equation for adiabatic flow,

$$\mathbf{u} \cdot \nabla b + wN^2 = 0, \quad \implies \quad W = \frac{UB}{LN^2} \quad (18.118)$$

Using (18.114) and (18.115) gives

$$\text{Mid-latitudes: } W = \frac{f_0 U^2}{HN^2} = \frac{f_0 L}{U} \frac{U^2}{N^2 H^2} \frac{UH}{L} = (Ro \ Ri)^{-1} \left(\frac{UH}{L} \right), \quad (18.119a)$$

$$\text{Tropics: } W = \frac{U^3}{LHN^2} = \frac{U^2}{N^2 H^2} \frac{UH}{L} = Ri^{-1} \left(\frac{UH}{L} \right), \quad (18.119b)$$

where $Ri \equiv N^2 H^2 / U^2$ is the Richardson number, with typical values of $\mathcal{O}(10)$ – $\mathcal{O}(100)$ for large-scale flow. Thus, again perhaps non-intuitively, the vertical velocity is, for adiabatic flow, *smaller* in the tropics than in mid-latitudes, by order of a mid-latitude Rossby number. In midlatitudes the scaling for W is more commonly written as

$$W = \frac{f_0 U^2}{HN^2} = Ro \frac{L^2}{L_d^2} \frac{UH}{L} = \frac{Ro}{Bu} \frac{UH}{L}, \quad (18.120)$$

so that for scales comparable to the mid-latitude deformation radius (i.e., with $Bu \sim 1$) the vertical velocity is order Rossby-number smaller than mass-continuity scaling suggests.

Vorticity

Cross-differentiating the horizontal momentum equation gives, as in (4.66) but without a baroclinic term, the vertical component of the vorticity equation with associated scalings,

$$\frac{D}{Dt}(\zeta + f) = -(\zeta + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left(\frac{\partial u}{\partial z} \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \frac{\partial w}{\partial x} \right), \quad (18.121a)$$

$$\text{Tropics:} \quad \left(\frac{U}{L} \right)^2 \sim \left(\frac{U}{L} + \frac{1}{Ro} \frac{U}{L} \right) \left(\frac{1}{Ri} \frac{U}{L} \right) \left(\frac{U}{H} \right) \left(\frac{1}{Ri} \frac{UH}{L^2} \right), \quad (18.121b)$$

$$\text{Mid-latitudes:} \quad \left(\frac{U}{L} \right)^2 \sim \left(\frac{U}{L} + \frac{1}{Ro} \frac{U}{L} \right) \left(\frac{Ro}{Bu} \frac{U}{L} \right) \left(\frac{U}{H} \right) \left(\frac{Ro}{Bu} \frac{UH}{L^2} \right). \quad (18.121c)$$

In the tropical case with $Ro = \mathcal{O}(1)$ or larger *all* the terms on the right-hand side are much smaller than the terms on the left-hand side, whereas in the midlatitude case vortex stretching by the Coriolis term, $f(\partial u/\partial x + \partial v/\partial y)$, is the same order (because f is big and the divergence is small). Thus, in the tropical case the vorticity equation simplifies severely and at lowest order becomes the two-dimensional vorticity equation,

$$\frac{D}{Dt}(\zeta + f) = 0. \quad (18.122)$$

The large-scale velocity is, at this order, purely rotational and is given by a streamfunction ψ such that

$$\nabla^2 \psi = \zeta, \quad (u, v) = \left(-\frac{\partial \psi}{\partial y}, \frac{\partial \psi}{\partial x} \right). \quad (18.123)$$

This equation nominally holds independently at each vertical level, although the assumptions that give rise to it assume that the depth scale is large. The pressure is not needed to step forward (18.123) but it may be obtained diagnostically. To do so we go back to the horizontal velocity equation written in the vector-invariant form,

$$\frac{\partial \mathbf{u}}{\partial t} - \mathbf{u} \times (\zeta + f) = -\nabla \left(\phi + \frac{1}{2} \mathbf{u}^2 \right), \quad (18.124)$$

neglecting the vertical and divergent and velocities. Taking the divergence gives the so-called non-linear balance or gradient-wind equation,

$$\nabla^2 \phi = \nabla \cdot \left[(f + \zeta) \nabla \psi - \frac{1}{2} \nabla (\nabla \psi)^2 \right], \quad (18.125)$$

which is similar to (2.238) and expresses a balance between the pressure gradient, Coriolis and centrifugal forces.

18.8.2 A few remarks

The above derivations and results require some comment.

- The relative weakness of large-scale horizontal gradients of pressure (or geopotential) and temperature in the tropics, for a given velocity field, is a robust result of the scaling analysis and borne out in observations.
- The smallness of the vertical velocity requires that the Richardson number, $N^2 H^2 / U^2$ be large; that is, stratification is strong. This is only true in regions that are not actively convecting; in convective regions N may be small.

- Relatedly, the scaling does not take into account diabatic sources, which may be expected to be particularly important in tropical regions.
- Equally tellingly, (18.123) tells us nothing about the vertical structure and so, unlike quasi-geostrophy in mid-latitudes, is not a sufficiently complete equation to be a useful prognostic, or even diagnostic, equation for tropical motion.

Let us now look at how diabatic effects might affect large-scale motion.

18.9† SCALING AND BALANCE FOR LARGE-SCALE FLOW WITH DIABATIC SOURCES

In the tropics we might expect that heat sources, for example condensational heating, would be important to the extent that they should be explicitly included in any development of reduced equations. Let us see if and how this is possible, using the shallow water equations for illustration. Using the standard nonlinear shallow water equations for the tropical atmosphere is rather ad hoc (and not justified by quasi-equilibrium except in the linear case) but our intention here is primarily illustrative. The underlying physical assumption is that in an air column adiabatic cooling associated with vertical motion balances diabatic heating, which in the shallow water equations becomes a balance between the heating and the divergence.

18.9.1 Diabatic balanced shallow water equations

On an f -plane and in conventional notation the equations may be written in vorticity-divergence form as

$$\frac{\partial h}{\partial t} + \nabla \cdot (\mathbf{u}h) = Q, \quad (18.126a)$$

$$\frac{\partial \zeta}{\partial t} + \nabla \cdot [\mathbf{u}(\zeta + f_0)] = -r\zeta, \quad (18.126b)$$

$$\frac{\partial \delta}{\partial t} + \nabla^2 \left(\frac{1}{2} \mathbf{u}^2 + gh \right) - \mathbf{k} \cdot \nabla \times [\mathbf{u}(\zeta + f_0)] = -r\delta. \quad (18.126c)$$

where $\delta = \partial u / \partial x + \partial v / \partial y$ is the divergence, Q is the mass or heating source, r is a frictional coefficient and other notation is standard. The height field is a proxy for both pressure and temperature and we will assume that horizontal gradients are weak, by which we mean that the dominant balance in (18.126a) is characterized by the scaling

$$H\Delta = Q_0 \quad (18.127)$$

where H is the mean thickness of the layer, Δ is a scaling for the divergence and Q_0 is the magnitude of the heating. We then choose the velocity scale, U and the vorticity scale, Z , to be

$$U = \frac{Q_0 L}{H}, \quad Z = \Delta = \frac{Q_0}{H}. \quad (18.128)$$

Finally, the magnitude of horizontal deviations in the height field, $\mathcal{H}$, are determined from the divergence equation. Depending on whether rotation is or is not important we deduce

$$\mathcal{H} = \frac{f_0 U L}{g} = \frac{Q_0 f_0 L^2}{gH}, \quad \text{or} \quad \mathcal{H} = \frac{U^2}{g} = \frac{Q_0^2 L^2}{gH^2}. \quad (18.129a,b)$$

The height field may then be separated into a mean and deviation, $h = H + \eta$, where $\eta = \mathcal{H}\hat{\eta}$. These scalings involve the heating in an essential way and so are fundamentally different from the adiabatic scaling of the previous section.

Using the above scalings, with rotation, (18.126) may be written in nondimensional form as,

$$Bu^{-1} \left[\frac{1}{f_0 T} \frac{\partial \hat{\eta}}{\partial \hat{t}} + Ro \nabla \cdot (\hat{\mathbf{u}} \hat{\eta}) \right] + \hat{\delta} = \hat{Q}, \quad (18.130a)$$

$$\frac{1}{f_0 T} \frac{\partial \hat{\zeta}}{\partial \hat{t}} + \nabla \cdot [\mathbf{u}(\hat{\zeta} + \hat{f}_0)] = -\frac{r}{f_0} \hat{\zeta}, \quad (18.130b)$$

$$\frac{1}{f_0 T} \frac{\partial \hat{\delta}}{\partial \hat{t}} + \nabla^2 \left(\frac{1}{2} \hat{\mathbf{u}}^2 + \hat{\eta} \right) - \mathbf{k} \cdot \nabla \times [\hat{\mathbf{u}}(\hat{\zeta} + \hat{f}_0)] = -\frac{r}{\hat{f}_0} \hat{\delta}. \quad (18.130c)$$

where T is the scaling for time, $Bu = (L_d/L)^2$ with $L_d = \sqrt{gH}/f_0$, and $\hat{f}_0 = 1$. The Rossby number is given by $Ro = Q_0/f_0 H$ and is not necessarily small. In the nonrotating case a similar set of equations can be derived but with somewhat different coefficients.

Reduced equations

Let us suppose that the mass source determines the divergence in (18.130a). The condition for this is that

$$\max \left(\frac{1}{f_0 T}, Ro \right) Bu^{-1} \ll 1, \quad (18.131)$$

which means that the scale of motion cannot be too large and the time scale cannot be too short. If (18.131) is satisfied then the dimensional height equation, (18.126a), becomes

$$\nabla \cdot \mathbf{u} = \frac{Q}{H}. \quad (18.132a)$$

This value of the divergence is used in (18.126b) and (18.126c) which, retaining all terms since none are obviously small, become

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla (\zeta + f_0) + (\zeta + f)Q = -r\zeta, \quad (18.132b)$$

$$g\nabla^2 h = \mathbf{k} \cdot \nabla \times [\mathbf{u}(\zeta + f_0)] - \frac{\partial Q}{\partial t} - r\delta - \nabla^2 \frac{\mathbf{u}^2}{2}. \quad (18.132c)$$

The equation set (18.132) has but one prognostic equation, namely (18.132b), and so is truly balanced and may be thought of as a generalization of (18.122) and (18.125) to the case with nonzero heating. The divergence equation is a nonlinear balance equation, similar to (18.125), except now with a diabatic term on the right-hand side. The divergent flow itself is computed using the height equation, by an assumed balance between adiabatic cooling and diabatic heating. The relationship between velocity and geopotential (or pressure) is the same as in the adiabatic case, because this arises through the momentum equation. Thus, even in the presence of a heating, gradients of geopotential and temperature remain relatively weak, a result that ultimately arises from the smallness of the Coriolis parameter. The arguments that lead to (18.132), along with the scaling of section 18.8, provide us with the *weak temperature-gradient approximation*.¹⁵ The importance of the result lies in what it implies about the response of the atmosphere to a localized heating: without making any linear approximation, the equations provide a scaling for the response of the velocity, and suggest that the response may become spread out over a sufficient area to keep the temperature gradients small.

Gravity wave adjustment and weak temperature gradients

Although convection certainly constrains the lapse rate where it is occurring, it does not directly do so elsewhere. Nevertheless, the observed tropical atmosphere has an *average* lapse rate close

to the saturated adiabat. Why should this be so? The reason is that a process akin to geostrophic adjustment (section 3.9) brings the atmosphere close to a state with weak horizontal temperature gradients. Suppose that a particular column undergoes moist convection and, perhaps in a matter of hours, adjusts to a moist neutral profile. Away from the cloud the buoyancy profile will in general be different from that, and so there will be unbalanced horizontal pressure (and hence temperature) gradients. Gravity waves, initiated by the motion surrounding the convection ('compensating subsidence'), will spread from the cloud and will adjust the environmental buoyancy to the same profile as that of the convecting region. The timescale for the adjustment is determined by the time that gravity waves take to propagate horizontally between convecting regions. Internal gravity wave speeds are typically about 10 m s^{-1} or somewhat faster and so will spread a distance 50 km from a cloud in a couple of hours, and the adjustment time on this scale will be of this order. This buoyancy adjustment time is much less than the time it would take a passive tracer to homogenize between clouds by advective or mixing processes. The actual process of gravity wave initiation and subsequent adjustment is complex and beyond our scope, and the reader is referred to the literature.¹⁶

18.9.2 ♦ Weak temperature gradient and stratified flow

The weak temperature gradient approximation is formally independent of any quasi-equilibrium argument, so in this section we briefly discuss its application to the stratified, three-dimensional equation of motion. The thermodynamic equation in pressure coordinates (see equation (P.5) on page 81) may be written,

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T + \omega \frac{\partial s}{\partial p} = Q, \quad (18.133)$$

where $s = T + gz/c_p$ is the dry static energy divided by c_p and Q represents heating terms. In the weak temperature gradient approximation this equation becomes

$$\omega \frac{\partial s}{\partial p} = Q. \quad (18.134)$$

If the stratification s is known (e.g., moist neutral) the above equation becomes a diagnostic for the vertical velocity, namely $\omega(p) = Q/\partial_p s$. Even with this approximation the remaining stratified equations (e.g., momentum equation) are somewhat complex, with both nonlinearity and continuous vertical structure.

If we are willing in addition to make the quasi-equilibrium assumption then further simplifications are possible because all the fields are assumed to have a simple vertical structure and the flow may be described, as before, by a system similar to the shallow water equations. The vertical and horizontal velocities are related by the mass continuity equation, (18.101). The barotropic flow has zero horizontal divergence so that the vertical (pressure) velocity and the horizontal are related by

$$\omega(x, y, p, t) = - \int_{p_s}^p \nabla \cdot \mathbf{u} \, dp = -n(p) \nabla \cdot \mathbf{u}_1, \quad (18.135)$$

where $n(p) = \int m(p) \, dp$ and we take $\omega = 0$ at the bottom boundary. Thus, we may write $\omega = n(p)\omega_1(x, y, t)$ and $\omega_1 = -\nabla \cdot \mathbf{u}_1$. If we divide (18.134) and vertically integrate we obtain

$$\bar{s} \nabla \cdot \mathbf{u}_1 = \bar{Q}, \quad (18.136)$$

where $\bar{Q}$ is a vertically integrated heating term, weighted by $1/n(p)$. This equation is the direct analog of (18.132a) and it provides a predictive equation for the divergence of the baroclinic flow if the heating and stratification are known. Moisture may be added to the mix, in which case a moist static energy appears, and the effects of evaporation and condensation must be included in

Q. The precise form of (18.136) and other particulars of implementation will depend on the basis functions chosen for the vertical structure, but the form is generically $\nabla \cdot \mathbf{u}_1 = [Q]/[s]$, where $[Q]$ is a measure of heating and $[s]$ a measure of stratification.¹⁷ Equation (18.136) may be used to close the momentum equation (18.108b). If we take the curl of that equation we obtain a vorticity equation analogous to (18.132b), and if we take its divergence we obtain a diagnostic equation for $\nabla^2 \phi_1$ analogous to (18.132c). The combination of a weak temperature gradient approximation and shallow-water-like equations arising from a constrained vertical stratification may be as close as the tropical atmosphere allows us to get to tractable and understandable equations of motion.

18.10† CONVECTIVELY COUPLED GRAVITY WAVES AND THE MJO

We now look at some of the manifestations of the theoretical development of this chapter and chapter 8, and three phenomena suggest themselves, namely the Walker circulation, monsoons, and the Madden–Julian Oscillations (MJO). We leave monsoons for another day, in part because the subject is large and its theoretical development is a moving target. And we defer discussion of the Walker circulation — an atmospheric overturning circulation largely in the zonal (i.e., y – z) plane — to chapter 22 because the phenomenon is closely tied to the equatorial ocean. So let us briefly discuss the MJO, although we cannot give it a wholly crisp explanation because at the time of writing none exists.¹⁸

What is the MJO? It is a pattern of precipitation and winds that takes shape across the western tropical Indian Ocean and drifts eastward at about 4 – 8 m s^{-1} into the western Pacific before dying out over the cooler waters of the eastern tropical Pacific. It often recurs roughly every 30–60 days or so, although it does not oscillate like a conventional wave. Rather, it is more like a somewhat coherent, drifting pattern several thousand kilometres across, consisting of a wet, rainy region of ascending air flanked by dryer regions on either side.

18.10.1 The observations

The MJO has a characteristic spectral signature that can be obtained from satellite measurements of cloud brightness. Figure 18.23 shows the symmetric (Northern Hemisphere plus Southern Hemisphere) and antisymmetric power spectrum (that is, the intensity of the field, Fourier analysed in zonal wavenumber and frequency) of tropical satellite brightness temperature, after filtering away some background noisiness.¹⁹ Power spectra of other fields, including velocity fields from re-analysis, show similar features. The main shaded regions in the figure fall nicely on the theoretical dispersion relations for Rossby, Kelvin and mixed Rossby-gravity waves as derived in section 8.2. This is perhaps somewhat startling to see, but evidently equatorial waves do exist!

The theoretical curves match the observations best if the unadorned gravity wave speed, c , is in region of 10 to 20 m s^{-1} . This speed is given by $c = \sqrt{gH_e}$ where H_e is the equivalent depth (section 3.4.2), so theoretical values match the observations with $H_e \approx 10$ – 50 m , perhaps with the best match around 25 m, somewhat smaller than the first equivalent depth computed for the atmosphere in section 3.4.²¹ Part of the difference may come from the fact that the tropopause is not a rigid lid, and part from the fact that the presence of moisture may reduce the effective static stability of the atmosphere below that implied by the value of N^2 computed using dry potential temperature ($g/\theta \partial \theta / \partial z$). In a saturated atmosphere the (smaller) value of $g/\theta_e \partial \theta_e / \partial z$ may be more relevant for the generation and propagation of gravity waves. The Rossby wave speed is given by $c_R \approx -\beta k / [(2m+1)\beta/c + k^2]$, so it too depends, albeit more weakly, on c .

In addition to the relatively well understood gravity and Rossby waves, there appears to be a prominent spectral signature in Fig. 18.23 at a small positive zonal wavenumber (between 1 and 5 in the left-hand panel) and a timescale of about 40 days, and this is a signature of the MJO. If we look in physical space, we get the sense that the MJO resembles a drifting Matsuno–Gill pattern more than an oscillating wave — compare Fig. 18.24 with Fig. 8.11 on page 325. Consistently, the

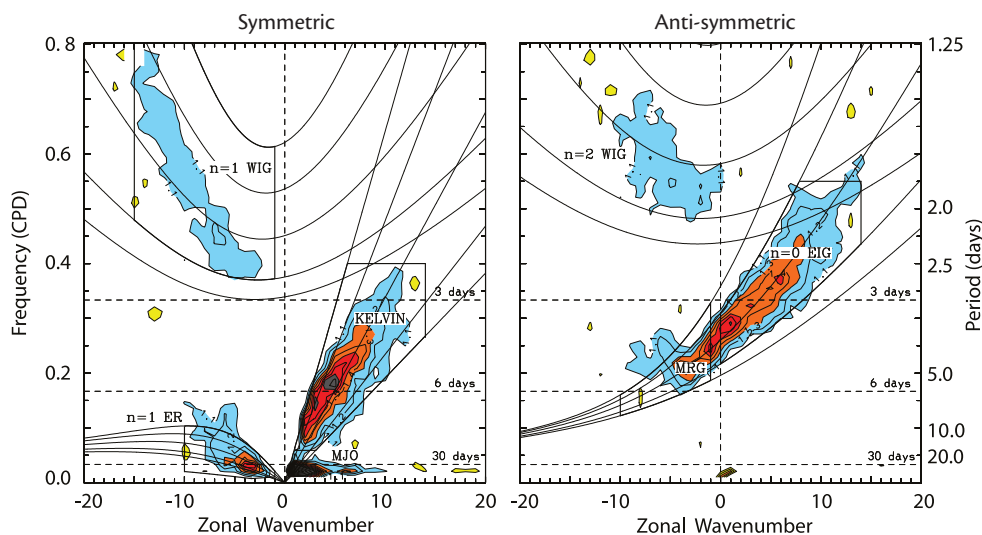


Fig. 18.23 A power spectrum of cloud brightness (colouring) from 15°S to 15°N , measured from satellite and Fourier-transformed into frequency-wavenumber space. Left panel is the symmetric spectrum (Northern plus Southern Hemispheres) and right panel the anti-symmetric spectrum. The solid lines are the corresponding dispersion relations with multiple equivalent depths ranging from 8 m to 90 m, with the best fit at about 25 m. Compare with Fig. 8.6 and Fig. 8.7.²⁰

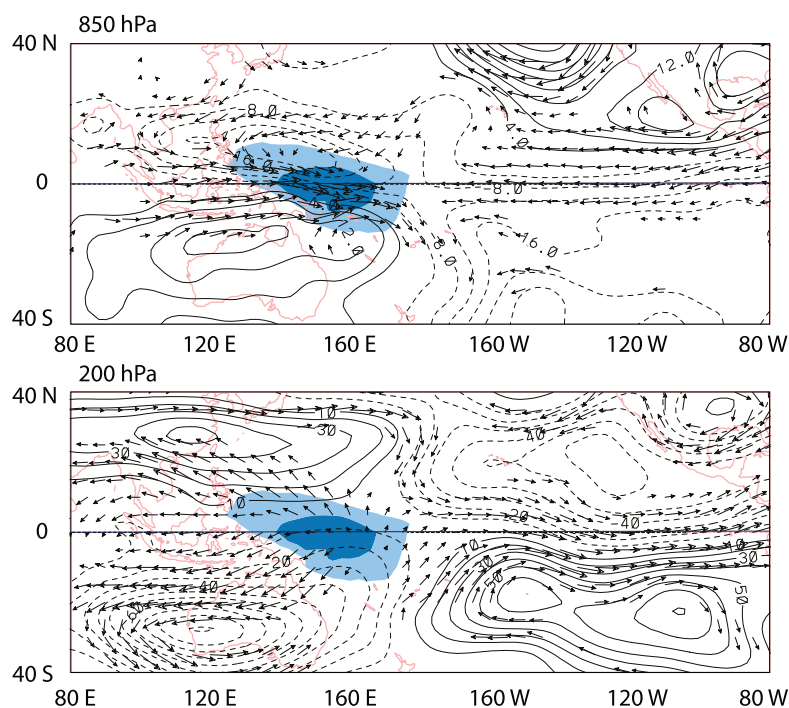


Fig. 18.24 Composite of an observed pattern during an MJO period. Shading indicates anomalously low outgoing IR radiation (less than 16 and 32 W m^{-2} below normal) and indicates the presence of high clouds and precipitation. The solid lines are streamfunctions and the arrows indicate velocities.²²

observed vertical structure largely has a first baroclinic mode structure, as expected if the lapse rate is constrained to be nearly constant by convection — note the oppositely directed velocity fields in Fig. 18.24, just as in Fig. 8.12. The shaded region corresponds to a region of heating in the Matsuno–Gill model and without too much imagination one may see a Kelvin wave response to its east, along the equator, and off-equatorial Rossby lobes to the west over Australia and South East Asia. A defining characteristic of the MJO is that this pattern drifts eastward at a speed of a few metres per second, much lower than the Kelvin wave speed at that equivalent depth.

18.10.2 † The mechanism

Precisely why this pattern should move eastward remains unclear, and there is little certainty as to the underlying mechanism except in so far as it likely has to do with the interaction of moist convection with the large-scale circulation — perhaps moist convection *producing* a large-scale circulation that then modulates the convection. Thus, a locally warm region near the equator (initiated by a sea-surface temperature anomaly, for example) will produce a pattern that evolves towards that of the Matsuno–Gill solution (Fig. 8.11), with a low-level convergence of moisture and condensation amplifying the initial pattern. Furthermore, the convergence will produce a dryer region on either side of the heat source (as observed). However, the timescales of condensation are fast and the process is unsteady, and it is unlikely that the condensation can produce a self-sustained stationary pattern — in the initial value problem it takes several days for a heating anomaly to settle into a steady Matsuno–Gill pattern in the tropical atmosphere.²³

Thus, instead of a truly steady pattern, the convection will initiate an eastward moving Kelvin wave, along with slower westward moving Rossby waves trapped nearer the source. Now, the initial moisture convergence arises because the Kelvin wave draws air in from the west, and the Rossby wave similarly draws air from the east. If the Kelvin waves moves east the region of low-level moisture convergence, and thus the region of condensational heating, will also move eastward, and the process begins again. However, the convergence region will move eastward much more slowly than the Kelvin wave, since it is the advective convergence that provides the heat source that is the source of both the Kelvin and Rossby waves. A Kelvin wave that breaks free of the convergence will eventually decay since it will not produce the moisture convergence to feed itself. Rather, the MJO is a self-sustained interaction between moisture convergence and the forced-dissipative Kelvin and Rossby waves it produces, with the requirement for moisture convergence greatly slowing the Kelvin wave. Radiative effects tied to the variable cloud field also almost certainly play a role in the way the pattern is damped.

Although wavelike, in this picture the MJO is not a conventional linear wave with a dispersion relation; rather it is related to the translation of a forced quasi-steady pattern with a longer timescale than either of those waves themselves, with the forcing *maintained by the pattern it itself produces*. Evidently, this will be a delicate balance for the interaction of moist convection with Kelvin and Rossby waves is a complex process, so it is not surprising that even the most sophisticated numerical models have trouble reproducing the phenomenon properly, even if the underlying mechanism can be described in relatively simple terms. Note also that the convection is not in quasi-equilibrium with a more slowly and independently evolving external environment — the convection helps produce its own external environment and the two evolve in synchrony.

18.A APPENDIX A: MOIST THERMODYNAMICS AND THE GIBBS FUNCTION

Many of the thermodynamic quantities of interest for moist air can be obtained from knowledge of the Gibbs function, in a manner analogous to that followed for dry air in the appendix on page 47. The benefits are that the approach is systematic, explicit formulae for thermodynamic variables can be given, and approximations can be made consistently as needed. Here we outline the methodology and provide some examples.²⁴ We use superscripts d , v and l to denote dry air, water

vapour, and liquid water, respectively, and subscripts of g denote derivatives with other variables held constant, except that a subscript 0 always denotes a constant, for any quantity. Quantities with multiple superscripts are mixtures of the quantities, but the superscripts are occasionally omitted. A subscript s denotes the saturated value.

Given (1.216), the specific Gibbs functions for dry and water vapour are, respectively,

$$g^d(p, T) = R^d T \ln(p^d/p_0) + c_p^d T [1 - \ln(T/T_0)], \quad (18.137a)$$

$$g^v(p, T) = R^v T \ln(p^v/p_0) + c_p^v T [1 - \ln(T/T_0)] - L_0^v \frac{T}{T_0} + L_0^v, \quad (18.137b)$$

where p^d and p^v are the partial pressures of dry air and water vapour. A constant, L_0^v , appears in the expression for the Gibbs function of water vapour because we will be concerned with water in its liquid phase. We take the analogous constant to be zero for dry air.

The two fluids are at the same temperature, T and the total pressure, p , is given by the sum of the partial pressures. Let q be the mass fraction of moist air in the mixture (i.e., the specific humidity), and let ϵ be the ratio of the molecular weights of water vapour and dry air, μ^v/μ^d . The two partial pressures are given by

$$p^v = \frac{qp}{q + \epsilon(1 - q)}, \quad p^d = \frac{\epsilon(1 - q)p}{q + \epsilon(1 - q)}, \quad (18.138)$$

satisfying $p = p^d + p^v$, and p^v is often denoted e in the literature. The Gibbs function for the mixture is

$$g^{dv}(p, T, q) = (1 - q)g^d + qg^v, \quad (18.139)$$

whence

$$\begin{aligned} g^{dv}(p, T, q) = & (qc_p^v + (1 - q)c_p^d) T [1 - \ln(T/T_0)] \\ & + T \left[qR^v \ln \left(\frac{qp/p_0}{q + \epsilon(1 - q)} \right) + (1 - q)R^d \ln \left(\frac{\epsilon(1 - q)p/p_0}{q + \epsilon(1 - q)} \right) \right] \\ & + qL_0^v \left(1 - \frac{T}{T_0} \right). \end{aligned} \quad (18.140)$$

This expression is symmetric in dry air and water vapour, and valid for any mass fraction of moisture in the air (given that $q \ln q \rightarrow 0$ as $q \rightarrow 0$). We will also need the Gibbs function for liquid water, which for our purposes may be written

$$g^l(p, T) = c^l T [1 - \ln(T/T_0)] - \eta_0^l T + \alpha^l p + g_0^l. \quad (18.141)$$

This expression is a simplified version of (1.145) with slightly different notation, with c^l being the heat capacity and α^l the inverse density.

Density and the thermal equation of state

The inverse density of the gas mixture is given by

$$\alpha^{dv} = \left(\frac{\partial g}{\partial p} \right)_{q, T} = \frac{T}{p} [qR^v + (1 - q)R^d] \quad (18.142)$$

or equivalently $p = \rho RT$ where $R = qR^v + (1 - q)R^d$ is the specific gas 'constant' for the mixture (which of course varies with q).

Entropy

The entropy is given by $\eta = -(\partial g / \partial T)_{p,q}$, giving

$$\eta^{dv} = (qc_p^v + (1-q)c_p^d) \ln T + \left[qR^v \ln \left(\frac{qp/p_0}{q + \epsilon(1-q)} \right) + (1-q)R^d \ln \left(\frac{\epsilon(1-q)p/p_0}{q + \epsilon(1-q)} \right) \right] + \frac{qL_0^v}{T_0}. \quad (18.143)$$

The entropy of the mixture may also be written as

$$\eta^{dv} = (1-q)\eta^d + q\eta^v, \quad (18.144)$$

where the specific entropies of dry air and water vapour are

$$\begin{aligned} \eta^d &= c_p^d \ln(T/T_0) - R^d \ln(p/p_0) - R^d \ln(1 - p^v/p), \\ \eta^v &= c_p^v \ln(T/T_0) - R^v \ln(p/p_0) - R^v \ln(1 - p^d/p) + L_0^v. \end{aligned} \quad (18.145)$$

The third terms on right-hand sides are called the entropies of mixing.

The entropy of liquid water is given by

$$\eta^l = \eta_0^l + c^l \ln(T/T_0). \quad (18.146)$$

The entropy of a mixture of dry air, liquid water and water vapour is given by

$$\eta^{dvl} = (1 - q^{vl})\eta^d + q^v\eta^v + q^l\eta^l \quad (18.147)$$

where q^v and q^l are the mass concentrations of vapour and liquid in the total mixture and $q^{vl} = q^v + q^l$.

Heat capacities

The heat capacity at constant pressure of moist air (i.e., dry air and water vapour, with no liquid content) is given by

$$c_p^{dv} \equiv T \left(\frac{\partial \eta}{\partial T} \right)_{p,q} = qc_p^v + (1-q)c_p^d. \quad (18.148)$$

That is, it is the mass weighted heat capacity of water vapour and dry air, as expected. The heat capacity at constant volume is given by

$$c_v^{dv} \equiv T \left(\frac{\partial \eta}{\partial T} \right)_{\alpha,q} = \frac{T(g_{pT}^2 - g_{pp}g_{TT})}{g_{pp}} = qc_v^v + (1-q)c_v^d. \quad (18.149)$$

where $c_v^v = c_p^v - R^v$ and $c_v^d = c_p^d - R^d$, and details are left to the reader. Alternatively, eliminate p in favour of α in (18.143) using (18.142) and then directly evaluate $T(\partial \eta / \partial T)_{\alpha,q}$.

Potential temperature

The potential temperature is the temperature that a parcel has if moved adiabatically and at constant composition to a reference pressure, and so satisfies

$$\eta(p_R, \theta, q) = \eta(p, T, q). \quad (18.150)$$

For moist air we use (18.143) for the entropy and (18.150) becomes

$$\begin{aligned} c_p \ln \theta - \left[qR^v \ln \left(\frac{qp_R/p_0}{q + \epsilon(1-q)} \right) + (1-q)R^d \ln \left(\frac{\epsilon(1-q)p_R/p_0}{q + \epsilon(1-q)} \right) \right] \\ = c_p \ln T - \left[qR^v \ln \left(\frac{qp/p_0}{q + \epsilon(1-q)} \right) + (1-q)R^d \ln \left(\frac{\epsilon(1-q)p/p_0}{q + \epsilon(1-q)} \right) \right]. \end{aligned} \quad (18.151)$$

With a couple of lines of algebra this expression simplifies to

$$c_p \ln \theta - [qR^v + (1 - q)R^d] \ln(p_R/p) = c_p \ln T \quad \text{or} \quad \theta = T \left(\frac{p_R}{p} \right)^{R/c_p}, \quad (18.152)$$

where $R = qR^v + (1 - q)R^d$ and c_p is given by (18.148).

Latent heat of evaporation and condensation

The latent heat of evaporation (or condensation) is the amount of energy that must be supplied to evaporate a unit mass of liquid to a vapour, or equivalently the amount of energy released when water vapour condenses. It is therefore equal to the difference between the specific enthalpy of water vapour and liquid water at a given temperature and also known as the enthalpy of vapourization. Using (18.141) the entropy and enthalpy of liquid water are given by

$$\eta^l = - \left(\frac{\partial g^l}{\partial T} \right)_p = \eta_0^l + c^l \ln(T/T_0), \quad h^l = g^l + T\eta = g_0^l + \alpha^l p + c^l T. \quad (18.153)$$

Using (18.137b) the enthalpy of water vapour is given by

$$h^v = g - T \left(\frac{\partial g^v}{\partial T} \right)_p = c_p^v T + L_0^v. \quad (18.154)$$

The enthalpy of vapourization, L , is therefore given by

$$L = h^v - h^l = L_0^v - (g_0^l + \alpha^l p) + (c_p^v - c^l)T. \quad (18.155)$$

The pressure term is negligibly small: $\alpha^l p \sim 100 \text{ J kg}^{-1}$ whereas, by experiment, $L = 2.501 \times 10^6 \text{ J kg}^{-1}$ at 0°C . Using this value for L_0^v (and setting $g_0^l = 0$), and with $c_p^v = 1859 \text{ J kg}^{-1} \text{ K}^{-1}$, $c^l = 4218 \text{ J kg}^{-1} \text{ K}^{-1}$, we obtain $L \approx (2.501 \times 10^6 - 2359T) \text{ J kg}^{-1}$, with T in Celsius. The temperature dependence of L arises because we have to expend the same amount of energy evaporating water at (say) 10°C and then raising the vapour temperature to 20°C as we do in raising the water temperature from 10° to 20° and then evaporating it.

Using the latent heat provides a convenient expression for the entropy of a saturated mixture of dry air, water vapour and liquid water. If the mixture is in equilibrium then $q^v = q_s^v$ where q_s^v is given by the Clausius–Clapeyron relation and is a function of p and T . Furthermore, the differences in entropy between vapour and liquid water are then related to the latent heat of vapourization by

$$T(\eta^v - \eta^l) = h^v - h^l = L, \quad (18.156)$$

and (18.147) becomes

$$\eta_s^{dvl} = (1 - q^{vl})\eta^d + q^{vl}\eta^l + \frac{Lq_s^v}{T}. \quad (18.157)$$

Equivalent potential temperature

There are various definitions of equivalent potential temperature, θ_{eq} , with different names and small quantitative differences.⁷ For qualitative uses the differences don't really matter as they are small, and for an exact calculation one rarely needs to use equivalent potential temperature anyway (entropy is better). Nevertheless, it is a commonly used thermodynamic variable. The definition used in the main text was that equivalent potential temperature, θ_{eq} , is the potential temperature a parcel reaches after it is lifted adiabatically and at constant composition to level at which it becomes saturated, and then all the water vapour is condensed and all of the latent heat released is used to

heat the parcel. To obtain an analytic expression for θ_{eq} so defined we assume the condensation occurs at constant temperature and so obtain (18.54), namely

$$\theta_{eq}^p = \theta \exp\left(\frac{Lq}{c_p T}\right) = T \left(\frac{p_R}{p}\right)^{R/c_p} \exp\left(\frac{Lq}{c_p T}\right). \quad (18.158)$$

Here we give it a superscript p and call it the pseudo-adiabatic equivalent potential temperature. It is related to entropy, but it is not a true measure of it. One problem is that the above process is not physically realizable. In order to condense all the water vapour the temperature must be taken to absolute zero (or in practice we must lift the parcel to a great height where the temperature is very low), but we cannot simply condense all the water at a constant temperature. Second, we have neglected the contribution of liquid water. To see this, begin with an expression for the entropy of a mixture of dry air, water vapour and liquid water in equilibrium, with an entropy given by (18.157). An equivalent potential temperature, θ_{eq} , may then be defined by

$$(1 - q^{vl})c_p^d \ln(\theta_{eq}/T_0) = (1 - q^{vl})\eta^d + q^{vl}\eta^l + \frac{Lq_s^v}{T}. \quad (18.159)$$

Using the expressions for η^d and η^l given in (18.145) the above expression becomes

$$(1 - q^{vl})c_p^d \ln\left(\frac{\theta_{eq}}{T_0}\right) = (1 - q^{vl})c_p^d \left[\ln\left(\frac{\theta^d}{T_0}\right) - \frac{R^d}{c_p^d} \ln\left(1 - \frac{p^v}{p}\right) \right] + q^{vl}c^l \ln\left(\frac{T}{T_0}\right) + \frac{Lq_s^v}{T}. \quad (18.160)$$

Solving this for θ_{eq} gives

$$\theta_{eq} = \theta^d \exp\left(\frac{Lw_s}{c_p^d T}\right) \left(\frac{T}{T_0}\right)^{w^{vl}c^l/c_p^d} \left(1 - \frac{p^v}{p}\right)^{-R^d/c_p^d}, \quad (18.161)$$

where $w = q/(1 - q^{vl})$ and $\theta^d = T(p_R/p)^{R^d/c_p^d}$. Equation (18.161) is a true measure of the entropy of a moist parcel, albeit a convoluted one. It differs slightly from (18.54) because θ_{eq}^p is not obtained by an adiabatic process, since the entropy of the liquid water is lost during condensation. If the water content is small ($q \ll 1$, $p^v/p \ll 1$) then (18.161) reduces to the more commonly used expression (18.158), with $q = q_s^v$.

Various other temperature-like quantities may be defined, notably the dew-point temperature, the wet-bulb temperature and wet-bulb potential temperature. Such quantities have intuitive and human appeal, and wet-bulb temperature is directly measurable. On the other hand, there is little objective reason to ever use a ‘potential’ temperature in a quantitative calculation — entropy itself is nearly always a more straightforward alternative. This comment also applies to seawater.

18.B APPENDIX B: EQUATIONS OF RADIATIVE TRANSFER

Consider a beam of radiation propagating through a thin slab of gas. Some of the incoming radiation may be absorbed, some may be scattered, and the slab may emit radiation of its own. Scattering is the change in direction of the radiation, so that it may reduce or — if radiation from other directions is scattered into the beam — amplify the beam’s intensity. The difference between the incoming and outgoing radiation is

$$dI_\nu \equiv I_\nu^{\text{out}} - I_\nu^{\text{in}} = -d\tau_\nu I_\nu + dJ_\nu. \quad (18.162)$$

In this expression I_ν is the spectral radiance (power per unit area, per unit solid angle, per unit frequency interval) of the radiation, the term $d\tau_\nu I_\nu$ is the extinction (the absorption plus the radiation scattered away) and dJ_ν is the emission plus the scattering into the beam. The quantity $d\tau_\nu$ is

the nondimensional *optical depth*; it may be written as $d\tau_v = k_v \rho ds$ where ds is the slab thickness, ρ is its density and k_v is the extinction coefficient, a property of the gas in question. The minus sign on $d\tau_v$ is appropriate when τ increases in the direction of the beam, and all of the above quantities depend on the frequency, ν , of the radiation.

Suppose there is no scattering, which is a good approximation for infra-red radiation. The emission of radiation is, in thermal equilibrium, then given by the Planck function, B_v , multiplied by the optical depth. Equation (18.162) becomes

$$dI_v = -d\tau_v(I_v - B_v) \quad \text{or} \quad \frac{dI_v}{d\tau_v} = -(I_v - B_v). \quad (18.163)$$

This equation is the foundation of much of radiative transfer. If radiation is propagating in all directions we must integrate over solid angle to obtain the upward and downward spectral irradiances (power per unit area per unit frequency interval). This is a complicated procedure in general but, to a good approximation for infra-red radiation in Earth's atmosphere, the simple upshot is the multiplication of the optical depth by a geometric, order-one (for example, 5/3) factor of γ — because most of the radiation is passing slantwise through the medium — and the multiplication of B by π because of the integration over a hemisphere, giving the two-stream approximation.²⁵ Equation (18.163) becomes

$$\frac{dF_v}{d\tau_v^*} = -F_v - \pi B_v, \quad (18.164)$$

where F_v is the spectral irradiance along a vertical path along which τ increases and τ_v^* is the 'scaled' optical depth given by $\tau^* = \gamma\tau_v$. We will drop the asterisk on τ_v^* and we will absorb the factor π into the definition of B_v .

In Earth's atmosphere it is common to choose τ increasing downwards, from 0 at the top of the atmosphere, although no physical result depends on this choice. The downwards (D_v) and upwards (U_v) irradiances are then

$$\frac{dD_v}{d\tau_v} = B_v - D_v, \quad \frac{dU_v}{d\tau_v} = U_v - B_v. \quad (18.165a,b)$$

These are the *two-stream equations*, without scattering, commonly known as the *Schwarzschild equations*. The downward and upward fluxes are uncoupled, because of the absence of scattering.

If there is no dependence of the optical depth on frequency then the medium is said to be *grey* and we may integrate (18.165) over frequency to give

$$\frac{dD}{d\tau} = B - D, \quad \frac{dU}{d\tau} = U - B, \quad (18.166a,b)$$

where U and D are the upward and downward (total) irradiances, $B = \sigma T^4$ (all three with units of W m^{-2}), and $\sigma = 5.6704 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ is Stefan's constant. The grey assumption is not accurate for Earth's atmosphere. Nevertheless, for conceptual or approximate calculations it is often useful to suppose that the atmosphere is grey in the infra-red, in which case (18.166) applies to infra-red radiation with separate equations (that might include scattering or reflection, but that might also be grey) for solar radiation; this is the *semi-grey* approximation.

18.C APPENDIX C: ANALYTIC APPROXIMATION OF TROPOPAUSE HEIGHT

Here we provide an approximate, analytic, expression for the height of the tropopause, given the optical depth and lapse rate. The idea is to solve for a self-consistent radiative-convective state, with a specified lapse rate extending upward to a tropopause and then transitioning to a radiative-equilibrium state, with the tropopause height being determined by the requirement of overall radiative balance.²⁶

Instead of trying to use the formal solutions to the Schwarzschild equations, it is easier to approximately solve the radiative-transfer equations for the upward long wave irradiance, U_L , *ab initio*. We make one other approximation, that the value of B/U_L varies linearly from the tropopause (where its value is 0.5) to its value at the surface (where $B/U_L = 1$). Thus,

$$\frac{B}{U_L} = 1 - \frac{z}{2H_T}. \quad (18.167)$$

Numerical calculations suggest this is a decent approximation. Proceeding, we write (18.65b) as

$$\frac{d \log U_L}{d\tau} = 1 - \frac{B}{U_L} = \frac{z}{2H_T}. \quad (18.168)$$

Using $\tau(z) = \tau_s \exp(-z/H_a)$ we obtain

$$\frac{d \log U_L}{dz} = -\frac{z}{2H_T H_a} \tau_s \exp(-z/H_a). \quad (18.169)$$

We can integrate this expression by parts to obtain a value of the upwelling radiation at the tropopause $U_L(H_T)$, namely

$$\log \left(\frac{U_L(H_T)}{U_L(0)} \right) = -\frac{\tau_s}{2H_T} \int_0^{H_T} \exp(-z/H_a) dz \approx -\frac{\tau_s H_a}{2H_T}. \quad (18.170)$$

for $H_T \gg H_a$. This is an expression for the upwelling longwave radiation, $U_L(H_T)$, and the unknowns in the equation are $U_L(0)$, the upwelling radiation at the surface, and the tropopause height, H_T . the value of $U_L(0)$ is given by the surface temperature, which is a function of the tropopause temperature, T_T , the specified lapse rate, Γ , and the tropopause temperature, T_T . Inverting the argument, if we are given $U_L(H_T)$ and T_T then we can calculate H_T .

Let us assume that the stratosphere is optically thin, in which case $U_L(H_T)$ is the outgoing longwave radiation, equal to $2\sigma T_T^4$, and this known because it equals the incoming solar radiation. Thus, we can in principle now solve (18.170) for H_T . To do this note that the upwelling radiation at the surface is given by $U_L(0) = \sigma T_g^4 = \sigma T_s^4$ where $T_s = T_T + \Gamma H_T$. The left-hand side of (18.170) then becomes

$$\log \left(\frac{2\sigma T_T^4}{\sigma T_s^4} \right) = \log 2 + 4 \log \frac{T_T}{T_s} = \log 2 + 4 \log \left(\frac{T_T}{T_T + \Gamma H_T} \right) \approx \log 2 - \frac{4\Gamma H_T}{T_T}. \quad (18.171)$$

The rightmost terms in (18.171) and (18.170) are approximately equal so that

$$\log 2 - \frac{4\Gamma H_T}{T_T} = -\frac{\tau_s H_a}{2H_T} \quad \text{or} \quad 8\Gamma H_T^2 - C H_T T_T - \tau_s H_a T_T = 0. \quad (18.172)$$

where $C = 2 \log 2 \approx 1.39$. The solution of this equation is

$$H_T = \frac{1}{16\Gamma} \left(C T_T + \sqrt{C^2 T_T^2 + 32\Gamma \tau_s H_a T_T} \right). \quad (18.173)$$

The tropopause height given by this equation is fairly close to the actual solution of the radiative-convective equations, obtained by numerically integrating the Schwarzschild equations and iterating to obtain the correct tropopause height following the algorithm of section 18.6.2, as seen in Fig. 18.21. The analytic approximation may be further improved with a bit of effort, but even in the form above it captures the essential aspects of the true solution.

Notes

- 1 I am grateful to Adam Sobel for a number of conversations and notes that informed the sections on convection and quasi-equilibrium. Will Beeson and his colleagues in Chicago also provided a number of useful comments. Brian Mapes provided a detailed critique of this chapter and although I was unable to address all of his concerns his point of view was salutary.
- 2 I use a formula given by Bolton (1980), which is a variant on the original Magnus formula, *aka* the August–Roche–Magnus formula or the Tetens formula — see Lawrence (2005) for discussion.
- 3 Measurements come from the hybrid advanced microwave sounding unit, Atmospheric Infrared Sounder (AIRS), as in Sherwood *et al.* (2010).
- 4 To read more about models of this form see Pierrehumbert *et al.* (2007), with extensions and applications by O’Gorman *et al.* (2011), Sukhatme & Young (2011), Tsang & Vanneste (2015) and others.
- 5 The simulations here used a model developed by Dr. Yue-Kin Tsang, and I am grateful to him for discussions and help.
- 6 The interested reader might start with Sherwood *et al.* (2010) or Schneider *et al.* (2010) and go forward and back from there.
- 7 This definition of θ_{eq} is sometimes called the ‘pseudo-equivalent potential temperature’, because the condensation product, liquid water, is assumed to fall out of the air parcel and the process is ‘pseudo-adiabatic’, not adiabatic. Other names with slightly different definitions exist (Betts 1973, Emanuel 1994, Ambaum 2010).
- 8 The notion of conditional instability has been with us for many years — it appears in Haurwitz (1941) for example — and a particular, and influential, form of it was introduced Ooyama (1963) (see Ooyama 1982) and Charney & Eliassen (1964). They proposed models of a cooperative mechanism between the convection and larger scale flow, with the convection producing a convergence at low levels, leading to ascent, more latent heat release and convection and so on. This mechanism (and variations about it) became known as ‘Conditional Instability of the Second Kind’, or CISK, to distinguish it from more conventional conditional instability (‘of the first kind’) which does not involve such a feedback with the large scale. The CISK mechanism tends to produce vortical up-draughts and may be important for hurricane growth, but this remains a topic of some debate (e.g., Raymond 1995, Emanuel 1994, Smith 1997). Other theories of hurricanes tend to de-emphasize the CISK mechanism in favour of model involving a feedback between wind speed and *evaporation*, called wind-induced surface heat exchange or WISHE, with stronger winds giving more evaporation, leading to saturation and thence convection. The convection then mainly serves to establish a moist adiabatic lapse rate (as in the quasi-equilibrium ideas of section 18.4.3) which ties the boundary layer to a warm core extending upward (e.g., Craig & Gray 1996).
- 9 Convective adjustment is a great simplification over what actually occurs, but nevertheless it a useful concept and the basis of many early parameterization schemes for numerical models of the atmosphere. Convective adjustment was introduced into modelling by Manabe & Strickler (1964) and a theoretical and observational discussion of the general problem was given by Ludlam (1966). A popular variation (a relaxation rather than an adjustment) was proposed by Betts (1986) and Betts & Miller (1986). These days (c. 2017) GCMs rarely use simple convective adjustment or relaxation schemes, although the underlying ideas endure.
- 10 The nature of a ‘quasi-equilibrium’ between convection and large-scale forcing was made explicit by Betts (1973) and Arakawa & Schubert (1974), with precedents to be found in Scorer & Ludlam (1951) and Ludlam (1966). Many papers have since followed, with, for example, an extended discussion in Emanuel *et al.* (1994), a counterpoint, examples and further discussion in Mapes (1997, 1998, 2000), and application to the boundary layer in Raymond (1997). More references can be found in the reviews by Arakawa (2004) and Emanuel (2007), and debates continue about applicability and efficacy.
- 11 This behaviour is seen in many comprehensive General Circulation Models of the atmosphere, going back to Manabe & Wetherald (1980) and beyond. The explanation given here follows Vallis *et al.* (2015) but has earlier roots.

- 12 Numerical calculations of the radiative constraint with more realistic treatments of radiation were carried out by Thuburn & Craig (2000).
- 13 We roughly follow Emanuel (1987). Neelin & Zeng (2000) and Zeng *et al.* (2000) give details of how a functional reduced model, including diabatic and frictional terms, may be constructed. Lindzen & Nigam (1987) have another, albeit related, take on the problem.
- 14 The tropical scaling was presented by Charney (1963).
- 15 The derivation of the weak temperature-gradient approximation given here is more-or-less that of Sobel *et al.* (2001), and may be regarded as an extension of Charney's (1963) ideas to include diabatic effects. A number of authors previously used the approximation, implicitly or explicitly, in one form or another (e.g., Neelin 1988, Browning *et al.* 2000), and various extensions and rigour have been added by Majda & Klein (2003) and others.
- 16 See Bretherton & Smolarkiewicz (1989) and Mapes (1997) and go from there.
- 17 Details are described in Bretherton & Sobel (2002) and Neelin & Zeng (2000).
- 18 A review of convectively coupled equatorial waves is provided by Kiladis *et al.* (2009). The MJO was first described by Madden & Julian (1971, 1972), and a review may be found in Zhang (2005). Schubert & Masarik (2006) discuss aspects of a moving Matsuno–Gill pattern and its relation to the MJO, and Raymond & Fuchs (2009), Majda & Stechmann (2009) and Sobel & Maloney (2013), among others, offer theoretical models of the MJO.
- 19 Diagrams such as these are known as Wheeler–Kiladis diagrams, after Wheeler & Kiladis (1999). Filtering the noise is required to obtain clean plots and requires some attention.
- 20 Adapted from Kiladis *et al.* (2009).
- 21 Dias & Kiladis (2014) discuss reasons why the value is smaller than might be expected.
- 22 Adapted from Kiladis *et al.* (2005).
- 23 Heckley & Gill (1984).
- 24 Feistel *et al.* (2010) and Thuburn (2017) give more derivations. I am grateful to John Thuburn for a number of conversations on this matter.
- 25 There are various versions of the two-stream approximation; see Goody & Yung (1995) or Pierrehumbert (2010).
- 26 The contents of this appendix are the results of a collaboration with Prof. Pablo Zurita-Gotor.

Part IV

LARGE-SCALE OCEANIC CIRCULATION

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